Written proofs for seven native obligations

J05, J06, J07, J21, J10, J15, and J14

Updated consolidated document

1 November 2026

Exact scope, completed proofs, and source limits

This standalone LaTeX source contains the complete written arguments for the seven requested jobs, with their exact hypotheses, proofs, source ledgers, and exceptional cases. It supersedes the mathematical status of the separately preserved earlier snapshot. That snapshot has not been modified. No Lean code is included.

The principal new seam result is stronger than the requested parity: for every pair of seam sizes and every pair of retained states,

$$\texttt{SeamCorrectionPath}\ k\ p\ q = \varnothing \qquad (k \geq 3).$$

If both retained factors change, the path type is empty at every internal length, including zero. The proof uses signed rectangle domains, nonnegative factor projections, and a reduction to at most five movable points. The finite table after that proved reduction lists every possible transition and its blocking points. No finite diagnostic is extrapolated to arbitrary sizes.

The J21 chapter retains the completed explicit second representative, including the correction chain, every boundary coefficient, the exact anchor-class identification, and the second-vertex support bound. The J14 and J15 chapters now include the missing root-covering construction and the comparison with every positive full-twist word whose closure is isotopic to the curve link.

Two distinctions must remain explicit. First, the displayed M405 API omits the multiplicity/transversality hypothesis in the user's geometric statement and is false. The J14 chapter proves a counterexample to that exact API and proves the full word theorem under the user's stated transverse hypotheses. Second, the literal J10 selected-grid instantiation requires two field equations and the primitive move definitions whose source files are absent from the combined attachment. Its complete switch-and-move proof is supplied for the exact marking equations that are reproduced, and the missing native implications are stated individually. They are not silently assumed.

Job	**Statement established**	**Status or precise limit**
J05	Every fixed-second correction of internal length at least three is absent. The long count is the integer zero.	Complete. M602–M604 are proved, with the arbitrary-endpoint M603 also covered by the combined truncation theorem.
J06	The one- and two-transition paths are in explicit bijection with the remaining second-factor rectangles. All longer paths are absent. The count equality holds in natural numbers.	Complete. M605 is proved.
J07	Every correction path changing both factors is absent, at every length.	Complete. M606 is proved. The synthesis at the end of this chapter proves M601 and M607.

Job	Statement established	Status or precise limit
J21	An explicit correction produces a degree-zero cycle in the exact anchor class with the full second-vertex bound.	Complete. M901–M903 are proved.
J10	One adjacent X switch in the two specified letter columns, unchanged O markings, and an explicit finite move sequence to every admissible shorter-word grid.	The selected-grid incidence and order equations, and primitive native move recognition, remain source checks. Their defining files are absent.
J15	The exponent equals the discriminant order and polar intersection. The branch and reduced-word length formulas hold. The full-twist comparison also proves M407's broader positive-word quantifier.	Complete as a written word-length proof. The reduced native sphere form retains the audit's M403(b) geometric realization input.
J14	For every reduced germ of multiplicity S with transverse projection, the root braid has the required positive full-twist prefix and exactly the branch cycles. No normalization input is left unproved.	The uncorrected M405 API is false, with an explicit counterexample. The native sphere realization retains the stated M403(b) and M419(B) dependencies.

All Floer chains, auxiliary cyclic-covering modules, and matrix coefficients use $\mathbb{F}_2$. Integer rectangle multiplicities,+braid exponents, analytic orders, and intersection multiplicities retain their ordinary integer meanings. Each new lemma is labelled as new and proved in the text. Existing Lean declarations are cited by their native names. Separate job chapters repeat some elementary domain lemmas so that their individual documents remain usable. The proof dependency is acyclic: J14 uses only the independently proved preparation and local argument-principle lemmas of J15, whereas J15's reduced full-twist conclusion then uses J14.

Contents

Exact scope, completed proofs, and source limits **1**

1 J05: First-sector seam parity **6**
1.1 Exact scope and status . . . 6
1.2 Native coordinates and rectangle conventions . . . 6
1.3 Classification and alternation of flat steps . . . 7
1.4 Persistent obstructions to the exit . . . 8
1.5 Consequences and exceptional cases . . . 11
1.6 Projected integral domains of a seam correction . . . 11
1.7 A finite order certificate on the five-point seam grid . . . 15
1.8 Fixed-second paths have at most two internal transitions . . . 17
1.9 Completion of the acceptance statement . . . 19
1.10 Lemma ledger . . . 20
1.11 Source and verification record . . . 21

2 J06: Second-sector seam parity **22**
2.1 Exact target and status . . . 22
2.2 Native conventions and notation . . . 22
2.3 The remaining rectangles . . . 23
2.4 The explicit inverse construction . . . 24
2.5 Every short path is one of these paths . . . 25
2.6 Projected integral domains of a seam correction . . . 27
2.7 A finite order certificate on the five-point seam grid . . . 31
2.8 Uniform reduction of the second sector to five points . . . 33
2.9 Exceptional cases and explicit coordinate checks . . . 36
2.10 Lemma and declaration table . . . 37
2.11 Remaining gap . . . 38

3 J07: Mixed-sector seam parity **39**
3.1 Exact statement and conventions . . . 39
3.2 Projected integral domains of a seam correction . . . 40
3.3 Removing two annuli . . . 43
3.4 Small sizes, orientations, and dependency ledger . . . 44
3.5 Remaining gap . . . 45
3.6 The combined seam conclusion: M601 and M607 . . . 45

4 J21: An explicit second torus representative **47**
4.1 The exact statement and the native conventions . . . 47
4.2 A relative-permutation formula for the Maslov grading . . . 49
4.3 The native rectangle coordinates . . . 51
4.4 Cyclic relabeling and the exact insertion index . . . 51
4.5 Permutation types occurring in the correction . . . 53

4.6 The first boundary correction and four-cycle deletion . . . 55
4.7 An explicit coefficient for the three-cycle–transposition layer . . . 57
4.7.1 The intrinsic order of the five moved columns . . . 57
4.7.2 The sixteen half-turn chambers . . . 57
4.7.3 Inversion and self-inverse targets . . . 58
4.7.4 The complete two-O rectangle coefficient . . . 58
4.8 The four-cycle correction defined by adjacent pairs . . . 60
4.8.1 The explicit coefficient and chain . . . 60
4.8.2 The coefficient identities used in the rectangle proof . . . 62
4.9 The complete lower boundary of the four-cycle correction . . . 63
4.9.1 The coefficient at a prescribed unit gap . . . 63
4.9.2 One-O rectangles ending at a triangle . . . 64
4.9.3 Cancellation at targets with two transpositions . . . 66
4.10 The complete Maslov minus six boundary of the explicit $(3, 2)$ correction . . . 67
4.10.1 Two elementary primitives for the coefficient table . . . 68
4.10.2 Which rectangles can contribute . . . 69
4.10.3 Cancellation of every four-cycle–transposition target . . . 70
4.10.4 The star coefficient . . . 70
4.10.5 The two-triangle coefficient . . . 71
4.10.6 The resulting native Alexander bound . . . 73
4.11 Two diagonal identities and one five-cycle cancellation . . . 74
4.11.1 The two diagonal identities . . . 74
4.11.2 A useful expression for a nonconvex four-cycle . . . 75
4.11.3 The five-cycle of circular type 01243 . . . 75
4.12 The remaining five-cycle coefficients . . . 77
4.12.1 Inversion and the circular five-cycles . . . 77
4.12.2 An exact sum over the adjacent pairs . . . 77
4.12.3 The exact residual and its Alexander grading . . . 80
4.13 The native second representative and the three acceptance statements . . . 81
4.13.1 The correction relative to the native first cycle . . . 81
4.13.2 The native bifiltration and weighted level . . . 82
4.13.3 The exact integer inequality of M903 . . . 83
4.13.4 The smallest parameters . . . 84
4.14 Catalogue of the proof inputs . . . 85
4.14.1 Existing Lean declarations . . . 85
4.14.2 New lemmas and explicit constructions . . . 87

5 J10: One X switch for a deleted braid letter 89
5.1 Exact statement and source status . . . 89
5.2 The row-separated word grids . . . 89
5.3 Elementary move lemmas . . . 90
5.4 The single switch . . . 92
5.5 The explicit sequence of grid moves . . . 92
5.6 Smallest cases and the role of the hypothesis . . . 94
5.7 Exact missing native identifications . . . 94
5.7.1 The precise selected-grid implication that is absent . . . 94
5.7.2 The precise native move implications that are absent . . . 95
5.8 Lemma table and remaining verification . . . 96

6 J15: Braid length, discriminants, and polar intersection 97
6.1 Exact statements and scope . . . 97
6.2 Conventions, continuous arguments, and crossing signs . . . 98
6.3 The boundary terms of all unordered pairs cancel . . . 99
6.4 The local argument principle and distinguished preparation . . . 101
6.5 Proof of the discriminant and polar equalities . . . 102
6.6 The branch word and the exact resolution-level conclusion . . . 104
6.7 The full-twist word . . . 105
6.8 The broader link-isotopy quantifier in M407 . . . 105
6.9 Exceptional cases and convention checks . . . 110
6.10 Lemma and declaration table . . . 111
6.11 Precise limits of the conclusion . . . 112

7 J14: Multi-branch positivity and the full-twist prefix 113
7.1 Statements and the distinction between the two hypotheses . . . 113
7.2 The displayed API is false . . . 114
7.3 Positive rotation for a finite collection of analytic differences . . . 115
7.4 Several tangent points and strict-transform sheets . . . 117
7.5 The full turn is the native full-twist word . . . 118
7.6 Proof of the factorization and the branch cycles . . . 119
7.7 Application to every transverse reduced germ . . . 120
7.8 Exceptional cases and source ledger . . . 125

Chapter 1

J05: First-sector seam parity

1.1 Exact scope and status

All coefficients in this document are in $\mathbb{F}_2$. Every cardinality before reduction modulo two is an ordinary nonnegative integer. Let $n, m \in \mathbb{N}$, put $N = n + m + 1$, and use the native grid states, rectangles, retained indices, and matching sources in the supplied seam listings. Write

$$\mathrm{CP}_k(p, q) = \texttt{SeamCorrectionPath}\ k\ p\ q.$$

The integer k counts internal transitions. An entry rectangle and an exit rectangle are also part of each such path. They are not counted in k.

The acceptance statement of J05 requires that, for every $n, m \in \mathbb{N}$, every $x, y \in \mathrm{GridState}(n+1)$, and every $z \in \mathrm{GridState}(m+1)$,

$$\texttt{seamLongCorrectionCount}((x, z), (y, z)) = 0 \quad \text{in } \mathbb{F}_2. \tag{1.1.1}$$

This document proves (1.1.1) for all n, m. In fact, Theorem 1.8.4 proves that every fixed-second correction path with at least three internal transitions is absent. The proof uses a signed domain of index one, two nonnegative factor projections, and deletion of fixed state points. Every putative long path compresses to at most five points, where an explicit order certificate excludes it. The following flat-path theorem handles the thinner compressed sizes and also proves M603 directly.

Theorem 1.1.1 (New lemma: exclusion of completely flat correction paths)**.** *Let $n, m, k \in \mathbb{N}$ with $k \geq 2$, and let p, q be native retained indices. There is no element of $\mathrm{CP}_k(p, q)$ whose every matching source has exactly one northeast column.*

Corollary 1.1.2 (New lemma: M603)**.** *Let $n, m \in \mathbb{N}$ and let p, q be retained indices. No element of $\mathrm{CP}_3(p, q)$ has northeast count one at both interior matching sources. Consequently, for every $x, y \in \texttt{GridState}(n+1)$ and $z \in \texttt{GridState}(m+1)$, the set $\mathrm{CP}_3((x, z), (y, z))$ is empty.*

Corollary 1.1.3 (New lemma: J05 for a factor of size at most two)**.** *If $\min(n, m) \leq 1$, then* (1.1.1) *holds for these n, m. In fact, every correction path with at least two internal transitions is absent.*

Theorem 1.1.1 concerns arbitrary retained endpoints. The later domain argument handles all northeast counts for the fixed-second acceptance statement. The failing module `GridConnectedSumSeamFlatThreeStepBarrier.lean` is not used as an existing result.

1.2 Native coordinates and rectangle conventions

Identify $\mathrm{Fin}(N)$ with $\{0, \ldots, N-1\}$, preserving its order. A grid state is a permutation s from columns to rows. Its northeast and southwest columns are

$$\mathrm{NE}(s) = \{a < n : s(a) > n\}, \qquad \mathrm{SW}(s) = \{P \geq n : s(P) \leq n\}.$$

These are the native sets characterized by `mem_seamNortheastColumns` and `mem_seamSouthwestColumns`, as used in the supplied proofs. The native identity `seamSouthwestColumns_card` gives $\operatorname{card}\mathrm{SW}(s) = \operatorname{card}\mathrm{NE}(s) + 1$.

For a matching source with northeast count one, write its two southwest columns as $P < Q$. Write

$$s(P) = p, \qquad s(Q) = q, \qquad p < q \le n.$$

The last inequality is the native increasing-source condition. The matching partner swaps the rows at P, Q.

For distinct indices a, b, let $(a, b)_{\mathrm{cyc}}$ be the open cyclic interval. It is (a, b) if $a < b$ and $\{c : c > a \text{ or } c < b\}$ if $a > b$. An oriented rectangle with left and right columns l, r has bottom and top rows $s(l), s(r)$. Its emptiness means

$$\text{there is no column } c \in (l, r)_{\mathrm{cyc}} \text{ with } s(c) \in (s(l), s(r))_{\mathrm{cyc}}. \tag{1.2.1}$$

This is `isEmpty_iff_forall_notMem_cIoo`, or equivalently the supplied `flatEmpty_source`. Thus a row interval from a larger row to a smaller row wraps. This orientation is essential below.

Suppose a correction path is completely flat. By `flatEntry_data`, its first matching source has a column $A < n$ and row $h = s(A) > n$. Each flat step preserves this point by `flatApply_high`. The same A, h therefore serves at every matching source. Because the northeast count is one, this is its unique northeast point.

The native entry lemma `flatEntry_data` says that one of the following two rectangles is empty, expressed in the first matching source's coordinates:

$$E_L = (A, P) \times (q, h), \qquad E_R = (Q, A)_{\mathrm{cyc}} \times (h, p)_{\mathrm{cyc}}. \tag{1.2.2}$$

The native exit lemma `flatExit_data` says that one of the following is empty in the last matching source:

$$F_L = (A, P) \times (h, p)_{\mathrm{cyc}}, \qquad F_R = (Q, A)_{\mathrm{cyc}} \times (q, h). \tag{1.2.3}$$

The letters P, Q, p, q in (1.2.2) and (1.2.3) belong to their respective endpoint matching sources. Formula (1.2.2) already accounts for the inverse matching move. It is not the rectangle obtained by using the source rows $s(A), s(P)$ for the entry.

1.3 Classification and alternation of flat steps

The existing declarations `flatStep_cases` and `flatStep_apply` classify every flat internal transition. Table 1.1 writes their conclusions without Lean notation. Here C is the exchanged column, $c = s(C)$, and s' is the next matching source. Values at columns not displayed remain unchanged. In the last column, the ordered pair gives the new southwest columns.

Type	Required range	Rectangle sides	$(s'(P), s'(Q), s'(C))$	New columns
L_+	$C < n$, $q < c \le n$	(C, P)	(q, c, p)	(P, Q)
L_-	$C < n$, $c < p$	(Q, C)	(c, p, q)	(P, Q)
H_+	$C > Q$, $c > n$	(C, P)	(c, p, q)	(Q, C)
H_-	$n \le C < P$, $c > n$	(Q, C)	(q, c, p)	(C, P)

Table 1.1: The native four flat transitions.

Lemma 1.3.1 (New lemma: the northeast point separates the four types)**.** *In the notation above, an empty flat transition satisfies the following additional inequalities:*

$$L_+ :\ A < C, \qquad L_- :\ C < A, \qquad H_+ :\ h < c, \qquad H_- :\ c < h.$$

Proof. For L_+, the row interval is $(c,p)_{\mathrm{cyc}}$ and contains h because $h > n \geq c$. If $C < A$, the column A lies between C and P, contrary to (1.2.1). The equality $C = A$ contradicts $c \leq n < h$.

For L_-, the column interval is $(Q,C)_{\mathrm{cyc}}$ and the row interval $(q,c)_{\mathrm{cyc}}$ contains h. If $A < C$, the point at A is inside the rectangle. Equality is again excluded by its row. Thus $C < A$.

For H_+, $A < P < C$, so $A \in (C,P)_{\mathrm{cyc}}$. The row interval $(c,p)_{\mathrm{cyc}}$ contains h if $h > c$. Emptiness and injectivity therefore give $h < c$.

For H_-, $A < C < Q$, so $A \in (Q,C)_{\mathrm{cyc}}$. The row interval (q,c) contains h if $h < c$. Emptiness and injectivity give $c < h$. □

Lemma 1.3.2 (New lemma: successive flat types alternate)**.** *For two consecutive empty flat transitions, the ordered pair of types is one of*

$$(L_+, L_-), \quad (L_-, L_+), \quad (H_+, H_-), \quad (H_-, H_+).$$

Proof. Use C, c for the first exchange and D, d for the second exchange. In this proof d is the row of D immediately before the second transition. Whenever the proof tests D against the first rectangle, D differs from P, Q, C and its row has not changed. For equal low types, $D \neq C$ follows from the row range in Table 1.1. For equal high types it follows from the column range. For a low type followed by a high type, or the reverse, it follows from the quadrant restrictions and Lemma 1.3.1.

Suppose the first type is L_+. A second L_+ has $d > c$. The first rectangle excludes $D > C$, and equality is impossible, so $D < C$. The second rectangle then contains the point at C, whose row is $p < q$. A second H_+ has left endpoint $D > Q$ and right endpoint P. Its row interval ends at q, and it contains the point at C of row p. A second H_- uses a column $n \leq D < P$ of high row. That point already lies inside the first rectangle $(C,P) \times (c,p)_{\mathrm{cyc}}$. These contradictions leave only L_-.

Suppose the first type is L_-. A second L_- has $d < c$. The first rectangle excludes $D < C$, so $D > C$. The second rectangle contains the point at C, now of row $q > p$. A second H_+ uses a column $D > Q$ of high row. That point is inside the first rectangle $(Q,C)_{\mathrm{cyc}} \times (q,c)_{\mathrm{cyc}}$. A second H_- has sides (Q,D) in the unchanged southwest columns P, Q. Its row interval starts at p and ends at a row above n. The point at $C < D$ has row q, so it lies inside this rectangle. Only L_+ remains.

Suppose the first type is H_+. The new southwest columns are Q, C, with rows p, q, and the point at the old P has row $c > h$. A second H_+ uses $D > C$. The first rectangle forces $d < c$. The second rectangle has sides (D,Q) and row interval $(d,p)_{\mathrm{cyc}}$. It contains the point at the old P, whose row is $c > d$. A second L_+ has sides (D,Q) and starts at a row $d \leq n$. It again contains the point at the old P, whose row exceeds n. A second L_- uses a low column with row $d < p$. That point is inside the first rectangle $(C,P)_{\mathrm{cyc}} \times (c,p)_{\mathrm{cyc}}$. Only H_- remains.

Suppose the first type is H_-. The new southwest columns are C, P, with rows p, q, and the point at the old Q has row $n < c < h$. A second H_- uses $D < C$. The first rectangle forces $d > c$. The second rectangle has sides (P,D) and row interval (q,d), so it contains the point at the old Q of row c. A second L_+ uses a low column of row $d > q$. That point is inside the first rectangle $(Q,C)_{\mathrm{cyc}} \times (q,c)$. A second L_- has sides (P,D) and wraps vertically from q to a row below p. It contains the point at the old Q, whose row exceeds n. Only H_+ remains. This proves all four assertions. □

1.4 Persistent obstructions to the exit

Proof of Theorem 1.1.1. Assume a completely flat correction path has $k \geq 2$ internal transitions. Use the fixed northeast point (A,h), and write $P < Q$ and $p < q$ for the initial southwest columns and rows. By Lemma 1.3.2, there are four possible alternating sequences, determined by the first type. We treat each sequence and prove all inequalities used in the obstruction.

First type L_+

Write C_j, c_j for the column and entering row of the jth L_+ exchange, and D_j, d_j for those of the jth L_- exchange. Only indices corresponding to transitions present in the finite path are used. Put $d_0 = p$. The southwest columns stay equal to P, Q. Table 1.1 gives

last transition	$(s(P), s(Q))$	row left at the exchanged column
$L_+(C_j)$	(q, c_j)	$s(C_j) = d_{j-1}$
$L_-(D_j)$	(d_j, q)	$s(D_j) = c_j$.

Lemma 1.3.1 gives $A < C_j < n$, $D_j < A$, $d_j < q < c_j$. The first entry cannot be E_L, because the point at C_1 has row $q < c_1 < h$ and column $A < C_1 < P$. Hence it is E_R. The second exchange has $D_1 < A$ and row $d_1 < q$. If $d_1 < p$, this point is inside E_R. Equality is excluded by injectivity. Therefore

$$p < d_1 < q < c_1. \tag{1.4.1}$$

The subsequent exchanges satisfy

$$C_{j+1} > C_j, \quad c_{j+1} < c_j, \qquad D_{j+1} < D_j, \quad d_{j+1} > d_j, \tag{1.4.2}$$

whenever the relevant transitions occur. To prove the first two inequalities, suppose the last two transitions were $L_+(C_j), L_-(D_j)$ and $d_{j-1} < d_j$. The next L_+ rectangle goes from C_{j+1} to P and wraps vertically from c_{j+1} to d_j. The point at C_j has row $d_{j-1} < d_j$. Its emptiness forces $C_{j+1} > C_j$. Test C_{j+1} against the earlier $L_+(C_j)$ rectangle. Its row has not changed between that earlier source and its exchange, since the only intervening changed columns were P, Q, C_j, D_j, all different from C_{j+1}. The earlier rectangle forces $c_{j+1} < c_j$.

For the next L_- rectangle, the point at D_j has row $c_j > c_{j+1}$. If $D_j < D_{j+1}$, this point lies in its wrapping column and row intervals. Thus $D_{j+1} < D_j$. Test D_{j+1} against the earlier $L_-(D_j)$ rectangle. The intervening L_+ transition did not change its row. That earlier rectangle forces $d_{j+1} > d_j$. These arguments start with (1.4.1) and prove (1.4.2) by induction. Equalities are excluded either by the current row range or by injectivity of the earlier state.

In particular, the original exchanged columns C_1, D_1 are never exchanged again. Their rows remain p, c_1, respectively. At every source after at least two transitions, the current southwest rows p', q' satisfy $p < p'$ and $q' < c_1$. The point at C_1 lies in F_L, and the point at D_1 lies in F_R. This contradicts (1.2.3).

First type L_-

Write C_j, c_j for the successive L_- exchanges and D_j, d_j for the successive L_+ exchanges. Put $d_0 = q$. The southwest columns stay P, Q, and the row formulas are

$L_-(C_j)$	(c_j, p)	$s(C_j) = d_{j-1}$
$L_+(D_j)$	(p, d_j)	$s(D_j) = c_j$.

Here $C_j < A < D_j < n$ and $c_j < p < d_j$. The point at C_1 blocks E_R, since its row is below p. Thus the entry is E_L. Its emptiness at D_1 forces $d_1 < q$. Consequently

$$c_1 < p < d_1 < q.$$

The inductive inequalities are

$$C_{j+1} < C_j, \quad c_{j+1} > c_j, \qquad D_{j+1} > D_j, \quad d_{j+1} < d_j.$$

For their proof, immediately before $L_-(C_{j+1})$ the point at C_j has row $d_{j-1} > d_j$. The new rectangle has sides (Q, C_{j+1}) and wraps vertically from d_j to c_{j+1}. Testing C_j forces $C_{j+1} < C_j$. Testing C_{j+1} against the preceding $L_-(C_j)$ rectangle then forces $c_{j+1} > c_j$. Immediately before

$L_+(D_{j+1})$, the point at D_j has row $c_j < c_{j+1}$. Testing D_j against the new rectangle from D_{j+1} to P forces $D_{j+1} > D_j$. Testing D_{j+1} against the preceding $L_+(D_j)$ rectangle forces $d_{j+1} < d_j$. Each test uses an unchanged row at a column distinct from the intervening exchanged column, as in the first case. These arguments prove the induction starting at $d_1 < d_0 = q$.

The original C_1 retains row q, and D_1 retains row c_1. After at least two transitions, the current southwest rows satisfy $c_1 < p'$ and $q' < q$. The point at D_1 blocks F_L, and the point at C_1 blocks F_R. Neither exit is empty.

First type H_+

In this case the southwest rows stay equal to the original p, q. Write $P_0 = P$, $Q_0 = Q$. Write C_j, c_j for the successive H_+ exchanges and D_j, d_j for the successive H_- exchanges, with $D_0 = P_0$. The southwest columns and the row left behind are

$$\begin{array}{l|l|l} H_+(C_j) & (Q_0, C_j) & s(D_{j-1}) = c_j \\ H_-(D_j) & (D_j, Q_0) & s(C_j) = d_j. \end{array}$$

The type inequalities give $D_j < Q_0 < C_j$ and $n < d_j < h < c_j$. The point at C_1 blocks E_R, so the entry is E_L. Testing D_1 against this entry forces $P_0 < D_1$. Thus

$$P_0 < D_1 < Q_0 < C_1.$$

The inductive inequalities are

$$C_{j+1} < C_j, \quad c_{j+1} > c_j, \qquad D_{j+1} > D_j, \quad d_{j+1} < d_j.$$

Suppose first that $D_{j-1} < D_j$. The rectangle of $H_+(C_{j+1})$ has sides (C_{j+1}, D_j) and wraps vertically from c_{j+1} to p. The point at D_{j-1} has row c_j and lies in its column interval. Emptiness forces $c_{j+1} > c_j$. Testing C_{j+1} against the preceding $H_+(C_j)$ rectangle now forces $C_{j+1} < C_j$. The tested row is unchanged because $C_{j+1} > Q_0$ and $C_{j+1} \neq C_j$, while the intervening transition changed only Q_0, C_j, D_j.

The rectangle of $H_-(D_{j+1})$ has sides (C_{j+1}, D_{j+1}) and row interval (q, d_{j+1}). The point at $C_j > C_{j+1}$ has row d_j. Its emptiness forces $d_{j+1} < d_j$. Testing D_{j+1} against the preceding $H_-(D_j)$ rectangle forces $D_{j+1} > D_j$. Again its row is unchanged between these tests. This proves the induction.

The original P_0 retains row $c_1 > h$, and C_1 retains row $d_1 < h$. Every later southwest pair P', Q' after at least two transitions satisfies $P_0 < P'$ and $Q' < C_1$. The point at P_0 blocks F_L, and the point at C_1 blocks F_R.

First type H_-

Keep $P_0 = P$, $Q_0 = Q$, and write C_j, c_j for the successive H_- exchanges and D_j, d_j for the successive H_+ exchanges, with $D_0 = Q_0$. The southwest rows stay p, q, and the formulas are

$$\begin{array}{l|l|l} H_-(C_j) & (C_j, P_0) & s(D_{j-1}) = c_j \\ H_+(D_j) & (P_0, D_j) & s(C_j) = d_j. \end{array}$$

Here $C_j < P_0 < D_j$ and $n < c_j < h < d_j$. The point at C_1 blocks E_L, so the entry is E_R. Testing D_1 against E_R forces $D_1 < Q_0$. Hence

$$C_1 < P_0 < D_1 < Q_0.$$

The inductive inequalities are

$$C_{j+1} > C_j, \quad c_{j+1} < c_j, \qquad D_{j+1} < D_j, \quad d_{j+1} > d_j.$$

Suppose that $D_j < D_{j-1}$. The rectangle of $H_-(C_{j+1})$ has sides (D_j, C_{j+1}) and row interval (q, c_{j+1}). The point at D_{j-1} has row c_j and belongs to its column interval. Emptiness forces $c_{j+1} < c_j$. Testing C_{j+1} against the preceding $H_-(C_j)$ rectangle then forces $C_{j+1} > C_j$. The rectangle of $H_+(D_{j+1})$ has sides (D_{j+1}, C_{j+1}) and wraps vertically from d_{j+1} to p. The point at $C_j < C_{j+1}$ has row d_j, so emptiness forces $d_{j+1} > d_j$. Testing D_{j+1} against the preceding $H_+(D_j)$ rectangle forces $D_{j+1} < D_j$. In both earlier-rectangle tests, the tested column is different from all columns changed in the intervening transition. Its row is therefore the same in the two tests.

The original Q_0 retains row $c_1 < h$, and C_1 retains row $d_1 > h$. After at least two transitions, every southwest pair satisfies $C_1 < P'$ and $Q' < Q_0$. The point at C_1 blocks F_L, and the point at Q_0 blocks F_R.

All four possible alternating sequences contradict the required empty exit. This proves the theorem. □

1.5 Consequences and exceptional cases

Proof of Corollary 1.1.2. The declarations `seamMatchingEntry_northeast_card` and `seamMatchingExit_northeast_card` give northeast count one at the first and last matching sources of any correction path. For a three-transition path whose two interior sources also have count one, Theorem 1.1.1 applies. For fixed second factor, the existing `seamCorrectionThree_middle_eq_one` supplies exactly these two interior equalities. No proved three-step exclusion is repeated here. □

Lemma 1.5.1 (New lemma: cardinality bound for the northeast set)**.** *For every state s of size $n + m + 1$,*

$$\operatorname{card} \mathrm{NE}(s) \leq \min(n, m).$$

Proof. The inclusion into the n columns $0, \ldots, n-1$ bounds the cardinality by n. The restriction of the permutation s to $\mathrm{NE}(s)$ is an injection into the m rows $n+1, \ldots, n+m$, which bounds the cardinality by m. □

Proof of Corollary 1.1.3. Every matching source has nonempty northeast set by `seamNortheastColumns_nonempty_iff`. If $n = 0$ or $m = 0$, Lemma 1.5.1 rules out every matching source, and therefore every correction path. If $\min(n, m) = 1$, every matching source has northeast count one. Theorem 1.1.1 excludes every correction path with $k \geq 2$. In both cases every summand in the native `seamLongCorrectionCount` is zero, proving the corollary. □

The sizes $(n+1, m+1) = (1,1), (1,2), (2,1), (2,2)$ are covered by this proof. For the smallest nonempty complement, $n = m = 1$, there is one matching source, namely the state $(2, 0, 1)$ on columns $0, 1, 2$. Its southwest columns are $1, 2$ with rows $0, 1$, and its partner is $(2, 1, 0)$. The positively oriented matching rectangle has sides $(1, 2)$ and row interval $(0, 1)$. There is no internal transition because distinct matching sources are required. This checks the orientation and the strict versus weak inequalities at the shared index n directly. It is an exceptional-case check, not the argument for arbitrary sizes.

More generally, if the number M of matching sources is at most three, the native long sum is empty by its definition. This is already recorded by the audit and does not require a new parity argument.

1.6 Projected integral domains of a seam correction

All path coefficients remain in $\mathbb{F}_2$. Integral domains in this section record the multiplicities of grid squares before any parity is taken. Their use does not change the coefficient field of the chain complex.

For a grid of size h, an integral domain is an integer-valued function $D(c,r)$ on the h^2 squares. All cell and vertex indices in the next two lemmas are cyclic modulo h. Define its mixed difference and corner average by

$$(\partial D)(c,r) = D(c,r) + D(c-1,r-1) - D(c-1,r) - D(c,r-1),$$
$$(\mathcal{A}D)(c,r) = \tfrac{1}{4}\big(D(c,r) + D(c-1,r-1) + D(c-1,r) + D(c,r-1)\big).$$

For a state u, let $[u]$ be the indicator function of its points. A domain from u to v satisfies $\partial D = [u] - [v]$. Its index is

$$\iota(D;u,v) = \sum_{c=0}^{h-1} \big((\mathcal{A}D)(c,u(c)) + (\mathcal{A}D)(c,v(c))\big).$$

This is a definition by finite sums. In particular it makes sense for domains with negative coefficients.

Lemma 1.6.1 (New lemma: domain index identities)**.** *The following statements hold for integral domains on a toroidal grid.*

1. *If D is a domain from u to v and E is a domain from v to w, then*

$$\iota(D+E;u,w) = \iota(D;u,v) + \iota(E;v,w).$$

2. *A domain with mixed difference zero has the form $D(c,r) = a_c + b_r$, with integers a_c, b_r. Its index between any equal endpoint states is $2\sum_c a_c + 2\sum_r b_r$.*

3. *An empty native rectangle has index one. For any auxiliary grid with marking permutation O and any domain from u to v,*

$$\iota(D;u,v) = M_O(u) - M_O(v) + 2\sum_c D(c,O(c)). \tag{1.6.1}$$

   Here M_O is the native integer Maslov grading.

Proof. Finite cyclic reindexing gives

$$\sum_{c,r} (\partial D)(c,r)(\mathcal{A}E)(c,r) = \sum_{c,r} (\partial E)(c,r)(\mathcal{A}D)(c,r).$$

For completeness, after moving the differences from D to E, the coefficient of $D(c,r)$ on the left is

$$\tfrac{1}{4}\big(E(c+1,r+1) - E(c+1,r-1) - E(c-1,r+1) + E(c-1,r-1)\big).$$

Its coefficients are unchanged when both offsets are negated. This proves the displayed symmetry. Substituting $\partial D = [u]-[v]$ and $\partial E = [v]-[w]$ proves the first assertion by expanding the definition of ι.

If $\partial D = 0$, the difference $D(c,r) - D(c-1,r)$ is independent of r. Fixing one row and one column and summing these differences gives $D(c,r) = a_c + b_r$. Each state occupies every row and every column once. Substitution into the corner average therefore gives index $2\sum a_c + 2\sum b_r$.

The domain of a rectangle with sides (a,b) and source u is the indicator of

$$[a,b)_{\mathrm{cyc}} \times [u(a),u(b))_{\mathrm{cyc}}.$$

Its mixed difference is $[u] - [u \circ (a\ b)]$ by the native declaration `GridRectangleBetween.target_eq_swapColumns`. If the rectangle is empty, its four endpoint corners each contribute $1/4$ to the index and every other state point contributes zero. The native declaration `laurentMaslovDegree_add_card_OColumns` then proves (1.6.1) for an empty rectangle.

To obtain the formula for an arbitrary domain, connect u to v by adjacent-column transpositions. Such a sequence exists by moving the desired value into the first incorrect column and repeating on the remaining columns. Use the rectangle of width one for each transposition. It is empty because there is no interior column. Their sum D_0 has the required boundary, and the formula for D_0 follows by additivity and telescoping of the native grading differences. The difference $D - D_0$ has zero mixed difference. For a row or column annulus the index is two and the sum of the O multiplicities is one. The second assertion, with arbitrary integer coefficients, now gives the formula for $D - D_0$ and hence for D. □

Lemma 1.6.2 (New lemma: nonnegative domains of index one)**.** *Let D be a nonnegative integral domain from distinct states u to v. Then $\iota(D; u, v) \geq 1$. If its index is one, then D is the indicator domain of an empty native rectangle from u to v.*

Proof. Suppose that the states differ in e columns. Then $e \geq 2$, and the mixed difference has a coefficient 1 or -1 at each of their $2e$ distinct changed points. At each such point at least one adjacent cell has a positive integer coefficient. Its corner average is therefore at least $1/4$. All remaining contributions are nonnegative, so $\iota(D; u, v) \geq e/2 \geq 1$.

If the index is one, this bound gives $e = 2$. Write the changed columns as A, B, and put $R = u(A)$ and $S = u(B)$. The target exchanges these two rows. Let V and W be the nonempty proper cyclic cell intervals $[A, B)$ and $[R, S)$, and let D_0 be the indicator of $V \times W$. By Lemma 1.6.1,

$$D(c, r) = \mathbf{1}_V(c)\mathbf{1}_W(r) + a_c + b_r.$$

Replace a_c by its minimum separately on V and its complement, and replace b_r by its minimum separately on W and its complement. The resulting domain D_b is nonnegative, because on each of the four products of intervals its value is the minimum of D there. The difference $D - D_b$ is a sum of row and column annuli with nonnegative integer coefficients. The domain D_b has the same boundary, so its index is at least one by the preceding paragraph. Each annulus has index two. Additivity and $\iota(D) = 1$ force every annulus coefficient to be zero. Thus $D = D_b$ is constant on each of the four products of intervals.

Let the four constants be a, b, c, d, with a on $V \times W$ and d on its opposite product. They are nonnegative integers and satisfy $a + d - b - c = 1$. The four changed points contribute exactly $a + b + c + d$ to the index. Hence this sum is at most one, and the only possibilities are $(a, b, c, d) = (1, 0, 0, 0)$ and $(0, 0, 0, 1)$. These are the two oriented rectangle domains from u to v. The four corners already account for the full index. Every other source-state corner average is zero, so there is no source point in the rectangle's interior. This is the native emptiness condition `isEmpty_iff_forall_notMem_cIoo`. □

Now let $n, m \geq 1$ and $N = n + m + 1$. For a correction path from retained state $p = \texttt{seamState}(x, z)$ to retained state $q = \texttt{seamState}(y, w)$, define D to be the sum of its entry, internal, and exit rectangle domains, minus the domains of the selected matching rectangles at all its matching sources.

Lemma 1.6.3 (New lemma: the signed correction domain)**.** *The signed domain D of a correction path satisfies $\partial D = [p] - [q]$ and*

$$\iota(D; p, q) = 1. \tag{1.6.2}$$

Proof. The matching rectangle at a source with southwest columns $P < Q$ and rows $a < b \leq n$ is $[P, Q) \times [a, b)$. It is empty, because every column strictly between the first two southwest columns has a high row. Telescoping the rectangle boundaries gives $\partial D = [p] - [q]$. There is one more positive rectangle than subtracted matching rectangles. Each has index one, so additivity in Lemma 1.6.1 gives (1.6.2). □

Define two restrictions, with the indicated factor indices, by

$$\begin{aligned} D_1(i,j) &= D(i,j) && (0 \le i < n,\ 0 \le j \le n), \\ D_1(n,j) &= D(N-1,j) && (0 \le j \le n), \\ D_2(i,j) &= D(n+i,n+j) && (0 \le i,j \le m). \end{aligned} \tag{1.6.3}$$

Put $d = D(N-1,n)$.

Lemma 1.6.4 (New lemma: positive projections and the index relation)**.** *For every correction path as above, D_1 is a nonnegative domain from x to y, and D_2 is a nonnegative domain from z to w. Their indices satisfy*

$$\iota(D_1;x,y) + \iota(D_2;z,w) = 1 + 2d. \tag{1.6.4}$$

Proof. A matching rectangle uses only columns from n through $N-2$ and rows from 0 through $n-1$. It therefore misses both restrictions in (1.6.3). All remaining summands of D are nonnegative rectangle indicators, proving positivity.

For the boundary assertion, restricting a domain to cyclic lists of cell columns c_i and cell rows r_j replaces its mixed difference at (i,j) by

$$\sum_{c \in (c_{i-1},c_i]_{\mathrm{cyc}}} \ \sum_{r \in (r_{j-1},r_j]_{\mathrm{cyc}}} (\partial D)(c,r).$$

This formula follows by telescoping in both coordinates. For D_1, the last column collects the original columns $n,\dots,N-1$, and row zero collects the original rows $n+1,\dots,N-1,0$. Under these collections the retained state p becomes $[x] + m[(n,0)]$. The extra m copies are the high points in the second columns. The retained state q gives $[y] + m[(n,0)]$, so the extra terms cancel. For D_2, column zero collects columns $0,\dots,n$, and row zero collects rows $0,\dots,n$. The retained states become $[z] + n[(0,0)]$ and $[w] + n[(0,0)]$. Again the extra terms cancel. These statements use precisely the three supplied coordinate formulas `seamState_left_castSucc`, `seamState_right_of_zero`, and `seamState_right_of_ne_zero`.

We prove the index relation by explicit auxiliary markings. Let $O_1(i) = i$ on the first factor, let $O_2(j) = j+1 \pmod{m+1}$ on the second factor, and let

$$O(c) = c \quad (c < n), \qquad O(n+j) = n + O_2(j) \quad (0 \le j \le m).$$

Each is a permutation and can be made into a valid auxiliary grid by choosing X to be its row successor. The Maslov grading depends only on the specified O markings. The full marking set is the union of the two projected marking sets with their common square $(N-1,n)$ counted once. Consequently

$$\sum_c D(c,O(c)) = \sum_i D_1(i,O_1(i)) + \sum_j D_2(j,O_2(j)) - d. \tag{1.6.5}$$

The native Maslov point-count formula gives

$$M_O(\texttt{seamState}(x,z)) = M_{O_1}(x) + M_{O_2}(z). \tag{1.6.6}$$

Here is a direct verification, so no condition on the auxiliary X markings is being suppressed. For finite planar point sets A, B, write $I(A,B)$ for the number of ordered pairs that increase strictly in both coordinates. State points have integral coordinates and marks have both coordinates shifted by $1/2$. The supplied formula `GridDiagram.maslovOâĎď_eq_card` is

$$M_O(s) = I(s,s) - I(s,O) - I(O,s) + I(O,O) + 1.$$

Splitting each point set into its first n and last $m+1$ columns gives

$$\begin{aligned} I(p,p) &= I(x,x) + I(z,z) + nm, \\ I(O,O) &= I(O_1,O_1) + I(O_2,O_2) + nm, \\ I(p,O) &= I(x,O_1) + I(z,O_2) + nm - 1, \\ I(O,p) &= I(O_1,x) + I(O_2,z) + nm. \end{aligned}$$

To check the first equality, put $t = x(n)$. Removing the last column from x removes t increasing pairs. The cross pairs with second-column points contribute $nm + t$. The second equality is the same computation for the marking permutations. For the third equality, removing the last marking of O_1, whose row is n, removes $n + 1$ pairs. All $n(m + 1)$ pairs from first-column state points to second-column markings contribute. The pairs within the second columns agree with $I(z, O_2)$ because the low state row t is at most the lowest marking row $n + 1/2$. For the fourth equality, removing the last state point of x removes t pairs, whereas the cross pairs contribute $nm + t$. Substituting the four identities into the native formula proves (1.6.6).

Apply (1.6.1) to the three domains. Equations (1.6.2), (1.6.5), and (1.6.6) give (1.6.4). □

1.7 A finite order certificate on the five-point seam grid

Lemma 1.7.1 (New lemma: the five-point exclusion)**.** *Let $n = m = 2$ and let p, q be arbitrary retained indices. Every native complete correction path from p to q has at most two internal transitions.*

Proof. A state is written as its five rows, in column order. For example 03142 denotes the permutation $(0, 3, 1, 4, 2)$. All digits in this proof are single row indices, rather than an integer written in decimal notation. An arrow annotated (a, b) denotes the actual empty rectangle with ordered column sides a, b, followed by the selected inverse matching. In particular the orientation of every rectangle in the certificate is specified.

We first give an order calculation that makes the certificate exhaustive. Every matching source with northeast count one has the following data:

$$A \in \{0, 1\}, \quad h \in \{3, 4\}, \quad L = 1 - A, \quad s(A) = h, \quad s(L) = t,$$
$$\{P, Q, C\} = \{2, 3, 4\}, \quad P < Q, \quad s(P) = v < s(Q) = w, \quad s(C) = 7 - h, \quad \{v, w, t\} = \{0, 1, 2\}.$$

Thus there are exactly $2 \cdot 2 \cdot 3 \cdot 3 = 36$ such sources. The native theorem `flatStep_cases` shows that a transition which preserves northeast count is one of the four flat types below. A birth can only exchange (L, t) with $(C, 7 - h)$. Its target has a decreasing first southwest pair only if either $C < P$ and $t > v$, or $P < C < Q$ and $t < v$. In the first case the wrapping-column orientation contains (P, v). In the second case the nonwrapping-column orientation contains (P, v). These facts leave the two birth orientations in the table. A death from a source with northeast count one reaches a retained state, so it is an exit rather than an internal transition. Consequently the following six rows include every possible internal transition from any of the 36 sources.

type	ordered columns	necessary and sufficient conditions
L_+	(L, P)	$t > w, \quad A = 0, \quad C > P$
L_-	(Q, L)	$t < v, \quad A = 1, \quad C < Q$
H_+	(C, P)	$C > Q, \quad h = 3, \quad v = 0$
H_-	(Q, C)	$C < P, \quad h = 4, \quad w = 2$
B_+	(L, C)	$C < P, \quad t > v, \quad (A = 0 \text{ or } h = 4)$
B_-	(C, L)	$P < C < Q, \quad t < v, \quad (A = 1 \text{ or } h = 3)$

(J05.F5.1)

Here is the emptiness check for every row of this table, including its excluded cases. For L_+, the row data must be $(v, w, t) = (0, 1, 2)$. The cyclic row interior consists of rows $3, 4$. The northeast point blocks precisely when $A > L$, and the southeast point blocks precisely when $C < P$. This gives $A = 0$ and $C > P$. For L_-, the row data must be $(t, v, w) = (0, 1, 2)$, and the same two high rows form the cyclic interior. Its northeast blocker occurs when $A < L$, and its southeast blocker occurs when $C > Q$. This gives $A = 1$ and $C < Q$. For H_+, the column data are $(P, Q, C) = (2, 3, 4)$. The only interior columns are $0, 1$. Their two points avoid the wrapping row interval exactly when $h < 7 - h$ and $t > v$. These conditions are $h = 3$ and $v = 0$. For H_-, the column data are $(C, P, Q) = (2, 3, 4)$, again with interior columns $0, 1$. The necessary and

sufficient inequalities are $h > 7 - h$ and $t < w$, namely $h = 4$ and $w = 2$. For B_+, the column data are $(C, P, Q) = (2, 3, 4)$. If $L = 1$ there is no interior column. If $L = 0$, its only interior column is $A = 1$, whose high row blocks exactly when $h < 7 - h$. For B_-, the column data are $(P, C, Q) = (2, 3, 4)$ and $t = 0$. Column 4 contains the low row $w = 2$, which is outside the wrapping row interval. The only possible further interior point is at column $A = 0$ when $L = 1$, and it blocks exactly when $h > 7 - h$. These checks prove all six equivalences in (J05.F5.1). The other orientation of each flat move is already excluded by `flatStep_cases`.

An entry can reach only a matching source with northeast count one, by `seamMatchingEntry_northeast_card`. Two of the 36 sources admit no entry. For 04312, its possible entry rectangles are blocked respectively by the points $(2, 3)$ and $(0, 0)$. For 32014, they are blocked respectively by $(1, 2)$ and $(4, 4)$. These are the two candidates in `flatEntry_data`. Let E_0 be the 34 other northeast-count-one sources. Every actual entry source belongs to E_0. No assertion that every member is reached is required for the argument.

Substitution of the three choices of (v, w, t), the three choices of (P, Q, C), and the two choices each of A, h in (J05.F5.1) gives the following complete list of nonempty outgoing internal-arrow sets for sources in E_0. The rows not displayed have no internal arrow.

03142	$(3,0) : 43012,\ (4,0) : 23041$	03412	$(4,0) : 23401$
04132	$(3,0) : 34012,\ (4,0) : 24031$	13024	$(4,2) : 13402$
14302	$(0,2) : 34012,\ (4,2) : 14023$	23014	$(4,2) : 23401$
24301	$(0,2) : 34021$	30142	$(3,1) : 34012$
31024	$(4,2) : 31402$	31402	$(1,2) : 34012$
32041	$(1,2) : 30142$	32401	$(1,2) : 34021$
40312	$(4,2) : 40123$	41302	$(1,2) : 43012,\ (4,2) : 41023$
42013	$(1,2) : 40123$	42031	$(1,2) : 40132$
42301	$(1,2) : 43021$		

(J05.F5.2)

The target in each cell is obtained by swapping the two displayed columns and then swapping the first two southwest columns. For example the row 24301 uses the birth with sides $(0, 2)$. The actual target is 34201, whose southwest rows at columns $2, 3$ are $2, 0$. Its selected partner is 34021, as displayed. Thus (J05.F5.2) records state maps as well as all the emptiness and source tests in (J05.F5.1).

Only four sources with northeast count two occur among these targets: 34012, 34021, 43012, and 43021. Their complete internal-arrow sets are

34012	$\varnothing$
43012	$\varnothing$
34021	$(1,2) : 30412$
43021	$(0,2) : 03412,\ (1,2) : 40312.$

(J05.F5.3)

We verify this table without leaving any orientation unchecked. There is no northwest or southeast point, so there is no birth and no flat exchange with either of those quadrants. Swapping the two northeast points leaves the first southwest pair increasing. Swapping the first two southwest points is the selected matching and is excluded from an internal arrow. An interchange of the second and third southwest points also leaves the first pair increasing. An interchange of the first and third can yield a decreasing target pair only for southwest row order 012. Its nonwrapping rectangle then contains the second southwest point, and its wrapping rectangle contains a northeast point. Thus no flat internal exchange remains.

A death must remove column 2, since removing either of the other two southwest columns leaves an increasing first pair. Its remaining pair is decreasing exactly when the southwest order is 021. The wrapping-column orientation contains a southwest point. For the nonwrapping orientation $(A, 2)$, $A = 1$ has no interior column and is empty. For $A = 0$, column 1 blocks exactly when its high row is greater than the high row in column 0. This gives precisely the three arrows in (J05.F5.3) and their displayed targets.

Let E_j be the set of sources reached by j successive internal arrows starting from E_0. The two explicit arrow tables now give

$$\begin{aligned} E_1 &= \{13402, 14023, 23041, 23401, 24031, 30142, 31402,\\ &\qquad 34012, 34021, 40123, 40132, 41023, 43012, 43021\},\\ E_2 &= \{03412, 30412, 34012, 40312\},\\ E_3 &= \{23401, 40123\}, \qquad E_4 = \varnothing. \end{aligned} \tag{J05.F5.4}$$

These are finite set substitutions in the displayed tables, rather than an inference from tests on further sizes. For completeness neither member of E_3 has an exit. For 23401, the two exit rectangles in `flatExit_data` contain respectively $(2, 4)$ and $(0, 2)$. For 40123, they contain respectively $(1, 0)$ and $(4, 3)$. Thus a path with exactly three internal transitions cannot exit. A path with four or more internal transitions would have to pass through E_4, which is empty. This proves the lemma. □

1.8 Fixed-second paths have at most two internal transitions

Use the signed path domain D, its nonnegative projections D_1, D_2, and the shared cell $d = D(N-1, n)$ from Lemma 1.6.4. The next statements concern the first sector, so the retained endpoints are $p = \texttt{seamState}(x, z)$ and $q = \texttt{seamState}(y, z)$.

Lemma 1.8.1 (New lemma: the fixed-second projected domain)**.** *Assume $n, m \geq 1$ and $x \neq y$. For any complete correction path from p to q, D_1 is the indicator of an empty first-factor rectangle, $d \in \{0, 1\}$, and*

$$D_2(i, j) = d\,\mathbf{1}_{\{m\}}(i) \qquad (0 \leq i, j \leq m). \tag{1.8.1}$$

For every west cell column $c < n$ and every high cell row $r \geq n$, one also has

$$D(c, r) = D_1(c, n). \tag{1.8.2}$$

Proof. The endpoints of D_2 agree. By Lemma 1.6.1, write $D_2(i, j) = a_i + b_j$. Normalize this expression by subtracting $\min_i a_i$ from each a_i and adding it to each b_j. Then all a_i are nonnegative. At a column where $a_i = 0$, positivity of D_2 implies $b_j \geq 0$ for every j. Therefore

$$\iota(D_2; z, z) = 2\sum_i a_i + 2\sum_j b_j \geq 2(a_m + b_0) = 2d.$$

The endpoints of D_1 differ, so Lemma 1.6.2 gives $\iota(D_1) \geq 1$. The index identity (1.6.4) forces equality in both inequalities. Hence D_1 is an empty rectangle and $d = D_1(n, n) \in \{0, 1\}$. Every a_i with $i \neq m$ and every b_j with $j \neq 0$ vanishes. Put $\alpha = a_m$ and $\beta = b_0$. Then

$$D_2(i, j) = \alpha\mathbf{1}_{\{m\}}(i) + \beta\mathbf{1}_{\{0\}}(j), \qquad \alpha + \beta = d.$$

There are no retained endpoint points in the northeast region. For $c < n$ and $1 \leq j \leq m$, sum $\partial D = 0$ over columns $0, \ldots, c$ and vertex rows $n+1, \ldots, n+j$. The two telescoping sums give

$$D(c, n+j) - D(N-1, n+j) - D(c, n) + D(N-1, n) = 0.$$

By the projection formulas this is

$$D(c, n+j) = D_1(c, n) + D_2(m, j) - d = D_1(c, n) - \beta.$$

Matching rectangles miss these cells. Each signed multiplicity in this region is therefore nonnegative. It follows that $D_1(c, n) \geq \beta$ for every $c < n$. If $\beta = 1$, these inequalities and $D_1(n, n) = d = 1$ would make the empty rectangle D_1 cover row n in every cell column of the first-factor torus. A rectangle's cyclic column interval is proper, which excludes this possibility. Consequently $\beta = 0$ and $\alpha = d$. This proves (1.8.1). The displayed telescoping equality now proves (1.8.2) for $r > n$. At $r = n$ it is the definition of D_1. □

Lemma 1.8.2 (New lemma: freezing the unchanged state points)**.** *Under the hypotheses of Lemma 1.8.1, the following original state points of p remain fixed at every stage of the correction path, including the intermediate matching targets.*

1. *A west point $(c, x(c))$, with $c < n$, whose column is not an endpoint column of the first-factor rectangle D_1.*

2. *A southeast point $(c, p(c))$ with $n < c < N - 1$ and $p(c) > n$.*

Proof. The two endpoint states of an empty rectangle agree outside its two endpoint columns. At such a common point, the four adjacent cell multiplicities of the rectangle are zero. Indeed the point is not on an edge, because a state has only one point in each row and each column, and it cannot be in the rectangle's interior by emptiness.

Let $(c, x(c))$ be a point in item 1. Its four adjacent cells in the full seam grid have the same multiplicities as the four adjacent cells at $(c, x(c))$ in D_1. For rows $1, \dots, n$ and columns $1, \dots, n-1$, this is the restriction defining D_1. At west column zero, the left adjacent cell uses global column $N - 1$, which is precisely the last-column restriction defining D_1. At state row zero, the lower adjacent cell uses global row $N - 1$. Equation (1.8.2) identifies its multiplicity with that at factor row n. If both boundary identifications occur, $D(N-1, N-1) = d = D_1(n, n)$ follows from (1.8.1). Thus all four full-grid multiplicities are zero. Each of these cells is missed by every selected matching rectangle: a matching rectangle uses only cell columns $n, \dots, N-2$ and cell rows $0, \dots, n-1$. Consequently the multiplicity at each of these four cells is the sum of nonnegative contributions of the actual path rectangles. Every actual rectangle therefore misses each of the four cells.

For a point in item 2, all four adjacent cells lie in the second projection. Their column indices are different from its last cell column m. Their multiplicities are zero by (1.8.1). They also lie outside every selected matching rectangle, because their cell rows are at least n. Every actual rectangle again misses all four adjacent cells.

In either case, suppose the stated point first changed at some stage. An inverse matching cannot cause that first change. A point which has not yet changed is either in a west column, or has a high row, whereas both matching endpoints are southwest points. Thus the first change would be an actual rectangle using the point as a corner. Such a rectangle covers one of its four adjacent cells. This contradicts the vanishing just proved. □

Lemma 1.8.3 (New lemma: deletion of fixed points preserves correction paths)**.** *Let a complete correction path on a seam of parameters n, m be given. Delete any collection of original northwest points and original southeast points which remain fixed throughout the path. Delete their rows and columns, and relabel the remaining rows and columns in increasing order. If n' west columns and m' high rows remain, the result is a complete correction path, with the same number of internal transitions, on the native seam of parameters n', m'.*

Proof. A retained original state has n west low points, one east low point, and m east high points. Deleting a northwest point removes one west column and one low row. Deleting a southeast point removes one east column and one high row. Thus the remaining columns consist of n' west columns and $m' + 1$ east columns, while the remaining rows consist of $n' + 1$ low rows and m' high rows. Increasing relabelling identifies these precisely with the native cuts at n'.

All states on the path restrict to bijections on the remaining rows and columns, since the deleted point values never change. Every actual rectangle endpoint remains present, since an actual rectangle changes both of its endpoint points. The increasing injections of the remaining row and column sets preserve cyclic order. A surviving point inside a compressed cyclic rectangle would therefore have been inside the original rectangle. Every compressed rectangle is consequently empty, has the prescribed ordered sides, and exchanges its two endpoint values.

A deleted point is never southwest. Thus the first two southwest columns of every intermediate matching state are preserved by restriction. Their row order is preserved as well. The compressed

involution is exactly the native partner which swaps those first two columns, and its sources have increasing rows there. Two consecutive matching sources cannot become equal after deletion, because they agreed on all deleted points before deletion already. The nonmatching condition on internal transitions is therefore preserved. The original retained endpoints remain retained, and the entry, internal rectangles, and exit supply the asserted complete path of the same length. □

Theorem 1.8.4 (New lemma: all fixed-second long paths are absent)**.** *Let $n, m, k \in \mathbb{N}$, let $x, y \in \texttt{GridState}(n+1)$, and let $z \in \texttt{GridState}(m+1)$. If $k \geq 3$, then*

$$\texttt{SeamCorrectionPath}\ k\ (x, z)\ (y, z) = \varnothing.$$

Proof. The northeast count is at most $\min(n, m)$, since the northeast columns inject both into the n west columns and into the m high rows. If $\min(n, m) = 0$, there is no matching source, so there is no correction path. If $\min(n, m) = 1$, every matching source on a path has northeast count one. Theorem 1.1.1 excludes a complete path with at least two internal transitions. We may therefore assume $n, m \geq 2$.

If $x = y$, the signed path domain has zero mixed difference. Its index is even by Lemma 1.6.1, whereas the signed-domain index of any correction path is one. This is impossible. Hence assume $x \neq y$. By Lemma 1.8.1, the first projection is an empty rectangle. At most two of its endpoint columns are west columns. Lemma 1.8.2 freezes every other original west point, and freezes every original southeast point except possibly the points in the two boundary columns $n, N-1$. Keep the at most two west endpoint points, the original east low point, and those at most two southeast boundary points. Delete all other original points.

Lemma 1.8.3 gives a complete correction path of the same internal length on a seam with $n' \leq 2$ and $m' \leq 2$. If one parameter is zero, a matching source cannot exist. If $\min(n', m') = 1$, every matching source is flat and Theorem 1.1.1 excludes the path. The remaining case is $n' = m' = 2$, which is excluded by Lemma 1.7.1 for $k \geq 3$. All possible compressed sizes have been considered. □

Corollary 1.8.5 (New lemma: full J05 acceptance)**.** *Let $n, m \in \mathbb{N}$ and let $x, y \in \texttt{GridState}(n+1)$. For every $z \in \texttt{GridState}(m+1)$, one has*

$$\texttt{seamLongCorrectionCount}((x, z), (y, z)) = 0$$

as an ordinary integer, and hence also after coercion to $\mathbb{F}_2$. The first conjunct of `SeamPositive PathComparisons` *n m follows.*

Proof. Every summand in the native definition of `seamLongCorrectionCount` is the cardinality of a path type with at least three internal transitions. Theorem 1.8.4 makes every such type empty. The finite sum is therefore zero, including an empty index range. The existing declaration `seam PositiveCorrectionCount_first_eq_remaining_add_long` identifies the positive correction count with the number of remaining first-factor rectangles plus this long count. Substitution gives exactly the required first comparison. □

1.9 Completion of the acceptance statement

Theorem 1.8.4 and Corollary 1.8.5 discharge M602 and M604. Corollary 1.1.2 discharges M603. There is no remaining gap in the stated J05 acceptance statement. The long count vanishes as an integer, which is stronger than its required parity vanishing over $\mathbb{F}_2$.

The exceptional cases are explicit. A zero seam parameter gives an empty complement. A parameter equal to one gives only flat matching sources. Equal retained endpoints are ruled out by the even index of a periodic domain. A first-factor rectangle with a terminal endpoint leaves at most one west point after compression, so it belongs to the thin case. If the original east low point is in a boundary column, at most one east high point is retained in the compressed grid, again giving the thin case. The only remaining compressed parameters are $n' = m' = 2$, covered in full by Lemma 1.7.1. The truncated long-count sum also vanishes when its indexing range is empty.

1.10 Lemma ledger

In the table, "existing" refers to native declarations in the supplied verified Lean material. A declaration used by a supplied proof may reside in one of its imported modules. The failed three-step module is excluded from this category. All new mathematical assertions used above have proofs in this document.

Declaration or statement	Status	Use
`SeamCorrectionPath`, `SeamInternalPath`	Existing	Actual rectangles and the internal length convention.
`mem_seamNortheastColumns`, `mem_seamSouthwestColumns`	Existing	The native quadrant inequalities.
`seamSouthwestColumns_card`	Existing	Exactly two southwest columns in a flat source.
`seamComplementFirst_lt_second`, `SeamComplementSource`	Existing	Ordered columns and increasing southwest rows.
`flatEmpty_source`, `isEmpty_iff_forall_notMem_cIoo`	Existing	Empty-rectangle point tests.
`flatStep_cases`, `flatStep_apply`	Existing	The four transition formulas in Table 1.1.
`flatApply_high`	Existing	Persistence of the northeast point.
`flatEntry_data`, `flatExit_data`	Existing	The two possible entry and exit rectangles.
Lemma 1.3.1	New	The northeast point separates low exchanges by column and high exchanges by row.
Lemma 1.3.2	New	The twelve forbidden successive type pairs.
Theorem 1.1.1	New	Persistent blockers exclude every completely flat path of length at least two.
`seamMatchingEntry_northeast_card`, `seamMatchingExit_northeast_card`	Existing	Endpoint counts equal one.
`seamCorrectionThree_middle_eq_one`	Existing	A fixed-second three-step path is completely flat.
Corollary 1.1.2	New	M603 and the fixed-second length-three vanishing.
Lemma 1.5.1	New	Northeast count is at most $\min(n, m)$.
`seamNortheastColumns_nonempty_iff`	Existing	Positive northeast count in the complement.
Corollary 1.1.3	New	J05 when one factor has size at most two.
`seamLongCorrectionCount`, `seamFirstPositiveComparison_iff_long_paths`	Existing	Exact long-count definition and the acceptance equivalence.
`GridRectangleBetween.target_eq_swapColumns`, `laurentMaslovDegree_add_card_OColumns`	Existing	Rectangle boundary and native index calculation.
`GridDiagram.maslovÔâD̃d̃_eq_card`	Existing	Explicit Maslov point-count formula for the projected index identity.
`seamState_left_castSucc`, `seamState_right_of_zero`, `seamState_right_of_ne_zero`	Existing	Boundary projection to the native retained factors.
Lemmas 1.6.1 and 1.6.2	New	Index additivity, periodic domains, and positive index-one classification.
Lemmas 1.6.3 and 1.6.4	New	The signed path domain and its nonnegative factor projections.

Declaration or statement	Status	Use
Lemma 1.8.1	New	An empty first-factor rectangle and the last-column second-factor annulus.
Lemma 1.8.2	New	All unchanged west points and all interior southeast points stay fixed.
Lemma 1.8.3	New	Deleting fixed points preserves native correction paths and their length.
Lemma 1.7.1	New	Complete symbolic five-point certificate for arbitrary retained endpoints.
Theorem 1.8.4	New	All fixed-second paths with at least three internal transitions are absent.
`seamPositiveCorrectionCount_first_eq_remaining_add_long`	Existing	Conversion of long-count vanishing to the first positive comparison.
Corollary 1.8.5	New	Full J05 acceptance, with integer vanishing.

1.11 Source and verification record

The source is the supplied `combined_01-10.md`, specifically the audit entry J05, records M602–M604, the mathematical writing rules, and the seam listings under item 07, together with the native Maslov point-count formula used in item 08. The definitions and proofs used for flat transitions belong to `GridConnectedSumSeamFlatTwoStepBarrier.lean`. The fixed-second three-step classification belongs to `GridConnectedSumSeamThreeStepSectors.lean`. The source snapshot is treated as read-only. No Lean code is written or modified. The supplied material does not include `snapshot-sources.sha256`, so the audit's repository hash comparison cannot be performed from these attachments.

No holomorphic Floer theorem is a proof step in this document. The five-point calculation is a finite written order certificate whose candidate types, blocker inequalities, state maps, and reachability sets are all displayed. Its role is justified by the proved support compression from arbitrary sizes. Diagnostic computer enumerations are not used as proof steps. All other arguments use finite permutations, integral square domains, the native rectangle conventions, and the cited existing declarations.

Chapter 2

J06: Second-sector seam parity

2.1 Exact target and status

The acceptance statement of J06 is the following assertion. For every $n, m \in \mathbb{N}$, every $x \in \mathrm{GridState}(n+1)$, and every $z, w \in \mathrm{GridState}(m+1)$,

$$\begin{aligned} &\texttt{seamPositiveCorrectionCount}((x,z),(x,w)) \\ &\quad = |\texttt{SeamSecondRemainingRectangle}(x,z,w)| \quad \text{in } \mathbb{F}_2. \end{aligned} \tag{2.1.1}$$

The counts in this formula are the native counts in the supplied listings.

This document proves the acceptance statement for all parameters. In fact the following stronger equality holds over the natural numbers:

$$\begin{aligned} &\texttt{seamPositiveCorrectionCount}((x,z),(x,w)) \\ &\quad = |\texttt{SeamSecondRemainingRectangle}(x,z,w)|. \end{aligned} \tag{2.1.2}$$

The proof first constructs an explicit bijection between the one-transition and two-transition correction paths and the remaining second-factor rectangles. It then proves that every correction with three or more internal transitions is absent.

The long-path proof uses signed integral domains and their two nonnegative projections. Its index calculation forces the second projection to be an empty rectangle. All but at most five state points must then remain fixed throughout the path. Deleting those fixed points reduces to four explicitly described small seam grids, whose complete transition certificates are proved below. This is a uniform reduction and a finite written proof, with no inference from diagnostic computations.

All chain coefficients and parity statements use $\mathbb{F}_2$. The integer cardinalities, domain multiplicities, and gradings retain their native integer meanings. The proof uses no holomorphic Floer result and no unproved assertion from the failing seam module.

2.2 Native conventions and notation

Put $N = n + m + 1$ and identify $\mathrm{Fin}(N)$ with $\{0, \ldots, N-1\}$. A state is a permutation from columns to rows. For distinct $a, b \in \mathrm{Fin}(N)$, write

$$I_N(a,b) = \begin{cases} \{c : a < c < b\}, & a < b, \\ \{c : c > a \text{ or } c < b\}, & b < a. \end{cases}$$

This is the open cyclic interval used by `Grid.cIoo`. For a state s, an oriented rectangle with side columns (a, b) has bottom row $s(a)$, top row $s(b)$, and target $s \circ (a\ b)$. It is empty precisely when

$$\text{there is no } c \in I_N(a,b) \text{ with } s(c) \in I_N(s(a), s(b)). \tag{2.2.1}$$

The supplied proofs use these statements as `bottom_def`, `top_def`, `target_eq_swapColumns`, and `isEmpty_iff_forall_notMem_cIoo`. A rectangle is determined by its source and its ordered side columns.

The retained state $p = \texttt{seamState}(x, z)$ has coordinates

$$p(c) = x(c) \quad (0 \le c < n), \qquad p(n+j) = \begin{cases} x(n), & z(j) = 0, \\ n + z(j), & z(j) > 0. \end{cases} \tag{2.2.2}$$

These are the supplied declarations `seamState_left_castSucc`, `seamState_right_of_zero`, and `seamState_right_of_ne_zero`.

A northeast point has column less than n and row greater than n. A southwest point has column at least n and row at most n. For a complementary state s, let $P < Q$ be its first two southwest columns. Its partner, denoted s^*, exchanges the values at P, Q. A matching source is a complementary state with $s(P) < s(Q)$. These are the coordinate properties of the native matching used in the supplied proofs, including `seamComplementPartner_apply_first`, `seamComplementPartner_apply_second`, and `seamComplementPartner_firstBlock_apply`. The definitions of the foundational sector and matching modules are imported by the supplied listings rather than reproduced there. The coordinate properties just listed are also stated in Section 2.3 of the supplied seam notes and used explicitly throughout the supplied proofs. No assertion from the failing flat-three-step module is used.

For $k \in \mathbb{N}$, abbreviate

$$\mathcal{P}_k(x, z, w) = \texttt{SeamCorrectionPath}(k, (x, z), (x, w)),$$
$$\mathcal{R}(x, z, w) = \texttt{SeamSecondRemainingRectangle}(x, z, w).$$

A path in $\mathcal{P}_k$ consists of matching sources $s_0, \dots, s_k$, an empty entry rectangle $p \longrightarrow s_0^*$, empty internal rectangles $s_{j-1} \longrightarrow s_j^*$ for $1 \le j \le k$, and an empty exit rectangle $s_k \longrightarrow q$, where $q = \texttt{seamState}(x, w)$. The consecutive matching sources must differ. This is exactly the definition of `SeamCorrectionPath`, together with `SeamInternalRectangle` and `SeamInternalPath`. Thus k counts internal rectangles, and there are $k + 2$ rectangles in all.

2.3 The remaining rectangles

Lemma 2.3.1 (New lemma: coordinate description of the remaining rectangles)**.** *Let $n \ge 1$, let $x \in \mathrm{GridState}(n+1)$, and let $z, w \in \mathrm{GridState}(m+1)$. A rectangle from z to w belongs to $\mathcal{R}(x, z, w)$ if and only if it is empty and its ordered side columns ℓ, r and endpoint rows satisfy*

$$r < \ell, \qquad 0 < z(r) < z(\ell).$$

For such a rectangle, if $c = z^{-1}(0)$, then $r < c < \ell$.

Proof. By the definition of `SeamSecondRemainingRectangle`, the factor rectangle is empty, its lifted rectangle is obstructed, and both endpoint rows are nonzero. The supplied declaration `seamSecondRemainingRectangle_wraps` gives $r < \ell$. If $z(\ell) < z(r)$, the rows in the interior of the lifted rectangle are strictly greater than n. No first-block point of (2.2.2) can lie there. The second-block points are excluded by emptiness of the factor rectangle. This contradicts obstruction. Thus $0 < z(r) < z(\ell)$.

Conversely, suppose the displayed inequalities hold. Every first-block column lies in $I_N(n + \ell, n + r)$, and its row is at most n. Such a row lies in $I_N(n + z(\ell), n + z(r))$. Since $n \ge 1$, a first-block point exists, and the lifted rectangle is obstructed. The nonzero-row conditions in the definition of $\mathcal{R}$ hold.

Finally, 0 lies in $I_{m+1}(z(\ell), z(r))$. Therefore factor emptiness excludes c from $I_{m+1}(\ell, r)$. It is distinct from both endpoint columns, so $r < c < \ell$. □

Fix a rectangle as in Lemma 2.3.1. For the rest of the construction put

$$R = n + r, \quad C = n + c, \quad L = n + \ell, \quad K = n + z(r), \quad H = n + z(\ell), \quad t = x(n). \tag{2.3.1}$$

Then

$$n \le R < C < L < N, \qquad n < K < H < N, \qquad 0 \le t \le n. \tag{2.3.2}$$

The retained source has $p(R) = K$, $p(C) = t$, and $p(L) = H$. For every second-block column D with $D < R$ or $D > L$, factor emptiness also gives

$$K < p(D) < H. \tag{2.3.3}$$

Indeed, D is not C, so its row is high, and a row outside the open interval (K, H) would be an interior point of the factor rectangle. Equality with K or H is excluded by injectivity of the state.

2.4 The explicit inverse construction

Lemma 2.4.1 (New lemma: boundary and interior correction paths)**.** *Under* (2.3.1)–(2.3.2)*, the following rules associate a native correction path to the chosen remaining rectangle. If $t = 0$ or $t = n$, the path belongs to $\mathcal{P}_1(x, z, w)$. If $0 < t < n$, it belongs to $\mathcal{P}_2(x, z, w)$.*

Proof. Only the columns in the following tables change. Every omitted column retains its value from p. In each table, the last row is $q = \texttt{seamState}(x, w)$.

Suppose first that $t = 0$. Let $B = x^{-1}(n)$. Since $n \ge 1$, one has $B < n$. The values are

	B	R	C	L
p	n	K	0	H
s_0	K	0	n	H
s_1	K	H	0	n
q	n	H	0	K

and the ordered rectangle sides are

$$(B, R), \qquad (L, R), \qquad (L, B). \tag{2.4.1}$$

The first two southwest columns of s_0 are R, C with values $0, n$. Those of s_1 are C, L with values $0, n$. Swapping at the indicated rectangle sides and then applying the matching inverse gives exactly the next table row. The internal sources differ.

Suppose next that $t = n$. Let $A = x^{-1}(0)$, so $A < n$. The values are

	A	R	C	L
p	0	K	n	H
s_0	H	K	0	n
s_1	H	0	n	K
q	0	H	n	K

and the ordered rectangle sides are

$$(L, A), \qquad (L, R), \qquad (A, R). \tag{2.4.2}$$

The matching pairs are C, L in s_0 and R, C in s_1. Their values are again $0, n$ in increasing column order.

Finally, suppose $0 < t < n$. Let $A = x^{-1}(0)$ and $B = x^{-1}(n)$. Both columns are less than n, and they are distinct. No relative order between A and B is assumed. The values are

	A	B	R	C	L
p	0	n	K	t	H
s_0	H	n	K	0	t
s_1	H	K	0	n	t
s_2	0	K	H	t	n
q	0	n	H	t	K

(2.4.3)

and the ordered rectangle sides are

$$(L, A), \qquad (B, R), \qquad (A, R), \qquad (L, B). \tag{2.4.4}$$

The southwest columns of s_0 are C, L, with values $0, t$. Those of s_1 are R, C, L, with values $0, n, t$. Those of s_2 are C, L, with values t, n. In each source the first pair is increasing. The first internal step increases the northeast count from one to two, and the second decreases it to one. In particular, the consecutive matching sources differ. The transpositions in (2.4.4), followed by the specified matching inverses, give exactly (2.4.3).

It remains to verify emptiness of every rectangle. The open row interval of each rectangle is either $I_N(H, 0)$ or $I_N(n, K)$. The former consists of rows greater than H, and the latter consists of rows strictly between n and K. Every unchanged first-block row is at most n, so it belongs to neither interval. The exceptional high first-block value is K for a rectangle with row interval $I_N(H, 0)$, or H for a rectangle with row interval $I_N(n, K)$. It also lies outside the relevant open row interval. Endpoint values do not count as interior points.

For second-block columns, each rectangular column interior is contained in $\{D : D < R\} \cup \{D : D > L\}$. The changed low columns in the tables are endpoints or lie strictly between R and L, and hence do not provide an interior point. Every other second-block point in one of these rectangles has row strictly between K and H by (2.3.3). Such a row lies in neither of the two open row intervals. Equation (2.2.1) proves emptiness in all cases. The tables therefore define the required native paths. □

2.5 Every short path is one of these paths

Lemma 2.5.1 (New lemma: uniqueness for one transition)**.** *Let $n \geq 1$. For fixed x, z, w, the one-transition paths in $\mathcal{P}_1(x, z, w)$ are in bijection with $\mathcal{R}(x, z, w)$ when $x(n)$ is 0 or n. There are no such paths when $0 < x(n) < n$. Under the bijection, the inverse is* (2.4.1) *or* (2.4.2)*, respectively.*

Proof. If $z = w$, the supplied declaration `seamTransferPath_self_isEmpty` excludes every correction path, and a rectangle from z to itself cannot exist. Thus assume $z \neq w$. The declaration `seamOneStep_all_orientations` assigns one of four neutral types to a path. The declaration `seamSecondOneStep_kind_high` excludes the two low types because the second factor changes. For the high-first type, `seamSecondOneStep_high_first_boundary` gives $x(n) = 0$. The declarations `seamSecondOneStep_high_first_row_zero`, `seamSecondOneStep_high_first_row_last`, and `seamSecondOneStep_high_first_pivot` identify the two southwest rows as $0, n$ and the first-block pivot as $x^{-1}(n)$. The corresponding high-second declarations identify $x(n) = n$, the same extremal southwest rows, and pivot $x^{-1}(0)$. These declarations give the interior nonexistence assertion, also stated explicitly as `seamCorrectionOne_second_interior_isEmpty`.

The supplied definitions `seamSecondOneStepHighFirstRemaining` and `seamSecondOneSte pHighSecondRemaining` project these two cases to a member of $\mathcal{R}(x, z, w)$. They include proofs of emptiness, obstruction, and nonzero endpoint rows. In the first case, the projected rectangle has left column equal to the external high column and right column equal to the first southwest

column of the initial matching source. Its zero column is the second southwest column of that source. The source-value declarations `seamSecondOneStepHighFirst_source_values` therefore identify these columns as L, R, C of (2.3.1). The orientation declaration gives the three sides $(B, R), (L, R), (L, B)$, and its state values are the first table in Lemma 2.4.1. In the second case, the corresponding source-value declaration `seamSecondOneStepHighSecond_source_values` identifies the columns L, R, C and gives the sides $(L, A), (L, R), (A, R)$ and the second table there.

The projected rectangle and x thus determine every side and every state of the original path. Lemma 2.4.1 proves that these recovered data form a path for every remaining rectangle. The projection and construction are inverse maps. □

Lemma 2.5.2 (New lemma: normal form of a two-transition path)**.** *Let p, q be retained states, and let a two-transition correction join them. Write its matching sources as a_0, a_1, a_2. The initial source has a unique northeast column $A < n$, of row $H > n$, and two southwest columns $P < Q$, of rows $a < b \le n$. There are columns $C < n$ and $D \ge n$ such that*

$$C \neq A, \qquad D < P < Q, \qquad a < b < c \le n, \qquad a_0(C) = c, \qquad a_0(D) = K > n.$$

The first internal rectangle has sides (C, D). The second has sides (B, D), where B is either A or C. The entry has sides (Q, A), and the exit has sides (Q, E), where E is the remaining northeast column. If $B = C$, the first retained factors differ. If the first retained factors are equal, then $B = A$, $H > K$, and the retained endpoints differ by exchanging the high values in columns D, Q while fixing every other column.

Proof. The supplied declarations `seamMatchingEntry_northeast_card` and `seamMatchingExit_northeast_card` give one northeast point at each end. The declarations `seamMatchingEntry_southwest_card` and `seamMatchingExit_southwest_card` give two southwest points there. The supplied theorem `seamCorrectionTwo_middle_eq_two` gives two at the middle source. Thus the first step exchanges a low first-block column C with a high second-block column D, and the second exchanges one of the two high first-block columns $B \in \{A, C\}$ with a low second-block column. This assertion follows by counting the first-block high rows before and after a transposition. A transposition inside either block preserves that count, as formalized by `seamRectangle_northeast_card_eq_of_same_block`.

Before the first matching inverse, the low second-block columns are P, Q, D, with values a, b, c at those respective columns. If $Q < D$, the first pair has the increasing values a, b. After the matching inverse this pair would be decreasing, contrary to the matching-source condition for a_1. If $P < D < Q$, that condition requires $c < a$, and the values of a_1 in increasing low-column order are c, a, b. Deleting any one of these three columns leaves an increasing pair. The second matching inverse would make that pair decreasing. Thus this case admits no death step. The only possibility is $D < P < Q$. The values of a_1 in these columns are a, c, b, with $a < c$. Deleting P or Q again leaves an increasing pair. Hence the death removes D, and the remaining pair must satisfy $c > b$ before its inverse. This proves $a < b < c$.

The reversed orientation (D, C) for the birth would contain the point at P, whose row a is less than c. The reversed orientation (D, B) for the death would contain the point at P, whose row c is between a and the high value at B. Emptiness therefore forces (C, D) and (B, D).

Consider $B = C$ first. If $C < A$, the point at A obstructs the birth when $H < K$ and obstructs the death when $K < H$. The two values are distinct, so $A < C$. An entry of sides (A, P) would contain the point at C, whose retained row is $c \in (b, H)$. An exit of sides (A, P) would contain the point at C, whose final matching-source row is $a \in I_N(H, b)$. Hence both outer rectangles have their first-block endpoint on the right, namely (Q, A) and (Q, A). The retained source has rows a, c at A, C, and the retained target has rows c, a there. Since $a < c$, their first factors differ.

Now consider $B = A$. If $H < K$ and $C < A$, the point at A obstructs the birth. If $H < K$ and $A < C$, the point at C obstructs the death. Thus $K < H$. An entry (A, P) would contain D, whose row K lies strictly between b and H. An exit (C, P) would contain D, whose row H lies in the cyclic interval from K to b. The outer sides are therefore (Q, A) and (Q, C).

The values on the five relevant columns are now

	A	C	D	P	Q
p	a	c	K	b	H
a_0	H	c	K	a	b
a_1	H	K	a	c	b
a_2	a	K	H	b	c
q	a	c	H	b	K.

(2.5.1)

Every other column stays fixed. This proves the last assertion and completes the normal form. □

Lemma 2.5.3 (New lemma: uniqueness for two transitions)**.** *Let $n \geq 1$. For fixed x, z, w, the two-transition paths in $\mathcal{P}_2(x, z, w)$ are in bijection with $\mathcal{R}(x, z, w)$ when $0 < x(n) < n$. There are no such paths when $x(n)$ is 0 or n. The inverse bijection is* (2.4.4)*.*

Proof. Apply Lemma 2.5.2 to a path with first factor x at both endpoints. Its second case gives (2.5.1) and the sides

$$(Q, A), \qquad (C, D), \qquad (A, D), \qquad (Q, C).$$

The only low second-block value of the retained source is b, so $x(n) = b$.

We first prove $a = 0$. If a row less than a existed, its preimage in p would be a first-block column V, distinct from A, C. For $V < A$, the entry (Q, A) would contain that point. For $A < V$, the death (A, D) would contain that point. Its value is unchanged at these steps. Both alternatives contradict emptiness. Since the state contains row 0, one has $a = 0$. Similarly, if a row greater than c but at most n existed, its preimage would be a first-block column V distinct from A, C. For $C < V$, the birth (C, D) would contain it. For $V < C$, the exit (Q, C) would contain it. Thus $c = n$. In particular, $0 < x(n) < n$, $A = x^{-1}(0)$, and $C = x^{-1}(n)$.

The retained endpoints differ in columns D, Q, with rows $K < H$. Project the rectangle of sides (Q, D) to the second factor. Its endpoint rows are nonzero and its two intervals wrap. To prove factor emptiness, consider a second-block column $V < D$. A row greater than H there would obstruct the death, and a row less than K would obstruct the birth. For a second-block column $V > Q$, a row greater than H would obstruct the entry, and a row less than K would obstruct the exit. The low second-block column P is strictly between D and Q. Thus the projected rectangle is empty. Lemma 2.3.1 makes it a member of $\mathcal{R}(x, z, w)$.

This projection identifies D, P, Q with R, C, L of (2.3.1). Replacing the temporary first-block column C by $B = x^{-1}(n)$ transforms (2.5.1) into (2.4.3). It also transforms its four sides into (2.4.4). Hence the projection determines every state and rectangle of the original path. Lemma 2.4.1 proves that every remaining rectangle supplies that path. These maps are inverse. □

Theorem 2.5.4 (New lemma: complete second-sector short-path bijection)**.** *For every $n, m \in \mathbb{N}$, every $x \in \mathrm{GridState}(n+1)$, and every $z, w \in \mathrm{GridState}(m+1)$, there is a bijection*

$$\mathcal{P}_1(x, z, w) \sqcup \mathcal{P}_2(x, z, w) \longrightarrow \mathcal{R}(x, z, w).$$

Its inverse uses one transition when $x(n)$ is a boundary row and two transitions when $0 < x(n) < n$.

Proof. For $n \geq 1$, Lemmas 2.5.1 and 2.5.3 give mutually exclusive cases according to the value of $x(n)$. When $n = 0$, there are no first-block columns. Every state is retained, so there are no matching sources or correction paths. The seam state is z, its second-factor rectangle lift is the original rectangle, and no remaining rectangle is obstructed. All three sets are empty. □

2.6 Projected integral domains of a seam correction

All path coefficients remain in $\mathbb{F}_2$. Integral domains in this section record the multiplicities of grid squares before any parity is taken. Their use does not change the coefficient field of the chain complex.

For a grid of size h, an integral domain is an integer-valued function $D(c,r)$ on the h^2 squares. All cell and vertex indices in the next two lemmas are cyclic modulo h. Define its mixed difference and corner average by

$$(\partial D)(c,r) = D(c,r) + D(c-1,r-1) - D(c-1,r) - D(c,r-1),$$
$$(\mathcal{A}D)(c,r) = \tfrac{1}{4}\big(D(c,r) + D(c-1,r-1) + D(c-1,r) + D(c,r-1)\big).$$

For a state u, let $[u]$ be the indicator function of its points. A domain from u to v satisfies $\partial D = [u] - [v]$. Its index is

$$\iota(D;u,v) = \sum_{c=0}^{h-1} \big((\mathcal{A}D)(c,u(c)) + (\mathcal{A}D)(c,v(c))\big).$$

This is a definition by finite sums. In particular it makes sense for domains with negative coefficients.

Lemma 2.6.1 (New lemma: domain index identities)**.** *The following statements hold for integral domains on a toroidal grid.*

1. *If D is a domain from u to v and E is a domain from v to w, then*

$$\iota(D+E;u,w) = \iota(D;u,v) + \iota(E;v,w).$$

2. *A domain with mixed difference zero has the form $D(c,r) = a_c + b_r$, with integers a_c, b_r. Its index between any equal endpoint states is $2\sum_c a_c + 2\sum_r b_r$.*

3. *An empty native rectangle has index one. For any auxiliary grid with marking permutation O and any domain from u to v,*

$$\iota(D;u,v) = M_O(u) - M_O(v) + 2\sum_c D(c,O(c)). \tag{2.6.1}$$

   Here M_O is the native integer Maslov grading.

Proof. Finite cyclic reindexing gives

$$\sum_{c,r} (\partial D)(c,r)(\mathcal{A}E)(c,r) = \sum_{c,r} (\partial E)(c,r)(\mathcal{A}D)(c,r).$$

For completeness, after moving the differences from D to E, the coefficient of $D(c,r)$ on the left is

$$\tfrac{1}{4}\big(E(c+1,r+1) - E(c+1,r-1) - E(c-1,r+1) + E(c-1,r-1)\big).$$

Its coefficients are unchanged when both offsets are negated. This proves the displayed symmetry. Substituting $\partial D = [u]-[v]$ and $\partial E = [v]-[w]$ proves the first assertion by expanding the definition of ι.

If $\partial D = 0$, the difference $D(c,r) - D(c-1,r)$ is independent of r. Fixing one row and one column and summing these differences gives $D(c,r) = a_c + b_r$. Each state occupies every row and every column once. Substitution into the corner average therefore gives index $2\sum a_c + 2\sum b_r$.

The domain of a rectangle with sides (a,b) and source u is the indicator of

$$[a,b)_{\mathrm{cyc}} \times [u(a),u(b))_{\mathrm{cyc}}.$$

Its mixed difference is $[u] - [u \circ (a\ b)]$ by the native declaration `GridRectangleBetween.target_eq_swapColumns`. If the rectangle is empty, its four endpoint corners each contribute $1/4$ to the index and every other state point contributes zero. The native declaration `laurentMaslovDegree_add_card_OColumns` then proves (2.6.1) for an empty rectangle.

To obtain the formula for an arbitrary domain, connect u to v by adjacent-column transpositions. Such a sequence exists by moving the desired value into the first incorrect column and repeating on the remaining columns. Use the rectangle of width one for each transposition. It is empty because there is no interior column. Their sum D_0 has the required boundary, and the formula for D_0 follows by additivity and telescoping of the native grading differences. The difference $D - D_0$ has zero mixed difference. For a row or column annulus the index is two and the sum of the O multiplicities is one. The second assertion, with arbitrary integer coefficients, now gives the formula for $D - D_0$ and hence for D. □

Lemma 2.6.2 (New lemma: nonnegative domains of index one)**.** *Let D be a nonnegative integral domain from distinct states u to v. Then $\iota(D; u, v) \geq 1$. If its index is one, then D is the indicator domain of an empty native rectangle from u to v.*

Proof. Suppose that the states differ in e columns. Then $e \geq 2$, and the mixed difference has a coefficient 1 or -1 at each of their $2e$ distinct changed points. At each such point at least one adjacent cell has a positive integer coefficient. Its corner average is therefore at least $1/4$. All remaining contributions are nonnegative, so $\iota(D; u, v) \geq e/2 \geq 1$.

If the index is one, this bound gives $e = 2$. Write the changed columns as A, B, and put $R = u(A)$ and $S = u(B)$. The target exchanges these two rows. Let V and W be the nonempty proper cyclic cell intervals $[A, B)$ and $[R, S)$, and let D_0 be the indicator of $V \times W$. By Lemma 2.6.1,

$$D(c, r) = \mathbf{1}_V(c)\mathbf{1}_W(r) + a_c + b_r.$$

Replace a_c by its minimum separately on V and its complement, and replace b_r by its minimum separately on W and its complement. The resulting domain D_b is nonnegative, because on each of the four products of intervals its value is the minimum of D there. The difference $D - D_b$ is a sum of row and column annuli with nonnegative integer coefficients. The domain D_b has the same boundary, so its index is at least one by the preceding paragraph. Each annulus has index two. Additivity and $\iota(D) = 1$ force every annulus coefficient to be zero. Thus $D = D_b$ is constant on each of the four products of intervals.

Let the four constants be a, b, c, d, with a on $V \times W$ and d on its opposite product. They are nonnegative integers and satisfy $a + d - b - c = 1$. The four changed points contribute exactly $a + b + c + d$ to the index. Hence this sum is at most one, and the only possibilities are $(a, b, c, d) = (1, 0, 0, 0)$ and $(0, 0, 0, 1)$. These are the two oriented rectangle domains from u to v. The four corners already account for the full index. Every other source-state corner average is zero, so there is no source point in the rectangle's interior. This is the native emptiness condition `isEmpty_iff_forall_notMem_cIoo`. □

Now let $n, m \geq 1$ and $N = n + m + 1$. For a correction path from retained state $p = \texttt{seamState}(x, z)$ to retained state $q = \texttt{seamState}(y, w)$, define D to be the sum of its entry, internal, and exit rectangle domains, minus the domains of the selected matching rectangles at all its matching sources.

Lemma 2.6.3 (New lemma: the signed correction domain)**.** *The signed domain D of a correction path satisfies $\partial D = [p] - [q]$ and*

$$\iota(D; p, q) = 1. \tag{2.6.2}$$

Proof. The matching rectangle at a source with southwest columns $P < Q$ and rows $a < b \leq n$ is $[P, Q) \times [a, b)$. It is empty, because every column strictly between the first two southwest columns has a high row. Telescoping the rectangle boundaries gives $\partial D = [p] - [q]$. There is one more positive rectangle than subtracted matching rectangles. Each has index one, so additivity in Lemma 2.6.1 gives (2.6.2). □

Define two restrictions, with the indicated factor indices, by

$$\begin{aligned} D_1(i,j) &= D(i,j) && (0 \le i < n,\ 0 \le j \le n),\\ D_1(n,j) &= D(N-1,j) && (0 \le j \le n),\\ D_2(i,j) &= D(n+i,n+j) && (0 \le i,j \le m). \end{aligned} \tag{2.6.3}$$

Put $d = D(N-1,n)$.

Lemma 2.6.4 (New lemma: positive projections and the index relation)**.** *For every correction path as above, D_1 is a nonnegative domain from x to y, and D_2 is a nonnegative domain from z to w. Their indices satisfy*

$$\iota(D_1;x,y) + \iota(D_2;z,w) = 1 + 2d. \tag{2.6.4}$$

Proof. A matching rectangle uses only columns from n through $N-2$ and rows from 0 through $n-1$. It therefore misses both restrictions in (2.6.3). All remaining summands of D are nonnegative rectangle indicators, proving positivity.

For the boundary assertion, restricting a domain to cyclic lists of cell columns c_i and cell rows r_j replaces its mixed difference at (i,j) by

$$\sum_{c \in (c_{i-1},c_i]_{\mathrm{cyc}}} \ \sum_{r \in (r_{j-1},r_j]_{\mathrm{cyc}}} (\partial D)(c,r).$$

This formula follows by telescoping in both coordinates. For D_1, the last column collects the original columns $n,\dots,N-1$, and row zero collects the original rows $n+1,\dots,N-1,0$. Under these collections the retained state p becomes $[x] + m[(n,0)]$. The extra m copies are the high points in the second columns. The retained state q gives $[y] + m[(n,0)]$, so the extra terms cancel. For D_2, column zero collects columns $0,\dots,n$, and row zero collects rows $0,\dots,n$. The retained states become $[z] + n[(0,0)]$ and $[w] + n[(0,0)]$. Again the extra terms cancel. These statements use precisely the three supplied coordinate formulas `seamState_left_castSucc`, `seamState_right_of_zero`, and `seamState_right_of_ne_zero`.

We prove the index relation by explicit auxiliary markings. Let $O_1(i) = i$ on the first factor, let $O_2(j) = j+1 \pmod{m+1}$ on the second factor, and let

$$O(c) = c \quad (c < n), \qquad O(n+j) = n + O_2(j) \quad (0 \le j \le m).$$

Each is a permutation and can be made into a valid auxiliary grid by choosing X to be its row successor. The Maslov grading depends only on the specified O markings. The full marking set is the union of the two projected marking sets with their common square $(N-1,n)$ counted once. Consequently

$$\sum_c D(c,O(c)) = \sum_i D_1(i,O_1(i)) + \sum_j D_2(j,O_2(j)) - d. \tag{2.6.5}$$

The native Maslov point-count formula gives

$$M_O(\texttt{seamState}(x,z)) = M_{O_1}(x) + M_{O_2}(z). \tag{2.6.6}$$

Here is a direct verification, so no condition on the auxiliary X markings is being suppressed. For finite planar point sets A, B, write $I(A,B)$ for the number of ordered pairs that increase strictly in both coordinates. State points have integral coordinates and marks have both coordinates shifted by $1/2$. The supplied formula `GridDiagram.maslovO`$\mathbb{Z}$`_eq_card` is

$$M_O(s) = I(s,s) - I(s,O) - I(O,s) + I(O,O) + 1.$$

Splitting each point set into its first n and last $m+1$ columns gives

$$\begin{aligned} I(p,p) &= I(x,x) + I(z,z) + nm,\\ I(O,O) &= I(O_1,O_1) + I(O_2,O_2) + nm,\\ I(p,O) &= I(x,O_1) + I(z,O_2) + nm - 1,\\ I(O,p) &= I(O_1,x) + I(O_2,z) + nm. \end{aligned}$$

To check the first equality, put $t = x(n)$. Removing the last column from x removes t increasing pairs. The cross pairs with second-column points contribute $nm + t$. The second equality is the same computation for the marking permutations. For the third equality, removing the last marking of O_1, whose row is n, removes $n + 1$ pairs. All $n(m + 1)$ pairs from first-column state points to second-column markings contribute. The pairs within the second columns agree with $I(z, O_2)$ because the low state row t is at most the lowest marking row $n + 1/2$. For the fourth equality, removing the last state point of x removes t pairs, whereas the cross pairs contribute $nm + t$. Substituting the four identities into the native formula proves (2.6.6).

Apply (2.6.1) to the three domains. Equations (2.6.2), (2.6.5), and (2.6.6) give (2.6.4). □

Lemma 2.6.5 (New lemma: the fixed-first-factor domain)**.** *Suppose that $x = y$ and $z \neq w$. Then D_2 is the indicator of an empty second-factor rectangle, $d \in \{0, 1\}$, and*

$$D_1(i, j) = d\,\mathbf{1}_{\{n\}}(j). \tag{2.6.7}$$

Furthermore, for every first column $c < n$ and every row r,

$$D(c, r) = D(N - 1, r). \tag{2.6.8}$$

Proof. Since D_1 has equal endpoints, write $D_1(i, j) = a_i + b_j$. Subtract the minimum of the a_i from every a_i and add it to every b_j. Then all a_i and all b_j are nonnegative, because D_1 is nonnegative. Lemma 2.6.1 gives

$$\iota(D_1) = 2\sum_i a_i + 2\sum_j b_j \geq 2(a_n + b_n) = 2d.$$

Lemma 2.6.2 gives $\iota(D_2) \geq 1$. The index relation forces equality in both inequalities. Thus D_2 is an empty rectangle and $d \in \{0, 1\}$, and every a_i with $i < n$ and every b_j with $j < n$ vanishes. Write $\alpha = a_n$ and $\beta = b_n$, so that $\alpha + \beta = d$.

For a first column $c < n$, the retained endpoints agree at that column. Hence $\partial D(c, r) = 0$ for every row r. Summing this equality over columns $0, \ldots, c$ shows that $D(c, r) - D(N - 1, r)$ is independent of r. At row n this difference equals $\beta - (\alpha + \beta) = -\alpha$. For $r \geq n$, matching rectangles contribute zero, so $D(c, r) \geq 0$. It follows that

$$D_2(m, j) = D(N - 1, n + j) \geq \alpha \qquad (0 \leq j \leq m).$$

If $\alpha = 1$, the empty rectangle D_2 would cover its last column in every row. Its cyclic row interval is proper, which makes this impossible. Therefore $\alpha = 0$, $\beta = d$, and both conclusions follow. □

2.7 A finite order certificate on the five-point seam grid

Lemma 2.7.1 (New lemma: the five-point exclusion)**.** *Let $n = m = 2$ and let p, q be arbitrary retained indices. Every native complete correction path from p to q has at most two internal transitions.*

Proof. A state is written as its five rows, in column order. For example 03142 denotes the permutation $(0, 3, 1, 4, 2)$. All digits in this proof are single row indices, rather than an integer written in decimal notation. An arrow annotated (a, b) denotes the actual empty rectangle with ordered column sides a, b, followed by the selected inverse matching. In particular the orientation of every rectangle in the certificate is specified.

We first give an order calculation that makes the certificate exhaustive. Every matching source with northeast count one has the following data:

$$A \in \{0, 1\}, \quad h \in \{3, 4\}, \quad L = 1 - A, \quad s(A) = h, \quad s(L) = t,$$
$$\{P, Q, C\} = \{2, 3, 4\}, \quad P < Q,$$
$$s(P) = v < s(Q) = w, \quad s(C) = 7 - h, \quad \{v, w, t\} = \{0, 1, 2\}.$$

Thus there are exactly $2 \cdot 2 \cdot 3 \cdot 3 = 36$ such sources. The native theorem `flatStep_cases` shows that a transition which preserves northeast count is one of the four flat types below. A birth can only exchange (L, t) with $(C, 7-h)$. Its target has a decreasing first southwest pair only if either $C < P$ and $t > v$, or $P < C < Q$ and $t < v$. In the first case the wrapping-column orientation contains (P, v). In the second case the nonwrapping-column orientation contains (P, v). These facts leave the two birth orientations in the table. A death from a source with northeast count one reaches a retained state, so it is an exit rather than an internal transition. Consequently the following six rows include every possible internal transition from any of the 36 sources.

type	ordered columns	necessary and sufficient conditions
L_+	(L, P)	$t > w, \quad A = 0, \quad C > P$
L_-	(Q, L)	$t < v, \quad A = 1, \quad C < Q$
H_+	(C, P)	$C > Q, \quad h = 3, \quad v = 0$
H_-	(Q, C)	$C < P, \quad h = 4, \quad w = 2$
B_+	(L, C)	$C < P, \quad t > v, \quad (A = 0 \text{ or } h = 4)$
B_-	(C, L)	$P < C < Q, \quad t < v, \quad (A = 1 \text{ or } h = 3)$

(J06.F5.1)

Here is the emptiness check for every row of this table, including its excluded cases. For L_+, the row data must be $(v, w, t) = (0, 1, 2)$. The cyclic row interior consists of rows $3, 4$. The northeast point blocks precisely when $A > L$, and the southeast point blocks precisely when $C < P$. This gives $A = 0$ and $C > P$. For L_-, the row data must be $(t, v, w) = (0, 1, 2)$, and the same two high rows form the cyclic interior. Its northeast blocker occurs when $A < L$, and its southeast blocker occurs when $C > Q$. This gives $A = 1$ and $C < Q$. For H_+, the column data are $(P, Q, C) = (2, 3, 4)$. The only interior columns are $0, 1$. Their two points avoid the wrapping row interval exactly when $h < 7 - h$ and $t > v$. These conditions are $h = 3$ and $v = 0$. For H_-, the column data are $(C, P, Q) = (2, 3, 4)$, again with interior columns $0, 1$. The necessary and sufficient inequalities are $h > 7 - h$ and $t < w$, namely $h = 4$ and $w = 2$. For B_+, the column data are $(C, P, Q) = (2, 3, 4)$. If $L = 1$ there is no interior column. If $L = 0$, its only interior column is $A = 1$, whose high row blocks exactly when $h < 7 - h$. For B_-, the column data are $(P, C, Q) = (2, 3, 4)$ and $t = 0$. Column 4 contains the low row $w = 2$, which is outside the wrapping row interval. The only possible further interior point is at column $A = 0$ when $L = 1$, and it blocks exactly when $h > 7 - h$. These checks prove all six equivalences in (J06.F5.1). The other orientation of each flat move is already excluded by `flatStep_cases`.

An entry can reach only a matching source with northeast count one, by `seamMatchingEntry_northeast_card`. Two of the 36 sources admit no entry. For 04312, its possible entry rectangles are blocked respectively by the points $(2, 3)$ and $(0, 0)$. For 32014, they are blocked respectively by $(1, 2)$ and $(4, 4)$. These are the two candidates in `flatEntry_data`. Let E_0 be the 34 other northeast-count-one sources. Every actual entry source belongs to E_0. No assertion that every member is reached is required for the argument.

Substitution of the three choices of (v, w, t), the three choices of (P, Q, C), and the two choices each of A, h in (J06.F5.1) gives the following complete list of nonempty outgoing internal-arrow sets for sources in E_0. The rows not displayed have no internal arrow.

03142	$(3,0) : 43012,\ (4,0) : 23041$	03412	$(4,0) : 23401$
04132	$(3,0) : 34012,\ (4,0) : 24031$	13024	$(4,2) : 13402$
14302	$(0,2) : 34012,\ (4,2) : 14023$	23014	$(4,2) : 23401$
24301	$(0,2) : 34021$	30142	$(3,1) : 34012$
31024	$(4,2) : 31402$	31402	$(1,2) : 34012$
32041	$(1,2) : 30142$	32401	$(1,2) : 34021$
40312	$(4,2) : 40123$	41302	$(1,2) : 43012,\ (4,2) : 41023$
42013	$(1,2) : 40123$	42031	$(1,2) : 40132$
42301	$(1,2) : 43021$		

(J06.F5.2)

The target in each cell is obtained by swapping the two displayed columns and then swapping the first two southwest columns. For example the row 24301 uses the birth with sides $(0, 2)$. The

actual target is 34201, whose southwest rows at columns $2,3$ are $2,0$. Its selected partner is 34021, as displayed. Thus (J06.F5.2) records state maps as well as all the emptiness and source tests in (J06.F5.1).

Only four sources with northeast count two occur among these targets: 34012, 34021, 43012, and 43021. Their complete internal-arrow sets are

$$\begin{array}{l|l} 34012 & \varnothing \\ 43012 & \varnothing \\ 34021 & (1,2) : 30412 \\ 43021 & (0,2) : 03412,\ \ (1,2) : 40312. \end{array} \tag{J06.F5.3}$$

We verify this table without leaving any orientation unchecked. There is no northwest or southeast point, so there is no birth and no flat exchange with either of those quadrants. Swapping the two northeast points leaves the first southwest pair increasing. Swapping the first two southwest points is the selected matching and is excluded from an internal arrow. An interchange of the second and third southwest points also leaves the first pair increasing. An interchange of the first and third can yield a decreasing target pair only for southwest row order 012. Its nonwrapping rectangle then contains the second southwest point, and its wrapping rectangle contains a northeast point. Thus no flat internal exchange remains.

A death must remove column 2, since removing either of the other two southwest columns leaves an increasing first pair. Its remaining pair is decreasing exactly when the southwest order is 021. The wrapping-column orientation contains a southwest point. For the nonwrapping orientation $(A,2)$, $A=1$ has no interior column and is empty. For $A=0$, column 1 blocks exactly when its high row is greater than the high row in column 0. This gives precisely the three arrows in (J06.F5.3) and their displayed targets.

Let E_j be the set of sources reached by j successive internal arrows starting from E_0. The two explicit arrow tables now give

$$\begin{aligned} E_1 &= \{13402, 14023, 23041, 23401, 24031, 30142, 31402, \\ &\qquad 34012, 34021, 40123, 40132, 41023, 43012, 43021\}, \\ E_2 &= \{03412, 30412, 34012, 40312\}, \\ E_3 &= \{23401, 40123\}, \qquad E_4 = \varnothing. \end{aligned} \tag{J06.F5.4}$$

These are finite set substitutions in the displayed tables, rather than an inference from tests on further sizes. For completeness neither member of E_3 has an exit. For 23401, the two exit rectangles in `flatExit_data` contain respectively $(2,4)$ and $(0,2)$. For 40123, they contain respectively $(1,0)$ and $(4,3)$. Thus a path with exactly three internal transitions cannot exit. A path with four or more internal transitions would have to pass through E_4, which is empty. This proves the lemma. □

2.8 Uniform reduction of the second sector to five points

The domains and indices in this section are those of Lemmas 2.6.1–2.6.5. In particular the cells are indexed by their lower left corners, and the value at a state point is the average of the four adjacent cells.

Lemma 2.8.1 (New lemma: points forced to remain fixed)**.** *Let $n, m \geq 1$, and let a correction path have retained endpoints $p = \texttt{seamState}(x, z)$ and $q = \texttt{seamState}(x, w)$ with $z \neq w$. Put $t = x(n)$, and let R be the empty second-factor rectangle whose domain is D_2. Then the following points of p remain in their original columns throughout every matching source, matching target, and retained endpoint of the path.*

1. *Every first-column point whose row lies strictly between 0 and n.*

2. *Every high second-column point whose second-factor point is not one of the two source corners of R.*

Proof. Equation (2.6.7) gives

$$D(c, r) = 0 \qquad (c \in \{0, \dots, n-1, N-1\},\ 0 \le r < n).$$

A first-column state point of row strictly between 0 and n has all four adjacent cells in this set. This includes column zero, whose left adjacent cell column is $N-1$. Every matching rectangle avoids these cell columns. Thus each actual entry, internal, or exit rectangle has coefficient zero at all four cells.

Consider next a high second-column point (c, r), with $c \ge n$ and $r > n$. If $c > n$, all four adjacent cells lie in the second projection. If $c = n$, the adjacent cell column $n-1$ is identified with column $N-1$ by (2.6.8). In both cases the four full-domain coefficients equal the four adjacent coefficients of D_2 at the corresponding second-factor point. For an unchanged source point of the empty rectangle R, all four of those coefficients are zero. Indeed the four changed rectangle corners already contribute its entire index one, so every other source point has zero corner average, as proved in Lemma 2.6.2. The coefficients are nonnegative. Matching rectangles avoid every row of these four cells, because the adjacent cell rows are at least n. Hence every actual rectangle again has coefficient zero at all four cells.

In either case the point cannot change for the first time at an actual rectangle. Such a change would make it a source corner, and at least one of its adjacent cells would occur in that rectangle. A matching inverse does not change a first-column point or a high-row point. Induction over the finite ordered list of rectangles and matching inverses proves that each stated point remains fixed throughout the path. □

Lemma 2.8.2 (New lemma: compression of a correction path)**.** *Under the hypotheses of Lemma 2.8.1, every correction path of length k induces a correction path of the same length for seam parameters*

$$n' \in \{1, 2\}, \qquad m' \in \{1, 2\},$$

with equal first retained factors at its two endpoints.

Proof. Select every first column whose original row is 0 or n. There are $a \in \{1, 2\}$ such columns. There is only one when $t = x(n)$ equals 0 or n, and there are two otherwise. Select the second column C carrying row t, together with the source-corner columns of R. Their union has $b \in \{2, 3\}$ columns. It has two when one source-corner row of R is zero, and three otherwise.

Lemma 2.8.1 fixes every original point outside these $a + b \le 5$ selected columns. Since each intermediate state is a permutation, its restriction to the selected columns is a bijection onto their original row values. Those values comprise $a + 1$ low rows and $b - 1$ high rows. The low rows are $0, n$, together with t if $0 < t < n$. The high rows are the nonzero source rows of R, translated by n.

Relabel the selected first columns in increasing order by $0, \dots, a-1$, and the selected second columns in increasing order by $a, \dots, a+b-1$. Relabel the low rows in increasing order by $0, \dots, a$, and the high rows in increasing order by $a+1, \dots, a+b-1$. Each intermediate state becomes a state of a seam grid with $n' = a$ and $m' = b - 1$.

Every actual rectangle has both endpoint columns among the selected columns, since otherwise it would change a point that has been proved fixed. Its two endpoint rows belong to the selected set as well. An order-preserving cyclic relabeling preserves membership of selected indices in the cyclic open interval between two selected endpoints. This statement includes a wrapping interval: its members are exactly those after the left endpoint or before the right endpoint in the ordinary order, before and after relabeling. Therefore the restricted rectangle is empty, because an interior point after restriction would have been an interior point before restriction. Its target is the restricted transposed state, by `GridRectangleBetween.target_eq_swapColumns`.

Every southwest column belongs to the selected set. The remaining second-column points are high and fixed. Thus the first two southwest columns are preserved by restriction, and restriction commutes with matching partners. It preserves the strict inequality between their low rows. The matching sources remain complementary, because all northeast points lie in selected first columns. Consecutive restricted matching sources remain distinct, since every unselected column is fixed.

The restricted retained endpoints are seam states for the new parameters. Their first-factor permutations agree, because their values on all selected first columns agree and hence their missing low row also agrees. The restricted entry, internal rectangles, and exit form the required correction path of the same length. □

Lemma 2.8.3 (New lemma: the three smaller reduced grids)**.** *If (n, m) is $(1, 1)$, $(1, 2)$, or $(2, 1)$, then no native internal path between matching sources has two or more transitions.*

Proof. Every complementary state has exactly one northeast point, since its count is positive and at most $\min(n, m) = 1$. Write that point as (A, H) and the southwest points as $(P, a), (Q, b)$, where $P < Q$ and $a < b$. The existing declarations `flatStep_cases` and `flatStep_apply` classify each internal transition by an additional column C and its row c:

type	range	ordered sides	new southwest data
L_+	$C < n,\ c > b$	(C, P)	$(P, b), (Q, c)$
L_-	$C < n,\ c < a$	(Q, C)	$(P, c), (Q, a)$
H_+	$C > Q,\ c > n$	(C, P)	$(Q, a), (C, b)$
H_-	$n \le C < P,\ c > n$	(Q, C)	$(C, a), (P, b)$.

The row left at C in the low cases is respectively a and b. The row left at the removed southwest column in the high cases is c. Testing (A, H) against the rectangle interior gives the further necessary inequalities

$$L_+ : A < C, \quad L_- : C < A, \quad H_+ : H < c, \quad H_- : c < H.$$

For the two low types the wrapping row interval contains H, so the column of the northeast point must be outside the wrapping or nonwrapping column interval, respectively. For the two high types the wrapping column interval contains A, so its row must be outside the specified row interval. These are exact applications of `isEmpty_iff_forall_notMem_cIoo`.

For $(n, m) = (1, 1)$ there is no additional column, so no internal transition is possible. For $(n, m) = (1, 2)$ the additional column is the unique southeast column, so only H_+ or H_- can occur. After H_+, the only southeast column is the old P, with row $c > H$. It lies before the new first southwest column, so a second H_+ is impossible by its column range. A second H_- would require $c < H$, a contradiction. After H_- the only southeast column is the old Q, with row $c < H$. It lies after the new second southwest column. A second H_- is excluded by its column range, and H_+ would require $H < c$.

For $(n, m) = (2, 1)$ the additional column is the unique northwest column, so only L_+ or L_- can occur. After L_+ its row is a, and the southwest rows are b, c with $a < b < c$. A second L_+ fails its row range. A second L_- would require $C < A$, contradicting the first transition's inequality $A < C$. After L_- the additional row is b, and the southwest rows are c, a with $c < a < b$. A second L_- fails its row range, while L_+ would require $A < C$, contradicting $C < A$. This exhausts all possibilities for two consecutive transitions. □

Theorem 2.8.4 (New lemma: complete second-sector path truncation)**.** *For every $n, m \in \mathbb{N}$, every first-factor state x, every second-factor states z, w, and every $k \ge 3$, the native type*

$$\texttt{SeamCorrectionPath}(k, (x, z), (x, w))$$

has no element.

Proof. If $n = 0$, there is no first column and hence no northeast point. If $m = 0$, there is no high row. In either case every state is retained, so no correction path can have a matching source. Assume $n, m \geq 1$.

If $z = w$, the full domain D has equal endpoints. Its mixed difference is zero, so its index is an even integer by Lemma 2.6.1. This contradicts (2.6.2). Assume $z \neq w$, and let $R = D_2$ be the empty second-factor rectangle from Lemma 2.6.5.

Lemma 2.8.2 produces a correction path of the same length with $n', m' \in \{1, 2\}$. If $n' = m' = 2$, Lemma 2.7.1 excludes $k \geq 3$. Each of the other three parameter pairs is excluded by Lemma 2.8.3. The compression includes both $t = 0, n$ and zero rectangle-endpoint rows, as specified in its proof. This exhausts every case. □

Corollary 2.8.5 (New lemma: J06 acceptance, with integer equality)**.** *For every n, m, x, z, w,*

$$\texttt{seamPositiveCorrectionCount}((x, z), (x, w)) = \left|\texttt{SeamSecondRemainingRectangle}(x, z, w)\right|$$

as an equality of natural numbers. In particular this equality holds in $\mathbb{F}_2$, which is the J06 acceptance statement and record M605.

Proof. Theorem 2.5.4 identifies the disjoint union of the one-transition and two-transition path sets with the remaining second-factor rectangles. Theorem 2.8.4 excludes every longer path. Substitute these facts into the existing declaration `seamPositiveCorrectionCount_one_two_long`. □

2.9 Exceptional cases and explicit coordinate checks

The case $n = 0$ is proved in Theorem 2.5.4 and in fact proves (2.1.1), because there are no matching sources. If $m = 0$, every row is at most n. Again every state is retained, there are no matching sources, and the second factor has no rectangle. Thus (2.1.1) also holds.

If $z = w$, all correction paths are excluded by `seamTransferPath_self_isEmpty`, and $\mathcal{R}(x, z, z)$ is empty. More generally, if z, w have the same permutation sign, the supplied `seamTransferPath_factor_sign` excludes correction paths with fixed first factor. The supplied `gridRectangle_sign` excludes factor rectangles between them. Hence (2.1.1) holds in this entire case.

The cases $x(n) = 0$ and $x(n) = n$ are included explicitly in Lemma 2.8.2. They reduce to a seam with one first column. A zero source-corner row of the projected rectangle reduces to a seam with only one high row. Lemma 2.8.3 proves the required exclusions in both situations. When $n = 1$, the interior alternative $0 < x(n) < n$ is empty. When $m \leq 1$, two distinct nonzero second-factor endpoint rows do not exist, so $\mathcal{R}$ is empty. These conclusions agree with the explicit short-path construction.

For a concrete boundary example, let $n = 1$, $m = 2$, $x = (1, 0)$, $z = (1, 0, 2)$, and $w = (2, 0, 1)$, where tuples list row values in increasing column order. The only remaining rectangle has factor sides $(2, 0)$. Its seam path has matching sources

$$(2, 0, 1, 3), \qquad (2, 3, 0, 1)$$

and rectangle sides

$$(0, 1), \qquad (3, 1), \qquad (3, 0).$$

These values follow by substitution into (2.4.1). The emptiness proof in Lemma 2.4.1 applies to this example as a proof, without a numerical inference.

For the first possible interior example, take $n = m = 2$, $x = (0, 2, 1)$, $z = (1, 0, 2)$, and $w = (2, 0, 1)$. The retained endpoints are $(0, 2, 3, 1, 4)$ and $(0, 2, 4, 1, 3)$. The matching sources in (2.4.3) are

$$(4, 2, 3, 0, 1), \qquad (4, 3, 0, 2, 1), \qquad (0, 3, 4, 1, 2),$$

and the four ordered sides are

$$(4,0), \qquad (1,2), \qquad (0,2), \qquad (4,1).$$

This checks explicitly the orientation, the use of the first two southwest columns at the middle source, and the meaning of internal length two.

2.10 Lemma and declaration table

All existing declarations below occur in the supplied listings, either as definitions and proved theorems or as imported declarations explicitly used in those proofs. No theorem in this document is a Lean declaration.

Declaration or result	Status	Use
`seamState_left_castSucc`, `seamState_right_of_zero`, `seamState_right_of_ne_zero`	Existing	Retained coordinates (2.2.2).
`SeamMatchingSource`, `SeamInternalRectangle`, `SeamInternalPath`, `SeamCorrectionPath`	Existing	Native path types and their length convention.
`bottom_def`, `top_def`, `target_eq_swapColumns`, `isEmpty_iff_forall_notMem_cIoo`	Existing	Coordinate interpretation and emptiness.
`SeamSecondRemainingRectangle`, `seamSecondRemainingRectangle_wraps`	Existing	Remaining-rectangle definition and wrapping columns.
`seamOneStep_all_orientations`, `seamSecondOneStep_kind_high`	Existing	Exhaustion of second-sector one-transition types.
`seamSecondOneStep_high_first_boundary`, `seamSecondOneStep_high_second_boundary`	Existing	Boundary values of $x(n)$.
`seamSecondOneStep_high_first_row_zero`, `seamSecondOneStep_high_first_row_last`, `seamSecondOneStep_high_second_row_zero`, `seamSecondOneStep_high_second_row_last`	Existing	Extremal southwest rows for one transition.
`seamSecondOneStep_high_first_pivot`, `seamSecondOneStep_high_second_pivot`	Existing	First-block pivot reconstruction.
`seamSecondOneStepHighFirstRemaining`, `seamSecondOneStepHighSecondRemaining`, `seamSecondOneStepHighFirst_source_values`, `seamSecondOneStepHighSecond_source_values`	Existing	One-transition projection and its coordinates.
`seamMatchingEntry_northeast_card`, `seamMatchingExit_northeast_card`, `seamMatchingEntry_southwest_card`, `seamMatchingExit_southwest_card`, `seamCorrectionTwo_middle_eq_two`, `seamRectangle_northeast_card_eq_of_same_block`	Existing	Birth followed by death in every two-transition path.
`seamPositiveCorrectionCount_one_two_long`, `seamLongCorrectionCount`, `seamInternalPath_length_lt`	Existing	Exact splitting of the bounded count.
`seamTransferPath_self_isEmpty`, `seamTransferPath_factor_sign`, `gridRectangle_sign`	Existing	Equal endpoints and equal-sign cases.
Lemma 2.3.1	New, proved	Coordinate characterization of all remaining rectangles.

Declaration or result	Status	Use
Lemma 2.4.1	New, proved	Explicit inverse paths and all emptiness checks.
Lemma 2.5.1	New, proved	One-transition inverse and uniqueness.
Lemma 2.5.2	New, proved	Exhaustive two-transition normal form.
Lemma 2.5.3	New, proved	Interior second-sector projection and uniqueness.
Theorem 2.5.4	New, proved	Complete short-path bijection.
`flatStep_cases`, `flatStep_apply`, `flatEntry_data`, `flatExit_data`	Existing	Exhaustive flat transitions and finite-grid entry and exit tests.
`laurentMaslovDegree_add_card_OColumns`, `GridDiagram.maslov0ℤ_eq_card`	Existing	Native rectangle index and Maslov point-count formulas.
Lemma 2.6.1	New, proved	Additivity of the corner-average index and periodic-domain formula.
Lemma 2.6.2	New, proved	Every nonnegative domain of index one is an empty rectangle.
Lemma 2.6.3	New, proved	The signed domain of every correction has index one.
Lemma 2.6.4	New, proved	Positive factor projections and the exact index sum.
Lemma 2.6.5	New, proved	Fixed-first-factor projection and the boundary-row annulus.
Lemma 2.7.1	New, proved	Complete five-point transition certificate.
Lemmas 2.8.1 and 2.8.2	New, proved	Fixed points and a path-preserving reduction to at most five points.
Lemma 2.8.3	New, proved	The three smaller reduced grids.
Theorem 2.8.4 and Corollary 2.8.5	New, proved	Absence of all longer paths and the full J06 acceptance.

2.11 Remaining gap

There is no remaining mathematical gap in the J06 acceptance statement under the supplied native coordinate conventions. Theorem 2.8.4 excludes every individual correction with at least three internal transitions. Theorem 2.5.4 identifies every shorter positive correction with exactly one remaining second-factor rectangle. Corollary 2.8.5 therefore proves the stronger integer equality (2.1.2) and removes M605. This is a written proof and does not assert that the new lemmas have been formalized in Lean.

Chapter 3

J07: Mixed-sector seam parity

3.1 Exact statement and conventions

All chain coefficients in this document are in $\mathbb{F}_2$. Integral multiplicities will be used only to record unions and differences of rectangle domains before any path cardinality is reduced modulo two. No change of the coefficient field is involved.

Theorem 3.1.1 (New lemma: mixed-sector exclusion at every length)**.** *For every $n, m, k \in \mathbb{N}$ and every two retained indices $p = (p_1, p_2)$ and $q = (q_1, q_2)$, if $p_1 \neq q_1$ and $p_2 \neq q_2$, then the native type*

$$\mathcal{P}_k(p, q) = \texttt{SeamCorrectionPath}\ k\ p\ q$$

is empty. Consequently the exact acceptance statement of J07 holds:

$$(\texttt{seamPositiveCorrectionCount}(p, q) : \mathbb{F}_2) = 0. \tag{3.1.1}$$

The theorem proves catalogue record M606, including all long-path contributions in the mixed sector. It does not assert the universal truncation statement M607 when a factor is fixed. The proof applies to each chosen path, so it proves that the integer number of mixed correction paths is zero before reduction to $\mathbb{F}_2$. It does not require an involution pairing paths or a bound on their length.

Put $N = n + m + 1$ and identify the native index set $\mathrm{Fin}(N)$ with $\{0, \ldots, N-1\}$. A state is a permutation from columns to rows. The native declarations `seamState_left_castSucc`, `seamState_right_of_zero`, and `seamState_right_of_ne_zero` give the following retained state:

$$s_p(c) = \begin{cases} p_1(c), & 0 \leq c < n, \\ p_1(n), & c = n + d \text{ and } p_2(d) = 0, \\ n + p_2(d), & c = n + d \text{ and } p_2(d) \neq 0. \end{cases} \tag{3.1.2}$$

Thus every retained point in a column $c < n$ has row at most n.

For a complementary state u, its southwest columns are the columns $c \geq n$ with $u(c) \leq n$. If its first two southwest columns are $P < Q$, the native matching partner τu exchanges their rows. This is the operation `seamComplementPartner`. A matching source has $u(P) = a < b = u(Q)$. Its matching rectangle has cell domain $[P, Q) \times [a, b)$. These are the coordinate meanings of `SeamMatchingSource`, `seamComplementFirst`, `seamComplementSecond`, and `seamComplementRectangle` used explicitly in the supplied seam listings and described in Section 2.3 of the supplied I5 notes. The two defining modules `GridConnectedSumSeamSector` and `GridConnectedSumSeamMatching` are referenced but not reproduced in the combined attachment. The proof uses the coordinate descriptions just stated, which also appear in the supplied proofs involving those objects.

For distinct indices a, b, the notation $[a, b)_{\mathrm{cyc}}$ means the cyclic interval containing a and ending immediately before b. Its open version is denoted by $(a, b)_{\mathrm{cyc}}$. An ordered rectangle from a state u with column sides (a, b) has cell domain

$$[a, b)_{\mathrm{cyc}} \times [u(a), u(b))_{\mathrm{cyc}}$$

and target obtained by exchanging the rows at a, b. It is empty when no other state point lies in the product of the two open intervals. These are the supplied conventions of `GridRectangleBetween.target_eq_swapColumns`, `GridRectangleBetween.bottom_def`, `GridRectangleBetween.top_def`, and `isEmpty_iff_forall_notMem_cIoo`. The path definitions impose no restriction on marking squares.

By the native definitions `SeamInternalRectangle`, `SeamInternalPath`, and `SeamCorrectionPath`, a correction with k internal transitions consists of matching sources $u_0, \dots, u_k$, an entry rectangle $s_p \longrightarrow \tau u_0$, internal rectangles $u_j \longrightarrow \tau u_{j+1}$, and an exit rectangle $u_k \longrightarrow s_q$. All these rectangles are empty, and successive internal sources are unequal. The argument below uses the selected rectangles themselves.

3.2 Projected integral domains of a seam correction

All path coefficients remain in $\mathbb{F}_2$. Integral domains in this section record the multiplicities of grid squares before any parity is taken. Their use does not change the coefficient field of the chain complex.

For a grid of size $h \geq 2$, an integral domain is an integer-valued function $D(c, r)$ on the h^2 squares. All cell and vertex indices in the next two lemmas are cyclic modulo h. Define its mixed difference and corner average by

$$(\partial D)(c, r) = D(c, r) + D(c-1, r-1) - D(c-1, r) - D(c, r-1),$$
$$(\mathcal{A}D)(c, r) = \tfrac{1}{4}\big(D(c, r) + D(c-1, r-1) + D(c-1, r) + D(c, r-1)\big).$$

For a state u, let $[u]$ be the indicator function of its points. A domain from u to v satisfies $\partial D = [u] - [v]$. Its index is

$$\iota(D; u, v) = \sum_{c=0}^{h-1} \big((\mathcal{A}D)(c, u(c)) + (\mathcal{A}D)(c, v(c))\big).$$

This is a definition by finite sums. In particular it makes sense for domains with negative coefficients.

Lemma 3.2.1 (New lemma: domain index identities)**.** *The following statements hold for integral domains on a toroidal grid.*

1. *If D is a domain from u to v and E is a domain from v to w, then*

$$\iota(D + E; u, w) = \iota(D; u, v) + \iota(E; v, w).$$

2. *A domain with mixed difference zero has the form $D(c, r) = a_c + b_r$, with integers a_c, b_r. Its index between any equal endpoint states is $2\sum_c a_c + 2\sum_r b_r$.*

3. *An empty native rectangle has index one. For any auxiliary grid with marking permutation O and any domain from u to v,*

$$\iota(D; u, v) = M_O(u) - M_O(v) + 2\sum_c D(c, O(c)). \tag{3.2.1}$$

Here M_O is the native integer Maslov grading.

Proof. Finite cyclic reindexing gives

$$\sum_{c,r}(\partial D)(c,r)(\mathcal{A}E)(c,r)=\sum_{c,r}(\partial E)(c,r)(\mathcal{A}D)(c,r).$$

For completeness, after moving the differences from D to E, the coefficient of $D(c,r)$ on the left is

$$\tfrac{1}{4}(E(c+1,r+1)-E(c+1,r-1)-E(c-1,r+1)+E(c-1,r-1)).$$

Its coefficients are unchanged when both offsets are negated. This proves the displayed symmetry. Substituting $\partial D=[u]-[v]$ and $\partial E=[v]-[w]$ proves the first assertion by expanding the definition of ι.

If $\partial D=0$, the difference $D(c,r)-D(c-1,r)$ is independent of r. Fixing one row and one column and summing these differences gives $D(c,r)=a_c+b_r$. Each state occupies every row and every column once. Substitution into the corner average therefore gives index $2\sum a_c+2\sum b_r$.

The domain of a rectangle with sides (a,b) and source u is the indicator of

$$[a,b)_{\mathrm{cyc}}\times[u(a),u(b))_{\mathrm{cyc}}.$$

Its mixed difference is $[u]-[u\circ(a\ b)]$ by the native declaration `GridRectangleBetween.target_eq_swapColumns`. If the rectangle is empty, its four endpoint corners each contribute $1/4$ to the index and every other state point contributes zero. The native declaration `laurentMaslovDegree_add_card_OColumns` then proves (3.2.1) for an empty rectangle.

To obtain the formula for an arbitrary domain, connect u to v by adjacent-column transpositions. Such a sequence exists by moving the desired value into the first incorrect column and repeating on the remaining columns. Use the rectangle of width one for each transposition. It is empty because there is no interior column. Their sum D_0 has the required boundary, and the formula for D_0 follows by additivity and telescoping of the native grading differences. The difference $D-D_0$ has zero mixed difference. For a row or column annulus the index is two and the sum of the O multiplicities is one. The second assertion, with arbitrary integer coefficients, now gives the formula for $D-D_0$ and hence for D. □

Lemma 3.2.2 (New lemma: a positive domain joining distinct states has index at least one)**.** *Let $h\geq 2$, let u,v be distinct states of size h, and let D be a nonnegative integral domain from u to v. Then $\iota(D;u,v)\geq 1$.*

Proof. Suppose that u and v differ in e columns. They cannot differ in exactly one column, because both are permutations, so $e\geq 2$. Their $2e$ changed points are distinct vertices. At each such vertex the mixed difference of D equals 1 or -1. Since the four adjacent cell multiplicities are nonnegative integers, at least one of them is positive. The corner average at that vertex is consequently at least $1/4$. Each changed vertex occurs once in the defining sum for $\iota(D;u,v)$, and every remaining summand is nonnegative. Therefore $\iota(D;u,v)\geq 2e/4\geq 1$. □

Now let $n,m\geq 1$ and $N=n+m+1$. For a correction path from retained state $p=$ `seamState`(x,z) to retained state $q=$ `seamState`(y,w), define D to be the sum of its entry, internal, and exit rectangle domains, minus the domains of the selected matching rectangles at all its matching sources. The matching rectangle at a source with southwest columns $P<Q$ and rows $a<b\leq n$ is $[P,Q)\times[a,b)$. It is empty, because every column strictly between the first two southwest columns has a high row and therefore lies outside the open row interval. Telescoping the rectangle boundaries gives $\partial D=[p]-[q]$. There is one more positive rectangle than subtracted matching rectangles, so Lemma 3.2.1 gives

$$\iota(D;p,q)=1. \tag{3.2.2}$$

Define two restrictions, with the indicated factor indices, by

$$\begin{aligned} D_1(i,j) &= D(i,j) && (0 \le i < n,\ 0 \le j \le n), \\ D_1(n,j) &= D(N-1,j) && (0 \le j \le n), \\ D_2(i,j) &= D(n+i,n+j) && (0 \le i,j \le m). \end{aligned} \tag{3.2.3}$$

Put $d = D(N-1,n)$.

Lemma 3.2.3 (New lemma: positive projections and the index relation)**.** *For every correction path as above, D_1 is a nonnegative domain from x to y, and D_2 is a nonnegative domain from z to w. Their indices satisfy*

$$\iota(D_1;x,y) + \iota(D_2;z,w) = 1 + 2d. \tag{3.2.4}$$

Proof. A matching rectangle uses only columns from n through $N-2$ and rows from 0 through $n-1$. It therefore misses both restrictions in (3.2.3). All remaining summands of D are nonnegative rectangle indicators, proving positivity.

For the boundary assertion, restricting a domain to cyclic lists of cell columns c_i and cell rows r_j replaces its mixed difference at (i,j) by

$$\sum_{c \in (c_{i-1},c_i]_{\mathrm{cyc}}} \sum_{r \in (r_{j-1},r_j]_{\mathrm{cyc}}} (\partial D)(c,r).$$

This formula follows by telescoping in both coordinates. For D_1, the last column collects the original columns $n,\dots,N-1$, and row zero collects the original rows $n+1,\dots,N-1,0$. Under these collections the retained state p becomes $[x] + m[(n,0)]$. The extra m copies are the high points in the second columns. The retained state q gives $[y] + m[(n,0)]$, so the extra terms cancel. For D_2, column zero collects columns $0,\dots,n$, and row zero collects rows $0,\dots,n$. The retained states become $[z] + n[(0,0)]$ and $[w] + n[(0,0)]$. Again the extra terms cancel. These statements use precisely the three supplied coordinate formulas `seamState_left_castSucc`, `seamState_right_of_zero`, and `seamState_right_of_ne_zero`.

We prove the index relation by explicit auxiliary markings. Let $O_1(i) = i$ on the first factor, let $O_2(j) = j+1 \pmod{m+1}$ on the second factor, and let

$$O(c) = c \quad (c < n), \qquad O(n+j) = n + O_2(j) \quad (0 \le j \le m).$$

Each is a permutation and can be made into a valid auxiliary grid by choosing X to be its row successor. The Maslov grading depends only on the specified O markings. The full marking set is the union of the two projected marking sets with their common square $(N-1,n)$ counted once. Consequently

$$\sum_c D(c,O(c)) = \sum_i D_1(i,O_1(i)) + \sum_j D_2(j,O_2(j)) - d. \tag{3.2.5}$$

The native Maslov point-count formula gives

$$M_O(\texttt{seamState}(x,z)) = M_{O_1}(x) + M_{O_2}(z). \tag{3.2.6}$$

Here is a direct verification, so no condition on the auxiliary X markings is being suppressed. For finite planar point sets A, B, write $I(A,B)$ for the number of ordered pairs that increase strictly in both coordinates. State points have integral coordinates and marks have both coordinates shifted by $1/2$. The supplied formula `GridDiagram.maslovOℤ_eq_card` is

$$M_O(s) = I(s,s) - I(s,O) - I(O,s) + I(O,O) + 1.$$

Splitting each point set into its first n and last $m+1$ columns gives

$$\begin{aligned} I(p,p) &= I(x,x) + I(z,z) + nm, \\ I(O,O) &= I(O_1,O_1) + I(O_2,O_2) + nm, \\ I(p,O) &= I(x,O_1) + I(z,O_2) + nm - 1, \\ I(O,p) &= I(O_1,x) + I(O_2,z) + nm. \end{aligned}$$

To check the first equality, put $t = x(n)$. Removing the last column from x removes t increasing pairs. The cross pairs with second-column points contribute $nm + t$. The second equality is the same computation for the marking permutations. For the third equality, removing the last marking of O_1, whose row is n, removes $n + 1$ pairs. All $n(m + 1)$ pairs from first-column state points to second-column markings contribute. The pairs within the second columns agree with $I(z, O_2)$ because the low state row t is at most the lowest marking row $n + 1/2$. For the fourth equality, removing the last state point of x removes t pairs, whereas the cross pairs contribute $nm + t$. Substituting the four identities into the native formula proves (3.2.6).

Apply (3.2.1) to the three domains. Equations (3.2.2), (3.2.5), and (3.2.6) give (3.2.4). □

3.3 Removing two annuli

Lemma 3.3.1 (New lemma: the unoccupied northeast region)**.** *Let $n, m \geq 1$, let D be the signed domain of a native correction between retained states, and let D_1, D_2, d be defined by* (3.2.3)*. For every $0 \leq c < n$ and every $0 \leq j \leq m$,*

$$D(c, n + j) = D_1(c, n) + D_2(m, j) - d \geq 0. \tag{3.3.1}$$

If

$$a = \min_{0 \leq i \leq n} D_1(i, n), \qquad b = \min_{0 \leq j \leq m} D_2(m, j),$$

then a, b are nonnegative integers and $a + b \geq d$.

Proof. At every vertex (c, r) with $c < n$ and $r > n$, both retained endpoint indicator functions vanish by (3.1.2). Hence $\partial D(c, r) = 0$ there. Fix $c < n$. Summing over columns $0, \ldots, c$ gives, for $n < r < N$,

$$D(c, r) - D(N - 1, r) = D(c, r - 1) - D(N - 1, r - 1).$$

Telescoping from row n to row $n+j$ proves (3.3.1), because $D(c, n) = D_1(c, n)$ and $D(N-1, n+j) = D_2(m, j)$. All the subtracted matching rectangles have rows less than n and columns at least n. Therefore $D(c, n + j)$ is a sum of nonnegative rectangle indicators. This proves the inequality in that equation.

The two minima exist because their finite index sets are nonempty. They are nonnegative integers by Lemma 3.2.3. If a minimum for a occurs at $i = n$, then $a = D_1(n, n) = d$, so $a + b \geq d$. Otherwise choose $c < n$ attaining a, and choose j attaining b. Equation (3.3.1) gives $a + b - d \geq 0$. These alternatives also include the case in which the first minimum is attained both at n and at another column. □

Lemma 3.3.2 (New lemma: normalization of the factor domains)**.** *Under the hypotheses and notation of Lemma 3.3.1, define*

$$E_1(i, j) = D_1(i, j) - a\,\mathbf{1}_{\{n\}}(j), \qquad E_2(i, j) = D_2(i, j) - b\,\mathbf{1}_{\{m\}}(i).$$

Then E_1 and E_2 are nonnegative integral domains with the same respective endpoints as D_1 and D_2, and

$$\iota(E_1; p_1, q_1) + \iota(E_2; p_2, q_2) = 1 + 2d - 2a - 2b \leq 1. \tag{3.3.2}$$

Proof. Only the last row is altered in D_1, and a is the minimum multiplicity on that row. Only the last column is altered in D_2, and b is the minimum on that column. Thus every cell of E_1, E_2 is nonnegative. A complete cell row or cell column has mixed difference zero, so the endpoints are unchanged. Each such annulus has index two by Lemma 3.2.1. Additivity of that lemma and (3.2.4) give the equality in (3.3.2). The inequality follows from $a + b \geq d$, proved in Lemma 3.3.1. □

Proof of Theorem 3.1.1. If $n = 0$, the first-factor grid state is the unique permutation of a singleton. The hypothesis $p_1 \neq q_1$ is impossible. If $m = 0$, the same argument applies to $p_2 \neq q_2$. It remains to consider $n, m \geq 1$.

Suppose that a chosen correction path of some length $k \geq 0$ exists. Form its signed domain and the nonnegative normalized domains E_1, E_2 in Lemma 3.3.2. Their endpoint pairs are distinct in both factors by hypothesis. Lemma 3.2.2 therefore gives

$$\iota(E_1; p_1, q_1) \geq 1, \qquad \iota(E_2; p_2, q_2) \geq 1.$$

Their sum is at least two, contradicting (3.3.2). This proves that every $\mathcal{P}_k(p, q)$ is empty.

The native definition `seamPositiveCorrectionCount` is the finite sum of the cardinalities of $\mathcal{P}_{j+1}(p, q)$ for $0 \leq j < \#\texttt{SeamMatchingSource}(n, m) - 1$, with natural-number subtraction in the upper bound. Every summand is zero by the first assertion. The sum is therefore the integer zero, and its image in $\mathbb{F}_2$ is zero. This proves (3.1.1), including M606 exactly as stated. □

3.4 Small sizes, orientations, and dependency ledger

The cases $n = 0$ and $m = 0$ were treated in the proof and require no auxiliary grid of size one. For $n = m = 1$, the first projection uses columns $0, 2$ and rows $0, 1$ of the size-three seam grid. The second uses columns $1, 2$ and rows $1, 2$. Their common square is $(2, 1)$. Every matching rectangle is contained in the remaining square $(1, 0)$, so the positivity argument and the shared-square subtraction have exactly the stated signs in this smallest nonvacuous case. The same projection formulas apply when just one of n, m equals one.

On a size-two factor, the two cyclic orientations of a rectangle can both occur. The signed-domain argument treats each chosen rectangle separately and uses its half-open cell interval. The four endpoint corner contributions still total one. A row or column annulus still has index two. No orientation or multiplicity is discarded in the passage from a path to its signed domain. The argument also treats $k = 0$ and arbitrary larger k without a separate classification of short paths.

All exact declaration names in the table refer to the supplied Lean listings. The main seam declarations lie in `OnLeConjecture.KnotFloer.GridFull`. The rectangle and point-count declarations are cited with the namespaces displayed in those listings.

Lemma or declaration	Status and use
`seamState_left_castSucc`, `seamState_right_of_zero`, `seamState_right_of_ne_zero`	Existing Lean declarations. They give the retained coordinates and both factor boundary projections.
`SeamInternalRectangle`, `SeamInternalPath`, `SeamCorrectionPath`, `seamPositiveCorrectionCount`	Existing Lean declarations. They specify the exact path choices, the count of internal transitions, and the finite acceptance sum.
`seamComplementRectangle_isEmpty`, `seamComplementRectangle_left`, `seamComplementRectangle_right`	Existing Lean declarations used in the supplied matching proofs. They identify the selected empty southwest rectangle and its column interval. The increasing source condition supplies its row interval.
`GridRectangleBetween.target_eq_swapColumns`, `GridRectangleBetween.bottom_def`, `GridRectangleBetween.top_def`, `isEmpty_iff_forall_notMem_cIoo`	Existing Lean declarations. They fix the corner orientation, rectangle boundary, and native emptiness convention.
`laurentMaslovDegree_add_card_OColumns`	Existing Lean declaration. It gives the Maslov identity for an empty rectangle in each explicitly constructed auxiliary grid.
`GridDiagram.maslovO$\mathbb{Z}$_eq_card`	Existing Lean declaration. Its finite point-count formula proves the Maslov sum for the chosen auxiliary markings.

Lemma or declaration	Status and use
Lemma 3.2.1	New lemma, proved here. It proves index additivity, the structure of zero-boundary domains, and the arbitrary-domain Maslov formula.
Lemma 3.2.2	New lemma, proved here. It bounds the index of a nonnegative domain between distinct states using its changed corners.
Lemma 3.2.3	New lemma, proved here. It proves both positive factor projections and their exact index relation.
Lemma 3.3.1	New lemma, proved here. It derives the northeast interpolation formula and the inequality for the two annulus multiplicities.
Lemma 3.3.2	New lemma, proved here. It produces two nonnegative domains with total index at most one.
Theorem 3.1.1	New lemma, proved here. It excludes every mixed correction path and proves the exact J07 acceptance statement.

3.5 Remaining gap

There is no remaining gap in the written proof of J07 or M606 under the native coordinate definitions recorded above. The statements in this document are written proofs and are not claims that their new lemmas have already been formalized in Lean. The proof uses no holomorphic Heegaard Floer result and no computation as a proof step. It makes no assertion here about the fixed-factor long-path counts in J05 or J06, or about universal truncation M607.

3.6 The combined seam conclusion: M601 and M607

Theorem 3.6.1 (New lemma: universal truncation at internal length two)**.** *For every $n, m \in \mathbb{N}$, every retained indices p, q, and every $k \geq 3$, the native type* `SeamCorrectionPath` *k p q is empty.*

Proof. If $p_2 = q_2$, apply Theorem 1.8.4. If $p_2 \neq q_2$ and $p_1 = q_1$, apply Theorem 2.8.4. If both factors differ, apply Theorem 3.1.1. These cases exhaust all retained endpoint pairs, including equal pairs. Each cited theorem includes the parameters zero and the smallest positive sizes. This is precisely M607. □

Theorem 3.6.2 (New lemma: the exact seam positive-path comparisons)**.** *For every $n, m \in \mathbb{N}$, the native proposition* `SeamPositivePathComparisons` *n m holds. In fact its three cardinality comparisons hold before reduction to $\mathbb{F}_2$:*

$$\texttt{seamPositiveCorrectionCount}((x,z),(y,z)) = \#\texttt{SeamFirstRemainingRectangle}(x,y,z),$$
$$\texttt{seamPositiveCorrectionCount}((x,z),(x,w)) = \#\texttt{SeamSecondRemainingRectangle}(x,z,w),$$
$$\texttt{seamPositiveCorrectionCount}(p,q) = 0 \quad \textit{if } p_1 \neq q_1 \textit{ and } p_2 \neq q_2.$$

All variables in each displayed line range over the corresponding native factor state sets.

Proof. The first equality is Corollary 1.8.5, using the existing integer identity `seamPositiveCorrectionCount_first_eq_remaining_add_long`. The second equality is Corollary 2.8.5, which uses the explicitly proved short-path bijection and the absence of every longer path. The third equality follows because every summand is the cardinality of an empty type by Theorem 3.1.1. Reducing these equalities modulo two gives exactly the three conjuncts of the supplied definition of `SeamPositivePathComparisons`. This is M601. □

The universal theorem also implies the arbitrary-endpoint M603 statement, since it excludes every three-transition path outright. It implies the sufficient emptiness form of M604 for all lengths at least four. Thus all records M601–M607 named in the seam jobs are settled by the displayed written proofs.

Lemma or declaration	**Status and role**
Theorems 1.8.4, 2.8.4, and 3.1.1	New, proved in the preceding chapters. They cover every retained endpoint pair.
`seamPositiveCorrectionCount_first_eq_remaining_add_long`	Existing Lean declaration. It identifies the first short count.
`SeamPositivePathComparisons`	Existing Lean definition. It is the exact three-sector target.
Theorems 3.6.1 and 3.6.2	New, proved here. They assemble M607 and M601 without additional assumptions.

There is no remaining mathematical gap in these combined seam statements. They are not assertions that the new written lemmas have already been entered into the Lean development.

Chapter 4

J21: An explicit second torus representative

Chapter overview

For every $n \geq 2$, we construct a finite Laurent correction of the native first torus cycle over $\mathbb{F}_2$. We compute its complete boundary by empty-rectangle cancellation. The resulting cycle represents exactly `reflectedTorusAnchorClass` and lies in the bifiltration rectangle

$$\left(3, \frac{(n-2)(n-1)}{2}\right).$$

This proves the strict-range acceptance statement of J21 and the explicit correction and displacement assertions M902 and M903. Every new combinatorial identity used in the construction is proved below. The argument uses the native full differential, including rectangles containing X markings. It uses no identification with a holomorphic complex.

4.1 The exact statement and the native conventions

Fix a natural number $n \geq 2$, and put

$$p = n + 1, \qquad N = 2p + 1 = 2n + 3, \qquad g = \frac{p(p-1)}{2} = \frac{n(n+1)}{2}.$$

All chains in this document have coefficients in $\mathbb{F}_2[U, U^{-1}]$. Columns and rows are elements of $\mathbb{Z}/N\mathbb{Z}$, with positive cyclic order induced by $0, 1, \ldots, N-1$. Whenever ordinary inequalities or absolute values are used, the corresponding residues are represented in this range. For columns a, b, put

$$d(a, b) = (b - a) \bmod N \in \{0, 1, \ldots, N-1\}.$$

For distinct endpoints, $[a, b)$ and (a, b) denote the positive half-open and open cyclic intervals of columns. An indicator $[P]$ is one when the proposition P holds and is zero otherwise. Indicators in coefficient identities belong to $\mathbb{F}_2$. A bar over an integer denotes its reduction modulo two. Counts, gradings, and displacement weights are ordinary integers or rational numbers, before any explicitly indicated reduction.

The native grid $G = \texttt{reflectedTorusGrid}\, n$ has markings

$$O(c) = -c - 1, \qquad X(c) = p - c - 1 \pmod{N}. \tag{4.1.1}$$

The marking with index c occupies the square with center $(c + \frac{1}{2}, O(c) + \frac{1}{2})$ or $(c + \frac{1}{2}, X(c) + \frac{1}{2})$. For a permutation σ of the columns, define the state

$$x_\sigma(c) = -\sigma(c) \pmod{N}.$$

Every state has exactly one such relative permutation, since negation is a bijection of the rows. The anchor is $x_0 = x_{\mathrm{id}}$. These identifications are the existing declarations `reflectedTorusGrid_O_apply`, `reflectedTorusGrid_X_apply`, `reflectedTorusAnchor_val`, and `reflectedTorusRelativeColumns`. A cycle word $(a_0\, a_1 \cdots a_{q-1})$ denotes the permutation sending a_i to a_{i+1} with indices modulo q and fixing every unlisted column. Products of disjoint cycles have their usual meaning. A word such as 01342 denotes a cycle on ordered labels, not a one-line permutation.

Write $M(x)$ for the native O-Maslov grading and $A(x)$ for the native Alexander grading. The monomial $U^k x$ has Maslov degree $M(x) - 2k$ and bifiltration coordinates $(-k, A(x) - k)$. The grading convention is `laurentMaslovDegree`. The support criterion for these two filtration coordinates is `mem_laurentMaslovBifiltrationLevel_iff`. The native full Laurent differential, denoted by D, is

$$Dx = \sum_{y} \sum_{R \in \mathrm{Rect}^\circ(x,y)} U^{o(R)} y = \sum_{j \geq 0} U^j d_j x. \tag{4.1.2}$$

Here $\mathrm{Rect}^\circ(x, y)$ is the set of directed rectangles whose open interior contains no source-state point, and $o(R)$ is the number of covered O squares. Rectangles containing X markings are included. The operators d_j act on the finite $\mathbb{F}_2$-span of the states. Equation (4.1.2) is the existing `laurentCoefficient_eq`. Its Laurent linear extension is `laurentDifferential_monomialGenerator`. For every state-empty rectangle from x to y, the existing `laurentMaslovDegree_add_card_OColumns` gives

$$M(y) = M(x) - 1 + 2o(R). \tag{4.1.3}$$

We will use the following exact native displacement identity. For $0 \leq d < N$, define

$$W(d) = pd - N \max\{0, d - p - 1\}.$$

This is `reflectedTorusDisplacementWeight`. The declarations `alexanderTwo_reflectedTorus_eq_sum_displacement`, `alexander_reflectedTorusAnchor`, and the native conversion `GridDiagram.two_mul_alexander_eq_intCast` give

$$A(x_\sigma) = g - \frac{1}{N} \sum_{c \in \mathbb{Z}/N\mathbb{Z}} W(d(c, \sigma(c))). \tag{4.1.4}$$

Indeed, the first declaration gives the equality after multiplying both sides by $2N$. The conversion replaces twice the rational Alexander grading with its integer version, and $N > 0$ permits cancellation. The imported conversion is used explicitly in the supplied proof of `alexander_reflectedTorus_le_of_displacement_weight`. In particular $W(0) = 0$, and for $1 \leq d \leq p$ one has

$$W(d) = pd, \qquad W(N - d) = (p + 1)d. \tag{4.1.5}$$

Substitution in the defining formula for W proves these two equalities.

Let $z_1 = \texttt{firstTorusCycle}\, n\, \mathbb{F}_2$ and let $\xi = \texttt{reflectedTorusAnchorClass}\, n\, \mathbb{F}_2$. For a chain z, the notation $[U^k x]z$ denotes its coefficient at that Laurent monomial. For a chain with Laurent exponent zero, $[x]z$ denotes the coefficient of the state x.

Theorem 4.1.1 (New theorem: the exact J21 acceptance statement)**.** *For every natural number $n \geq 2$, there is an explicitly specified degree-one Laurent chain w_2 such that $z_2 = z_1 + Dw_2$ is a degree-zero cycle, $[z_2] = \xi$, and every monomial $U^k y$ with $[U^k y]z_2 = 1$ satisfies*

$$-k \leq 3, \qquad A(y) - k \leq g - 2p + 3. \tag{4.1.6}$$

For every integer $m > 0$ with $2m < p$, the same monomials satisfy

$$A(y) - mk \leq g + 3m - 2p \tag{4.1.7}$$

and the exact displacement inequality of M903,

$$N(2p - m(3 + k)) \le \sum_{c \in \mathbb{Z}/N\mathbb{Z}} W(d(c, \sigma_y(c))), \qquad y = x_{\sigma_y}. \tag{4.1.8}$$

Consequently ξ is represented below `Staircase.consecutiveWeight`$(n, 2/m, 2)$ *in the weighted filtration at $t = 2/m$.*

The formulas for the correction chains and the complete coefficient calculation occupy the intervening sections. The proof of Theorem 4.1.1 is completed in Section 4.13. Equation (4.1.6) is the parameter-free strengthening recorded in M901. The target of J21 keeps precisely its stated hypotheses.

The smallest native convention check. For $n = 2$, one has $p = 3$, $N = 7$, and the three native coordinate lists are

c	0	1	2	3	4	5	6
$O(c)$	6	5	4	3	2	1	0
$X(c)$	2	1	0	6	5	4	3
$x_0(c)$	0	6	5	4	3	2	1

These lists follow by substitution in (4.1.1). The existing `maslov0_reflectedTorusAnchor` gives $M(x_0) = 0$, and `alexander_reflectedTorusAnchor` gives $A(x_0) = 3$. The proofs below keep the half-open square intervals and open state interiors that these coordinates require.

4.2 A relative-permutation formula for the Maslov grading

The following calculation uses the native state coordinates and the native half-integer marking coordinates. It does not use a Floer-theoretic identification. Let $N \ge 1$, let $O(c) = N - 1 - c$, and let σ be a permutation of $\{0, \dots, N - 1\}$. Define

$$x_\sigma(c) = (-\sigma(c)) \bmod N, \qquad \mathrm{fr}(\sigma) = \sum_{c=0}^{N-1} |\sigma(c) - c|.$$

Let $\mathrm{inv}(\sigma)$ denote the number of pairs $c < d$ for which $\sigma(c) > \sigma(d)$.

For finite sets of points P, Q in the coordinate square, define $I(P, Q) = \#\{(s, t) \in P \times Q : s_1 < t_1 \text{ and } s_2 < t_2\}$. Thus the inequalities in this pair count are strict.

Lemma 4.2.1 (New lemma: relative-permutation Maslov formula)**.** *For every such N and σ, the native O-Maslov grading satisfies*

$$M_O(x_\sigma) = \mathrm{inv}(\sigma) - \mathrm{fr}(\sigma).$$

Proof. Put $a = \sigma^{-1}(0)$. For $c \ne a$ the row of the state point is $N - \sigma(c)$, whereas the row at column a is zero. Therefore

$$I(x_\sigma, x_\sigma) = \mathrm{inv}(\sigma) + N - 1 - 2a.$$

Indeed, deleting the value zero removes exactly a inversions, and the state point in row zero contributes exactly $N-1-a$ increasing pairs. The marking points have coordinates $(d+\frac{1}{2}, N-d-\frac{1}{2})$. For a state point in column $c \ne a$, its contribution to $I(x_\sigma, O)$ is $\max\{\sigma(c) - c, 0\}$. The state point in column a contributes $N - a$. Thus

$$I(x_\sigma, O) = \sum_c \max\{\sigma(c) - c, 0\} + N - a.$$

Likewise the contribution of the state point in column $c \neq a$ to $I(O, x_\sigma)$ is $\max\{c - \sigma(c), 0\}$, and the contribution of column a is zero. Hence

$$I(O, x_\sigma) = \sum_c \max\{c - \sigma(c), 0\} - a.$$

The markings form a decreasing sequence, so $I(O, O) = 0$. Substitution in the existing finite-count formula `GridDiagram.maslov0ℤ_eq_card`

$$M_O(x) = I(x, x) - I(x, O) - I(O, x) + I(O, O) + 1$$

proves the identity. Every equality above includes the case $N = 1$. □

Lemma 4.2.2 (New lemma: deletion of fixed columns)**.** *Suppose t is a fixed point of σ. Delete t from both the domain and the range and reindex the remaining ordered sets increasingly. The resulting permutation $\bar{\sigma}$ satisfies*

$$\mathrm{inv}(\bar{\sigma}) - \mathrm{fr}(\bar{\sigma}) = \mathrm{inv}(\sigma) - \mathrm{fr}(\sigma).$$

Proof. Let r be the number of columns $c < t$ with $\sigma(c) > t$. The number of columns $c > t$ with $\sigma(c) < t$ also equals r, since σ is a bijection fixing t. Deleting t removes exactly $2r$ inversions. It also reduces the absolute displacement of each of these $2r$ crossing arrows by one and leaves every other absolute displacement unchanged. Thus the inversion number and the footrule sum decrease by the same quantity. □

Lemma 4.2.3 (New lemma: circular rotations)**.** *Let $S = \{s_0 < \cdots < s_{q-1}\}$ be a subset of columns, with $q \geq 2$. For $1 \leq j < q$, let σ fix the complement of S and satisfy $\sigma(s_i) = s_{(i+j) \bmod q}$. Then*

$$M_O(x_\sigma) = -j(q - j).$$

Proof. Delete every fixed column using the preceding lemma. The resulting permutation is the cyclic shift by j on q elements. Its inversion number is $j(q - j)$, since its two increasing blocks have respective lengths $q - j$ and j. Its absolute displacement is $(q - j)j + j(q - j) = 2j(q - j)$. The first lemma gives the result. □

Corollary 4.2.4 (New lemma: small permutation types)**.** *Every transposition state has Maslov value -1. Every three-cycle state has value -2, and every four-cycle state has value -3. On four columns $a < b < c < d$ in circular order, the two-transposition states $(a\,b)(c\,d)$ and $(a\,d)(b\,c)$ have value -2, while $(a\,c)(b\,d)$ has value -4.*

Proof. Delete fixed columns by Lemma 4.2.2. The transposition and the two three-cycles are the rotations covered by Lemma 4.2.3. For the remaining permutations, the following table evaluates the two counts in Lemma 4.2.1 on the four ordered labels.

cycle word or pairing	fr	inv
$(0123), (0132), (0231), (0321)$	6	3
$(0213), (0312)$	8	5
$(01)(23)$	4	2
$(03)(12)$	8	6
$(02)(13)$	8	4

These six words are every four-cycle, and the last three entries are every pairing. Subtracting the second column from the third proves all the assertions. □

4.3 The native rectangle coordinates

Lemma 4.3.1 (New lemma: rectangle coordinates)**.** *For a rectangle with ordered column sides a, b and source x_σ, its number of O markings is*

$$o(a,b) = \big|[a,b) \cap [\sigma(b), \sigma(a))\big|.$$

It is empty if and only if there is no $c \in (a,b)$ such that $\sigma(c) \in (\sigma(b), \sigma(a))$.

Proof. The native vertical interval runs from $-\sigma(a)$ to $-\sigma(b)$. Negating reverses its orientation. An interior state point in column c therefore lies in the rectangle precisely when $c \in (a,b)$ and $\sigma(c) \in (\sigma(b), \sigma(a))$. An O marking in column c has row $-c-1$. Its membership in the half-open vertical interval is equivalent to $c \in [\sigma(b), \sigma(a))$. Intersecting this condition with the half-open column interval gives the asserted count. The convention that the lower and left edges are included is the convention used by `OColumns` and `laurentCoefficient_eq`. □

4.4 Cyclic relabeling and the exact insertion index

For the native grid with $N = 2p+1$ and $p \geq 3$, retain the relative states $x_\sigma(c) = -\sigma(c)$ and put

$$\ell(\sigma) = -M_O(x_\sigma) = \sum_{c=0}^{N-1} |\sigma(c) - c| - \operatorname{inv}(\sigma). \tag{IG.1}$$

The equality is the proved new Maslov formula in Section 4.2. The following lemmas concern the native toroidal rectangle conventions. They do not use any coefficient assigned to a correction chain.

Lemma 4.4.1 (New lemma: cyclic conjugation)**.** *Let $\rho(c) = c+1 \bmod N$. For every permutation σ,*

$$\ell(\rho\sigma\rho^{-1}) = \ell(\sigma).$$

Proof. If $\sigma(N-1) = N-1$, the last fixed point moves to the first fixed point. Deleting these fixed points gives the same increasingly reindexed permutation. The fixed-column deletion lemma in Section 4.2 proves the assertion.

Suppose $\sigma(N-1) \neq N-1$. Put $v = \sigma(N-1)$ and $b = \sigma^{-1}(N-1)$, so $v, b < N-1$. Moving the last position to the first changes the inversion number by $2v - (N-1)$. Then moving the largest value to the smallest, in its new position $b+1$, changes that number by $2(b+1) - (N-1)$. Their total is $2(v+b-N+2)$. All absolute displacements except the two incident to the last column or last value are unchanged. The changes of these two absolute displacements are $2v - N + 2$ and $2b - N + 2$. Their sum is again $2(v+b-N+2)$. Subtracting the two changes in (IG.1) proves the equality. □

Lemma 4.4.2 (New lemma: cycle-count lower bound)**.** *For every permutation σ, one has*

$$\ell(\sigma) \geq N - \#\{\textit{cycles of } \sigma\},$$

where fixed points are included among the cycles. In particular a single k-cycle has $\ell \geq k-1$.

Proof. If the permutation is not the identity, choose its least nonfixed column a. Put $c = \sigma(a)$ and $b = \sigma^{-1}(a)$. All columns smaller than a are fixed, so $b, c > a$. Swapping the values in columns a, b fixes a and deletes it from its old permutation cycle. Thus the number of cycles increases by one. The footrule sum decreases by $2\min\{b-a, c-a\}$. The inversion number decreases by $2t+1$, where

$$t = \#\{i : a < i < b \text{ and } a < \sigma(i) < c\}.$$

Both $b-a-1$ eligible columns and $c-a-1$ eligible values bound t. Hence $t \leq \min\{b-a, c-a\} - 1$, and ℓ decreases by at least one. Iterating this operation reaches the identity, whose ℓ is zero, after exactly $N - \#\{\text{cycles of } \sigma\}$ operations. The assertion follows by adding these inequalities. □

Lemma 4.4.3 (New lemma: exact insertion index)**.** *Let a, b, q be pairwise distinct columns. Suppose $\sigma(q) = q$ and $\sigma(a) = b$, and put $\tau = \sigma \circ (a\ q)$. Thus τ replaces the arrow $a \mapsto b$ by $a \mapsto q \mapsto b$. Exactly one of the two candidate rectangles from x_σ to x_τ avoids all O markings. If its number of interior source points is r, then*

$$\ell(\tau) - \ell(\sigma) = 1 + 2r. \tag{IG.2}$$

In particular this rectangle is state-empty if and only if the left side equals one.

Proof. Use Lemma 4.4.1 to make $a = 0$. There are two possible orders of b, q. If $0 < q < b$, swapping the two values changes the footrule sum by zero and decreases the inversion number by

$$1 + 2\#\{i : 0 < i < q \text{ and } q < \sigma(i) < b\}.$$

The rectangle from column zero to column q has covered column interval $[0, q)$ and possible O-column interval $[q, b)$. These are disjoint, so it avoids O. Its source interior is precisely the displayed set. The complementary rectangle has $N - b > 0$ O markings. Equation (IG.2) follows in this case.

If $0 < b < q$, the footrule sum increases by $2(q - b)$ and the inversion number increases by

$$1 + 2\#\{i : 0 < i < q \text{ and } b < \sigma(i) < q\}.$$

There are $q - b - 1$ values strictly between b and q. None is the image of column zero or column q. Subtracting the displayed count from $q - b - 1$ therefore gives

$$\#\{i : q < i < N \text{ and } b < \sigma(i) < q\}.$$

This is the interior count for the rectangle from column q to column zero. Its column interval $[q, 0)$ is disjoint from its possible marking interval $[b, q)$, so it avoids O. The complementary rectangle has $b > 0$ O markings. Subtracting the inversion change from the footrule change proves (IG.2) in the second case.

These assertions about the intervals are exactly the native half-open-square and open-state-interior conventions used in `GridRectangle.mem_coveredSquares` and `isEmpty_iff_forall_notMem_cIoo`. The two cases exhaust all orders of the three distinct columns. □

Lemma 4.4.4 (New lemma: deletion from a minimal cycle)**.** *Let $3 \le k \le N$, and suppose τ is a single k-cycle with $\ell(\tau) = k - 1$. For every vertex q of its support, delete q from its cyclic word and then fix it. Let $\sigma = \tau \setminus q$ be the resulting permutation. Then*

$$\ell(\sigma) = k - 2,$$

and exactly one zero-O rectangle goes from x_σ to x_τ. That rectangle is state-empty.

Proof. The permutation σ is a $(k-1)$-cycle, so Lemma 4.4.2 gives $\ell(\sigma) \ge k - 2$. Since $k \ge 3$, the two neighbors a, b of q in the deleted cycle are distinct, and a, b, q meet the hypotheses of Lemma 4.4.3. That lemma gives $\ell(\tau) - \ell(\sigma) = 1 + 2r$ for a nonnegative integer r. Together with $\ell(\tau) = k - 1$, the two inequalities force $\ell(\sigma) = k - 2$ and $r = 0$. The same lemma gives uniqueness. □

Lemma 4.4.5 (New lemma: the minimal-cycle boundary coefficient)**.** *Let $3 \le k \le N$, and let H be any chain supported on single $(k-1)$-cycle states. Write $h(\sigma) = [x_\sigma]H$. For every single k-cycle τ with $\ell(\tau) = k - 1$,*

$$[x_\tau]d_0 H = \sum_{q \in \operatorname{supp}\tau} h(\tau \setminus q). \tag{IG.3}$$

Proof. A swap taking a single $(k-1)$-cycle to a single k-cycle must insert a previously fixed vertex into the cycle. Swapping two moved columns splits the old cycle, and swapping two fixed columns creates an additional transposition, so neither can produce the target. Thus all possible sources are exactly the vertex deletions in (IG.3). Each contributes one empty zero-O rectangle by Lemma 4.4.4. The native coefficient formula `laurentCoefficient_eq` now gives (IG.3). □

Small cases. The exact insertion lemma requires a, b, q distinct. Accordingly the deletion and coefficient lemmas start at $k = 3$. For $k = 2$, the source obtained by deleting a vertex is the anchor, and its two complementary zero-O rectangles cancel. No assertion of uniqueness is made in that case. The applications below use $k = 5$, and the converse insertion applications use source type $(3, 2)$, so all their neighbors are distinct on the smallest permitted grid $N = 7$.

4.5 Permutation types occurring in the correction

The standing assumptions are $p \geq 3$, $N = 2p + 1$, and coefficients in $\mathbb{F}_2$. For a permutation σ of $\mathbb{Z}/N\mathbb{Z}$, write $x_\sigma(c) = -\sigma(c)$ and $\ell(\sigma) = -M_O(x_\sigma)$. The relative-permutation grading formula gives

$$\ell(\sigma) = \sum_i |\sigma(i) - i| - \mathrm{inv}(\sigma).$$

We use the previously proved invariance of this expression under deletion of fixed columns and simultaneous cyclic translation of all labels.

For a permutation σ, define $r(\sigma) = N - \#\{\text{cycles of } \sigma\}$, including fixed points in the count. Lemma 4.4.2 states $\ell(\sigma) \geq r(\sigma)$.

Lemma 4.5.1 (New lemma: the change in cycle count under a swap)**.** *Multiplying a permutation by a transposition changes r by exactly one. In particular, a target of a single swap from a four-cycle or from a permutation of type $(3, 2)$ has $r \geq 2$.*

Proof. If the swapped columns belong to distinct permutation cycles, swapping their outgoing arrows joins those two cycles into one cycle. If they belong to the same cycle, write its cyclic word as (a, A, b, B), where A, B are the lists of entries between the two columns. Swapping the arrows from a, b separates this word into the two cycles (a, B) and (b, A). An empty list gives a fixed point. Thus the number of cycles changes by one in either case. A four-cycle and a permutation of type $(3, 2)$ both have $r = 3$, which gives the last assertion. □

Lemma 4.5.2 (New lemma: insertion of a fixed column)**.** *Suppose that q is fixed by σ, that $a \neq q$, and that $b = \sigma(a) \neq q$. Let τ be the permutation obtained by replacing the arrow $a \mapsto b$ with $a \mapsto q \mapsto b$. Then $\ell(\tau) - \ell(\sigma)$ is a positive odd integer. Every state-empty rectangle from x_σ to x_τ avoids all O markings.*

Proof. Translate the labels cyclically so that $q = 0$. Then $a, b > 0$. The change in the sum of absolute displacements is

$$a + b - |a - b| = 2\min\{a, b\}.$$

Put

$$t = \#\{i : 0 < i < a \text{ and } 0 < \sigma(i) < b\}.$$

Swapping the values $0, b$ in columns $0, a$ increases the inversion number by $2t + 1$. There are at most $a - 1$ eligible columns and at most $b - 1$ eligible values, so $t \leq \min\{a, b\} - 1$. Consequently

$$\ell(\tau) - \ell(\sigma) = 2\min\{a, b\} - 2t - 1 \geq 1.$$

This difference is odd. For an empty rectangle with o markings, the existing native rectangle grading identity gives

$$\ell(\tau) - \ell(\sigma) = 1 - 2o.$$

Since $o \geq 0$, the preceding positive lower bound forces $o = 0$. □

Lemma 4.5.3 (New lemma: swapping two fixed columns)**.** *For every permutation σ and every distinct fixed columns a, b, the coefficient of $x_{\sigma\circ(a\ b)}$ in Dx_σ is zero.*

Proof. The two candidate rectangles have complementary half-open column and marking intervals in the relative rectangle formula, so both have O count zero. Let $I = (a, b)$ and $J = (b, a)$ be the two open cyclic intervals. Their emptiness conditions are respectively that no arrow of σ goes from I to J and that no arrow goes from J to I. The bijection σ preserves $I \cup J$, since it fixes a, b. The two numbers of crossing arrows are therefore equal. Thus both rectangles are empty or neither is empty. In the former case their identical terms cancel in $\mathbb{F}_2$. □

Lemma 4.5.4 (New lemma: inversion of relative permutations)**.** *The linear map $\iota(x_\sigma) = x_{\sigma^{-1}}$ commutes with the full native Laurent differential and preserves O counts of individual rectangles.*

Proof. On the grid torus, use the coordinate map

$$(c, r) \longmapsto (-r, -c).$$

It sends the point $(c, -\sigma(c))$ to $(\sigma(c), -c)$, which belongs to $x_{\sigma^{-1}}$. Every O square has center $(c + \frac{1}{2}, -c - \frac{1}{2})$, which this map fixes. The map interchanges the southwest and northeast source corners of a directed rectangle, and likewise interchanges its two target corners. It therefore preserves the direction of the rectangle. It is a bijection on covered squares and on state interiors. Thus it pairs the native empty rectangles in the two differentials and preserves their weights. □

Lemma 4.5.5 (New lemma: the relevant three-cycle and transposition states)**.** *Let σ be a permutation whose nontrivial cycles consist of one three-cycle and one transposition. Then $M_O(x_\sigma)$ is either -3 or -5. It equals -5 precisely when the five moved columns have a unique positive circular ordering*

$$u, a, v, b, c$$

in which the transposition is $(u\ v)$ and the three-cycle is either $(a\ b\ c)$ or $(a\ c\ b)$.

Proof. The two endpoints of the transposition split the other three moved columns into two circular intervals. If all three belong to the same interval, the endpoints are adjacent in the circular order of the five moved columns. Delete the fixed columns and rotate the labels so that the pair is (0 1). The other cycle is (2 3 4) or its inverse. For both permutations the footrule sum is 6 and the inversion number is 3. Hence $\ell = 3$.

Otherwise one interval contains one triangle vertex and the other contains two. Orient the pair uniquely so that its positive interval contains the single vertex, and call that vertex a. Call the two vertices in the remaining interval b, c in positive order. After deleting fixed columns, the two one-line permutations are

$$(2, 3, 0, 4, 1), \qquad (2, 4, 0, 1, 3).$$

Both have footrule sum 10 and inversion number 5. Hence $\ell = 5$. These alternatives exhaust the distributions of three vertices between the two intervals, and the construction proves uniqueness of the intrinsic ordering in the second case. □

Lemma 4.5.6 (New lemma: five-cycle Maslov values)**.** *On any five cyclically ordered columns, the two permutations that rotate these columns by two or three positions have Maslov value -6. Every other five-cycle on those columns has Maslov value -4.*

Proof. Delete fixed columns and label the remaining columns $0, 1, 2, 3, 4$ in positive circular order. The following table gives representatives of the orbits under simultaneous cyclic translation of

these five labels. The last two columns are the footrule sum and the inversion number.

cycle word	one-line permutation	orbit size	footrule	inv
$(0,1,2,3,4)$	$(1,2,3,4,0)$	1	8	4
$(0,1,2,4,3)$	$(1,2,4,0,3)$	5	8	4
$(0,1,3,4,2)$	$(1,3,0,4,2)$	5	8	4
$(0,1,4,3,2)$	$(1,4,0,2,3)$	5	8	4
$(0,2,1,4,3)$	$(2,4,1,0,3)$	5	10	6
$(0,2,4,1,3)$	$(2,3,4,0,1)$	1	12	6
$(0,3,1,4,2)$	$(3,4,0,1,2)$	1	12	6
$(0,4,3,2,1)$	$(4,0,1,2,3)$	1	8	4

The four orbits of size one are the four nonidentity circular rotations. For each other row, conjugating by the five label rotations gives five distinct permutations. Comparing the displayed cycle words up to adding a constant to every entry shows that these four orbits are distinct. Their sizes, together with the four singleton orbits, sum to 24, which is the number of five-cycles on five labels. Thus the table exhausts the permutations under consideration. Subtracting its last column from its preceding column and using cyclic-label invariance of ℓ proves the assertion. □

Lemma 4.5.7 (New lemma: the possible differentials of these states)**.** *Let H be a chain supported on the states of Lemma 4.5.5 with Maslov value -5, and suppose that $\iota H = H$. Write $D = \sum_{j\geq 0} U^j d_j$. Then $d_j H = 0$ for $j \geq 3$. The chain $d_2 H$ is supported on three-cycle states. The chain $d_1 H$ is supported on single five-cycle states of Maslov value -4. The chain $d_0 H$ is supported on the following permutation types of Maslov value -6: a single five-cycle, a four-cycle and a transposition, or two three-cycles.*

Proof. A rectangle swaps two columns. If both are fixed, its contribution cancels by Lemma 4.5.3. If exactly one is fixed, Lemma 4.5.2 gives O count zero. The new vertex extends either the three-cycle to a four-cycle or the transposition to a three-cycle. These are the two listed six-column types in $d_0 H$.

If both columns are moved, a swap within the transposition leaves the three-cycle, a swap within the three-cycle leaves two transpositions, and a swap between the two cycles joins them to one five-cycle. The possible Maslov values are -2 for a three-cycle, -2 or -4 for two transpositions, and -4 or -6 for a five-cycle by Lemma 4.5.6.

The rectangle grading identity gives target Maslov value $-6+2j$ for d_j. Thus $j \leq 2$, and the listed types have exactly the indicated degrees. A two-transposition target is fixed by ι. Its possible sources in H occur in distinct inverse pairs, since a three-cycle is not equal to its inverse. Lemma 4.5.4 and $\iota H = H$ make their contributions equal. All such terms cancel. This proves the assertions. □

4.6 The first boundary correction and four-cycle deletion

Lemma 4.6.1 (New lemma: the transposition differential)**.** *Let $a \neq b$, put $T_{ab} = x_{(a\ b)}$, and let C^+_{abc} and C^-_{abc} denote the two three-cycle states on $\{a,b,c\}$. The plus sign denotes the positive ambient circular orientation, and the minus sign denotes its inverse. Then*

$$DT_{ab} = U\,[d(a,b) = 1 \text{ or } d(b,a) = 1]x_{\mathrm{id}} + \sum_{c\notin\{a,b\}} (C^+_{abc} + C^-_{abc}).$$

Proof. Every target is obtained by one column transposition, by the definition of `GridRectangleBetween.emptyRectangles` appearing in `laurentCoefficient_eq`. If that transposition is $(a\ b)$, the two possible rectangles have O counts $d(a,b)$ and $d(b,a)$, respectively. Every strictly intermediate column on either cyclic interval is fixed by $(a\ b)$ and contributes an interior state

point. Hence the corresponding rectangle is empty exactly when that interval has length one. This proves the anchor coefficient.

Suppose that the transposition uses one of a, b and a third column c. To exhibit the rectangles without assuming a linear cut, first arrange a, c, b in this cyclic order. For the swap $(a\ c)$, the rectangle with ordered sides (a, c) has disjoint half-open intervals $[a, c)$ and $[c, b)$ in Lemma 4.3.1. It is empty and has O count zero. For the swap $(b\ c)$, the rectangle with ordered sides (c, b) has disjoint half-open intervals $[c, b)$ and $[a, c)$. It is empty and has O count zero. These two swaps produce the two cyclic orientations on the three columns. The alternative rectangle in each case has positive O count. It cannot be empty, since the existing declaration `maslov0_swap_reflectedTorusAnchor` gives source Maslov grading -1, Corollary 4.2.4 gives target Maslov grading -2, and `laurentMaslovDegree_add_card_0Columns` would then force its O count to be zero. Interchanging a and b covers the other cyclic order of the three columns.

Finally suppose that the swapped columns c, d are disjoint from $\{a, b\}$. The two half-open intervals for the rectangle with sides (c, d) are complementary, so its O count is zero. Every fixed state point misses its interior. The rectangle is obstructed precisely when one of a, b belongs to (c, d) and the other belongs to (d, c). The other rectangle is obstructed under the same condition. Thus the two rectangles are either both empty or both nonempty. Their contributions cancel over $\mathbb{F}_2$. This treats all possible target transpositions. □

Lemma 4.6.2 (New lemma: a balanced-triangle representative)**.** *For an unordered pair let* $e(a, b) = \min\{d(a, b), d(b, a)\} \bmod 2$*, and define*

$$H_e = \sum_{\{a,b\}} e(a, b)T_{ab}.$$

Then

$$x_{\mathrm{id}} + U^{-1}DH_e = U^{-1} \sum_{\substack{\{a,b,c\} \text{ with cyclic gaps } \alpha,\beta,\gamma \\ \max\{\alpha,\beta,\gamma\} \le p}} (C^+_{abc} + C^-_{abc}).$$

The right-hand side is a degree-zero cycle in the exact anchor class. Writing $g = p(p-1)/2$*, every positive cyclic orientation in this sum has Alexander grading* $g - p$*, and every negative cyclic orientation has Alexander grading* $g - p - 1$*.*

Proof. There are N adjacent unordered pairs. Their coefficients in H_e are one, and N is odd. Lemma 4.6.1 therefore makes the anchor coefficient of DH_e equal to U. The coefficient of either orientation on a triple is $e(a, b) + e(b, c) + e(c, a)$. If all three cyclic gaps are at most p, this is the parity of their sum N, which is one. Otherwise exactly one gap exceeds p. Replacing that gap by the sum of the other two makes the sum of the three shorter pair distances twice the sum of those two gaps. Its parity is zero. This proves the chain identity.

The existing declaration `laurentDifferential_reflectedTorusAnchor` and the square-zero declaration `laurentDifferential_sq_eq_zero` prove the cycle assertion. The displayed boundary proves equality with the exact anchor homology class. Corollary 4.2.4 proves degree zero term by term.

Use `alexanderTwo_reflectedTorus_eq_sum_displacement`. On the positive orientation the three nonzero displacements are $\alpha, \beta, \gamma \le p$. Their displacement weights are $p\alpha, p\beta, p\gamma$, with total pN. On the inverse orientation the displacements are $N - \alpha, N - \beta, N - \gamma \ge p + 1$. Their weights are $(p + 1)\alpha, (p + 1)\beta, (p + 1)\gamma$, with total $(p + 1)N$. The stated Alexander gradings follow from the native anchor normalization `alexander_reflectedTorusAnchor`. □

Lemma 4.6.3 (New lemma: deleting a vertex of a four-cycle)**.** *Let* σ *be a four-cycle on four distinct columns. Let* v *be one of its columns, let* $a = \sigma^{-1}(v)$ *and* $b = \sigma(v)$*, and let* τ *be the three-cycle obtained by deleting* v *from this cyclic word. The coefficient of* Ux_τ *in* Dx_σ *is*

$$[d(v, a) = 1 \text{ or } d(a, v) = 1] + [d(v, b) = 1 \text{ or } d(b, v) = 1].$$

Consequently the three-cycle part with O count one in the differential of a four-cycle is the sum, over its ambient-adjacent cycle edges, of the two three-cycles obtained by deleting either endpoint.

Proof. The target is obtained by swapping columns v, a. Lemma 4.3.1 gives the two possible O counts

$$\min\{d(v,a), d(v,b)\}, \qquad \min\{d(a,v), d(b,v)\}.$$

The first count is one precisely when a or b is the next column after v. The second is one precisely when a or b is the previous column. In either case the corresponding rectangle has column width one or row height one, so its open interior contains no state point. The source and target Maslov gradings are -3 and -2 by Lemma 4.2.1. The existing rectangle grading identity forces every empty rectangle between them to have O count one. These are therefore all contributions. Adding the two indicators over $\mathbb{F}_2$ proves the coefficient formula. □

4.7 An explicit coefficient for the three-cycle–transposition layer

All coefficient sums in this section take values in $\mathbb{F}_2$. Indicators used in coordinate formulas are the ordinary integers zero and one. Fix an integer $p \geq 3$ and put $N = 2p + 1$. We identify the columns with $\mathbb{Z}/N\mathbb{Z}$ and use the native markings $O(c) = N - 1 - c$. For distinct columns r, s, write

$$d(r,s) = (s - r) \bmod N \in \{1, \dots, N-1\}.$$

For a permutation σ of the columns, put $x_\sigma(c) = -\sigma(c) \bmod N$. The state x_{id} is the native anchor. These formulas agree with the attached declarations `reflectedTorusGrid_O_apply` and `reflectedTorusAnchor_val`.

4.7.1 The intrinsic order of the five moved columns

By Lemma 4.5.5, every permutation of type $(3,2)$ and Maslov value -5 has a unique positive circular ordering

$$(u, a, v, b, c) \tag{H32.2}$$

in which its transposition is $(u\,v)$ and its three-cycle is either $(a\,b\,c)$ or $(a\,c\,b)$. This is the intrinsic ordering used in the coefficient formula below.

4.7.2 The sixteen half-turn chambers

Take a tuple (H32.2), and use u as the origin. For $z \in \{a, v, b, c\}$, define

$$r_z = d(u,z), \qquad \chi_z = [r_z > p] \in \{0,1\}, \qquad \theta_z = r_z - \chi_z \left(p + \frac{1}{2}\right). \tag{H32.3}$$

In (H32.3), χ_z denotes the integer representative zero or one of the indicator, so the formula for θ_z is an equality of real numbers. The four numbers θ_z belong to $(0, p + \frac{1}{2})$ and are pairwise distinct. Distinct numbers from the same half-circle remain distinct after the translation in (H32.3). Numbers from opposite half-circles cannot coincide because one translation differs from the other by a half-integer. Arrange these four numbers in increasing order and record their χ values. This gives a word

$$w = w_1 w_2 w_3 w_4 \in \{0,1\}^4. \tag{H32.4}$$

For each of the four vertices let $t_z = \overline{r_z} \in \mathbb{F}_2$ be the parity of its distance from u.

New definition. Define $h_{32}(u,a,v,b,c) \in \mathbb{F}_2$ by the following complete table.

w	h_{32}	w	h_{32}
0000	0	1000	$1 + t_v$
0001	$1 + t_c$	1001	$t_v + t_c$
0010	$1 + t_c$	1010	$t_a + t_v$
0011	$t_b + t_c$	1011	0
0100	$1 + t_a$	1100	0
0101	$t_a + t_c$	1101	$1 + t_b$
0110	0	1110	$t_b + t_c$
0111	0	1111	0

(H32.5)

Every entry in this table is an explicit affine expression in four parities. No choice of a solution of a linear system enters the definition.

New definition. The raw correction chain is

$$H_{32} = \sum_{\substack{(u,a,v,b,c) \\ \text{in positive circular order}}} h_{32}(u,a,v,b,c) \left(x_{(u\,v)(a\,b\,c)} + x_{(u\,v)(a\,c\,b)} \right). \tag{H32.6}$$

In this sum all five columns are distinct. The distinguished first entry u is part of the indexing tuple. By the intrinsic-order lemma, each permutation of cycle type $(3,2)$ and Maslov value minus five occurs exactly once. Thus $U^{-3}H_{32}$ has Maslov degree one by the existing declaration `laurentMaslovDegree`.

Lemma 4.7.1 (New lemma: coefficient symmetries)**.** *The coefficient in (H32.6) is preserved by simultaneous cyclic translation of all columns and by replacing the relative permutation with its inverse.*

Proof. Every quantity in (H32.3)–(H32.5) uses a cyclic distance from the distinguished column u. Simultaneous cyclic translation preserves these quantities. Inversion changes $(a\,b\,c)$ into $(a\,c\,b)$ and preserves the transposition and its intrinsic order. The same coefficient is assigned to the two summands in (H32.6). □

4.7.3 Inversion and self-inverse targets

Corollary 4.7.2 (New lemma: inverse pairing of the sources)**.** *Every self-inverse relative permutation has coefficient zero in DH_{32}.*

Proof. Pair every source in (H32.6) with its inverse. The pair has equal coefficients by the definition, and its two permutations are distinct because their three-cycles are inverses of each other. For a self-inverse target, the differential coefficients of the two sources are equal by Lemma 4.5.4. They therefore cancel in $\mathbb{F}_2$. □

4.7.4 The complete two-O rectangle coefficient

Write

$$D = \sum_{j \geq 0} U^j d_j, \tag{H32.8}$$

where d_j counts native state-empty rectangles containing exactly j O markings. This is the coefficient expansion of the attached `laurentCoefficient_eq`.

Lemma 4.7.3 (New lemma: the two-O rectangle list)**.** *Let τ be a three-cycle with support T. The coefficient of x_τ in d_2H_{32} is*

$$\sum_{\substack{a\in T\\ a-1\notin T,\ a+1\notin T}} h_{32}(a-1,a,a+1,b,c), \tag{H32.9}$$

where b,c are the other two elements of T in positive circular order after $a+1$. This formula is valid for both orientations of the three-cycle.

Proof. A rectangle swaps two columns. In a source with one three-cycle and one transposition, swapping columns from different cycles joins them to a five-cycle. Swapping two columns of the three-cycle gives two transpositions. The only swap producing a single three-cycle is therefore the swap that deletes the transposition. Every possible source has the form $\tau(u\,v)$ with $u,v\notin T$.

For this source, the rectangle restoring the fixed columns u,v has the same positive cyclic interval $[u,v)$ for its column interval and for the columns of its covered O markings. Its O count is $d(u,v)$. The complementary rectangle has O count $d(v,u)$. Since $N\geq 7$, at most one of these counts can equal two. For a contribution to d_2, one of them does equal two. Relabel its endpoints as $u=a-1$ and $v=a+1$.

This rectangle has width two and height two. Its only interior column is a, and its only interior grid row is the anchor row $-a$. It is state-empty exactly when the source does not contain the anchor point $(a,-a)$. The transposition fixes the column a, so this condition is exactly $a\in T$. The requirement that the transposition be disjoint from τ is $a-1,a+1\notin T$. Thus each summand in (H32.9) contributes exactly one rectangle, and there are no other contributing rectangles. Its intrinsic order is $(a-1,a,a+1,b,c)$. The two orientations of the source three-cycle have the same coefficient by (H32.6), which proves the last assertion. □

New definition. For positive integers r,s with $r+s\leq 2p$, define $L_p(r,s)\in\mathbb{F}_2$ as follows. Set $L_p(r,s)=0$ if $r=1$ or $s=1$. For $r,s\geq 2$, use the complete table

Condition	$L_p(r,s)$
$s\geq p+2$	0
$s=p+1$	1
$s=p$	$[r=p]$
$s\leq p-1,\ r\leq p-1$	$1+\overline{s}$
$s\leq p-1,\ r=p$	$\overline{s}$
$s\leq p-1,\ r=p+1$	$1+\overline{s}$
$s\leq p-1,\ r\geq p+2$	$1+\overline{r}+\overline{s}$

(H32.10)

Here bars denote integer parity. The conditions in the table are disjoint and exhaustive. Some rows have no instances for the smallest values of p.

Lemma 4.7.4 (New lemma: the complete two-O gap coefficient)**.** *Let a,b,c be in positive circular order and put*

$$\alpha=d(a,b),\qquad \beta=d(b,c),\qquad \gamma=d(c,a).$$

The coefficient of either $x_{(a\,b\,c)}$ or $x_{(a\,c\,b)}$ in d_2H_{32} is

$$L_p(\alpha,\gamma)+L_p(\beta,\alpha)+L_p(\gamma,\beta). \tag{H32.11}$$

Proof. Consider the contribution centered at a in (H32.9). It is absent if $\alpha=1$ or $\gamma=1$, which agrees with the first clause of (H32.10). Suppose $\alpha,\gamma\geq 2$. Translate the column $a-1$ to zero. The five intrinsic coordinates become

$$(u,a,v,b,c)=(0,1,2,B,C),\qquad B=1+\alpha,\qquad C=2p+2-\gamma. \tag{H32.12}$$

They satisfy $3 \le B < C \le 2p$. In particular $t_a = 1$, $t_v = 0$, $t_b = 1 + \overline{\alpha}$, and $t_c = \overline{\gamma}$.

The following table obtains the half-turn word directly from (H32.12). It lists every possible location of B, C.

Condition	w	h_{32}
$C \le p$	0000	0
$C = p+1$	1000	1
$C = p+2,\ B \le p$	0100	0
$C = p+2,\ B = p+1$	1010	1
$C \ge p+3,\ B \le p$	0010 or 0001	$1+\overline{\gamma}$
$C \ge p+3,\ B = p+1$	1001	$\overline{\gamma}$
$C \ge p+3,\ B = p+2$	0101	$1+\overline{\gamma}$
$C \ge p+3,\ B \ge p+3$	0011	$1+\overline{\alpha}+\overline{\gamma}$

(H32.13)

For example, a point at coordinate $p+1$ projects to $\frac{1}{2}$, a point at coordinate $p+2$ projects to $\frac{3}{2}$, and every point at coordinate at least $p+3$ projects beyond 2. The two fixed projected coordinates from a, v are $1, 2$. These facts determine the displayed order in every row. In the fifth row, the projected coordinate from C lies either before or after the projected coordinate from B. The two words have the same entry in (H32.5). Substitution of the four parities into (H32.5) gives the last column.

The equalities in (H32.12) transform (H32.13) exactly into $L_p(\alpha, \gamma)$ in (H32.10). Applying the same argument with the vertices cyclically relabelled gives $L_p(\beta, \alpha)$ and $L_p(\gamma, \beta)$. Their sum is (H32.9), proving (H32.11). □

Corollary 4.7.5 (New lemma: no further positive-O layers)**.** *The chain $d_2 H_{32}$ is supported on three-cycle states, and $d_j H_{32} = 0$ for every $j \ge 3$.*

Proof. The chain (H32.6) satisfies every hypothesis of Lemma 4.5.7: each source has Maslov value -5, and its coefficient agrees with that of its inverse. That lemma gives both assertions. □

4.8 The four-cycle correction defined by adjacent pairs

Fix $p \ge 3$ and put $N = 2p + 1$. All coefficients in this section belong to $\mathbb{F}_2$. Columns are elements of $\mathbb{Z}/N\mathbb{Z}$. For distinct columns u, v, the symbol $d(u, v)$ denotes the representative of $v - u$ in $\{1, \dots, N-1\}$. A bar denotes integer parity. For a relative permutation σ, write $x_\sigma(c) = -\sigma(c)$, as in the native reflected-grid conventions established above.

4.8.1 The explicit coefficient and chain

New definition. Define $h_p : \{2, \dots, 2p\} \longrightarrow \mathbb{F}_2$ by

$$h_p(d) = \begin{cases} \overline{d}, & 2 \le d \le p, \\ 1 + \overline{p}, & d = p+1, \\ 1, & p+2 \le d \le 2p. \end{cases} \tag{H4D.1}$$

For integers $2 \le A < B \le 2p$, put

$$C_p^0(A, B) = h_p(A) + h_p(B). \tag{H4D.2}$$

Define $T_p^0(A,B)$ by the following disjoint and exhaustive table.

condition	$T_p^0(A,B)$
$B \le p$	$1+\overline{B}$
$A \le p, B \ge p+2, B-A \ge p+1$	$1+\overline{B}$
$A \le p, B \ge p+2, B-A \le p$	0
$B = p+1$	$1+\overline{A}+\overline{p}$
$A = p+1$	1
$A \ge p+2$	$1+\overline{A}$

(H4D.3)

To verify exhaustiveness, first separate $B \le p$, $B = p+1$, and $B \ge p+2$. In the last case, separate $A \le p$, $A = p+1$, and $A \ge p+2$, and split $A \le p$ according to whether $B-A$ is at most p. Thus the definition covers all stated arguments, including $p = 3$.

New definition. Let S be a four-element subset of the columns and let σ be a four-cycle with support S. For every $u \in S$ with $u+1 \in S$, there are unique integers $2 \le A < B \le 2p$ such that

$$S = \{u, u+1, u+A, u+B\} \tag{H4D.4}$$

with the displayed columns in positive circular order. Label these four columns by $0,1,2,3$ in that order. The contribution $w_p(\sigma,u)$ is given by the following complete table. A word in the first column is the cycle word beginning with label zero.

cycle word	w_p	cycle word	w_p
0123	$C_p^0(A,B)$	0321	0
0132	$C_p^0(A,B)$	0231	0
0213	$T_p^0(A,B)$	0312	$T_p^0(A,B)+C_p^0(A,B)$

(H4D.5)

Define

$$f_4(\sigma) = \sum_{\substack{u \in S \\ u+1 \in S}} w_p(\sigma,u), \qquad H_4 = \sum_{\substack{S \subset \mathbb{Z}/N\mathbb{Z} \\ |S|=4}} \sum_{\substack{\sigma \text{ a four-cycle} \\ \operatorname{supp}(\sigma)=S}} f_4(\sigma) x_\sigma. \tag{H4D.6}$$

All indexing sets are finite. For each summand, the adjacent pair determines its normalization (H4D.4) uniquely, so these formulas involve no auxiliary choice.

Equivalently, the complete raw correction chain is the explicit sum

$$\begin{aligned} H_4 = \sum_{u \in \mathbb{Z}/N\mathbb{Z}} \sum_{2 \le A < B \le 2p} \big[& C_p^0(A,B)\big(x_{(u\,u+1\,u+A\,u+B)} + x_{(u\,u+1\,u+B\,u+A)} + x_{(u\,u+B\,u+1\,u+A)}\big) \\ & + T_p^0(A,B)\big(x_{(u\,u+A\,u+1\,u+B)} + x_{(u\,u+B\,u+1\,u+A)}\big)\big]. \end{aligned} \tag{H4D.7}$$

Repeated states in this expression are added in $\mathbb{F}_2$. Equation (H4D.7) follows by distributing the six entries of (H4D.5) and then interchanging the two finite sums in (H4D.6).

Lemma 4.8.1 (New lemma: grading of the four-cycle correction)**.** *Every state in H_4 has Maslov value -3. Thus $U^{-2}H_4$ has Maslov degree one.*

Proof. Delete fixed columns and apply the proved new relative-permutation Maslov formula. The six cycle words in (H4D.5) have the following footrule sums and inversion numbers.

cycle words	$\sum_i \lvert\sigma(i)-i\rvert$	$\operatorname{inv}(\sigma)$
$0123, 0321, 0132, 0231$	6	3
$0213, 0312$	8	5

For each word, subtracting the footrule sum from the inversion number gives -3. Fixed-column deletion and simultaneous cyclic translation preserve that difference by the new grading lemmas already proved. Finally the existing declaration `laurentMaslovDegree` gives $-3-2(-2) = 1$ for every term of $U^{-2}H_4$. □

4.8.2 The coefficient identities used in the rectangle proof

Lemma 4.8.2 (New lemma: support on a unit gap)**.** *If a four-element support contains no pair $u, u+1$, then f_4 is zero on every four-cycle with that support.*

Proof. The indexing set in the coefficient sum (H4D.6) is empty. □

Lemma 4.8.3 (New lemma: common inverse sum)**.** *Let S be any four-element subset of the columns. Let σ_+ be its positive circular four-cycle and let $\sigma_- = \sigma_+^{-1}$. For every four-cycle σ on S, one has*

$$f_4(\sigma_-) = 0, \qquad f_4(\sigma) + f_4(\sigma^{-1}) = f_4(\sigma_+). \tag{H4D.8}$$

Proof. Fix an adjacent pair $u, u+1$ in S and normalize as in (H4D.4). The inverse pairs of cycle words are precisely the pairs of entries in the three rows of (H4D.5). In each row their sum is $C_p^0(A, B)$. The positive circular cycle has word 0123 in every such normalization, so its contribution is also $C_p^0(A, B)$. The negative circular cycle has word 0321, whose contribution is zero. Sum these three identities over every adjacent pair of S. Equation (H4D.6) gives exactly (H4D.8). □

New definition. For a three-element subset T of the columns, define

$$P_3(T) = \sum_{\substack{u \in T \\ u+1 \in T}} h_p(d(u, z)), \qquad \{z\} = T \setminus \{u, u+1\}. \tag{H4D.9}$$

The third vertex z is distinct from the adjacent endpoints, so $2 \le d(u, z) \le 2p$ and every summand belongs to the domain of (H4D.1).

Lemma 4.8.4 (New lemma: a primitive for the convex coefficient)**.** *For every four-element subset S, let $C(S)$ denote f_4 of its positive circular cycle. Then*

$$C(S) = \sum_{q \in S} P_3(S \setminus \{q\}). \tag{H4D.10}$$

For every five-element subset T, one consequently has

$$\sum_{q \in T} C(T \setminus \{q\}) = 0. \tag{H4D.11}$$

Proof. Expand the right side of (H4D.10) by (H4D.9) and group terms by their adjacent pair $u, u+1$. If the other two columns of S are $u+A, u+B$ in the normalization (H4D.4), that pair occurs in exactly two of the three-element subsets. Its total contribution is $h_p(A) + h_p(B) = C_p^0(A, B)$. This is its contribution to the positive circular four-cycle by (H4D.5). Summing over all adjacent pairs proves (H4D.10).

Expand the left side of (H4D.11) using (H4D.10). Each three-element subset of T occurs twice, once for each order of deleting its two complementary vertices. Its coefficient is therefore zero in $\mathbb{F}_2$. This proves (H4D.11). □

Exceptional cases. The construction applies for every $p \ge 3$. It includes adjacent pairs crossing the cyclic cut because all normalizations use the cyclic distance. When a four-element support contains several adjacent pairs, (H4D.6) includes each pair once, including the case of four consecutive columns. The definition and the proofs above require no nonempty interval between the distinguished values $2, p, p+1, p+2, 2p$, so they apply at the smallest allowed value $p = 3$.

4.9 The complete lower boundary of the four-cycle correction

Fix an integer $p \geq 3$ and put $N = 2p + 1$. All coefficients and all displayed polynomial identities in this section are in $\mathbb{F}_2$. A bar denotes reduction of an integer modulo two. The function f_4 and the chain H_4 are the explicit four-cycle coefficient and chain defined in (H4D.5)–(H4D.7). The chain H_{32} is defined in (H32.6). We use the decomposition

$$D = \sum_{j \geq 0} U^j d_j$$

of the native Laurent differential, where d_j counts state-empty rectangles with exactly j covered O markings. This is the existing declaration `laurentCoefficient_eq` with coefficients grouped by their nonnegative O counts.

4.9.1 The coefficient at a prescribed unit gap

For positive integers a, b, c with $a + b + c = 2p$, define

$$K_p(a, b, c) = f_4\big((0\,1\,1 + a + b\,1 + a)\big). \tag{H4L.1}$$

The four columns in positive circular order are $0, 1, 1 + a, 1 + a + b$. For every integer u with $1 \leq u \leq 2p - 2$, define

$$F_p(u) = \begin{cases} 1 + [u = p - 1], & 1 \leq u \leq p, \\ \overline{u}, & p + 1 \leq u \leq 2p - 2. \end{cases} \tag{H4L.2}$$

Also define

$$B_p(a, b, c) = \begin{cases} [a \leq p]\,\overline{a}, & c < p, \\ \overline{a + p}, & c = p, \\ \overline{b}, & c > p. \end{cases} \tag{H4L.3}$$

Lemma 4.9.1 (New lemma: the unit-gap coefficient)**.** *For all positive a, b, c with $a + b + c = 2p$,*

$$K_p(a, b, c) = B_p(a, b, c) + [a = 1]F_p(b) + [c = 1]F_p(a). \tag{H4L.4}$$

Proof. Put $A = 1 + a$ and $B = 1 + a + b = N - c$. The support in (H4L.1) is $(0, 1, A, B)$ in positive circular order. Apply the explicit adjacent-pair density (H4D.5) to the cycle $(0\,1\,B\,A)$. The pair $(0, 1)$ is a forward cycle edge, with contribution $C_p^0(A, B)$. The pair $(1, A)$ occurs precisely when $A = 2$, or equivalently $a = 1$. After normalizing this pair to $(0, 1)$, the cycle word is 0213, so its contribution is $T_p^0(B - 1, 2p)$. The pair (A, B), when it occurs, is a reverse cycle edge and has zero contribution. The pair $(B, 0)$ occurs precisely when $B = 2p$, or equivalently $c = 1$. Its normalized cycle word is 0312, so its contribution is $T_p^0(2, A + 1) + C_p^0(2, A + 1)$. These are all possible ambient adjacent pairs in the support. Consequently

$$\begin{aligned} K_p(a, b, c) = {} & C_p^0(A, B) + [a = 1]T_p^0(B - 1, 2p) \\ & + [c = 1]\big(T_p^0(2, A + 1) + C_p^0(2, A + 1)\big). \end{aligned} \tag{H4L.5}$$

This formula also holds when more than one of these unit pairs occurs, because the coefficient is defined as the sum of their separate contributions.

The definition of h_p in (H4D.1) gives

$$C_p^0(1 + a, N - c) = B_p(a, b, c). \tag{H4L.6}$$

Indeed, if $c < p$, then $h_p(N - c) = 1$. Adding $h_p(1 + a)$ gives $\overline{a}$ for $a \leq p$ and zero for $a > p$. If $c = p$, then $a + b = p$ and the sum is $\overline{a + 1} + 1 + \overline{p} = \overline{a + p}$. If $c > p$, both $1 + a$ and $N - c$ are at most p. The sum is $\overline{1 + a + N - c} = \overline{a + c} = \overline{b}$. These are the three rows of (H4L.3).

For $1 \le u \le 2p-2$, the table (H4D.3) also gives the two identities

$$T_p^0(u+1,2p) = F_p(u), \qquad T_p^0(2,u+2) + h_p(u+2) = F_p(u). \tag{H4L.7}$$

Here are all their cases.

range of u	$T_p^0(u+1,2p)$	$T_p^0(2,u+2)+h_p(u+2)$
$1 \le u \le p-2$	1	1
$u = p-1$	0	0
$u = p$	1	1
$p+1 \le u \le 2p-2$	$\overline{u}$	$\overline{u}$

(H4L.8)

In the first column of values, the first range has $2p-(u+1) \ge p+1$, the next range has $u+1 = p$, and the third has $u+1 = p+1$. The final range has $u+1 \ge p+2$. These respectively use the second, third, fifth, and sixth rows of (H4D.3). For the second column of values, the subrange $u \le p-2$ has $u+2 \le p$, so its terms are $1+\overline{u+2}$ and $\overline{u+2}$. At $u = p-1$ they are $1+\overline{p}$ and $1+\overline{p}$. At $u = p$ they are zero and one. In the final range they are $1+\overline{u+2}$ and one. This proves the entire table and (H4L.7).

If $a = 1$, then $B - 1 = b + 1$, so the second summand of (H4L.5) is $[a = 1]F_p(b)$. If $c = 1$, then $A + 1 = a + 2$ and $h_p(2) = 0$, so the last summand is $[c = 1]F_p(a)$. Using (H4L.6) proves (H4L.4). □

4.9.2 One-O rectangles ending at a triangle

Lemma 4.9.2 (New lemma: the complete triangle coefficient)**.** *Let A, B, C be distinct columns in positive circular order, and put*

$$a = d(A,B), \qquad b = d(B,C), \qquad c = d(C,A).$$

Thus $a, b, c > 0$ and $a+b+c = 2p+1$. The coefficient of either triangle orientation on this support in d_1H_4 is

$$J_p(a,b,c) = \sum_{\text{cyc }(a,b,c)} [a \ge 2]\big(K_p(a-1,b,c) + K_p(b,c,a-1)\big). \tag{H4L.10}$$

Proof. We use the proved new rectangle deletion formula, Lemma 4.6.3. If a four-cycle has consecutive cycle entries (r,q,s) and deleting q produces the target triangle, its one-O rectangle coefficient is

$$[d(r,q) = 1 \text{ or } d(q,r) = 1] + [d(s,q) = 1 \text{ or } d(q,s) = 1]. \tag{H4L.11}$$

This formula follows from the native empty-rectangle criterion and the O interval formula. In particular every contributing removed column is an ambient neighbor of one of the three target columns.

Group the terms of (H4L.11) by that adjacent target column. In the open gap from A to B, there are no such terms if $a = 1$. If $a \ge 2$, the neighbor after A is $q = A + 1$. For the positive triangle orientation, the two terms belonging to this neighbor are the four-cycles

$$(A\,q\,B\,C), \qquad (A\,B\,C\,q).$$

Their sum of coefficients equals the coefficient of $(A\,q\,C\,B)$. Indeed, Lemma 4.8.3 says that the sum of the coefficients of a four-cycle and its inverse is the forward convex coefficient, and the inverse of the forward convex cycle has coefficient zero by (H4D.5). After translating A to zero, the resulting coefficient is $K_p(a-1,b,c)$. For the negative triangle orientation, the same two source cycles are replaced by their inverses. One is the reverse convex cycle with coefficient zero, and the other has coefficient $K_p(a-1,b,c)$. Thus both target orientations give the same result.

The neighbor before B is $q' = B-1$. Starting the positive circular order at q' gives (q', B, C, A) with gaps $(1, b, c, a-1)$. The identical argument gives $K_p(b,c,a-1)$ for either target orientation.

If $a = 2$, then $q = q'$. Counting these as two neighbor incidences is required by (H4L.11). A source adjacent to both A and B is then counted twice and has zero coefficient, as (H4L.11) prescribes. Applying the same grouping to the other two gaps proves (H4L.10). □

Lemma 4.9.3 (New lemma: cancellation of the entire triangle layer)**.** *Let T_{bal} be the sum of both orientations of all three-cycles whose three positive circular gaps are at most p. The triangle coefficients satisfy*

$$d_1 H_4 + d_2 H_{32} = T_{\mathrm{bal}}. \tag{H4L.12}$$

Here the equality is restricted to three-cycle targets. The complete chain equality will follow below after the other targets are excluded.

Proof. For $a \geq 2$, set

$$T_p(a,b,c) = K_p(a-1,b,c) + K_p(b,c,a-1).$$

Formula (H4L.4) gives

$$\begin{aligned} T_p(a,b,c) = {} & [c<p]\,\overline{c} + [b>p]\,\overline{b} + [a=p+1]\,\overline{p} \\ & + [c=1]F_p(a-1) + [b=1]F_p(c). \end{aligned} \tag{H4L.13}$$

For completeness, put $x = a-1$, $y = b$, and $z = c$, so $x+y+z = 2p$. The two terms involving $[x=1]F_p(y)$ cancel. The sum $B_p(x,y,z) + B_p(y,z,x)$ is zero if $z \geq p$. For $z < p$, it is respectively

$$\overline{x} \quad \text{if } y > p, \qquad \overline{y} \quad \text{if } x = p, \qquad \overline{z} \quad \text{otherwise.}$$

These disjoint cases follow from the three rows of (H4L.3). Since $\overline{x+y+z} = 0$, their common indicator expression is the first line of (H4L.13). This proves (H4L.13) without any restriction on the sizes of the gaps.

Suppose first that all three gaps a, b, c are at least two. The second line of (H4L.13) vanishes. Summing the first line cyclically yields

$$J_p(a,b,c) = 1 + \overline{p}\left(\sum_{v\in\{a,b,c\}} [v=p] + \sum_{v\in\{a,b,c\}} [v=p+1]\right). \tag{H4L.14}$$

The initial one is $\overline{a+b+c}$. If exactly one gap is one, rotate the triple to $(1,b,c)$, where $b,c \geq 2$ and $b+c = 2p$. The two nonzero summands in (H4L.10) give

$$J_p(1,b,c) = 1 + \overline{p}\big([b=p+1] + [c=p+1]\big) + F_p(b-1) + F_p(b). \tag{H4L.15}$$

The terms $[c>p]\overline{c}$ and $[b<p]\overline{b}$ cancel because their indicators are equal and $\overline{b} = \overline{c}$. If two gaps are one, rotate them to $(1,1,2p-1)$. Equation (H4L.13), with (H4L.2), gives $J_p = 0$.

The coefficient of $d_2 H_{32}$ is the proved expression

$$L_p(a,c) + L_p(b,a) + L_p(c,b) \tag{H4L.16}$$

from (H32.10)–(H32.11). The following table displays all cases of (H4L.14)–(H4L.16). Write $\eta = \overline{p}$.

Gap condition	J_p	$d_2 H_{32}$	$[\max(a,b,c) \leq p]$
$a,b,c \geq 2,\ \max(a,b,c) < p$	1	0	1
$a,b,c \geq 2,\ \max(a,b,c) = p$	$1+\eta$	η	1
$a,b,c \geq 2,\ c = p+1$	$1+\eta$	$1+\eta$	0
$a,b,c \geq 2,\ c \geq p+2$	1	1	0
$(a,b,c) = (1,b,2p-b),\ 2 \leq b \leq p-2$	1	1	0
$(a,b,c) = (1,p-1,p+1)$	η	η	0
$(a,b,c) = (1,p,p)$	0	1	1
$(a,b,c) = (1,p+1,p-1)$	1	1	0
$(a,b,c) = (1,b,2p-b),\ p+2 \leq b \leq 2p-2$	0	0	0
$(a,b,c) = (1,1,2p-1)$	0	0	0

(H4L.17)

The third and fourth rows permit cyclic rotation to put the unique gap greater than p in the last position. All other rows are also understood up to the cyclic rotation already specified. No permutation of the gap order is used.

Here is a direct verification of the third column. If all gaps are less than p, its three terms are $1+\overline{c}$, $1+\overline{a}$, and $1+\overline{b}$, whose sum is zero. If all gaps are at least two and one equals p, exactly one does. Rotating it to $c=p$ gives the three terms 0, $1+\overline{a}$, and $\overline{b}$, whose sum is η. For $c=p+1$, the three terms are 1, $1+\overline{a}$, and $1+\overline{b}$. Since $a+b=p$, their sum is $1+\eta$. For $c\geq p+2$, the terms are 0, $1+\overline{a}$, and $1+\overline{c}+\overline{b}$, whose sum is one. When $a=1$, the first two terms of (H4L.16) vanish. Substitution of the five possible ranges of b into $L_p(c,b)$ gives respectively $1,\eta,1,1,0$. When two gaps are one, each term of (H4L.16) contains an argument equal to one and is zero. The second column follows from (H4L.14), (H4L.15), and the two rows of (H4L.2). Thus every row is an exact algebraic case, rather than an enumeration at finitely many grid sizes.

In each row, the sum of the second and third columns is the fourth column. Lemma 4.9.2 and (H32.11) hold for both orientations of a triangle, so this proves (H4L.12) for every three-cycle target. □

4.9.3 Cancellation at targets with two transpositions

Lemma 4.9.4 (New lemma: the two-transposition coefficients)**.** *Every relative permutation consisting of two disjoint transpositions has coefficient zero in DH_4.*

Proof. Let the four moved columns be $A<B<C<D$ after a cyclic cut. The three possible pairings are

$$(A\,B)(C\,D),\qquad (A\,D)(B\,C),\qquad (A\,C)(B\,D).$$

Their Maslov values, from the proved new inversion-minus-footrule formula, are respectively $-2,-2,-4$. A rectangle from a four-cycle can reach such a target only by cutting that four-cycle into its two transpositions. Conversely, joining two columns that belong to different target transpositions produces every possible four-cycle source. There are four sources, partitioned into two inverse pairs.

For the first pairing, these inverse pairs are

$$\{(A\,B\,C\,D),(A\,D\,C\,B)\},\qquad \{(A\,B\,D\,C),(A\,C\,D\,B)\}.$$

For either inverse pair, each source has one-O differential coefficient

$$e=[C-B=1]+[N-D+A=1]. \tag{H4L.18}$$

To verify (H4L.18), the source $(A\,B\,C\,D)$ swaps the columns B,D. Its two possible rectangle O counts are $C-B$ and $N-D+A$. The source $(A\,B\,D\,C)$ swaps B,C, with the same two O counts. When the relevant count is one, the open overlap of the column interval and the corresponding anchor-row interval contains no fixed state point. The two remaining moved points lie outside the rectangle interior. Thus that rectangle is state-empty. When the count is not one, it does not contribute to d_1. This proves the assertion for these two sources. The proved new inversion symmetry of the rectangle differential gives the same assertion for their inverses. By Lemma 4.8.3, the coefficient sum within each source pair is the same forward convex coefficient C_0. The total coefficient of the target is therefore $eC_0+eC_0=0$. The native rectangle Maslov relation excludes O counts other than one for this target.

For the second pairing, take the positive circular order (B,C,D,A). It becomes the first pairing in that order, so the same proof applies. Equivalently, its common rectangle coefficient is $[D-C=1]+[B-A=1]$, and it again multiplies C_0+C_0.

For the crossing pairing $(A\,C)(B\,D)$, the four sources again form two inverse pairs. Each source has exactly one state-empty rectangle with no O marking. For one inverse pair, choose the source $(A\,D\,B\,C)$ and swap A,B. The rectangle with column interval $[A,B)$ and anchor-row

interval $[C, D)$ has disjoint intervals. It contains no O marking and no state point in its interior. The complementary rectangle has positive O count. For the other inverse pair, choose $(A\,B\,D\,C)$ and swap A, D. The rectangle with column interval $[D, A)$ and anchor-row interval $[B, C)$ likewise has disjoint intervals and is state-empty. Its complement has positive O count. Inversion supplies the same rectangle count for the other two sources. The common inverse-sum lemma therefore gives total coefficient $C_0 + C_0 = 0$. The Maslov relation requires O count zero for this target. All three pairings have been treated, proving the lemma. □

Corollary 4.9.5 (New lemma: the full lower-layer identity)**.** *The following equalities of complete chains hold.*

$$d_1 H_4 + d_2 H_{32} = T_{\mathrm{bal}}, \qquad d_j H_4 = 0 \quad (j \geq 2). \tag{H4L.19}$$

Moreover $d_0 H_4$ has no target consisting of two transpositions.

Proof. Every source in H_4 is a four-cycle of Maslov value -3. Swapping two columns of its support cuts the cycle into a three-cycle with a fixed column or into two transpositions. Swapping a moved column with a fixed column inserts that column into the cycle and therefore produces a five-cycle. The proved new insertion inequality implies that an empty rectangle inserting a fixed column has O count zero. Swapping two fixed columns contributes zero by the proved new two-fixed-column rectangle pairing. These alternatives exhaust all swaps. Consequently the only possible targets in $d_1 H_4$ are three-cycles and two transpositions. The latter vanish by Lemma 4.9.4. The proved new inversion pairing for H_{32} and (H32.11) show that $d_2 H_{32}$ is supported entirely on three-cycles. Lemma 4.9.3 now proves the first complete chain equality in (H4L.19).

For the second equality, a four-cycle has $r = 3$. By Lemma 4.5.1, multiplication by one transposition leaves $r \geq 2$. The proved new inequality $-M(x_\sigma) \geq r(\sigma)$ therefore gives target Maslov value at most -2. The native empty-rectangle Maslov relation gives target Maslov value $-4 + 2j$ for an O count j. It follows that $j \leq 1$, as required. The last assertion is part of Lemma 4.9.4. □

Smallest sizes and the scope of the argument. All formulas above hold at $p = 3$, corresponding to the smallest allowed $n = 2$. Intervals such as $2 \leq b \leq p - 2$ are then empty and contribute no case. The values $F_p(p-1) = 0$ and $F_p(p) = 1$ are explicitly retained, including the extreme balanced triple $(1, p, p)$. The proof counts both incidences when a triangle gap equals two. It therefore also covers the coincidence of the two inserted neighbor columns. This section proves all lower-layer cancellations. The remaining Maslov-minus-four five-cycle coefficients are computed in Sections 4.11 and 4.12.

4.10 The complete Maslov minus six boundary of the explicit $(3, 2)$ correction

This section proves a support statement for the explicit chain H_{32} defined in Section 4.7. Every coefficient below belongs to $\mathbb{F}_2$. We retain the integers $p \geq 3$, $N = 2p + 1$, and $g = p(p-1)/2$, the native markings $O(c) = -c - 1$ and $X(c) = p - c - 1$, and $x_\sigma(c) = -\sigma(c)$, with all coordinates reduced modulo N. We write

$$S(a, b) = [d(a, b) \leq p] \qquad (a \neq b).$$

Thus $S(a, b) + S(b, a) = 1$. This identity follows because the two positive cyclic distances sum to $N = 2p+1$. The operators d_j are defined by $D = \sum_{j \geq 0} U^j d_j$ as in (H32.8). The raw chain $d_0 H_{32}$ has Maslov value minus six by the existing declaration `laurentMaslovDegree_add_card_OColumns`.

For three distinct vertices a, b, c in positive circular order, define

$$B(a, b, c) = S(a, b) S(b, c) S(c, a). \tag{M6.1}$$

We call their triangle balanced if $B(a,b,c)=1$. For five distinct vertices $c_0,\dots,c_4$ in positive circular order, we call their pentagon balanced if

$$S(c_i,c_{i+2})=1 \quad \text{for every } i \pmod 5. \tag{M6.2}$$

These are definitions by inequalities in the native column coordinates. They impose no condition on the gaps being equal.

4.10.1 Two elementary primitives for the coefficient table

A half-turn word for a positive circularly ordered tuple is defined by (H32.3)–(H32.4), with the first vertex as origin. The same definition applies to tuples of three or four vertices. For a four-tuple (u,a,v,z) put $t_x=\overline{d(u,x)}$ and define $F(u,a,v,z)$ by

w	F	w	F
000	0	100	$1+t_v$
001	$1+t_z$	101	0
010	$1+t_a$	110	$1+t_z$
011	0	111	0

(M6.3)

For a positive triple (u,a,v) define $K(u,a,v)$ by

w	00	01	10	11
K	0	0	t_v	t_a+t_v.

(M6.4)

The quantities in (M6.3)–(M6.4) are new explicitly defined functions.

Lemma 4.10.1 (New lemma: primitive identities)**.** *For every positive circularly ordered five-tuple (u,a,v,b,c),*

$$h_{32}(u,a,v,b,c)=F(u,a,v,b)+F(u,a,v,c). \tag{M6.5}$$

For every positive circularly ordered four-tuple (u,a,v,b),

$$F(u,a,v,b)+F(v,b,u,a)=K(u,a,v)+K(v,b,u). \tag{M6.6}$$

Moreover the function

$$G(u,a,v,b)=F(u,a,v,b)+F(a,v,b,u)$$

satisfies

$$\begin{aligned} G(u,a,v,b)=S(u,a)S(a,v)\,\overline{d(u,a)}+S(u,v)(1+S(a,b)) \\ +(1+S(v,b))\,\overline{d(a,b)}. \end{aligned} \tag{M6.7}$$

Proof. We give all finite algebraic cases used in this lemma. Write a tuple as $(c_0,\dots,c_{q-1})$ in positive circular order, with origin c_0, and put $t_i=\overline{d(c_0,c_i)}$, so $t_0=0$. If the origin changes to c_i, then

$$\overline{d(c_i,c_j)}=t_i+t_j+[j<i]. \tag{M6.8}$$

The last summand is present because a wrap adds the odd integer N. Consequently all the identities are identities in formal parity variables, with no assumption on their numerical values.

For (M6.5), use the indexing $(u,a,v,b,c)=(c_0,c_1,c_2,c_3,c_4)$. The next table lists the two four-point values obtained by deleting c or b. The sum of its last two columns is precisely the

corresponding entry of (H32.5).

w	$F(u,a,v,b)$	$F(u,a,v,c)$	w	$F(u,a,v,b)$	$F(u,a,v,c)$
0000	0	0	1000	0	$1+t_2$
0001	0	$1+t_4$	1001	$1+t_2$	$1+t_4$
0010	0	$1+t_4$	1010	$1+t_2$	$1+t_1$
0011	$1+t_3$	$1+t_4$	1011	0	0
0100	0	$1+t_1$	1100	$1+t_2$	$1+t_2$
0101	$1+t_1$	$1+t_4$	1101	$1+t_3$	0
0110	$1+t_1$	$1+t_1$	1110	$1+t_3$	$1+t_4$
0111	0	0	1111	0	0

(M6.9)

Deleting a point deletes its projected position from the ordered half-turn list. This describes directly how the entries in (M6.9) follow from (M6.3).

For the remaining identities use $(u,a,v,b)=(c_0,c_1,c_2,c_3)$. The following table substitutes (M6.3)–(M6.4), changing origins by (M6.8) where necessary.

w	$F(u,a,v,b)+F(v,b,u,a)$	$K(u,a,v)$	$K(v,b,u)$	$G(u,a,v,b)$
000	0	0	0	t_1
001	$1+t_3$	0	$1+t_3$	$1+t_3$
010	$1+t_2$	0	$1+t_2$	$1+t_1$
011	0	0	0	0
100	$1+t_2$	0	$1+t_2$	t_1
101	t_2	t_2	0	t_1
110	t_2	t_2	0	t_1
111	t_1+t_2	t_1+t_2	0	0

(M6.10)

This proves (M6.6). To verify (M6.7) without any unlisted geometric cases, the six indicators

$$(S(u,a),S(u,v),S(u,b),S(a,v),S(a,b),S(v,b))$$

have respective values

w	$(S_{01},S_{02},S_{03},S_{12},S_{13},S_{23})$	w	$(S_{01},S_{02},S_{03},S_{12},S_{13},S_{23})$
000	111111	100	110111
001	110100	101	100101
010	110101	110	100111
011	100001	111	000111

Substituting these eight strings in the right side of (M6.7), with $\overline{d(a,b)}=t_1+t_3$, gives its last column in (M6.10). The eight words exhaust the four-point half-turn orders. This proves all three identities. □

4.10.2 Which rectangles can contribute

The next arguments use the relative-permutation Maslov formula and fixed-column deletion lemma in Section 4.2, and the new exact insertion-index identity (IG.2) in Lemma 4.4.3. In the notation used there,

$$\ell(\sigma)=-M(x_\sigma)=\sum_c|\sigma(c)-c|-\mathrm{inv}(\sigma). \tag{M6.11}$$

An empty zero-O rectangle increases ℓ by one by `laurentMaslovDegree_add_card_OColumns`.

Corollary 4.10.2 (New lemma: the Maslov minus six types)**.** *Every nonzero term of d_0H_{32} has relative cycle type* (5)*,* $(4,2)$*, or* $(3,3)$ *and Maslov value* -6*. For type* (5)*, the only possibilities are the two star permutations $c_i\mapsto c_{i+2}$ and $c_i\mapsto c_{i+3}$ on five circularly ordered columns.*

Proof. Apply Lemma 4.5.7 to the chain (H32.6). Its inverse symmetry follows from Lemma 4.7.1. The five-cycle assertion is Lemma 4.5.6. □

For later use we record why deleting a vertex gives every contribution to the targets with two nontrivial cycles. If a target of type $(4,2)$ or $(3,3)$ has $\ell = 6$, delete a vertex from its four-cycle in the $(4,2)$ case, or from either three-cycle in the $(3,3)$ case. This gives a possible source of type $(3,2)$. It occurs in H_{32} exactly when its two transposition endpoints belong to different complementary arcs of its triangle. This is the intrinsic-order lemma of Section 4.7. Such a source has $\ell = 5$. Lemma 4.4.3 then states that its unique zero-O reinsertion rectangle has no interior state points, because $\ell(\text{target}) - \ell(\text{source}) = 1$. Thus each eligible deletion contributes its coefficient exactly once. No other source of type $(3,2)$ can have that target, since the swap must remove precisely the newly fixed vertex.

4.10.3 Cancellation of every four-cycle–transposition target

Lemma 4.10.3 (New lemma: the $(4,2)$ coefficient)**.** *Every relative permutation of cycle type $(4,2)$ has coefficient zero in $d_0 H_{32}$.*

Proof. It suffices to consider a target with $\ell = 6$. Let its transposition be $(u\ v)$. If all four other vertices belong to the same complementary arc of u, v, deleting any one of them leaves the transposition endpoints in the same arc of the remaining triangle. There are no eligible sources, so the coefficient is zero.

Suppose the two arcs contain one and three of the four-cycle vertices. Write their positive order as (u, a, v, b_1, b_2, b_3). Deleting a is ineligible. Deleting one of the three b vertices leaves the three eligible sources, and their total coefficient is

$$h_{32}(u,a,v,b_1,b_2) + h_{32}(u,a,v,b_1,b_3) + h_{32}(u,a,v,b_2,b_3) = 0.$$

Indeed, (M6.5) writes this sum as a sum in which each of $F(u,a,v,b_1), F(u,a,v,b_2), F(u,a,v,b_3)$ occurs twice. The orientation of the remaining three-cycle does not affect its coefficient by (H32.6).

Finally suppose the two arcs contain two vertices each, with order $(u, a_1, a_2, v, b_1, b_2)$. All four deletions are eligible. Their sum is

$$\begin{aligned} &h_{32}(u,a_1,v,b_1,b_2) + h_{32}(u,a_2,v,b_1,b_2) \\ &\qquad + h_{32}(v,b_1,u,a_1,a_2) + h_{32}(v,b_2,u,a_1,a_2) \\ &= \sum_{i,j=1}^{2} \big(F(u,a_i,v,b_j) + F(v,b_j,u,a_i)\big) \\ &= \sum_{i,j=1}^{2} \big(K(u,a_i,v) + K(v,b_j,u)\big) = 0. \end{aligned}$$

The first identity is (M6.5), the second is (M6.6), and every term in the last sum occurs twice. The three distributions exhaust the positions of the four-cycle vertices. □

4.10.4 The star coefficient

Lemma 4.10.4 (New lemma: the pentagon coefficient)**.** *Let $c_0, \dots, c_4$ be five vertices in positive circular order. The coefficient in $d_0 H_{32}$ of each of their two star states is one if the pentagon is balanced and is zero otherwise.*

Proof. Write the indices modulo five. The five possible sources are the permutations with transposition $(c_i\ c_{i+2})$ and a three-cycle on the other three vertices. Their intrinsic orders are $(c_i, c_{i+1}, c_{i+2}, c_{i+3}, c_{i+4})$. For the forward star, the source at $i = 0$ is

$$(c_0\ c_2)(c_1\ c_3\ c_4).$$

Its rectangle from column c_2 to column c_3 is empty and avoids O. There are no moved columns in that open column interval. The columns of its covered O markings would have to belong to both $[c_2, c_3)$ and $[c_4, c_0)$, which are disjoint. An interior fixed state point would force its adjacent O center inside the rectangle, so no fixed point is in its interior either. Its complementary rectangle contains O. For the inverse star the source is $(c_0\ c_2)(c_1\ c_4\ c_3)$, and the same argument applies to the rectangle from c_4 to c_0, whose possible marking columns belong to both $[c_4, c_0)$ and $[c_2, c_3)$. Cyclic reindexing proves the same statement for all five sources. A transposition splitting a five-cycle into lengths three and two has exactly five choices, so this list is exhaustive. Thus either coefficient is

$$\sum_{i=0}^{4} h_{32}(c_i, c_{i+1}, c_{i+2}, c_{i+3}, c_{i+4}). \tag{M6.12}$$

Changing the origin cyclically permutes the sixteen half-turn chambers in the following four orbits. The words in each row are listed in the order of successive origins.

Representative	Successive words
0000	0000, 0001, 0011, 0111, 1111
0010	0010, 0101, 1011, 1110, 1000
0100	0100, 1001, 0110, 1101, 1100
1010	1010, 1010, 1010, 1010, 1010

Here is the explicit rule producing the table. Adjoin the initial bit $b_0 = 0$ to a word $b_1b_2b_3b_4$. If there is another zero bit, let j be the least positive index with $b_j = 0$. Otherwise put $j = 1$. The vertex at projected position j is the next vertex in the original circular order. After changing the origin to this vertex, the new word is

$$(b_{j+1} + b_j, \ldots, b_4 + b_j, b_0 + b_j + 1, \ldots, b_{j-1} + b_j + 1).$$

The additions here are modulo two. Projected positions before the new origin wrap once through a half-circle, which accounts for the extra one. This rule gives the five words in each displayed row. The table lists all sixteen words, with no omissions or repetitions between rows. Using (M6.8), the four corresponding sums in (M6.12) are respectively

$$\begin{aligned}
&0 + t_1 + t_1 + 0 + 0 = 0,\\
&(1 + t_4) + (1 + t_2) + 0 + (t_1 + t_2) + (t_1 + t_4) = 0,\\
&(1 + t_1) + (1 + t_3) + 0 + (t_1 + t_3) + 0 = 0,\\
&(t_1 + t_2) + (t_2 + t_3) + (t_3 + t_4) + (1 + t_4) + t_1 = 1.
\end{aligned}$$

These are direct substitutions of all entries of (H32.5), so they prove that the coefficient is the indicator of chamber 1010.

It remains to identify this chamber geometrically. Put $L = N/2$ and use $c_0 = 0$ as origin. If (M6.2) holds, then $0 < c_1 < c_2 < L < c_3 < c_4 < N$. The other two-step inequalities give

$$0 < c_3 - L < c_1 < c_4 - L < c_2 < L.$$

Conversely these inequalities imply that all five positive two-step distances are less than L, and hence at most p because they are integers. This projected order has half-circle bits 1010. Therefore the indicator just obtained is exactly (M6.2). □

4.10.5 The two-triangle coefficient

Lemma 4.10.5 (New lemma: the $(3, 3)$ coefficient)**.** *Let τ consist of two disjoint three-cycles and have $M(x_\tau) = -6$. Let T_1, T_2 be their two unordered vertex triples. Then*

$$[x_\tau] d_0 H_{32} = B(T_1)B(T_2). \tag{M6.13}$$

Here $B(T)$ means (M6.1) after placing the triple in positive circular order. The formula holds for every orientation of the two cycles having the stated Maslov value.

Proof. Write the six vertices in positive circular order as $c_0, \dots, c_5$. For brevity, write $S(i,j) = S(c_i, c_j)$, and similarly suppress the c in F, G, h_{32}. Up to a cyclic change of origin and interchange of the two triples, the underlying partition has one of three forms:

$$\{0,1,2\} \mid \{3,4,5\}, \qquad \{0,1,3\} \mid \{2,4,5\}, \qquad \{0,2,4\} \mid \{1,3,5\}. \tag{M6.14}$$

For a direct enumeration, the ten unordered partitions into triples consist of the three cyclic placements of the first type, the six placements of the second type, and the last type. Eligibility of a deletion depends only on this partition, by the intrinsic-order criterion and the insertion-index discussion above. It does not depend on the orientations of the three-cycles.

For the first partition in (M6.14), every deletion leaves the two transposition endpoints in the same complementary arc of the other triangle. Therefore there are no sources and the coefficient is zero. Both triangles cannot be balanced: balance would require both $d(c_0, c_2) > p$ and $d(c_3, c_5) > p$, but these disjoint arcs cannot have total length greater than N. Thus (M6.13) holds in this case.

For the second partition, the four eligible intrinsic source orders are

$$(0,2,3,4,5), \quad (1,2,3,4,5), \quad (2,3,5,0,1), \quad (2,3,4,0,1).$$

Expanding their coefficients by (M6.5) and pairing each four-tuple with its one-step cyclic rotation gives the rectangular sum

$$\sum_{i \in \{0,1\}} \sum_{j \in \{4,5\}} G(i,2,3,j).$$

In (M6.7) the first term is independent of j, and the third term is independent of i. Each of those terms occurs twice. The sum is therefore

$$(S(0,3) + S(1,3))(S(2,4) + S(2,5)). \tag{M6.15}$$

Because $0,1,2,3,4,5$ are circularly ordered, the first factor equals one exactly when $S(0,3) = 0$ and $S(1,3) = 1$. The second equals one exactly when $S(2,4) = 1$ and $S(2,5) = 0$. Under these four conditions $S(0,1) = 1$ and $S(4,5) = 1$. Indeed, if $d(c_0, c_1) > p$, it and the disjoint arc from c_2 to c_5 would have total length at least $2p + 2 > N$. If $d(c_4, c_5) > p$, the same contradiction follows with the arc from c_0 to c_3. Consequently (M6.15) is one exactly when both triangles $(0,1,3)$ and $(2,4,5)$ are balanced. Conversely their balance implies the four displayed conditions immediately from (M6.1). This proves (M6.13) for the second partition.

For the last partition in (M6.14), every deletion is eligible. The six intrinsic orders are

$$(i, i+1, i+2, i+3, i+5), \qquad i \pmod 6.$$

By (M6.5), their sum is

$$\sum_{i=0}^{5} G(i, i+1, i+2, i+3). \tag{M6.16}$$

To see this equality, the first four-point term is the consecutive four-tuple starting at i, and the second is the one-step rotation of the consecutive four-tuple starting at $i+5$.

Let $\delta_i = d(c_i, c_{i+1})$, with indices modulo six, and put

$$A_i = [\delta_i \le p], \qquad B_i = [\delta_i + \delta_{i+1} \le p].$$

Formula (M6.7) gives

$$G(i, i+1, i+2, i+3) = A_i A_{i+1} \overline{\delta_i} + B_i(1 + B_{i+1}) + (1 + A_{i+2})(\overline{\delta_{i+1}} + \overline{\delta_{i+2}}).$$

In the sum (M6.16), the coefficient of $\overline{\delta_i}$ is $A_i A_{i+1} + A_i + A_{i+1}$. Two successive positive gaps cannot both exceed p, since their sum would exceed N. Thus this coefficient is one. Since $\sum_i \delta_i = N$ is odd, (M6.16) becomes

$$1 + \sum_i B_i + \sum_i B_i B_{i+1}. \tag{M6.17}$$

Two disjoint two-gap arcs cannot both have length greater than p. Their total would again be at least $2p+2 > N$. Therefore the set of indices with $B_i = 0$ is empty, a singleton, or two adjacent indices. A cycle on six indices has no three pairwise adjacent distinct indices, so these are all possibilities. If that set is empty, (M6.17) is $1+6+6=1$ in $\mathbb{F}_2$. For one zero it is $1+5+4=0$. For two adjacent zeros it is $1+4+3=0$. Hence (M6.17) equals $\prod_i B_i$. The latter product is $B(c_0,c_2,c_4)B(c_1,c_3,c_5)$, since the six positive two-step gaps are the successive gaps in those two triangles. This proves (M6.13) in the final case. □

4.10.6 The resulting native Alexander bound

Theorem 4.10.6 (New theorem: exact Maslov minus six support)**.** *The chain d_0H_{32} is the sum of the following distinct states, each with coefficient one:*

1. *The two star states on every balanced pentagon.*
2. *Every state consisting of two balanced three-cycles and having native Maslov value minus six.*

Every one of these states satisfies

$$A(x) \le g - 2p. \tag{M6.18}$$

Consequently, for every integer $m > 0$, every term of $U^{-3}d_0H_{32}$ satisfies the exact second-vertex bound

$$A(x) - m(-3) \le g + 3m - 2p. \tag{M6.19}$$

Proof. The exact coefficient statement follows from Lemmas 4.10.2, 4.10.3, 4.10.4, and 4.10.5. It remains to compute the native Alexander values, including the orientation signs. The attached declarations `alexanderTwo_reflectedTorus_eq_sum_displacement` and `alexander_reflecte dTorusAnchor` give

$$A(x_\sigma) = g - \frac{1}{N}\sum_c W(d(c,\sigma(c))), \qquad W(d) = pd - N\max\{0, d-p-1\}, \tag{M6.20}$$

where fixed columns contribute $W(0)=0$. In particular, if $1 \le d \le p$, then

$$W(d) = pd, \qquad W(N-d) = (p+1)d. \tag{M6.21}$$

For a balanced triangle, its three positive successive gaps are at most p and sum to N. The positive cycle therefore has total displacement weight pN, and its inverse has total weight $(p+1)N$ by (M6.21). Their Alexander drops from g are respectively p and $p+1$. For two disjoint balanced triangles the weights add. Their total drop is $2p$, $2p+1$, or $2p+2$, according to their orientations. This proves (M6.18) for the second family.

For a balanced pentagon, its five positive two-step distances are at most p and sum to $2N$. The star $c_i \mapsto c_{i+2}$ has total weight $2pN$, and the inverse star $c_i \mapsto c_{i+3}$ has total weight $2(p+1)N$. Their Alexander drops are exactly $2p$ and $2p+2$, respectively. This proves (M6.18) for the first family. Finally (M6.19) is obtained by adding $3m$ to (M6.18). No restriction stronger than $p \ge 3$ and $m > 0$ has been used. □

Exceptional cases and scope. The smallest allowed grid is $N = 7$, corresponding to $p = 3$. All tuple lengths in this section are at most six, so no proof requires a nonexistent column in this case. For example, the pentagon $(0,1,3,4,5)$ has successive gaps $(1,2,1,1,2)$ and is balanced. Its forward and inverse stars have Maslov value minus six and Alexander values -3 and -5, respectively, since $g = 3$. Thus the forward term $U^{-3}x$ attains the bound in (M6.19). This checks the exponent and chirality conventions on the smallest native grid. The proof above applies to every larger grid without changing the coefficient tables. Theorem 4.10.6 proves the complete d_0H_{32} part of the support calculation. The d_2H_{32} cancellation was proved in Corollary 4.9.5. The d_1H_{32} cancellation is proved in the next two sections.

4.11 Two diagonal identities and one five-cycle cancellation

Fix $p \geq 3$ and put $N = 2p + 1$. All coefficients in this section belong to $\mathbb{F}_2$. We use the functions h_p, C_p^0, and T_p^0 from (H4D.1)–(H4D.3), the four-cycle coefficient f_4 from (H4D.5)–(H4D.6), and the four-point function F from (M6.3). The primitive identity

$$h_{32}(u, a, v, b, c) = F(u, a, v, b) + F(u, a, v, c) \tag{M4A.1}$$

is the proved new Lemma 4.10.1.

4.11.1 The two diagonal identities

Lemma 4.11.1 (New lemma: first diagonal identity)**.** *For every pair of integers $2 \leq A < B \leq 2p$,*

$$T_p^0(A, B) + h_p(A) + h_p(B) + F(0, 1, A, B) = 1 + \overline{A} + [A = p + 1]. \tag{M4A.2}$$

Proof. Put $\eta = \overline{p}$. The following six disjoint cases are the cases defining T_p^0. The second column is the half-turn word for $(0, 1, A, B)$. The next two columns follow from the explicit definitions of F and h_p.

condition	w	$F(0, 1, A, B)$	$h_p(A) + h_p(B)$
$B \leq p$	000	0	$\overline{A} + \overline{B}$
$A \leq p,\ B \geq p + 2,\ B - A \geq p + 1$	001	$1 + \overline{B}$	$1 + \overline{A}$
$A \leq p,\ B \geq p + 2,\ B - A \leq p$	010	0	$1 + \overline{A}$
$B = p + 1$	100	$1 + \overline{A}$	$1 + \overline{A} + \eta$
$A = p + 1$	101	0	η
$A \geq p + 2$	011	0	0

(M4A.3)

To obtain the half-turn words, the projected position of the column 1 is 1, the projected position of $p + 1$ is $\frac{1}{2}$, and the projected position of every column at least $p + 2$ is greater than 1. When $A \leq p$ and $B \geq p + 2$, the projected position of B is greater than A precisely when $B - A \geq p + 1$. These comparisons give every word in the table, including the boundary cases.

Add the value of $T_p^0(A, B)$ in (H4D.3) to the last two columns. The result is $1 + \overline{A}$ in the first, second, third, fourth, and sixth rows. In the fifth row it is $1 + \eta$. Here $A = p + 1$, so $1 + \overline{A} + [A = p + 1] = 1 + (\eta + 1) + 1 = 1 + \eta$. This proves (M4A.2) in every case. □

Lemma 4.11.2 (New lemma: second diagonal identity)**.** *Let r, z be integers with*

$$1 \leq r, \qquad r + 2 \leq z \leq 2p.$$

Then

$$T_p^0(z - r, N - r) + F(0, r, r + 1, z) = \overline{r} + \overline{p}\,[r = p]. \tag{M4A.4}$$

Proof. The arguments of T_p^0 belong to its domain, because $2 \leq z - r < N - r \leq 2p$. The following table gives every case of the half-turn word and the two summands. Every parity is measured from the first column zero.

condition	w	$F(0, r, r + 1, z)$	$T_p^0(z - r, N - r)$
$r \leq p - 1,\ z \leq p$	000	0	$\overline{r}$
$r \leq p - 1,\ p + 1 \leq z \leq p + r$	100	$\overline{r}$	0
$r \leq p - 1,\ z = p + r + 1$	010	$1 + \overline{r}$	1
$r \leq p - 1,\ z \geq p + r + 2$	001	$1 + \overline{z}$	$1 + \overline{z} + \overline{r}$
$r = p$	110	$1 + \overline{z}$	$1 + \overline{z}$
$r \geq p + 1$	111	0	$\overline{r}$

(M4A.5)

In the first four rows, the columns $r, r + 1$ lie in the first half-circle. For $z \geq p + 1$, its projected position lies before r, strictly between r and $r+1$, or after $r+1$ according as $z \leq p+r$, $z = p+r+1$,

or $z \geq p + r + 2$. This gives the indicated words. For $r = p$, the projected order is $r + 1, z, r$, which gives 110. For $r \geq p + 1$, all three columns are in the second half-circle, and their projected order is unchanged. The entries for F now follow from (M6.3).

For the entries in the last column, put $A = z - r$ and $B = N - r$. In the first row, $A \leq p$ and $B - A = N - z \geq p + 1$, so $T_p^0 = 1 + \overline{B} = \overline{r}$. In the second row, $A \leq p$, $B \geq p + 2$, and $B - A \leq p$. In the third row, $A = p + 1$. In the fourth row, $A \geq p + 2$. In the fifth row, $B = p + 1$, so $T_p^0 = 1 + \overline{z - p} + \overline{p} = 1 + \overline{z}$. In the last row, $B \leq p$, so again $T_p^0 = 1 + \overline{B} = \overline{r}$. These deductions verify the entire last column.

Adding the two summands gives $\overline{r}$ in every row except $r = p$, where it gives zero. This is exactly the right side of (M4A.4). □

4.11.2 A useful expression for a nonconvex four-cycle

For a positive circularly ordered four-tuple (a, b, d, e), put

$$G'(a, b, d, e) = f_4((a\, d\, e\, b)).$$

Write $E(x, y) = [d(x, y) = 1]$ for a positive ambient adjacent pair. Define the two auxiliary functions on positive distances by

$$\psi_1(s) = 1 + \overline{s} + [s = p + 1], \qquad \psi_2(s) = \overline{s} + \overline{p}\,[s = p]. \tag{M4A.6}$$

Lemma 4.11.3 (New lemma: the nonconvex four-cycle formula)**.** *For every positive circularly ordered (a, b, d, e),*

$$\begin{aligned} G'(a, b, d, e) = {} & E(b, d)\big(F(b, d, e, a) + \psi_1(d(b, e))\big) \\ & + E(d, e)\big(h_p(d(d, a)) + h_p(d(d, b))\big) \\ & + E(e, a)\big(F(d, e, a, b) + \psi_2(d(d, e))\big). \end{aligned} \tag{M4A.7}$$

Each expression multiplied by an adjacent-pair indicator is used only when that indicator is one. Its arguments then belong to the stated domains of the corresponding functions.

Proof. There are four possible positive ambient adjacent pairs in this support, namely (a, b), (b, d), (d, e), and (e, a). Normalize each pair in turn to labels $(0, 1)$ and read its cycle word from $(a\, d\, e\, b)$. The four resulting words are respectively

$$0231, \qquad 0312, \qquad 0132, \qquad 0213.$$

By (H4D.5), their densities are respectively zero, $T_p^0 + C_p^0$, C_p^0, and T_p^0.

If $E(b, d) = 1$, translate b to zero. The four columns become $(0, 1, A, B)$ with $A = d(b, e)$ and $B = d(b, a)$. Lemma 4.11.1 expresses its density as $F(b, d, e, a) + \psi_1(d(b, e))$. If $E(d, e) = 1$, its two remaining coordinates relative to d are $d(d, a), d(d, b)$, so its density is the second line of (M4A.7). If $E(e, a) = 1$, use d as origin and put $r = d(d, e)$ and $z = d(d, b)$. The four columns are $(0, r, r + 1, z)$. Relative to the adjacent pair (e, a), the other two coordinates are $z - r$ and $N - r$. Lemma 4.11.2 gives its density as $F(d, e, a, b) + \psi_2(d(d, e))$. Summing the densities over all adjacent pairs gives (M4A.7) by the definition (H4D.6). □

4.11.3 The five-cycle of circular type 01243

Lemma 4.11.4 (New lemma: cancellation for the first nonconvex type)**.** *Let A, B, C, D, E be distinct columns in positive circular order and put*

$$\tau = (A\, B\, C\, E\, D).$$

Then the coefficient of x_τ in $d_0 H_4 + d_1 H_{32}$ is zero.

Proof. The new five-cycle grading table, Lemma 4.5.6, gives $M(x_\tau) = -4$. The proved new minimal-cycle deletion formula, Lemma 4.4.5, says that every four-cycle obtained by deleting one vertex of τ contributes through exactly one empty zero-O reinsertion rectangle. Thus the coefficient of x_τ in $d_0 H_4$ is

$$\begin{aligned} f_4((B\,C\,E\,D)) + f_4((A\,C\,E\,D)) + f_4((A\,B\,E\,D)) \\ + C(\{A,B,C,E\}) + C(\{A,B,C,D\}), \end{aligned} \tag{M4A.8}$$

where $C(S)$ is the forward convex coefficient on a four-element support S. The common inverse-sum lemma replaces each of the first three terms by the corresponding G' term plus its convex coefficient. The five convex coefficients then sum to zero by Lemma 4.8.4. Consequently (M4A.8) equals

$$G'(B,C,D,E) + G'(A,C,D,E) + G'(A,B,D,E). \tag{M4A.9}$$

Expand (M4A.9) using (M4A.7). The indicator $E(B,D)$ is zero because C lies strictly between B and D. Likewise $E(E,B) = 0$ because A lies strictly between E and B. The two terms with indicator $E(C,D)$ have the same $\psi_1(d(C,E))$, which cancels in their sum. Their remaining sum is

$$E(C,D)\big(F(C,D,E,B) + F(C,D,E,A)\big) = E(C,D)h_{32}(C,D,E,A,B)$$

by (M4A.1). The two terms with indicator $E(E,A)$ have the same $\psi_2(d(D,E))$, which also cancels. They leave

$$E(E,A)\big(F(D,E,A,C) + F(D,E,A,B)\big) = E(E,A)h_{32}(D,E,A,B,C).$$

Finally, each of $h_p(d(D,A))$, $h_p(d(D,B))$, and $h_p(d(D,C))$ occurs exactly twice with indicator $E(D,E)$. Those six terms cancel. We have proved

$$[x_\tau](d_0 H_4) = E(C,D)h_{32}(C,D,E,A,B) + E(E,A)h_{32}(D,E,A,B,C). \tag{M4A.10}$$

We now compute the one-O rectangles from H_{32} to this target. Cutting the five-cycle τ into a three-cycle and a transposition has exactly five possibilities. The following table lists the source permutation and its Maslov value, obtained from the proved new $(3,2)$ interlacing criterion.

swapped columns	source	M
A, C	$(A\,E\,D)(B\,C)$	-3
A, E	$(A\,D)(B\,C\,E)$	-5
B, D	$(A\,B)(C\,E\,D)$	-3
B, E	$(A\,B\,D)(C\,E)$	-5
C, D	$(A\,B\,C)(D\,E)$	-3

(M4A.11)

Only the two rows of Maslov value minus five belong to H_{32}. For the source $(A\,D)(B\,C\,E)$, the swap of columns A, E has O counts $d(B,D)$ and $d(E,A)$. The first is at least two because C lies strictly between B, D. The second is one exactly when $E(E,A) = 1$. In that event its column interval has width one, so the rectangle is state-empty. The intrinsic order of this source is (D,E,A,B,C). Its contribution is therefore the second term of (M4A.10).

For the source $(A\,B\,D)(C\,E)$, the swap of columns B, E has O counts $d(C,D)$ and $d(E,B)$. The second is at least two because A lies strictly between E, B. The first is one exactly when $E(C,D) = 1$. Its anchor-row interval then has height one, so the rectangle is state-empty. The intrinsic order of this source is (C,D,E,A,B). Its contribution is the first term of (M4A.10). The native relative-rectangle interval formula gives these counts directly, and the five cuts in (M4A.11) exhaust every source of cycle type $(3,2)$ reaching the target. Thus (M4A.10) is also the coefficient of x_τ in $d_1 H_{32}$. The two equal coefficients cancel in $\mathbb{F}_2$, proving the lemma. □

Smallest cases. The diagonal identities include $A = p + 1$, $B = p + 1$, and $r = p$ as explicit separate cases. They hold at $p = 3$ with any empty open ranges omitted. The four-cycle and five-cycle arguments permit every collection of adjacent pairs, including consecutive runs of three or more support columns. Such pairs are counted separately by (H4D.6), and the cancellations above are equalities of these finite sums.

4.12 The remaining five-cycle coefficients

Throughout this section $p \geq 3$, $N = 2p + 1$, and all sums of coefficients are taken in $\mathbb{F}_2$. Put

$$R_4 = d_0 H_4 + d_1 H_{32}.$$

The preceding boundary-type lemmas and the cancellation of the two-transposition targets show that R_4 is supported on single five-cycles of Maslov value -4. We determine these coefficients exactly.

4.12.1 Inversion and the circular five-cycles

Lemma 4.12.1 (New lemma: inversion invariance of the residual)**.** *For every five-cycle τ of Maslov value -4, its coefficient in R_4 equals that of τ^{-1}. The positive and negative circular five-cycles both have coefficient zero.*

Proof. By the five-cycle deletion formula, the coefficient in $d_0 H_4$ is the sum of the five four-cycle coefficients obtained by deleting a vertex. The common inverse-sum lemma therefore makes the sum of the coefficients of τ, τ^{-1} equal to

$$\sum_{q \in T} C(T \setminus \{q\}),$$

where T is their common support. This sum vanishes by Lemma 4.8.4. The chain H_{32} is invariant under inversion, which commutes with every d_j by the native rectangle bijection. Thus its d_1 coefficients are also invariant under inversion. This proves the first assertion.

For the positive circular five-cycle, the $d_0 H_4$ coefficient is the same vanishing sum of C coefficients. A transposition splitting this cycle into cycles of lengths three and two produces a pair whose two endpoints are adjacent in the circular order of the five moved columns. Such a source has Maslov value -3, by the three-cycle–transposition classification, and is absent from H_{32}. Hence its $d_1 H_{32}$ coefficient is zero. Inversion gives the assertion for the negative circular cycle. □

4.12.2 An exact sum over the adjacent pairs

For this subsection, fix five positively ordered columns and assume that one selected adjacent pair is $0, 1$. Write the remaining three columns as A, B, C, so

$$2 \leq A < B < C \leq 2p.$$

These coordinates are obtained by cyclic translation from any selected pair $u, u + 1$. Abbreviate

$$h(d) = h_p(d), \qquad T(A, B) = T_p^0(A, B), \qquad C^0(A, B) = h(A) + h(B).$$

A local contribution means the summands indexed by this one adjacent pair in the definition of H_4, together with the one-O rectangles from H_{32} whose unique covered O square is in this gap. The total coefficient is the sum of these local contributions over all adjacent pairs in the five-element support.

Lemma 4.12.2 (New lemma: two identities for the local contribution)**.** *For every* $2 \le A < B < C \le 2p$ *one has*

$$h_{32}(C,0,1,A,B) = T(A,C) + T(B,C), \tag{4.12.1}$$

$$h_{32}(0,1,A,B,C) = T(A,B) + T(A,C) + h(B) + h(C). \tag{4.12.2}$$

Proof. The first diagonal-primitive identity proved above states

$$T(A,Z) + h(A) + h(Z) + F(0,1,A,Z) = 1 + \overline{A} + [A = p+1].$$

Use this identity with $Z = B$ and $Z = C$, and add the two equations. The terms depending only on A cancel. The primitive formula for h_{32} then gives (4.12.2).

For (4.12.1), translate C to zero in the second diagonal-primitive identity. Its parameters are $r = N - C$ and $z = r + A$. They satisfy $1 \le r$, $r + 1 < z < N$. That identity gives

$$T(A,C) + F(C,0,1,A) = \overline{N - C} + \overline{p}\,[N - C = p],$$

whose right side is independent of A. Add the corresponding equation with A replaced by B and apply the primitive formula for $h_{32}(C,0,1,A,B)$. This proves (4.12.1). □

Lemma 4.12.3 (New lemma: the five local coefficients)**.** *Let the five columns* $0,1,A,B,C$ *be labelled* $0,1,2,3,4$ *in that order. For the following five cycle words, the local contribution of the selected adjacent pair is as stated.*

cycle word	*local contribution*
01342	0
02314	0
03412	0
01423	0
03124	$[\{0,1,A,B,C\}$ *is a balanced pentagon*$]$

Here a pentagon is balanced when each sum of two successive positive circular gaps is at most p*.*

Proof. The following table records the complete contributing terms. Its middle column lists the three adjacent-pair densities obtained by deleting one of labels $2,3,4$. Its last column gives the intrinsic orders of the sources in H_{32} whose restoring rectangle has its one O marking in the gap $0,1$. An empty entry is a zero sum.

word	four-cycle densities	three-cycle–transposition coefficients
01342	$C^0(B,C) + C^0(A,C) + C^0(A,B)$	0
02314	$T(B,C) + T(A,C)$	$h_{32}(C,0,1,A,B)$
03412	$T(A,C) + T(A,B) + h(B) + h(C)$	$h_{32}(0,1,A,B,C)$
01423	$C^0(B,C) + C^0(A,C) + C^0(A,B)$	0
03124	$T(B,C) + T(A,B) + h(B) + h(C)$	$h_{32}(1,A,B,C,0)$ $+ h_{32}(A,B,C,0,1)$ $+ h_{32}(B,C,0,1,A)$

(M4B.1)

We justify that this table is complete from the native rectangles. Deleting either endpoint of the selected pair gives no density for that pair. Deleting any of the other three columns gives exactly the middle-column words in the six-word density definition.

For the last column, a source splitting a five-cycle into lengths three and two is obtained by one of the five cuts separated by two positions in the cycle word. The relative rectangle formula says that its marking columns are the intersection of two half-open intervals whose endpoints are moved columns. If this intersection contains just the one marking in the gap $0,1$, it must be exactly $[0,1)$. Any interval containing two moved-column gaps has length at least two. Applying

this criterion to the five cuts gives precisely the following list. The source cycles and the ordered rectangle columns are written in the labels $0, \dots, 4$.

target word	source cycles	ordered columns	$[\sigma(j), \sigma(i))$
02314	(0 2 3)(1 4)	$(4, 3)$	$[0, 1)$
03412	(0 2)(1 3 4)	$(0, 1)$	$[3, 2)$
03124	(0 2 4)(1 3)	$(0, 1)$	$[3, 2)$
03124	(0 3 1)(2 4)	$(4, 1)$	$[0, 2)$
03124	(0 3)(1 2 4)	$(4, 3)$	$[0, 1)$

(M4B.2)

In the first and fifth rows the row interval has height one. In the second and third rows the column interval has width one. In the fourth row, the intersection $[4, 1) \cap [0, 2)$ is $[0, 1)$, and the only moved point in its column interior is in column zero. Its image is column three, outside the required row interval. Fixed points cannot lie in its interior because the open intersection contains no integer column. Thus all five listed rectangles are empty and have one O marking. Their source permutations all have the interlaced $(3, 2)$ type of Maslov value -5. The other cuts either have Maslov value -3 or have marking intersection different from $[0, 1)$, so none contributes to this selected pair. This proves (M4B.1) without any restriction on the other circular gaps.

The following table also records all excluded cuts of Maslov value -5. Let c_0, c_1, c_2, c_3, c_4 be the five support columns in positive order, and let $I_s = [c_s, c_{s+1})$, with indices modulo five. For a cut $i < j$, the last two entries list the sets of indices s whose intervals I_s form the marking-column intersections for the ordered rectangles (i, j) and (j, i) respectively. Thus an entry $\{0\}$ is precisely the selected marking square. An entry containing two indices contains at least two squares, because all five support gaps have positive integer length.

target	(i, j)	source	(i, j) gaps	(j, i) gaps
01342	$(0, 3)$	(0 4 2)(1 3)	$\{1, 2\}$	$\{4\}$
01342	$(0, 4)$	(0 2)(1 3 4)	$\{1\}$	$\{4\}$
01342	$(2, 3)$	(0 1 3)(2 4)	$\{2\}$	$\{4\}$
02314	$(1, 2)$	(0 2 4)(1 3)	$\{1\}$	$\{3\}$
02314	$(2, 4)$	(0 2)(1 4 3)	$\{3\}$	$\{0, 1\}$
02314	$(3, 4)$	(0 2 3)(1 4)	$\{3\}$	$\{0\}$
03412	$(0, 1)$	(0 2)(1 3 4)	$\{0\}$	$\{2\}$
03412	$(1, 3)$	(0 3 2)(1 4)	$\{2\}$	$\{0, 4\}$
03412	$(2, 3)$	(0 3)(1 2 4)	$\{2\}$	$\{4\}$
01423	$(0, 2)$	(0 3)(1 4 2)	$\{1\}$	$\{3, 4\}$
01423	$(0, 4)$	(0 2 3)(1 4)	$\{1\}$	$\{4\}$
01423	$(1, 2)$	(0 1 3)(2 4)	$\{1\}$	$\{3\}$
03124	$(0, 1)$	(0 2 4)(1 3)	$\{0\}$	$\{2\}$
03124	$(1, 4)$	(0 3 1)(2 4)	$\{2, 3\}$	$\{0\}$
03124	$(3, 4)$	(0 3)(1 2 4)	$\{3\}$	$\{0\}$

(M4B.2a)

Each row is obtained by substituting its displayed permutation in $[i, j) \cap [\sigma(j), \sigma(i))$ and in the complementary ordered rectangle formula. These are all three interlaced cuts of each target. The other two cuts have Maslov value -3 by Lemma 4.5.5. Thus the five entries $\{0\}$ are exactly the five rectangles in (M4B.2).

The first and fourth rows of (M4B.1) vanish because each h value appears twice. The second and third rows vanish by Lemma 4.12.2. For the final row, those same identities give

$$T(B, C) + T(A, B) + h(B) + h(C) = h_{32}(C, 0, 1, A, B) + h_{32}(0, 1, A, B, C).$$

Its total local contribution is therefore the sum of h_{32} over all five cyclic rotations of $(0, 1, A, B, C)$. The pentagon coefficient lemma, Lemma 4.10.4, proves that this sum is exactly the balanced-pentagon indicator. This proves the displayed five-row conclusion. □

4.12.3 The exact residual and its Alexander grading

Theorem 4.12.4 (New theorem: the Maslov minus four residual)**.** *Define, for $u \in \mathbb{Z}/N\mathbb{Z}$ and integers $2 \le b \le p$, $p+2 \le c \le p+b$, the five-cycle*

$$\tau_{u,b,c} = (u\, u+p+1\, u+1\, u+b\, u+c).$$

All its five entries are distinct. Then

$$R_4 = \sum_{u\in\mathbb{Z}/N\mathbb{Z}} \sum_{b=2}^{p} \sum_{c=p+2}^{p+b} (x_{\tau_{u,b,c}} + x_{\tau_{u,b,c}^{-1}}). \tag{M4B.3}$$

Every displayed state has Maslov value -4 and Alexander value $g-2p$.

Proof. The five-cycle orbit table shows that the cycles of Maslov value -4 are the two circular cycles and the circular relabellings of 01243, 01342, and their inverses. The circular-cycle coefficients vanish by Lemma 4.12.1. Lemma 4.11.4 shows that all its coefficients vanish. Inversion gives the same conclusion for its inverse type.

Consider the 01342 type and its five circular relabellings. For each adjacent pair in its support, normalize that pair to $0, 1$. The five possible relabelled words are exactly those in Lemma 4.12.3. Thus only word 03124 can contribute, and it contributes one precisely when the support is balanced. For a fixed cycle of this type, at most one adjacent pair gives word 03124. The circular-label orbit has size five, so two different choices of first vertex cannot yield that same word.

In the normalization $0, 1, A, B, C$, balance forces $B = p+1$. Indeed the two-step inequalities from zero and from one give $A \le p$ and $B-1 \le p$. The two-step inequality along the complementary arc from B to zero gives $N - B \le p$. Together these last two inequalities imply $B = p+1$. The remaining balance inequalities are exactly

$$2 \le A \le p, \qquad p+2 \le C \le p+A.$$

To check the converse, these inequalities give the five two-step distances A, p, $C-A$, p, and $N-C+1$, all at most p. Thus they are also sufficient for balance. The contributing cycle word 03124 is therefore $(0, p+1, 1, A, C)$. Translating back and using inversion invariance gives exactly (M4B.3).

The distinguished adjacent pair also proves that no state is repeated within the sum in (M4B.3). It determines u, and the increasing support then determines b, c. The five inverse words of the words in Lemma 4.12.3 are respectively 02431, 04132, 02143, 03241, and 04213. None belongs to the original five-word list. Hence the inverse states are distinct from the forward states as well. Every state in (M4B.3) therefore has coefficient one.

The distinctness of its five entries follows from $0 < 1 < b < p+1 < c < N$. Its compressed cycle word is 03124, which has Maslov value -4 by the five-cycle orbit table. The five positive displacements of $\tau_{u,b,c}$ in cycle order are

$$p+1, p+1, b-1, c-b, N-c.$$

The last three belong to $\{1, \dots, p\}$ and sum to $2p$. All five are at most $p+1$, and their total is $2N$. The native displacement formula therefore gives Alexander drop $2p$. For the inverse cycle, the two first displacements become p, and the remaining three become $N-d$, with d ranging over those last three positive displacements. The weight of each such inverse displacement is $(p+1)d$. Hence its total displacement weight is

$$2p^2 + (p+1)(2p) = 2pN.$$

Its drop is also $2p$. The anchor normalization $A(x_0) = g$ proves the stated Alexander values. □

4.13 The native second representative and the three acceptance statements

All coefficients in this section lie in $\mathbb{F}_2 = \mathbb{Z}/2$. Let $n \geq 2$, put

$$p = n + 1, \qquad N = 2n + 3 = 2p + 1, \qquad g = \frac{n(n+1)}{2} = \frac{p(p-1)}{2},$$

and let $G = \texttt{reflectedTorusGrid}\, n$. We write U^k for the native Laurent monomial `LaurentPolynomial.T` at exponent $k \in \mathbb{Z}$. Thus multiplication by U^k changes the Maslov degree by $-2k$, as specified by `laurentMaslovDegree`. Let $x_0 = \texttt{reflectedTorusAnchor}\, n$ and let $D = \texttt{laurentDifferential}\, G\, (\mathbb{Z}/2)$. The markings and the anchor are

$$O(c) = N - 1 - c, \qquad X(c) = N - 1 - ((c + p + 1) \bmod N), \qquad x_0(c) = -c \bmod N. \quad \text{(FB.1)}$$

These are exactly `reflectedTorusGrid_O_apply`, `reflectedTorusGrid_X_apply`, and `reflectedTorusAnchor_val`. The existing declarations `maslov0_reflectedTorusAnchor` and `alexander_reflectedTorusAnchor` give $M(x_0) = 0$ and $A(x_0) = g$.

4.13.1 The correction relative to the native first cycle

The native triangular chain has the following formula in the present notation:

$$H_{\triangle} = \sum_{a=0}^{p-2} \sum_{b=p-1}^{p-1+a} x_{(a\,b)}. \quad \text{(FB.2)}$$

Here a, b are ordinary representatives in $\{0, \ldots, N-1\}$, and $x_{(a\,b)}$ is the anchor with columns a, b interchanged. Indeed, `firstTorusTrianglePairs` uses precisely the inequalities $a < n$, $n \leq b$, and $b \leq n + a$. Substitution of $n = p - 1$ gives (FB.2), and `firstTorusTriangle` assigns coefficient one to every listed transposition. The lower bound $p - 1 > a$ shows that the two columns are distinct. The upper bound $b \leq 2p - 3 < N$ verifies that every pair is in the native column range. Thus

$$\texttt{firstTorusCycle}\, n\, (\mathbb{Z}/2) = x_0 + U^{-1} D H_{\triangle} \quad \text{(FB.3)}$$

is an equality with the native definition, with no change of cyclic cut.

Let H_e be the explicit odd-shorter-distance transposition sum of Lemma 4.6.2. Let H_4 be the explicit adjacent-pair sum (H4D.7), and let H_{32} be the intrinsic five-column sum (H32.6). Define

$$\boxed{w_2 = U^{-1}(H_e + H_{\triangle}) + U^{-2} H_4 + U^{-3} H_{32}} \quad \text{(FB.4)}$$

and

$$z_2 = \texttt{firstTorusCycle}\, n\, (\mathbb{Z}/2) + D w_2. \quad \text{(FB.5)}$$

Equations (FB.2), (H4D.7), (H32.6), and the displayed definition of H_e specify every term and coefficient of w_2 by finite sums.

Theorem 4.13.1 (New theorem: the explicit correction and exact class)**.** *For every $n \geq 2$, the chain w_2 in (FB.4) has native Maslov degree one. The chain z_2 in (FB.5) has native Maslov degree zero, is a cycle, and represents exactly* $\texttt{reflectedTorusAnchorClass}\, n\, (\mathbb{Z}/2)$*. Its full coefficient formula is*

$$z_2 = U^{-2} R_4 + U^{-3} R_6, \quad \text{(FB.6)}$$

where R_4 is the explicit sum in (M4B.3) and R_6 is the explicit sum in Theorem 4.10.6. In particular, the only Laurent exponents in its support are -2 and -3.

Proof. Every state of H_e and $H_\triangle$ has Maslov value -1. For $H_\triangle$ this is also the existing declaration `firstTorusTriangle_mem_maslovPiece`, together with `maslov0_reflectedTorusAnchor`. Every state of H_4 has Maslov value -3 by Lemma 4.8.1, and every state of H_{32} has Maslov value -5 by the intrinsic-order calculation in Section 4.7. The shifts in (FB.4) therefore give respectively

$$-1+2=1, \qquad -3+4=1, \qquad -5+6=1.$$

The exact native membership follows from `mem_laurentMaslovPiece` and `laurentMonomial_smul_mem_maslovPiece`.

Since D is Laurent-linear and the coefficients have characteristic two, (FB.3)–(FB.5) give

$$z_2 = x_0 + D(U^{-1}H_e + U^{-2}H_4 + U^{-3}H_{32}). \tag{FB.7}$$

The existing declaration `laurentDifferential_mem_maslovPiece` states that D lowers degree by one. Together with $M(x_0)=0$, this proves degree zero for z_2. The existing declarations `laurentDifferential_reflectedTorusAnchor` and `laurentDifferential_sq_eq_zero` then give

$$Dz_2 = Dx_0 + D^2(\cdots) = 0.$$

We next verify the exact native homology class. The existing map `nativeCornerCycleOfChain` takes this homogeneous cycle to a cycle in degree zero of `laurentMaslovComplex` $G\,(\mathbb{Z}/2)$. The degree-one chain (FB.4) determines a native degree-one element using the inverse of `laurentMaslovChainEquiv`. By `laurentMaslovChainEquiv_d`, its native boundary is exactly Dw_2 on underlying Laurent chains. Equation (FB.5) therefore identifies the native cycle of z_2 with `firstTorusNativeCycle` $n\,(\mathbb{Z}/2)$ plus that specific boundary. Ordinary homology kills this boundary. The existing declaration `firstTorusNativeCycle_homology` identifies the remaining class with `reflectedTorusAnchorClass` $n\,(\mathbb{Z}/2)$. This proves equality with the prescribed class itself. Equivalently, (FB.7) supplies a degree-one boundary witness directly from the anchor cycle used to define that same class.

It remains to compute the support of z_2. Write $D=\sum_{j\ge 0}U^j d_j$. Lemma 4.6.2 gives $x_0 + U^{-1}DH_e = U^{-1}T_{\mathrm{bal}}$. Corollary 4.9.5 gives

$$d_1H_4 + d_2H_{32} = T_{\mathrm{bal}}, \qquad d_jH_4 = 0 \quad (j\ge 2).$$

The two-O calculation in Section 4.7 gives $d_jH_{32}=0$ for $j\ge 3$. Substitute these identities into (FB.7). The coefficient of U^{-1} is $T_{\mathrm{bal}} + d_1H_4 + d_2H_{32} = 0$. The remaining terms are

$$U^{-2}(d_0H_4 + d_1H_{32}) + U^{-3}d_0H_{32}.$$

Theorem 4.12.4 and Theorem 4.10.6 identify these two raw chains with R_4 and R_6 respectively. This proves (FB.6), including every coefficient, and proves M902. □

4.13.2 The native bifiltration and weighted level

Theorem 4.13.2 (New theorem: the full second-vertex bound)**.** *For every $n\ge 2$, the chain z_2 satisfies the native coefficient bounds*

$$-k \le 3, \qquad A(y) - k \le g - 2p + 3 \tag{FB.8}$$

for every monomial $U^k y$ with nonzero coefficient. Thus it belongs to

$$\texttt{laurentBifiltrationLevel}\,G\,(\mathbb{Z}/2)\,3\left(\frac{(n-2)(n-1)}{2}\right). \tag{FB.9}$$

Its native degree-zero cycle belongs to the corresponding level of `laurentMaslovBifiltration`.

Proof. By Theorem 4.12.4, every term of R_4 has $A(y) = g - 2p$. Its term in (FB.6) has $k = -2$, so its two coordinates are

$$2, \qquad g - 2p + 2.$$

By Theorem 4.10.6, every term of R_6 has $A(y) \le g - 2p$. Its term in (FB.6) has $k = -3$, so its two coordinates are bounded by

$$3, \qquad g - 2p + 3.$$

This proves (FB.8). The identity $g-2p+3 = (n-2)(n-1)/2$ is obtained by substituting $p = n+1$ and $g = n(n+1)/2$. The existing declaration `mem_laurentMaslovBifiltrationLevel_iff` states exactly that the native homogeneous bifiltration is given by these two coefficient bounds. Its defining pullback along `laurentMaslovChainEquiv` gives the final assertion. □

Theorem 4.13.3 (New theorem: exact acceptance M901)**.** *Let n, m be natural numbers with $2 \le n$, $0 < m$, and $2m < n+1$. In degree zero of* `laurentMaslovComplex` `(reflectedTorusGrid` n`)` $(\mathbb{Z}/2)$*, the class* `reflectedTorusAnchorClass` $n\,(\mathbb{Z}/2)$ *is* `RealFiltration.RepresentedBelow` *the level*

$$\texttt{Staircase.consecutiveWeight}\, n \left(\frac{2}{m}\right) \langle 2\rangle = \frac{g + 3m - 2p}{m} \tag{FB.10}$$

for the weighted native bifiltration at $t = 2/m$. More generally, the same cycle represents the class at the second vertex for every $t \in [0, 2]$.

Proof. The index $\langle 2\rangle$ belongs to $\mathrm{Fin}(n+1)$ because $n \ge 2$. The exact native definitions `Staircase.consecutiveFirst` and `Staircase.consecutiveSecond` give its coordinates

$$3, \qquad \frac{(n-2)(n-1)}{2} = g - 2p + 3.$$

The definition `Bifiltration.weighted` includes every element of this bifiltration level in weighted level

$$\left(1 - \frac{t}{2}\right) 3 + \frac{t}{2}(g - 2p + 3) = 3 + \frac{t}{2}(g - 2p). \tag{FB.11}$$

Indeed, this bifiltered element is one of the generators in the span defining that weighted level. Theorem 4.13.2 supplies the required membership, and Theorem 4.13.1 supplies both a cycle and its exact homology class. These are precisely the two conditions in the existing definition `RealFiltration.RepresentedBelow`.

For $m > 0$, put $t = 2/m$. Equation (FB.11) becomes $(g + 3m - 2p)/m$, which proves (FB.10) and the acceptance statement. Equivalently, the weight of an individual monomial is

$$\left(1 - \frac{1}{m}\right)(-k) + \frac{1}{m}(A(y) - k) = \frac{A(y)}{m} - k,$$

and (FB.6) gives directly

$$A(y) - mk \le g + 3m - 2p. \tag{FB.12}$$

The proof supplies (FB.12) for every positive integer m. The additional strict inequality is therefore included without requiring a further construction. □

4.13.3 The exact integer inequality of M903

For a native state y, abbreviate the integer appearing on the right side of M903 by

$$\mathcal{W}_n(y) = \sum_{c \in \mathrm{Fin}(N)} W_p\big(d(c, \sigma_y(c))\big), \qquad W_p(d) = p\,d - N \max\{d - p - 1, 0\}, \tag{FB.13}$$

where $d(c, c) = 0$ and $\sigma_y =$ `reflectedTorusRelativeColumns` $n\,y$. For representatives in $\{0, \dots, N-1\}$, the function d is exactly `reflectedTorusCyclicDisplacement` n. The integer weight W_p is exactly `reflectedTorusDisplacementWeight` n, because $n + 2 = p + 1$ and

the subtraction inside the native natural-number expression is truncated at zero. The existing identities `alexanderTwo_reflectedTorus_eq_sum_displacement` and `alexander_reflectedTorusAnchor` give, after embedding the integer quantities in $\mathbb{Q}$,

$$\mathcal{W}_n(y) = N(g - A(y)). \tag{FB.14}$$

Theorem 4.13.4 (New theorem: exact acceptance M903)**.** *Under the hypotheses of Theorem 4.13.3, every nonzero monomial $U^k y$ of z_2 satisfies*

$$N\big(2(n+1) - m(3+k)\big) \le \mathcal{W}_n(y). \tag{FB.15}$$

Proof. For $k = -2$, Theorem 4.12.4 gives $\mathcal{W}_n(y) = 2pN$. The left side of (FB.15) is $N(2p - m)$, which is at most $2pN$ because $m > 0$. For $k = -3$, Theorem 4.10.6 gives $\mathcal{W}_n(y) \ge 2pN$, and the left side is exactly $2pN$. These two cases exhaust the support by (FB.6). Equation (FB.15) is therefore the native M903 inequality with all three of its named functions unchanged. Applying the existing declaration `alexander_reflectedTorus_le_of_displacement_weight` with integer parameter $2p - 3m - mk$ recovers (FB.12). □

Completion of the proof of Theorem 4.1.1. Take the finite chain w_2 in (FB.4). Theorem 4.13.1 proves its degree, the cycle condition, and the exact anchor-class identity. Theorem 4.13.2 proves (4.1.6). Theorems 4.13.3 and 4.13.4 prove (4.1.7), (4.1.8), and the required represented-below statement. These statements establish every assertion of Theorem 4.1.1. □

4.13.4 The smallest parameters

The following checks are direct specializations of the proved formulas. They are not used to infer any assertion for larger sizes.

For $n = 2$, one has $p = 3$, $N = 7$, and $g = 3$. The second vertex is $(3, 0)$. The only positive integer in the strict range $2m < 3$ is $m = 1$, and the requested weighted parameter is $t = 2$. The native marking rows, in increasing column order, are

$$O = (6, 5, 4, 3, 2, 1, 0), \qquad X = (2, 1, 0, 6, 5, 4, 3), \qquad x_0 = (0, 6, 5, 4, 3, 2, 1).$$

Formula (FB.2) gives $H_\triangle = x_{(0\,2)} + x_{(1\,2)} + x_{(1\,3)}$. Every term of $U^{-2}R_4$ has coordinates $(2, -1)$, and every term of $U^{-3}R_6$ has coordinates bounded by $(3, 0)$. For example, the positive circular order $(0, 1, 3, 4, 5)$ is a balanced pentagon. Its positive star is $(0, 3, 5, 1, 4)$. Its five displacements are $3, 2, 3, 3, 3$, of total $2N = 14$. It consequently has Alexander value $g - 2p = -3$ and occurs in $U^{-3}R_6$ with weighted level zero at $t = 2$. This meets the asserted bound at the smallest allowed grid.

If $m \ge 2$, the first parameters satisfying the strict inequality are $p = 5$, $n = 4$, and $m = 2$. Here $N = 11$, $g = 10$, the vertex is $(3, 3)$, and $t = 1$. The level (FB.10) is 3, or equivalently the monomial inequality is $A(y) - 2k \le 6$. The terms of $U^{-2}R_4$ have $A = 0$ and satisfy $A - 2k = 4$. The terms of $U^{-3}R_6$ have $A \le 0$ and satisfy $A - 2k \le 6$. The pentagon $(0, 2, 4, 6, 8)$ is balanced because its successive two-gap distances are $4, 4, 4, 5, 5$. Its positive star $(0, 4, 8, 2, 6)$ has displacements $4, 4, 5, 4, 5$, of total $2N = 22$, and Alexander value zero. The corresponding term of $U^{-3}R_6$ has weighted level exactly 3. The native triangle in this case is

$$H_\triangle = x_{(0\,4)} + x_{(1\,4)} + x_{(1\,5)} + x_{(2\,4)} + x_{(2\,5)} + x_{(2\,6)} + x_{(3\,4)} + x_{(3\,5)} + x_{(3\,6)} + x_{(3\,7)},$$

which checks the native index shift independently of the cyclic formulas used for the other correction terms.

Acceptance and remaining gap. Formula (FB.4) and Theorem 4.13.1 prove M902. Theorem 4.13.2 proves the stronger bifiltration statement accompanying M901, and Theorem 4.13.3 proves its exact weighted acceptance statement. Theorem 4.13.4 proves the stated native integer inequality. There is no remaining mathematical gap in J21.

4.14 Catalogue of the proof inputs

All coefficient calculations in this proof take place in $\mathbb{F}_2$. The following tables distinguish existing native declarations from the new lemmas proved in this document. No holomorphic Heegaard Floer theorem is an input.

Unqualified names in rows supplied in full belong to `OnLeConjecture.KnotFloer.GridFull`, unless a different namespace is displayed. The prefixes `Staircase.`, `Bifiltration.`, and `RealFiltration.` are relative to `OnLeConjecture.KnotFloer`. Source filenames are relative to `OnLeConjecture/KnotFloer/`. They match the supplied file headings, except for the explicitly marked location supplied only in the research notes. The rows marked imported explicitly distinguish imported primitive names from declarations whose defining statements are included in the attachment.

4.14.1 Existing Lean declarations

Existing declaration or definition	Verified supplied source	Use in this proof
`consecutiveTorusGridSize`, `reflectedTorusGrid`, `reflectedTorusAnchor`, `reflectedTorusGrid_O_apply`, `reflectedTorusGrid_X_apply`	`GridReflectedTorus Basic.lean`	The exact size and native O, X marking conventions.
`reflectedTorusAnchor_val`, `maslovO_reflectedTorusAnchor`	`GridReflectedTorus Anchor.lean`	The state $x_0(c) = -c$ and its Maslov value zero.
`LaurentChain`, `laurentDifferential`, `laurentCoefficient_eq`, `laurentDifferential_monomialGenerator`, `laurentDifferential_sq_eq_zero`	`GridFullLaurent.lean`	The native chain module, the sum over state-empty rectangles with weight $U^{\#O}$, and $D^2 = 0$.
`laurentMaslovDegree`, `mem_laurentMaslovPiece`, `laurentMaslovComplex`, `laurentMaslovDegree_add_card_OColumns`, `laurentMonomial_smul_mem_maslovPiece`, `laurentDifferential_mem_maslovPiece`	`GridFullLaurent Maslov.lean`	The degree $M_O(x) - 2k$, the coefficient criterion for homogeneity, and the differential's degree minus one.
`maslovO_swap_reflectedTorusAnchor`	`GridReflectedTorus FirstCycle.lean`	The Maslov value minus one of the transposition states used in H_e.
`laurentDifferential_reflectedTorusAnchor`	`GridReflectedTorus Basic.lean`	The anchor is a cycle of the full Laurent differential.
`alexander_reflectedTorusAnchor`	`GridReflectedTorus AnchorAlexander.lean`	The absolute normalization $A(x_0) = n(n+1)/2 = g$.
`reflectedTorusCyclicDisplacement`, `reflectedTorusRelativeColumns`, `reflectedTorusDisplacementWeight`, `alexanderTwo_reflectedTorus_eq_sum_displacement`, `alexander_reflectedTorus_le_of_displacement_weight`	`GridReflectedTorus SecondAlexander.lean`	The exact displacement formula and the native implication from the integer inequality M903 to the Alexander bound.
`laurentMaslovChainEquiv`, `laurentMaslovChainEquiv_d`	`GridFullLaurent Spectral.lean`	Identification of the native graded complex and its differential with homogeneous Laurent chains and D.

Existing declaration or definition	Verified supplied source	Use in this proof
`nativeCornerCycleOfChain`, `nativeCornerCycleOfChain_apply`, `reflectedTorusAnchorCycle`, `reflectedTorusAnchorClass`	`GridReflectedTorus FirstClass.lean`	Passage from the displayed degree-zero cycle to the native cycle module and the exact anchor class.
`mem_laurentMaslovBifiltrationLevel_iff`, `laurentMaslovBifiltration`	`GridFullLaurent Spectral.lean`	The coefficient bounds $-k \le a$ and $A(x) - k \le b$ are precisely the native bifiltration bounds.
`Bifiltration.weighted`, `RealFiltration.RepresentedBelow`	`SpectralThreshold.lean`	The weighted level and the statement that the specified homology class has a representative at that level.
`Staircase.consecutiveFirst`, `Staircase.consecutiveSecond`, `Staircase.consecutiveWeight`	`Consecutive Staircase.lean`	The exact vertex-two coordinates and its weighted value.
`firstTorusTrianglePairs`, `firstTorusTriangle`, `firstTorusCycle`, `firstTorusTriangle_mem_maslovPiece`	`GridReflectedTorus FirstCycle.lean`	The exact triangular index set, its degree, and the definition of the existing first representative used in the explicit correction.
`firstTorusNativeCycle`, `firstTorusNativeCycle_homology`	`GridReflectedTorus FirstClass.lean`	Identification of the native first cycle with the exact anchor homology class in the boundary comparison.
Imported native filtration definition `laurentBifiltrationLevel`	`GridFullLaurent Bifiltration.lean`, identified in the supplied notes and imported by `GridFullLaurent Spectral.lean`	The ungraded native bifiltration named in the stronger conclusion. Its defining listing is omitted. The fully supplied homogeneous coefficient criterion above supplies the needed support test.
Imported rectangle and monomial primitives `LaurentPolynomial.T`, `OColumns`, `GridRectangleBetween.emptyRectangles`, `GridState.swapColumns`, `GridRectangle.mem_coveredSquares`, `isEmpty_iff_forall_notMem_cIoo`	Use sites in `GridFull Laurent.lean`, `GridReflectedTorus FirstCycle.lean`	The Laurent monomial notation, native half-open marking intervals, open state interiors, and column-swap convention. Their defining files are not separately listed in the attachment.
Imported grading primitives `GridDiagram.maslov0ℤ_eq_card`, `GridDiagram.two_mul_alexander_eq_intCast`	Use sites in `GridReflectedTorus Anchor.lean`, `GridReflectedTorus AnchorAlexander.lean`, `GridReflectedTorus SecondAlexander.lean`	The four finite counts defining integer Maslov grading and the identity $2A = \text{alexanderTwo}$. The supplied uses verify these names, but do not identify their own defining filenames.

The imported grading identifier containing $\mathbb{Z}$ uses the native Unicode character `U+2124` in its Lean spelling. The table does not assign an unverified defining filename to an imported primitive. In particular, the supplied criterion `mem_laurentMaslovBifiltrationLevel_iff` in `GridFullLaurentSpectral.lean` is used for the filtration conversion, so the omitted `GridFullLaurentBifiltration.lean` listing is not needed as a separate citation for that conversion.

4.14.2 New lemmas and explicit constructions

Every entry in the following table is new and is proved in this document. The references point to the corresponding mathematical statements rather than to diagnostic calculations.

New lemma group and reference	**Conclusion used**
Rectangle coordinates and finite-count Maslov formula, Lemmas 4.3.1, 4.2.1, 4.2.2, 4.2.3, and Corollary 4.2.4	The exact relative interval test and $M(x_\sigma) = \mathrm{inv}(\sigma) - \sum_c \lvert\sigma(c) - c\rvert$, including deletion of fixed columns, cyclic-support rotation gradings, and the small permutation types.
Cyclic relabelling, cycle-count bound, and exact insertion index, Lemmas 4.4.1, 4.4.2, 4.4.3, 4.4.4, and 4.4.5	The unique zero-O insertion rectangle has interior count $(\ell(\tau) - \ell(\sigma) - 1)/2$. Deleting vertices computes the boundary coefficients of cycles with $\ell = k - 1$.
Transposition differential and balanced first correction, Lemmas 4.6.1 and 4.6.2	The explicit chain H_e satisfies $DH_e = Ux_0 + T_{\mathrm{bal}}$.
Structural rectangle identities and cycle types, Lemmas 4.5.1, 4.5.2, 4.5.3, 4.5.4, 4.5.5, 4.5.6, and 4.5.7	Two-fixed-column terms cancel, inversion preserves rectangle coefficients, and the possible boundary types and Maslov values are completely classified.
The explicit H_{32} coefficient and its two-O boundary, Section 4.7, Lemmas 4.7.1, 4.7.3, 4.7.4, and Corollaries 4.7.2 and 4.7.5	The intrinsic five-column indexing, the complete coefficient table, inversion pairing, and the exact triangle coefficient $L_p(\alpha,\gamma) + L_p(\beta,\alpha) + L_p(\gamma,\beta)$, together with the exclusion of higher O counts.
The explicit adjacent-pair correction H_4, Section 4.8 and Lemma 4.8.1	A finite formula for every correction coefficient and the degree one of $U^{-2}H_4$.
Adjacent-pair coefficient identities, Lemmas 4.8.2, 4.8.3, and 4.8.4	Support on adjacent pairs, a common coefficient sum for inverse cycles, and the three-point primitive that cancels the circular five-cycle boundary.
Four-cycle deletion and the full lower cancellation, Lemmas 4.6.3, 4.9.1, 4.9.2, 4.9.3, 4.9.4, and Corollary 4.9.5	The complete equality $d_1H_4 + d_2H_{32} = T_{\mathrm{bal}}$, the cancellation of every two-transposition target, and $d_jH_4 = 0$ for $j \geq 2$.
The four-point primitive identities, Lemma 4.10.1	The functions F, K, G express the coefficient of H_{32} in forms that telescope in the boundary calculation.
The Maslov-minus-six coefficient and support calculation, Lemmas 4.10.2, 4.10.3, 4.10.4, 4.10.5, and Theorem 4.10.6	The exact chain $R_6 = d_0H_{32}$ consists of the stated balanced pentagon and two-triangle states. Every term has Alexander drop at least $2p$.
Diagonal identities and the first nonconvex five-cycle type, Lemmas 4.11.1, 4.11.2, 4.11.3, and 4.11.4	The explicit T_p^0 identities and the cancellation of the five-cycle type 01243.
The remaining five-cycle coefficients, Lemmas 4.12.1, 4.12.2, 4.12.3, and Theorem 4.12.4	The exact residual $R_4 = d_0H_4 + d_1H_{32}$ and the Alexander value $g - 2p$ of each surviving term.
The explicit native cycle and acceptance statements, Theorems 4.1.1, 4.13.1, 4.13.2, 4.13.3, and 4.13.4	The degree-one correction w_2, its complete coefficient formula, the exact anchor class, the second-vertex bifiltration bound, the required weighted representation, and the precise native integer displacement inequality.

The proof of the requested representative uses these identities in the single chain equation

$$x_0 + D\big(U^{-1}H_e + U^{-2}H_4 + U^{-3}H_{32}\big) = U^{-2}R_4 + U^{-3}R_6.$$

The displayed boundary gives the exact anchor homology class. The existing square-zero identity gives the cycle condition. The new residual calculations give the support bound.

Chapter 5

J10: One X switch for a deleted braid letter

5.1 Exact statement and source status

The J10 acceptance statement is the following assertion. Let $m \geq 2$, let u, v be positive adjacent words on m strands, and let $i \in \{0, \ldots, m-2\}$. Suppose that every adjacent generator occurs in uv. Put $w = u\sigma_i v$, $w' = uv$, and $j = |u|$. Here the subscripts are the native zero-based indices, so σ_0 denotes the generator usually written σ_1. There exists a grid H of size $3|w|$ such that

$$\text{XSwitchSequence}(\text{wordGrid}(w), H, 1) \text{ is nonempty}, \qquad H \overset{\text{moves}}{\sim} \text{wordGrid}(w'). \tag{5.1.1}$$

The switch in the proof is in the consecutive columns $3j+1, 3j+2$. The grid H has exactly the same O marking as the initial grid.

We prove an explicit statement for every pair of word grids satisfying the marking and row-order equations of the supplied declaration `exists_positiveRectilinearWordGrid`. Thus the proof is independent of every choice of distinct row perturbations. The supplied audit defines wordGrid(a) as the grid field of `positiveRoundedWordSelectedData` a. Instantiating the proved construction at that field requires its incidence equation and its strict height-order property. The defining structure and selector are not reproduced in the attachment, and the audit's field inventory does not state those two equations. They are isolated precisely in Section 5.7.

There is a source limitation. The combined attachment does not reproduce `PositiveBraidGridSelectedRepresentative.lean`, where the selected data are defined. It also imports rather than reproduces the defining modules for `RowsNoninterleaving`, `ColumnsNoninterleaving`, and `IsStabilization`. The supplied move module describes its generators as the grid moves of Chapter 3 of the attached OSS book. The proof specifies every local pattern and every noninterleaving condition. A review of the omitted selected-data fields and primitive move definitions is still needed to certify the literal native instantiation. No equality of smooth link types is used as a substitute for a grid-move sequence.

This is a combinatorial proof. It uses no coefficient field other than $\mathbb{F}_2$ if one subsequently applies the supplied chain-map results.

5.2 The row-separated word grids

Fix a positive word a of length L. Its columns are the pairs (h, c) with $0 \leq h < L$ and $c \in \{0, 1, 2\}$, ordered by

$$(h, c) \longmapsto 3h + c. \tag{5.2.1}$$

This is the order proved by `positiveRectilinearWordColumn_index_lt`. For the letter in position h, let r be its generator index. Its three outgoing horizontal arcs have unperturbed

heights

$$r + \tfrac{1}{2}, \qquad r, \qquad r + 1$$

at columns $(h, 0), (h, 1), (h, 2)$, respectively. They are the O corners. The corresponding X corners have unperturbed heights

$$r, \qquad r + 1, \qquad r + \tfrac{1}{2}.$$

These formulas follow from `positiveRectilinearOutgoingArc_zero`, `positiveRectilinearOutgoingArc_one`, `positiveRectilinearOutgoingArc_two`, their incoming versions, and `positiveRectilinearHorizontalArcRow`.

Each horizontal arc has its own row after separation. If its unperturbed height is $h(\alpha)$, write $R(\alpha)$ for its finite grid row. Distinct arcs have distinct rows, and

$$h(\alpha) < h(\beta) \quad \Longrightarrow \quad R(\alpha) < R(\beta). \tag{5.2.2}$$

The row of the X corner at a column is the row of the incoming arc matched to that column. The marking equations in `exists_positiveRectilinearWordGrid_of_injective` are exactly

$$O(c) = R(\text{outgoing arc at } c), \qquad X(c) = R(\text{incoming arc at } c). \tag{5.2.3}$$

The strict order in (5.2.2) follows from `exists_positiveRectilinearWordGrid`.

For an integer height r, the occurrences are the letters involving rank r. At each occurrence the incoming corner precedes the outgoing corner in the column order. The outgoing arc at one occurrence goes to the incoming corner of the next occurrence, in cyclic word order. This is precisely `positiveRectilinearWordMatching_occurrence` and `positiveRectilinearSortedSuccessor_next`. At a half-integer height, the arc belonging to position h goes from $(h, 0)$ to $(h, 2)$, by `positiveRectilinearWordMatching_middle`.

Definition 5.2.1. An admissible grid of a word a is a grid with the column order (5.2.1), the incidence equations (5.2.3), and a total order of arc rows satisfying (5.2.2).

The definition records the part of the supplied word-grid construction used below. The actual magnitudes of the real row perturbations do not enter a finite grid diagram.

5.3 Elementary move lemmas

We use the following conventions for the elementary moves. Two adjacent rows or columns commute when their endpoint pairs do not alternate cyclically and have no common endpoint. Equivalently, after choosing a cut of the circle, the two closed intervals are disjoint or one is strictly inside the other. All commutations below have distinct endpoints. A destabilization replaces three marked corners of an adjacent 2×2 block by the single marking of the duplicated type at the fourth corner, and deletes the row and column through the corner of the other type. The fourth corner must be unmarked. These are the conventions of [1, Definitions 3.1.6 and 3.1.7, Section 3.2]. They are also the move types named in the supplied `GridDiagram.IsMove`.

Lemma 5.3.1 (New lemma: commutations of a short row or column)**.** *Suppose the two markings of a row lie in adjacent columns a, b. That row can be commuted with an adjacent row having neither endpoint in columns a, b. The analogous statement holds for a column whose two marking rows are adjacent.*

Proof. No other column lies in the cyclic open interval from a to b in their adjacent direction. Thus the endpoints of the other row are on the same side of that interval. Its two endpoints cannot alternate with a, b. They are distinct from a, b by hypothesis, so the commutation criterion holds. Interchanging rows and columns proves the second assertion. Each commutation gives `MovesTo` by `movesTo_swapRows` or `movesTo_swapColumns`. □

Lemma 5.3.2 (New lemma: same-height rows do not interleave)**.** *In an admissible word grid, any two different horizontal arcs with the same unperturbed height have distinct, nonalternating endpoints. Consequently, any two admissible grids of the same word are related by row commutations.*

Proof. At a half-integer height, the two arcs belong to different letters. Each runs from column $3h$ to column $3h+2$ of its letter. These intervals are disjoint for different letters, and their endpoints are distinct.

At an integer height, list the occurrences in cyclic order. The incoming and outgoing endpoints of each occurrence are distinct, with incoming first. An arc joins an outgoing endpoint to the next incoming endpoint. The interiors of the resulting arcs are mutually disjoint on the column circle. No incoming endpoint equals any outgoing endpoint, because the two columns at an occurrence are distinct and different letters occupy disjoint column triples. Thus all endpoint pairs are nonalternating and disjoint. This also follows from the explicit endpoint exclusion in `positiveRectilinearIntegerRow_arc_excludes_occurrence`.

Condition (5.2.2) makes the rows of each unperturbed height a consecutive group. Two admissible row orders differ only in the orders inside these groups. A finite permutation of a linearly ordered set is a product of adjacent exchanges. One proof is to move the first desired element left to its desired place, then repeat on the shorter remaining list. Every exchange used here is between same-height rows, so the preceding endpoint argument verifies its commutation condition. Their composition proves the final assertion by `MovesTo.trans`. □

Lemma 5.3.3 (New lemma: deleting a continuation column)**.** *Consider a grid with an integer-height group of horizontal arcs indexed in cyclic order. Assume the rows of this height group occupy a consecutive interval in the row order. Suppose one column c joins the end of an incoming arc α_- to the start of an outgoing arc α_+ at that height, with*

$$X(c) = R(\alpha_-), \qquad O(c) = R(\alpha_+), \qquad \alpha_- \neq \alpha_+.$$

Let d be the X endpoint of α_+ and e the O endpoint of α_-. Assume c, d, e are distinct. Suppose the other arcs in this height group have endpoint pairs disjoint from and nonalternating with the pairs of α_- and α_+. Suppose also that, in the chosen cyclic direction of α_+ from c to d, the column e is not passed.

Then grid moves delete column c and row $R(\alpha_+)$, and replace the two horizontal arcs by a single arc from e to d on the row of α_-. The moves use row commutations within this height group, column commutations involving only c, cyclic column permutations if needed, and one destabilization. The cyclic order of every surviving column is preserved.

Proof. Move the row of α_+ toward the row of α_- by adjacent row commutations, stopping when they are adjacent. All intervening rows belong to the same height group. The hypothesis verifies each noninterleaving test, and the two rows sharing column c are never commuted with one another.

The column c now has adjacent endpoint rows. Move it along the cyclic column order toward d, stopping immediately before d. A cyclic column permutation can first place this portion of the circle in a linear interval. No crossed column has a marking in either of the two endpoint rows, because their other marked columns are d, e, and the route crosses neither. Lemma 5.3.1 makes each column commutation legal.

The two adjacent rows and the adjacent columns c, d now contain

	c	d
$R(\alpha_-)$	X	$\varnothing$
$R(\alpha_+)$	O	X.

The fourth corner is empty because the other marking of the row of α_- is in column $e \neq d$. This is an X destabilization. It deletes the column through the displayed O and the row through that

O, and places the remaining X in column d on the row of α_-. Every deleted or moved column was c, so the cyclic order of the surviving columns is unchanged. The row and incidence claims follow from the displayed markings. □

5.4 The single switch

For the remainder of the proof let $w = u\sigma_i v$ and $w' = uv$ be as in the acceptance statement. Let G be any admissible grid of w. Write

$$a = 3j, \qquad b = 3j + 1, \qquad c = 3j + 2$$

for the columns of the deleted letter. Denote its middle arc by M, its outgoing lower arc by L_+, and its outgoing upper arc by U_+. Let L_- and U_- be the arcs incoming to the lower and upper corners of this occurrence. Since σ_i occurs in w', each of the two integer ranks $i, i+1$ has another occurrence. Its cyclic predecessor is not this deleted occurrence. Hence

$$L_- \neq L_+, \qquad U_- \neq U_+. \tag{5.4.1}$$

The outgoing and incoming arc descriptions give the following complete local marking table:

	a	b	c
O	$R(M)$	$R(L_+)$	$R(U_+)$
X	$R(L_-)$	$R(U_-)$	$R(M)$.

(5.4.2)

All lower rows are below $R(M)$, and all upper rows are above $R(M)$, by (5.2.2).

Lemma 5.4.1 (New lemma: the candidate switch is a grid)**.** *Define $H.O = G.O$ and let $H.X$ be $G.X$ with its values in columns b, c exchanged. Then H is a grid diagram of size $3|w|$, and there is a one-step native X-switch sequence from G to H.*

Proof. The two marking maps remain permutations. The only disjointness conditions that change are those in columns b, c. In column b the new X row is $R(M)$ and the O row is $R(L_+)$. Their unperturbed heights differ, so they are distinct. In column c the new X row is $R(U_-)$ and the O row is $R(U_+)$. These are distinct by (5.4.1) and injectivity of the row assignment. All other column conditions hold because G is a grid.

The indices b, c are distinct and consecutive, with $c = \texttt{finRotate}(3|w|)(b)$. Indeed, $j < |w|$ and $b = 3j + 1 < 3|w| - 1$. The constructor `XSwitchSequence.cons`, with the empty tail `XSwitchSequence.nil` on H, gives the required sequence using the two defining marking equalities. □

After the switch the local table is

	a	b	c
O	$R(M)$	$R(L_+)$	$R(U_+)$
X	$R(L_-)$	$R(M)$	$R(U_-)$.

(5.4.3)

The proof now removes exactly these three columns by grid moves.

5.5 The explicit sequence of grid moves

Theorem 5.5.1 (New lemma: deletion for every admissible word grid)**.** *Let $m \geq 2$, let u, v be positive words on m strands, and let $i \in \{0, \ldots, m-2\}$. Suppose every generator occurs in uv. For every admissible grid G of $u\sigma_i v$ and every admissible grid G' of uv, the grid H of Lemma 5.4.1 satisfies*

$$H \overset{\text{moves}}{\sim} G'.$$

The sequence from H uses exactly three destabilizations, together with row and column commutations and cyclic column permutations.

Proof. Removal of the middle arc. The row of M has its O and X in the adjacent columns a, b. Both $R(L_-)$ and $R(L_+)$ lie below it. Let T be whichever of these two rows is higher. Move the row of M downward until it is immediately above T. The only other rows with a marking in columns a, b are precisely $R(L_-)$ and $R(L_+)$, by (5.4.3). Neither is crossed, since the process stops at the higher of them. Lemma 5.3.1 therefore verifies every row commutation used.

If $T = R(L_+)$, the adjacent block has markings

	a	b
$R(M)$	O	X
$R(L_+)$	$\varnothing$	O.

Its fourth corner is empty because $L_- \neq L_+$. An O destabilization deletes row $R(M)$ and column b, and places O at $(a, R(L_+))$. If $T = R(L_-)$, the block is instead

	a	b
$R(M)$	O	X
$R(L_-)$	X	$\varnothing$.

An X destabilization deletes row $R(M)$ and column a, and places X at $(b, R(L_-))$. In either case one surviving column, denoted d_0, has

$$X(d_0) = R(L_-), \qquad O(d_0) = R(L_+). \tag{5.5.1}$$

All markings outside the deleted letter's columns are unchanged. The upper continuation is still column c, with rows $R(U_-), R(U_+)$.

Removal of the lower continuation. In cyclic occurrence order at rank i, the preceding occurrence, the deleted occurrence, and the next occurrence appear consecutively, with the preceding and next allowed to be the same other occurrence. Let e_- be the outgoing corner of the preceding occurrence and e_+ the incoming corner of the next occurrence. The arcs L_- and L_+ now run from e_- to d_0 and from d_0 to e_+, respectively. The three columns are distinct. In particular, if the preceding and next occurrences coincide, their incoming and outgoing corners are distinct by `positiveRectilinearIntegerRowIn_lt_out`. The directed arc from d_0 to e_+ does not pass e_-. Every other rank-i arc has disjoint endpoints and disjoint interior from these two arcs, by the cyclic successor description.

The rows of the integer-height group remain consecutive. Moving and deleting M has not changed the relative order of any surviving rows. Lemma 5.3.3 therefore applies to (5.5.1). Its moves delete d_0 and the outgoing row $R(L_+)$, keeping $R(L_-)$ and joining e_- directly to e_+. This is exactly the change to the rank-i cyclic successor when one occurrence is deleted.

Removal of the upper continuation. The same argument applies to rank $i+1$, with incoming and outgoing arcs U_-, U_+. The previous moves have not changed the cyclic order of its surviving columns or the relative order of its rows. Lemma 5.3.3 deletes the remaining local column c and the outgoing row $R(U_+)$, and bypasses the deleted occurrence in the upper cyclic successor.

Three columns and three arc rows have now been removed. They are exactly the columns of the deleted letter and its outgoing arcs M, L_+, U_+. The cyclic order of all other columns is the original order with this triple removed. A cyclic column permutation restores the conventional first column if needed. Every half-integer arc belonging to another letter is unchanged. At integer ranks other than $i, i+1$, every successor is unchanged. At ranks $i, i+1$, the preceding and next occurrences have been joined, as the definition of the successor for uv requires. Thus the resulting incidence is exactly that of the word uv.

The relative order of surviving rows has only been changed by commutations inside an integer-height group. The temporarily moved row M was deleted. Hence the remaining row order still satisfies (5.2.2) for uv. The resulting grid is admissible for uv. Lemma 5.3.2 relates it to the prescribed admissible grid G' by row commutations. Composition of these elementary moves using `MovesTo.trans` proves the theorem. □

The construction is an explicit finite algorithm. Each row is moved one position at a time toward its specified target row. Each short column is moved one position at a time along the specified arc until it is adjacent to its target column. The number of intervening rows or columns decreases at each such step, so the algorithm terminates. The proof records the relevant noninterleaving test before each loop and the three marked corners before each destabilization.

5.6 Smallest cases and the role of the hypothesis

The smallest allowed strand number is $m = 2$. The shorter word must contain its unique generator, so the shortest example is $w = \sigma_0\sigma_0$, $w' = \sigma_0$. Choose increasing occurrence order within each equal-height row group. The initial six-column grid has

$$O = (2, 0, 4, 3, 1, 5), \qquad X = (1, 5, 2, 0, 4, 3).$$

Deleting the first letter uses the switch in columns $1, 2$. The complete movie is recorded below. A tuple lists row values in increasing column order, and all subsequent indices are renumbered after each deletion.

operation	O	X
initial grid	$(2, 0, 4, 3, 1, 5)$	$(1, 5, 2, 0, 4, 3)$
switch $1, 2$	$(2, 0, 4, 3, 1, 5)$	$(1, 2, 5, 0, 4, 3)$
delete row 2 and column 0	$(0, 3, 2, 1, 4)$	$(1, 4, 0, 3, 2)$
commute columns $0, 1$	$(3, 0, 2, 1, 4)$	$(4, 1, 0, 3, 2)$
delete row 0 and column 1	$(2, 1, 0, 3)$	$(3, 0, 2, 1)$
commute columns $0, 1$	$(1, 2, 0, 3)$	$(0, 3, 2, 1)$
delete row 2 and column 1	$(1, 0, 2)$	$(0, 2, 1)$.

At the two commutations the vertical endpoint intervals are disjoint. The three deletions are the X destabilizations specified in the general proof. The final grid is the three-column grid of σ_0. Every row and every column in each displayed grid has one marking of each type at distinct positions, which can also be checked directly from the tuples. This is a finite convention check, not the proof for arbitrary words.

The assumption on the shorter word matters to the validity of the candidate switch. If one instead deletes the sole letter of $w = \sigma_0$, then $O = (1, 0, 2)$ and $X = (0, 2, 1)$. The candidate switch would give $X = (0, 1, 2)$, and column 2 would have coincident O and X markings. This excluded example explains why the proof verifies $U_- \neq U_+$ before constructing H.

If the deleted letter is the first or last letter, the cyclic predecessor or successor can wrap across the word boundary. The continuation lemma uses the cyclic order and permits a cyclic column permutation, so the same proof applies. If the shorter word has only one occurrence of one of the two involved ranks, the previous and next occurrences coincide, but their incoming and outgoing columns are distinct. The empty fourth corner in the final destabilization remains empty. No assumption on the number of link components is used.

5.7 Exact missing native identifications

The proved deletion movie requires only the following selected-grid and primitive-move identifications. Their source declarations are absent from the combined attachment. They are stated as unverified implications, not as additional proved lemmas or silently assumed hypotheses.

5.7.1 The precise selected-grid implication that is absent

Fix a word w of length L and write

$$G_w = (\texttt{positiveRoundedWordSelectedData}(w)).\text{grid},$$
$$C = \mathrm{Fin}(L) \times \mathrm{Fin}(3).$$

Let $\jmath : C \to \mathrm{Fin}(3L)$ be `finProdFinEquiv`, let M_w be `positiveRectilinearWordMatching` w, and let $h_w : C \to \mathbb{R}$ be `positiveRectilinearWordArcRow` w. The exact data required to instantiate the proved deletion algorithm are

$$G_w.X(\jmath a) = G_w.O(\jmath M_w^{-1}(a)) \qquad \text{for every } a \in C, \tag{5.7.1}$$

$$h_w(a) < h_w(b) \Longrightarrow G_w.O(\jmath a) < G_w.O(\jmath b) \qquad \text{for every } a, b \in C. \tag{5.7.2}$$

No separate row-bijection hypothesis is required. Indeed, $R(a) = G_w.O(\jmath a)$ is automatically a bijection because $G_w.O$ and $\jmath$ are bijections. The two displayed properties say exactly that this R has the incoming marking equation and preserves the strict unperturbed height order.

The reproduced theorem `exists_positiveRectilinearWordGrid` proves the existence of some grid satisfying these properties. Its witness is a quantified variable G, not the selected field G_w. The supplied audit record L411 gives the type `positiveRoundedWordSelectedData` w and states that its structure contains a grid, rows, row-offset data, and profiles. It does not reproduce a selector equation, a soundness theorem for that grid field, or the propositions stored in the structure. Thus L410 does not identify its existential witness with G_w. The same-word independence of L412 concerns smooth representatives, and L414 recovers a matching only under the marking-equation hypotheses. Neither supplies (5.7.1) or (5.7.2) for G_w.

The irreducible missing source is therefore the portion of `PositiveBraidGridSelectedRepresentative.lean` defining `PositiveRoundedWordRepresentativeData` and `positiveRoundedWordSelectedData`, or existing consequences of those definitions that imply the two displayed properties. The smallness of the perturbation, the smooth offset, the profiles, and the one-cycle hypothesis of the smooth representative are not needed by J10 once those two properties are known.

5.7.2 The precise native move implications that are absent

The supplied module `OnLeConjecture/External/TauCeti/KnotTheory/Grid/Move/Basic.lean` imports `Grid.Commutation.Move` and `Grid.Stabilization.Basic`. It proves that a hypothesis `RowsNoninterleaving` or `ColumnsNoninterleaving` yields a row or column move, and that a hypothesis `IsDestabilization` yields a move. The combined source does not reproduce definitions of these hypotheses or lemmas recognizing them from local marking inequalities. The exact implications required by the written proof are the following.

1. For two cyclically adjacent rows, if their four marked columns are distinct and their endpoint pairs do not alternate around the column circle, then the native predicate `RowsNoninterleaving` holds. The analogous implication is required for columns.

2. Suppose two adjacent rows and two adjacent columns have three marked corners, with equal types at opposite corners and the other type at a third corner. Suppose the fourth corner is empty. Deleting the row and column through the unique-type corner and placing the duplicated marking type at the fourth corner satisfies the native predicate `IsDestabilization`. The proof uses an O duplicated pattern in one possible first step, and X duplicated patterns in the other first step and the two continuation steps. Both orders of the adjacent rows are allowed in the continuation lemma.

The move-module docstring explicitly says that its presentation follows OSS Chapter 3. This is substantial evidence for the intended conventions. It does not, as a matter of deduction from the supplied declarations, provide the constructors or recognition implications above. No equality of grid-link classes or smooth isotopy can replace those missing implications in the literal acceptance statement.

5.8 Lemma table and remaining verification

Declaration or lemma	Status and role
`PositiveAdjacentWord`, `PositiveAdjacentWord.toSigned`	Existing Lean definitions. They fix the word and generator convention.
`positiveRectilinearWordColumn_index_lt`	Existing Lean declaration. It fixes the column order $3h + c$.
`positiveRectilinearOutgoingArc`, `positiveRectilinearIncomingArc`, `positiveRectilinearHorizontalArcRow`	Existing Lean definitions. They fix corner types and unperturbed heights.
`positiveRectilinearWordMatching_middle`, `positiveRectilinearWordMatching_occurrence`, `positiveRectilinearSortedSuccessor_next`	Existing Lean declarations. They identify the literal horizontal incidence.
`positiveRectilinearIntegerRowIn_lt_out`, `positiveRectilinearIntegerRow_arc_excludes_occurrence`	Existing Lean declarations. They give endpoint separation at integer heights.
`exists_positiveRectilinearWordGrid_of_injective`, `exists_positiveRectilinearWordGrid`	Existing Lean declarations. They give the admissible marking equations and row order.
`XSwitchSequence.cons`, `XSwitchSequence.nil`	Existing Lean constructors. They give the sequence of exactly one switch.
`GridDiagram.movesTo_swapRows`, `GridDiagram.movesTo_swapColumns`, `GridDiagram.cyclicColumns_movesTo`, `GridDiagram.IsDestabilization.movesTo`, `GridDiagram.MovesTo.trans`	Existing Lean declarations. They assemble the elementary move sequence.
Lemmas 5.3.1, 5.3.2, and 5.3.3	New lemmas. Their proofs verify every commutation and deletion used.
Lemma 5.4.1	New lemma. It proves validity of the specified X switch with unchanged O markings.
Theorem 5.5.1	New theorem. It gives the full finite deletion algorithm for all admissible word grids.
`positiveRoundedWordSelectedData`	Existing object identified in the supplied audit. Its defining file is absent from the combined source attachment.

The written combinatorial argument has no remaining parity or geometric isotopy step. It proves the stated move theorem for the explicit word-grid marking equations supplied in L410. The unresolved source check is the literal instantiation in (5.1.1): the omitted definition of `positiveRoundedWordSelectedData` must supply those equations and the row-order property, and the omitted primitive native move definitions must agree with the elementary patterns used above. The audit records the selected object, and the move-module documentation names the intended grid moves, but neither supplies the two selected-grid equations or the primitive recognition implications isolated above. The literal native instantiation therefore remains unverified from the provided sources. This document does not claim a Lean proof or completion of M702's separate chain-map and canonical-class transport obligations.

Chapter 6

J15: Braid length, discriminants, and polar intersection

6.1 Exact statements and scope

The analytic rings in this document are the native rings of complex analytic germs. The braid words are the native adjacent-generator words. No Floer homology result or fibre-surface argument is used. The exponent of a signed word is an integer, as required by the native type `SignedAdjacentWord`, whose signs belong to $\mathbb{Z}^\times = \{1, -1\}$. This arithmetic is not a change of coefficient field in a chain complex.

Theorem 6.1.1 (New lemma: M1155)**.** *Let $m \geq 1$, and let*

$$P(\xi, \zeta) = \zeta^m + a_1(\xi)\zeta^{m-1} + \cdots + a_m(\xi)$$

have coefficients holomorphic near 0, with $a_j(0) = 0$ for every j. Assume its discriminant $D(\xi) = \operatorname{Disc}_\zeta P(\xi, \zeta)$ is not identically zero. For every sufficiently small $r > 0$, the roots along the positively oriented circle $\xi = re^{\mathrm{i}\theta}$ define a closed braid of m distinct sheets. For every signed event word representing this braid, with the native positive-crossing convention,

$$\exp(w) = \operatorname{ord}_0 D = i_0(P, P_\zeta). \tag{6.1.1}$$

Here i_0 is the native complex dimension of the analytic local pair quotient. The word may be obtained by resolving simultaneous projected events.

The conditions $a_j(0) = 0$ are the distinguished condition in the term "Weierstrass polynomial." They are necessary for the last equality to be an intersection number at the single point $(0, 0)$. A counterexample without that condition is given in Section 6.9.

Theorem 6.1.2 (New lemma: reduced-germ length identity)**.** *Let $f \in \mathbb{C}\{\xi, \zeta\}$ be reduced, with $f(0, 0) = 0$, and suppose $f(0, \zeta)$ is not the zero germ. Put $m = i_0(f, \xi)$ and*

$$\mu(f) = \dim_{\mathbb{C}} \mathbb{C}\{\xi, \zeta\}/(f_\xi, f_\zeta).$$

For every signed word w representing the root braid of the projection ξ on a sufficiently small positively oriented base circle,

$$\exp(w) + 1 = \mu(f) + m. \tag{6.1.2}$$

If w is positive, then $\operatorname{length}(w) + 1 = \mu(f) + m$*. If f has multiplicity S and the projection is transverse, then $m = S$.*

For a branch, the theorem has the following consequence in the exact form needed for the existing analytic event words.

Theorem 6.1.3 (New lemma: branch word and J15(a))**.** *Let f be an irreducible plane curve germ, let $\beta : (\mathbb{C}, 0) \longrightarrow (\mathbb{C}^2, 0)$ have an injective analytic representative with pullback kernel (f), and suppose one coordinate of β has finite positive order a. Choose the positive event word w and the normalized equation-link representative supplied by* `PointedMapGerm.exists_equation_link_canonical_positive_braid`*. The choices can be made on a sufficiently small base circle so that*

$$\mathrm{length}(w) + 1 = 2\delta(f) + a. \tag{6.1.3}$$

The same word has one permutation cycle, uses every adjacent generator, and has closure ambient isotopic to the normalized equation link with the orientation in that existing declaration.

In particular, for a resolution level L and one of its singular points j, the existing data in `SingularSectionPointData.exists_branch_component_order` *provide a with*

$$2 \le a \le (g.\texttt{criticalJacobianFactorization}(hi)).\texttt{maxMultiplicity}.$$

For that a there is a native positive word satisfying all the word-level conjuncts of the acceptance statement `ResolutionSectionLevel.exists_branch_word`*, including* (6.1.3) *for the local section equation at j and the link identification for the same word.*

For a branch of multiplicity a with a transverse coordinate, the a in Theorem 6.1.3 is that multiplicity. The theorem also allows a finite projection of larger degree, which is useful because the catalogued branch-component-order declaration specifies a coordinate order.

Corollary 6.1.4 (New lemma: J15(b) for the J14 word)**.** *Let f be a reduced germ of multiplicity $S \ge 1$ with b irreducible branches, and use a transverse projection. Suppose the word of J14 is $d\,v$, where $d = \texttt{fullTwistWord}(S)$ and v is positive, and the word represents the same root braid through closed S-braids in the solid torus. Then*

$$\mathrm{length}(d\,v) + 1 = \mu(f) + S, \tag{6.1.4}$$

and

$$\mathrm{length}(v) + S^2 + b = 2\delta(f) + 2S. \tag{6.1.5}$$

This corollary first proves the length of the actual analytic word. Section 6.8 then proves the broader proposed API in M407 for every positive full-twist word with an ambiently isotopic closure, by an additional calculation over $\mathbb{F}_2$.

6.2 Conventions, continuous arguments, and crossing signs

For a signed native word $w = ((i_1, e_1), \ldots, (i_q, e_q))$, define $\exp(w) = \sum_{\nu=1}^{q} e_\nu$, where each e_ν is 1 or -1. The generator indices are $0, \ldots, m-2$, as in `SignedAdjacentWord` and `adjacentSwap`. A positive native word becomes a signed word by giving every letter sign 1. Therefore its exponent is its list length.

Fix a labelled lift $y_1(\theta), \ldots, y_m(\theta)$ of a braid over $[b, b + 2\pi]$. Labels may be permuted at the end. For a fixed nonzero complex scalar c, projection means ordering the real parts of cy_j. At a projected crossing of two sheets, put $d = c(y_j - y_k)$. Then $\Re d = 0$ and $\Im d \neq 0$. The positive crossing convention is

$$-\Im(d)\frac{d}{d\theta}\Re(d) > 0. \tag{6.2.1}$$

This is the positive angular-rotation convention in `exists_radius_planeBranchAngularDifference_projected_crossings_pos`, recorded in L4.16 of the supplied notes. Its analytic input, reproduced in the supplied source, is `exists_radius_planeBranchSheetDifference_re_pos`. It is consistent with the positive event words in `PointedMapGerm.exists_equation_link_canonical_positive_braid`.

Lemma 6.2.1 (New lemma: continuous argument and its derivative)**.** *Let $d : [u, v] \longrightarrow \mathbb{C} \setminus \{0\}$ be continuous. There is a continuous real function α with $d/|d| = e^{\mathrm{i}\alpha}$. If d is continuously differentiable, then α is continuously differentiable and*

$$\alpha' = \Im(d'/d). \tag{6.2.2}$$

At a projected crossing, the sign in (6.2.1) *equals the sign of α'.*

Proof. The path has a positive minimum modulus. Uniform continuity permits a finite subdivision into intervals on which the quotient by the value at the left endpoint lies in $\{z : |z - 1| < 1\}$. On each such interval, use the argument with values in $(-\pi/2, \pi/2)$ for that quotient, and add the argument already chosen at the left endpoint. Patching these finite choices gives a continuous argument on $[u, v]$.

If d is continuously differentiable, choose an initial argument $\alpha(u)$ and put

$$\alpha(\theta) = \alpha(u) + \int_u^\theta \Im(d'(s)/d(s))\, ds.$$

Differentiating the unit complex number $e^{-\mathrm{i}\alpha(\theta)} d(\theta)/|d(\theta)|$ gives zero, because its logarithmic derivative is $d'/d - \Re(d'/d) - \mathrm{i}\Im(d'/d) = 0$. Its initial value is 1. This proves the asserted argument formula. At a crossing $d = \mathrm{i}h$ with $h \neq 0$, one has

$$\alpha' = -\frac{\Re d'}{h} = \frac{-h\Re d'}{h^2}.$$

The denominator is positive, which proves the sign assertion. □

The elementary two-strand positive half-turn can be written $y_1(s) = -e^{\mathrm{i}\pi s}$, $y_2(s) = e^{\mathrm{i}\pi s}$ for $0 \leq s \leq 1$. The argument of the difference increases by π, its squared difference winds once positively, and (6.2.1) is positive at $s = 1/2$. Thus one positive letter contributes $+1$, rather than -1 or 2, to the winding of the squared-difference product.

For a closed nonzero complex loop q, define $\mathrm{wind}(q) = (\alpha(v) - \alpha(u))/(2\pi)$ using Lemma 6.2.1. This is an integer because $q(v) = q(u)$. The value is independent of the initial argument. It is additive under multiplication and concatenation, by addition of arguments.

Lemma 6.2.2 (New lemma: homotopy invariance of the winding)**.** *The winding of a nonzero closed complex loop is constant under a continuous homotopy through nonzero closed loops.*

Proof. If two loops q_0, q_1 are sufficiently uniformly close relative to the positive minimum of $|q_0|$, their pointwise quotient lies in the disc $\{z : |z - 1| < 1\}$. That disc admits a single-valued continuous argument, for example the argument in $(-\pi/2, \pi/2)$. This argument returns to its initial value on the closed quotient loop. Thus the two windings agree. For a homotopy on a compact parameter square, nonvanishing gives a positive minimum modulus and uniform continuity. Partition the homotopy parameter into finitely many intervals for which this closeness condition holds. Equality on consecutive intervals proves the assertion. □

6.3 The boundary terms of all unordered pairs cancel

Lemma 6.3.1 (New lemma: pairwise crossing count)**.** *Suppose $y_1, \ldots, y_m$ are continuously differentiable, pairwise distinct on $[b, b + 2\pi]$, and*

$$y_j(b + 2\pi) = y_{\tau(j)}(b)$$

for a permutation τ. Choose $c \neq 0$ so that no projected coincidence occurs at either endpoint and every pairwise projected crossing is transverse. Count simultaneous events separately for each unordered pair. The signed crossing count is

$$\sum_{j<k} \sum_{\substack{\theta \in (b, b+2\pi) \\ \Re(c(y_j - y_k)) = 0}} \operatorname{sgn}\Big(-\Im(c(y_j - y_k)) \frac{d}{d\theta} \Re(c(y_j - y_k)) \Big) = \mathrm{wind}\left(\prod_{j<k} (y_j - y_k)^2 \right). \tag{6.3.1}$$

Proof. Choose a continuous argument α_{jk} of $c(y_j - y_k)$ for each $j < k$, using Lemma 6.2.1. Its crossing levels are $\pi/2 + \pi\mathbb{Z}$. At an upward crossing the integer function

$$F_{jk}(\theta) = \left\lfloor \frac{\alpha_{jk}(\theta)}{\pi} - \frac{1}{2} \right\rfloor$$

increases by one, and at a downward crossing it decreases by one. It has no other jumps. The endpoints are not crossing levels, and the crossings are finite and transverse. Telescoping its jumps gives the signed count for this pair as

$$F_{jk}(b + 2\pi) - F_{jk}(b).$$

Let $\{u\} = u - \lfloor u \rfloor$ denote fractional part. Consequently the count for this pair is

$$\frac{\alpha_{jk}(b+2\pi) - \alpha_{jk}(b)}{\pi} - \left\{ \frac{\alpha_{jk}(b+2\pi)}{\pi} - \frac{1}{2} \right\} + \left\{ \frac{\alpha_{jk}(b)}{\pi} - \frac{1}{2} \right\}.$$

The endpoint permutation permutes the unordered pairs. It may reverse a chosen ordering of a pair, but reversal changes an argument by an integer multiple of π. Therefore the multiset of the endpoint fractional parts in this display is the same as the multiset of the initial fractional parts. They cancel after summation over all unordered pairs. The total count is

$$\frac{1}{\pi} \sum_{j<k} \big(\alpha_{jk}(b + 2\pi) - \alpha_{jk}(b)\big).$$

A continuous argument of the squared-difference product is $2\sum_{j<k} \alpha_{jk}$, up to a constant argument for $c^{-m(m-1)}$. Dividing its total change by 2π proves (6.3.1). This also proves that the total is independent of the starting labels and the nonzero constant c. □

Lemma 6.3.2 (New lemma: generic projection and multiple events)**.** *For a real-analytic collision-free braid on a compact period, a constant projection direction can be chosen so that all pairwise projected crossings are transverse and the endpoints are regular. Simultaneous crossings may be separated by an arbitrarily small collision-free perturbation supported away from the endpoints. The exponent of a word for the perturbed braid is the pairwise signed count in Lemma 6.3.1. If every pairwise crossing is positive, its positive event word has length equal to the sum of the numbers of coincident unordered pairs at all events.*

Proof. For each pair, the derivative of its argument is real analytic. Either it vanishes identically or its zero set on a period is finite. In the first case the argument is constant, and one excludes projection directions making that pair project to the same real coordinate. In the second case, one excludes the finitely many directions that make a critical argument a crossing level. Also exclude the finitely many directions producing an endpoint coincidence. There are only finitely many pairs, so a direction outside this finite set exists. Every crossing is then simple, and there are finitely many crossings on the compact period.

Take disjoint small intervals around the finitely many event times. At an event, add small, separately chosen real offsets to the projected coordinates of the labelled sheets, constant near that event and zero near the interval endpoints. A simple zero of the difference of two projected coordinates persists with its sign. The first variation of its time is

$$-\frac{h_j - h_k}{(\Re(c(y_j - y_k)))'(\theta_0)},$$

where h_j, h_k are the chosen offsets. For two distinct unordered pairs, the corresponding linear forms in the offsets are different. Their coincidence excludes a proper hyperplane of offset choices.

Choose one direction $(h_1, \dots, h_m)$ outside the finitely many such hyperplanes and use offsets τh_j. The first derivatives at $\tau = 0$ of the competing crossing times are then different. For every sufficiently small $\tau > 0$, those times are distinct, so all crossings in an event interval are separated. The positive minimum separation of the original complex sheets ensures that these perturbations and their straight-line interpolation create no complex collision. They produce one ordinary adjacent crossing for each original coincident pair, with the same sign.

Reading the real order between successive simple events gives a signed adjacent-generator word. Its exponent is the sum of these signs by definition. If all signs are positive, its length is the number of pairs counted with multiplicity over the events. The native branch theorem already supplies exactly this pair-count length formula for its chosen event words, so its event resolution does not need to be replaced. □

Corollary 6.3.3 (New lemma: exponent is preserved by braid homotopy)**.** *The exponent in Lemma 6.3.2 is unchanged under a homotopy of closed braids through configurations of distinct complex points.*

Proof. The squared-difference product is symmetric in the sheets, so it is a well-defined nonzero loop for an unordered configuration braid. A collision-free homotopy induces a nonzero homotopy of these products. Apply Lemmas 6.3.1 and 6.2.2. □

6.4 The local argument principle and distinguished preparation

Lemma 6.4.1 (New lemma: local argument principle)**.** *Let h be a holomorphic germ at $0 \in \mathbb{C}$ which is not zero, and put $d = \operatorname{ord}_0 h$. For every sufficiently small $r > 0$,*

$$\operatorname{wind}\big(\theta \longmapsto h(re^{\mathrm{i}\theta})\big) = d.$$

More generally, if a holomorphic function has no zero on the boundary of a closed disc, the winding of its boundary values is the sum of the orders of its interior zeros.

Proof. The Taylor series factors as $h(\xi) = \xi^d u(\xi)$, with $u(0) \neq 0$. Shrink the disc until u is nonzero there. The holomorphic function u'/u has a primitive on this disc, obtained by integrating its power series. If $L' = u'/u$, then differentiation gives $(ue^{-L})' = 0$. Thus u is a nonzero constant times e^L, and its argument has total change zero on every concentric circle. The factor ξ^d contributes $2\pi d$, which proves the first assertion.

For the second assertion, there are finitely many zeros in the disc, since they are isolated and there are none on its boundary. Factor their linear factors with their orders from the function. The remaining holomorphic function is nonzero on the disc and has a logarithm by the same primitive argument. Each factor $\xi - a$ with a in the interior has winding one. This follows by moving a to the centre through the disc and applying Lemma 6.2.2. Multiplication adds the windings, proving the stated sum. □

Lemma 6.4.2 (New lemma: preparation in the projection coordinate)**.** *Let $f(\xi, \zeta)$ be holomorphic near $(0, 0)$, with $f(0, 0) = 0$ and $f(0, \zeta)$ nonzero of order $m \geq 1$ at $\zeta = 0$. There are holomorphic germs $a_1, \dots, a_m$, all vanishing at zero, and a holomorphic unit u such that*

$$f(\xi, \zeta) = u(\xi, \zeta)\big(\zeta^m + a_1(\xi)\zeta^{m-1} + \cdots + a_m(\xi)\big). \tag{6.4.1}$$

Moreover $i_0(f, \xi) = m$.

Proof. Choose $R > 0$ so that $f(0, \zeta)$ has no zero other than 0 in $|\zeta| \leq R$ and has no zero on $|\zeta| = R$. Uniform continuity on that circle permits a base disc in which $f(\xi, \zeta)$ has no boundary zero. Its boundary winding is constant as ξ varies in that disc, by Lemma 6.2.2, and equals m at $\xi = 0$. Lemma 6.4.1 gives exactly m interior zeros counted with multiplicity for each such ξ.

Their power sums are holomorphic functions of ξ. Indeed, for $q = 1, \dots, m$, the power sum is

$$p_q(\xi) = \frac{1}{2\pi \mathrm{i}} \int_{|v|=R} v^q \frac{f_\zeta(\xi, v)}{f(\xi, v)}\, dv.$$

The contour identities needed here follow directly from power series. If g is holomorphic on a neighbourhood of $|v| \le R$, expand g on a slightly larger disc and integrate the series term by term on $|v| = R$. Expanding $(v - z)^{-1} = \sum_{k \ge 0} z^k v^{-k-1}$ for $|z| < R$ gives $\frac{1}{2\pi \mathrm{i}} \int g(v)/(v - z)\, dv = g(z)$. Termwise integration also gives zero for the integral of a holomorphic function without this denominator. These are all the contour formulas used below. For a fixed ξ, factor the interior zeros as in the proof of Lemma 6.4.1. Each logarithmic derivative term $\nu/(v - v_0)$ contributes νv_0^q by the power-series form of Cauchy's formula. The nonzero factor contributes zero. The integrand is holomorphic in ξ on a neighbourhood of the contour, so differentiating under this compact contour integral proves holomorphy of p_q.

Newton's identities recursively define the elementary symmetric functions $e_q(\xi)$ by

$$q e_q = \sum_{j=1}^{q} (-1)^{j-1} e_{q-j} p_j, \qquad e_0 = 1.$$

They are holomorphic. The monic polynomial with coefficients $a_q = (-1)^q e_q$ has precisely these interior zeros, with their multiplicities. At $\xi = 0$ all of them are zero, so $a_q(0) = 0$. Call this polynomial P.

For fixed ξ, the quotient f/P has removable singularities at all these roots and is holomorphic and nonzero on $|\zeta| < R$. Its jointly holomorphic representative is the contour integral

$$u(\xi, \zeta) = \frac{1}{2\pi \mathrm{i}} \int_{|v|=R} \frac{f(\xi, v)}{P(\xi, v)(v - \zeta)}\, dv.$$

Cauchy's formula identifies this with the quotient for every fixed ξ. Differentiation under the integral proves joint holomorphy. The quotient has no zero, so $u(0, 0) \neq 0$, and (6.4.1) follows. Finally, the quotient by (f, ξ) is isomorphic to $\mathbb{C}\{\zeta\}/(\zeta^m)$, with basis $1, \zeta, \dots, \zeta^{m-1}$. Its dimension is m. □

This proof of the required preparation statement is included because the supplied file imports the project's preparation modules without reproducing their exact declarations. It fixes the distinguished hypothesis explicitly.

6.5 Proof of the discriminant and polar equalities

Proof of Theorem 6.1.1. Factor $D(\xi) = \xi^d v(\xi)$ with $v(0) \neq 0$. For every sufficiently small $r > 0$, the polynomial has distinct roots for all $|\xi| = r$, since its discriminant does not vanish there. A simple polynomial root varies holomorphically locally in ξ by the analytic implicit function theorem. Applying this fact successively along the compact circle gives labelled roots on $[b, b + 2\pi]$. They cannot escape to infinity because they are roots of a monic polynomial with bounded coefficients. Their endpoints are permuted because the polynomial at the two endpoints is the same.

For completeness, the simple-root fact used here follows by writing a root equation near (ξ_0, ζ_0) as $\zeta - \zeta_0 = T(\xi, \zeta - \zeta_0)$ with $\partial_\zeta T(\xi_0, 0) = 0$. On a sufficiently small polydisc this is a contraction in the second variable. Iteration from zero converges uniformly on a smaller polydisc to its unique holomorphic fixed point. This proves the local analytic root assertion needed above.

For a monic polynomial, the algebraic discriminant identity is

$$D(\xi) = \prod_{j<k} (y_j(\theta) - y_k(\theta))^2, \qquad \xi = r e^{\mathrm{i}\theta}. \tag{6.5.1}$$

One can verify this identity by differentiating $P(\xi,\zeta)=\prod_j(\zeta-y_j)$ and taking the product of $P_\zeta(\xi,y_j)$ over all roots. Each unordered pair contributes the negative of its squared difference. Thus

$$\operatorname{Res}_\zeta(P,P_\zeta)=(-1)^{m(m-1)/2}D. \tag{6.5.2}$$

This is also the monic specialization of the existing algebraic declaration `Polynomial.resultant_deriv`, cited in M407 and M1155. The standard definition of the discriminant fixes the sign in (6.5.1).

Lemmas 6.3.1 and 6.3.2 identify the exponent with the winding of the right side of (6.5.1). Lemma 6.4.1 makes this winding d. Corollary 6.3.3 permits any event resolution representing the same braid. This proves the first equality in (6.1.1), including its sign and its factor of two.

For the second equality, take the polynomial degree parameter $d_0=m-1$ in the supplied declarations from `IntersectionResultant.lean`. The polynomial P is monic of degree d_0+1, and every lower coefficient belongs to the maximal ideal, by the hypotheses. Equation (6.5.2) and $D\neq 0$ make the resultant nonzero. Therefore `Germ.hasFiniteIntersection_of_resultant_ne_zero` supplies finite intersection, and `Germ.intersectionMultiplicity_eq_resultant_order_of_ne_zero` gives

$$i_0(P,P_\zeta)=\operatorname{ord}_0\operatorname{Res}_\zeta(P,P_\zeta)=\operatorname{ord}_0 D.$$

The last equality uses the nonzero constant in (6.5.2). When applying these native declarations, the polynomial variable is native coordinate 0 and the coefficient variable is coordinate 1. If the named coordinates are in the other order, exchange the coordinates, which identifies the two pair quotients and preserves their dimensions. □

Lemma 6.5.1 (New lemma: the unit factor preserves the polar pair ideal)**.** *If $f=uP$ with u a holomorphic unit, then*

$$(f,f_\zeta)=(P,P_\zeta)$$

in the local analytic ring. In particular their pair quotient dimensions agree whenever either one is finite.

Proof. The product rule gives $f_\zeta=u_\zeta P+uP_\zeta$. Because u is invertible, $(f)=(P)$. Modulo this common principal ideal, f_ζ and P_ζ differ by multiplication by u. Each of the two displayed ideals therefore contains the generators of the other, proving equality. □

Proof of Theorem 6.1.2. Use Lemma 6.4.2 to write $f=uP$ with P distinguished monic of degree $m=i_0(f,\xi)$. The curve is reduced and pointed, and (f,ξ) has finite colength by that lemma. These are exactly the hypotheses of the supplied theorem `PlaneCurve.exists_polar_intersection_identity` with parameter ξ. Its polar is f_ζ or $-f_\zeta$, according to the coordinate order. This sign is a unit and leaves the pair ideal unchanged. That existing theorem supplies finite polar intersection and the identity

$$i_0(f,f_\zeta)+1=\mu(f)+i_0(f,\xi)=\mu(f)+m. \tag{6.5.3}$$

Lemma 6.5.1 gives finite intersection for (P,P_ζ). The supplied `Germ.resultant_ne_zero_of_hasFiniteIntersection` then makes its resultant nonzero. Equation (6.5.2) makes D nonzero, so Theorem 6.1.1 applies. The roots of P are the sheets of $f=0$, since u does not vanish. Consequently

$$\exp(w)=\operatorname{ord}_0 D=i_0(P,P_\zeta)=i_0(f,f_\zeta).$$

Combining with (6.5.3) proves (6.1.2). For a positive word its exponent equals its length.

Finally, let S be the order of f, and let f_S be its first nonzero homogeneous part. Transversality of the projection means $f_S(0,1)\neq 0$ in the named coordinates (ξ,ζ). Thus $f(0,\zeta)$ has exact order S, and Lemma 6.4.2 gives $m=S$. □

6.6 The branch word and the exact resolution-level conclusion

Lemma 6.6.1 (New lemma: the normalized branch has exactly the required degree)**.** *Let f be irreducible, let an injective analytic representative F have pullback kernel (f), and suppose its projection coordinate is $\xi(F(t)) = t^a$ with $a \geq 1$. Then $i_0(f, \xi) = a$. For every sufficiently small nonzero ξ, the roots of its distinguished polynomial are exactly the complementary coordinates of $F(t)$ with $t^a = \xi$, each occurring once.*

Proof. The germ $f(0, \zeta)$ cannot be identically zero. Otherwise the Taylor series would make ξ divide f. Irreducibility would make f associated to ξ, and its pullback would force $t^a = 0$, a contradiction. Prepare $f = uP$ by Lemma 6.4.2, and let the degree of P be $M = i_0(f, \xi)$.

Write the complementary coordinate of $F(t)$ as $Y(t)$. The polynomial

$$Q(t^a, \zeta) = \prod_{\omega^a = 1} (\zeta - Y(\omega t))$$

has coefficients invariant under $t \mapsto \omega t$. In a convergent power series, this invariance makes every nonzero coefficient have exponent divisible by a. Thus the coefficients are convergent power series in $\xi = t^a$. They vanish at $\xi = 0$. The polynomial $Q(\xi, \zeta)$ is distinguished monic of degree a. Its pullback along F is zero, so the kernel hypothesis gives $Q = fh$ for an analytic germ h.

For small nonzero ξ, the a points $F(t)$ with $t^a = \xi$ are distinct by injectivity. Their complementary coordinates are distinct because their projection coordinates agree. They are roots of P, so $M \geq a$. On the other hand, the identity $Q = uPh$ shows that every zero of P in a sufficiently small fibre disc is a zero of Q with at least the same multiplicity. All M roots of P and all a roots of Q lie in this disc after shrinking the base. Hence $M \leq a$. It follows that $M = a$, and the a distinct displayed roots exhaust those of P. This also proves the intersection assertion. □

Proof of Theorem 6.1.3. Apply the supplied normalization and positive-braid construction to β and its chosen coordinate. The resulting representative F satisfies $\xi(F(t)) = t^a$, is injective, and has the same equation kernel after the source reparametrization. Lemma 6.6.1 identifies its sheet braid with the roots of the distinguished polynomial of f, and gives $i_0(f, \xi) = a$. The product polynomial Q constructed in that lemma is holomorphic for every $|\xi| < R^a$ on which the chosen representative is defined and injective. It has distinct roots for $0 < |\xi| < R^a$. Thus its discriminant has no zero in that punctured disc. The local-order winding formula applies to every circle there, including the radius selected by the existing link construction.

The positivity agrees with the convention used above. If $h(t)$ is a difference of two parameter sheets, the supplied declaration `planeBranchSheetDifference_exists_positive_order` and its radius consequence give $\Re(th'(t)/h(t)) > 0$ for every sufficiently small nonzero t. Along the base angle one has $t = r_0 e^{i\theta/a}$, up to a root of unity, and hence

$$\frac{d}{d\theta} \arg h(t) = \frac{1}{a} \Re \left(\frac{th'(t)}{h(t)} \right) > 0.$$

Multiplication by the constant scale used for the native sheets does not alter this derivative. Lemma 6.2.1 therefore makes every projected pair event positive. A sufficiently small common radius exists because only finitely many roots of unity and sheet pairs occur.

Retain the regular base angle and positive event words in `PointedMapGerm.exists_equation_link_canonical_positive_braid`. Denote their flattened word by w. The positive projected-crossing input for these same event data is the existing `PointedMapGerm.exists_equation_link_finite_braid_projection`, quoted as L404 in the supplied audit. The angular calculation above checks its sign convention. No different word or radius is substituted for the chosen native word. The canonical-word declaration states exactly that its length is the sum, over all event times, of the numbers of coincident unordered pairs. The use of final-position labels in

that formula does not change the number of unordered pairs, since each sorting is a permutation. Lemma 6.3.1 applies, and the positivity just established gives

$$\operatorname{length}(w) = \operatorname{ord}_0 D = i_0(f, f_\zeta) = \mu(f) + a - 1.$$

Equivalently, the same conclusion follows from Theorem 6.1.2.

The supplied Milnor identity `PlaneCurve.jacobianColength_add_irreducibleBranchCount` applies because f is pointed and irreducible, hence reduced. An irreducible equation has one branch, so it gives $\mu(f) = 2\delta(f)$. Substitution proves (6.1.3). The endpoint permutation equations are unchanged because the word and the event data have not been replaced. Apply `braid_word_cycleCount_eq_one_of_endpoint_rotation` and `braid_word_generatorIndices_eq_univ_of_endpoint_rotation`. They give one cycle and all adjacent generators for this same w. The normalized link and its diffeotopy to the canonical word model are also the ones in the existing construction. No new link identification is inferred from the numerical equality.

For the resolution-level assertion, fix j. The supplied notes record that `ResolutionSectionLevel.singular` supplies `SingularSectionPointData` at $L.\texttt{point}(j)$ with upper bound $T = (g.\texttt{criticalJacobianFactorization}(hi)).\texttt{maxMultiplicity}$. The quoted existing declaration `SingularSectionPointData.exists_branch_component_order` provides the injective parameter germ, the local equation kernel, the coordinate index, and $2 \le a \le T$. Use those data in the result just proved. It gives a word on these same a strands with the required length, cycle count, generator set, and normalized link. The inequalities for a are preserved from that declaration. The defining modules of the resolution level and singular-point data are catalogued in the attachment rather than reproduced there. This last assembly uses their quoted statements and does not assert a new unverified size bound. □

6.7 The full-twist word

Proof of Corollary 6.1.4. The multiplicity and transversality hypotheses give $m = S$ in Theorem 6.1.2. The J14 word represents the same root braid, and its letters are all positive. Consequently (6.1.4) follows from (6.1.2) and Corollary 6.3.3.

For the full rotation of a configuration of S points, each difference is multiplied by $e^{\mathrm{i}\theta}$ over one turn. Its squared-difference product is multiplied by $e^{\mathrm{i}S(S-1)\theta}$. Thus the full twist has exponent $S(S-1)$ by Lemma 6.3.1. The positive word d representing that rotation in J14 therefore has length $S(S-1)$. Subtracting this length from (6.1.4) gives

$$\operatorname{length}(v) = \mu(f) - S^2 + 2S - 1.$$

The supplied Milnor identity gives $\mu(f) + b = 2\delta(f) + 1$. Substitution yields (6.1.5). □

There is also a direct check on the strict-transform word used in J14. If $u_j = y_j/\xi$, then on the base circle

$$\prod_{j<k} (u_j - u_k)^2 = \xi^{-S(S-1)} \prod_{j<k} (y_j - y_k)^2.$$

Hence the strict-transform braid has exponent $\operatorname{ord}_0 D - S(S-1)$. This equals the tail length because J14 makes that word positive. Distinct tangent clusters cause no difficulty in this identity. The product is over all unordered pairs, including pairs from different clusters, and the endpoint permutation still permutes those pairs.

6.8 The broader link-isotopy quantifier in M407

The discriminant determines the exponent of the analytic root braid. The proposed M407 API also permits a positive word known only through an ambient isotopy of its closure. The following

algebraic argument supplies the additional comparison. Every coefficient in its modules and matrices belongs to $\mathbb{F}_2$. The indeterminate T below is an auxiliary variable for an infinite cyclic covering. It is unrelated to the Floer variable U.

Lemma 6.8.1 (New lemma: the elementary braid matrices)**.** *Let $s \geq 2$, put $R = \mathbb{F}_2[T, T^{-1}]$, and let B_i be the s by s identity matrix with its rows and columns $i, i+1$ replaced by the block*

$$\begin{pmatrix} 1+T & T \\ 1 & 0 \end{pmatrix}, \qquad 0 \leq i \leq s-2.$$

Write $v = (1, \dots, 1)^{\mathsf{t}}$ and $q = (1, T, \dots, T^{s-1})$. The matrices satisfy the positive braid relations, fix v, preserve the row vector q, and induce matrices $\overline{B}_i$ on the free quotient R^s/Rv of rank $s-1$. They have the following properties.

1. $\det \overline{B}_i = T$.
2. *Every minor of every $\overline{B}_i$ has degree at most one as a polynomial in T.*
3. *If $c_s = \sigma_0 \cdots \sigma_{s-2}$ and $d_s = c_s^s$, then $\overline{B}_{d_s} = T^s I$.*
4. *For a positive word w of length l, every minor of $\overline{B}_w$ has degree at most l, and $\det \overline{B}_w = T^l$.*

Proof. The determinant of the displayed block is T, which is invertible in R. Blocks on disjoint pairs commute. On three successive coordinates, the two products in the adjacent braid relation both equal

$$\begin{pmatrix} 1+T & T+T^2 & T^2 \\ 1+T & T & 0 \\ 1 & 0 & 0 \end{pmatrix}.$$

This verifies all relations by matrix multiplication. The identities $B_i v = v$ and $q B_i = q$ follow on the same two coordinates from

$$\begin{pmatrix} 1+T & T \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad (1, T) \begin{pmatrix} 1+T & T \\ 1 & 0 \end{pmatrix} = (1, T).$$

The quotient is free, with basis the classes of $e_0, \dots, e_{s-2}$, since the last basis vector equals their sum there. The invariant line has eigenvalue one, so the determinant on the quotient equals the determinant on the whole module, namely T.

On R^s, the difference $B_i - I$ is a rank-one matrix $u_i(T)\lambda_i$, where $u_i(T)$ has degree at most one and λ_i is a constant row vector supported in coordinates $i, i+1$. Since $\lambda_i(v) = 0$, it descends to the quotient with the same property. In the multilinear expansion of any minor of $I + u_i(T)\lambda_i$, a term using the rank-one summand in two columns vanishes because those columns are proportional. Every surviving term uses it at most once. Thus every minor has degree at most one. The Cauchy–Binet formula for a product expresses each minor as a sum of products of minors of the factors. Induction over the letters of w gives the degree bound l. Multiplying determinants gives T^l.

The product $C = B_0 \cdots B_{s-2}$ has first row

$$(1+T,\ T(1+T), \dots, T^{s-2}(1+T),\ T^{s-1})$$

and has ones immediately below the diagonal, with all its other entries zero. Expansion of $\det(\lambda I - C)$ along its last column, followed by induction on s, gives

$$\det(\lambda I - C) = (\lambda + 1)(\lambda^{s-1} + T\lambda^{s-2} + \cdots + T^{s-1}).$$

The invariant line contributes $\lambda + 1$, so the second factor is the characteristic polynomial of $\overline{C}$. The Cayley–Hamilton identity and multiplication by $\overline{C} + TI$ give

$$\overline{C}^{,s} + T^s I = 0.$$

Over $\mathbb{F}_2$ this says $\overline{B}_{d_s} = \overline{C}^{,s} = T^s I$. This argument also covers even s, when the characteristic polynomial can have repeated roots. □

Lemma 6.8.2 (New lemma: a closed-braid determinant is a link invariant)**.** *Let β be an s-strand braid, with $s \geq 2$, and suppose*

$$P_\beta(T) = \det(I + \overline{B}_\beta(T)) \neq 0.$$

Then

$$A_{\widehat{\beta}}(T) = \frac{P_\beta(T)}{1 + T + \cdots + T^{s-1}} \tag{6.8.1}$$

is a nonzero Laurent polynomial, defined up to multiplication by a power of T as an invariant of the oriented closure. Its breadth, meaning its highest exponent minus its lowest exponent, is unchanged by an orientation-preserving ambient isotopy. It is also unchanged if every component orientation is reversed.

Proof. We prove the formula from a cellular presentation, so no property of a fibre surface is used. Let X be the complement of the closed braid and let $\widetilde{X}$ be the infinite cyclic covering specified by sending each oriented meridian to $1 \in \mathbb{Z}$. The homology $H_1(\widetilde{X}, \mathbb{F}_2)$ is an R-module, where T is the deck transformation. An oriented link isotopy identifies these coverings and their modules. Simultaneously reversing all component orientations replaces T by T^{-1}, which preserves the breadth of every Laurent polynomial.

Here is a direct presentation of the module. Choose successive meridians $x_0, \ldots, x_{s-1}$ in a transverse punctured disc. Transport across a positive adjacent crossing replaces

$$x_i \longmapsto x_i x_{i+1} x_i^{-1}, \qquad x_{i+1} \longmapsto x_i.$$

This is obtained by carrying the two small meridian loops through the half twist while keeping their paths to the boundary base point. One loop acquires the path around x_i and its inverse, and the other acquires the original path of x_i. For the auxiliary module calculation, fix this transport convention. There is no sign ambiguity in the length conclusion. If the native positive half twist is drawn with the inverse meridian transport, all its matrices are $B_i(T)^{-1}$. Let J reverse the order of the s basis vectors. Direct inversion of the two by two block gives

$$J B_i(T)^{-1} J = B_{s-2-i}(T^{-1}).$$

Thus reversing the meridian order and the choice of deck generator reduces that convention to the displayed polynomial matrices as well. These changes preserve the module and its breadth. They replace the Coxeter product by its reversed product, which has the same full-twist scalar as checked below. The analytic exponent sign in the preceding discriminant proof is still the fixed native positive sign.

If ϕ_β is the resulting automorphism of the free group, identifying the top and bottom meridians in the closed diagram gives the presentation

$$\pi_1(X) = \left\langle x_0, \ldots, x_{s-1} \;\middle|\; \phi_\beta(x_i) x_i^{-1} = 1 \ (0 \leq i < s) \right\rangle.$$

To justify the presentation directly, apply the meridian presentation of each crossing in succession. Its relation eliminates the outgoing meridian in terms of the incoming ones by the displayed conjugation. After all crossings have been processed, closure identifies each outgoing meridian with the corresponding incoming meridian. There are no other relations in the diagram complement, by gluing the crossing neighborhoods and their arc neighborhoods along discs.

The boundary coefficients for the cyclic covering of this presentation have an elementary word description. A positive letter x_j after a prefix of exponent a contributes T^a in the jth edge coordinate. A negative letter x_j^{-1} after that prefix contributes T^{a-1} in that coordinate, since signs are equal in $\mathbb{F}_2$. These are the lifted directed edges of the word path. For a concatenation,

the coefficients of the second word are multiplied by the exponent of the first. This rule is the abelianized derivative rule

$$\partial_j(uv) = \partial_j u + T^{\exp u}\partial_j v.$$

Applied to the two transported meridians above, it gives exactly the block B_i. Applied successively to a braid word, it gives B_β, with matrix multiplication in the order of this transport convention. If word composition is recorded in the opposite order, use the reversed matrix product throughout. The minor degree bound is valid in either order. The reversed Coxeter product has the same characteristic polynomial: with $E = \operatorname{diag}(1, T, \dots, T^{s-1})$, every generator satisfies $B_i^{\mathsf{t}} = EB_iE^{-1}$, so that product is conjugate to the transpose of the original Coxeter product. It also fixes v. The same quotient characteristic polynomial and Cayley–Hamilton argument therefore give the same scalar T^sI for its full twist. No comparison of arbitrary word orders is required.

Each relation $\phi_\beta(x_i)x_i^{-1}$ therefore has lifted boundary equal to row i of $B_\beta - I$. The boundary of each one-cell is $T - 1$, so the one-cycles form

$$K = \{(a_0, \dots, a_{s-1}) \in R^s : \sum_i a_i = 0\}.$$

Here $T - 1 \neq 0$ and R has no zero divisors. Deleting the last coordinate identifies K with R^{s-1}. Thus the module is presented by the rows of $A = B_\beta - I$ with its last column deleted. This computation of H_1 depends only on the fundamental group and its homomorphism to $\mathbb{Z}$: the covering fundamental group is the kernel, and its abelianization with $\mathbb{F}_2$ coefficients is H_1. Consequently using the displayed presentation instead of a full cell decomposition of X does not change the module.

Work temporarily over the field $\mathbb{F}_2(T)$. The hypothesis $P_\beta \neq 0$ says that the induced operator $I - \overline{B}_\beta$ on R^s/Rv is invertible over this field. Hence $I - B_\beta$ has rank $s - 1$, right kernel spanned by v, and left kernel spanned by q. The last statement follows from $qB_\beta = q$ and the rank. Its adjugate therefore has the form

$$\operatorname{adj}(I - B_\beta) = a(T)vq.$$

Since the first entry of q and every entry of v equal one, the scalar $a(T)$ is one of the adjugate entries and belongs to R. The maximal minors in the module presentation just described are its last-row adjugate entries $a(T), a(T)T, \dots, a(T)T^{s-1}$. Their greatest common divisor in the principal ideal domain R is $a(T)$, up to a unit. It is nonzero.

For completeness, this greatest common divisor is an isomorphism invariant of the presented torsion module. Euclidean row and column operations diagonalize a full-rank presentation over R. The diagonal entries present cyclic summands, and the product of the nonunit diagonal entries generates the ideal of maximal minors. For each irreducible polynomial, the exponents in those summands are recovered from the dimensions of the successive quotients by its powers. The product is therefore independent of the chosen presentation. The only units of R are powers of T.

It remains to compute $a(T)$. The characteristic polynomial satisfies

$$\det(\lambda I - B_\beta) = (\lambda - 1)\det(\lambda I - \overline{B}_\beta).$$

Differentiate at $\lambda = 1$. Expansion of a determinant by its columns gives the derivative as the trace of its adjugate, including in characteristic two. Hence

$$P_\beta(T) = \operatorname{tr}\operatorname{adj}(I - B_\beta) = a(T)(1 + T + \cdots + T^{s-1}).$$

This proves (6.8.1) and all its asserted invariance properties. □

Lemma 6.8.3 (New lemma: length detected by a full-twist prefix)**.** *Let $s \geq 2$ and let w be a native positive adjacent word on s strands. Put $l = \operatorname{length}(w)$ and $\beta = d_s w$. Then the invariant in Lemma 6.8.2 is nonzero and has breadth*

$$\operatorname{br} A_{\widehat{\beta}} = s(s-1) + l - s + 1 = \operatorname{length}(\beta) - s + 1. \tag{6.8.2}$$

Proof. Lemma 6.8.1 gives

$$P_\beta(T) = \det(I + T^s \overline{B}_w).$$

Its constant coefficient is one, since the entries of $\overline{B}_w$ are polynomials. Expanding by principal minors, the term using all $s-1$ rows and columns is

$$T^{s(s-1)} \det \overline{B}_w = T^{s(s-1)+l}.$$

Every term using $k \le s-2$ rows and columns has degree at most $sk + l \le s(s-2) + l$, by the minor bound. Thus the displayed highest term cannot cancel. The lowest degree of P_β is zero and its highest degree is exactly $s(s-1)+l$. Both extreme coefficients are one in $\mathbb{F}_2$.

The denominator in (6.8.1) has lowest degree zero and highest degree $s-1$. Since the quotient belongs to R, additivity of the lowest and highest exponents under multiplication gives lowest degree zero and highest degree $s(s-1)+l-s+1$ for that quotient. This proves the formula. □

Corollary 6.8.4 (New lemma: isotopic positive full-twist words have equal length)**.** *Let $s \ge 1$ and let v, w be native positive words on s strands. If the oriented closures of $d_s v$ and $d_s w$ are ambiently isotopic, then* $\mathrm{length}(v) = \mathrm{length}(w)$*. The same conclusion holds if the isotopy is given without component orientations.*

Proof. For $s = 1$ there is no adjacent generator, so both words are empty. For $s \ge 2$, the oriented assertion follows by applying Lemma 6.8.3 to the two equal invariant breadths.

For the assertion without specified orientations, give each closure its braid orientation. If there is only one component, an isotopy either preserves or reverses it, and both cases preserve breadth. If there are several components, the linking number of every pair is strictly positive. Indeed the full twist has two positive crossings for each pair of its strands, as proved by the rotation calculation in J14. Two different closure components are unions of disjoint strand sets. Their crossings in the prefix alone therefore contribute a positive linking number. The remaining positive letters add only nonnegative contributions. The integer linking number is half the signed number of crossings between the two components.

Reversing exactly one member of a component pair reverses this integer, whereas an ambient isotopy preserves it. Thus an isotopy between the two unoriented links must either preserve the orientations of all components or reverse all of them, after its permutation of components. Both possibilities preserve the invariant breadth by Lemma 6.8.2. The same length equality follows. Linking numbers here are ordinary integer topological counts. Every homology and matrix coefficient remains in $\mathbb{F}_2$. □

Theorem 6.8.5 (New lemma: the full positive-word quantifier in M407)**.** *Let f be a reduced plane curve germ of multiplicity $S \ge 1$, let b be its irreducible branch count, and choose a transverse projection. Let w be any positive adjacent word on S strands whose full-twist closure $\widehat{d_S w}$ is ambiently isotopic to the small link of f. Assume the geometric identification of that link with its transverse root braid supplied by M403(b), whenever the native sphere formulation is intended. Then*

$$\mathrm{length}(w) + S^2 + b = 2\delta(f) + 2S. \tag{6.8.3}$$

The theorem does not require a braid isotopy between the specified word and the analytic root braid.

Proof. Theorem 7.7.6 produces a positive word v such that $d_S v$ is the same analytic root braid. Its geometric link is the specified small link under the stated M403(b) identification. The two closures $\widehat{d_S v}$ and $\widehat{d_S w}$ are therefore ambiently isotopic. Corollary 6.8.4 gives $\mathrm{length}(v) = \mathrm{length}(w)$. The discriminant, resultant, and polar-intersection argument already proved in Corollary 6.1.4 gives

$$\mathrm{length}(v) + S^2 + b = 2\delta(f) + 2S.$$

Substitution proves (6.8.3).

For the proposed resolution-assembly API, set

$$f = A.\texttt{imageEquation}, \qquad S = A.\texttt{squareSum}.$$

The field names here denote the native assembly fields, rather than additional analytic choices. The fields `imageEquation_mem` and `imageEquation_not_mem`, recorded in L431 of the supplied audit, say that S is the exact order of this equation. The proposed closure-isotopy predicate in M407 supplies precisely the closure comparison used above. Thus (6.8.3) is its full displayed conclusion, including the quantifier over every such positive word. The native definition of that proposed predicate still belongs to the geometric assembly, rather than to this word-length proof. □

This comparison uses only a cyclic covering, finite matrix algebra, and the discriminant calculation for the analytic word. It uses neither Stallings' theorem nor Rudolph's fibre-surface argument. The classical determinant formula is also stated in [2, Introduction]. The complete $\mathbb{F}_2$ derivation required here is Lemma 6.8.2, so that published formula is not an unproved step of this argument.

The smallest matrix cases agree with the analytic checks below. For $s = 2$ and w empty, the numerator is $1 + T^2$ and the denominator is $1 + T$, giving $A = 1 + T$ of breadth one. For $s = 2$ and $w = \sigma_0$, they give $A = 1 + T + T^2$ of breadth two. These are applications of the displayed matrices and quotient formula, without an appeal to a knot table. For $s = 1$ the adjacent word is empty, as handled separately in Corollary 6.8.4.

6.9 Exceptional cases and convention checks

If $m = 1$, the discriminant is 1, the signed root braid has exponent zero, and the positive word is empty. For a transverse smooth branch, $a = 1$, $\mu = \delta = 0$, and (6.1.3) reads $0 + 1 = 0 + 1$. Its generator set is the empty set of indices, which is the full set $\mathrm{Fin}(0)$, and its permutation has one fixed-point cycle. The lower bound $a \geq 2$ in the resolution-level acceptance comes from singular-point data and is not imposed on arbitrary smooth branches.

For the cusp $f(\xi, \zeta) = \zeta^2 - \xi^3$, the discriminant is $4\xi^3$. The two sheet differences increase in argument by 3π over one base turn. Its Jacobian quotient has basis $1, \xi$, so $\mu = 2$. The supplied Milnor identity and irreducibility give $\delta = 1$. There are three positive pair crossings and $\mathrm{length}(w) = 3 = 2\delta + 2 - 1$. For the transverse ordinary double point $f = (\zeta - \xi)(\zeta + \xi)$, the discriminant is $4\xi^2$. There are two positive pair crossings, giving the two-letter positive full twist. Here $S = b = 2$. The Jacobian ideal is (ξ, ζ), so $\mu = 1$, and the supplied Milnor identity gives $\delta = 1$. The tail is empty. These computations are exact substitutions in the proved identities, not evidence for an unproved general statement.

For an ordinary S-fold point with distinct tangent slopes, every pair has a difference of order one in ξ. Thus D has order $S(S - 1)$, which is the full-twist length, and (6.1.5) gives an empty tail. This agrees with the separated strict-transform clusters in J14.

For a simultaneous event involving $q \geq 3$ sheets with equal projected real part, the pair count contains exactly $\binom{q}{2}$ terms from that cluster. Lemma 6.3.2 resolves it into that many simple pair crossings. The endpoint cancellation in Lemma 6.3.1 already counts it pairwise, so no division by q or 2 occurs.

The distinguished condition cannot be omitted from the local-intersection assertion. For

$$P(\xi, \zeta) = \zeta\big((\zeta - 1)^2 - \xi\big),$$

one has $\mathrm{Disc}\, P = 4\xi(1 - \xi)^2$, of order one at zero. However $P_\zeta(0, 0) = 1$, so the local pair quotient at $(0, 0)$ is zero and $i_0(P, P_\zeta) = 0$. The other two roots coalesce at $(0, 1)$. Thus the discriminant order of a general monic polynomial measures all central collisions, not only the collision at the origin.

Transversality is necessary when the strand number is identified with the multiplicity of the germ. The smooth germ $f = \zeta^2 - \xi$ has multiplicity one and $\mu = 0$, but projection to ξ has degree two. Its discriminant has order one and its root braid has one positive letter. Formula (6.1.2) gives $1 = 0 + 2 - 1$, as required. Replacing the projection degree by the multiplicity would give a false equality. This is also the obstruction to asserting a positive full-twist prefix on two strands from regularity of projection degree two alone.

6.10 Lemma and declaration table

Declaration or result	Status	Exact use
`SignedAdjacentWord`, `adjacentSwap`	Existing Lean	Signs $\mathbb{Z}^\times$ and zero-based generator indices.
`PointedMapGerm.exists_equation_link_canonical_positive_braid`	Existing Lean	Chosen branch word, pair-count length, endpoint equations, and unchanged link identification.
`planeBranchSheetDifference_exists_positive_order`, `exists_radius_planeBranchSheetDifference_re_pos`	Existing Lean	Positive angular derivative for branch pairs on a common radius.
`exists_radius_planeBranchAngularDifference_projected_crossings_pos`, `PointedMapGerm.exists_equation_link_finite_braid_projection`	Existing Lean, catalogued	Native positive-crossing convention and positive determinants for the chosen event data. Their modules are cited in L4.16 and L404 rather than reproduced.
`braid_word_cycleCount_eq_one_of_endpoint_rotation`, `braid_word_generatorIndices_eq_univ_of_endpoint_rotation`	Existing Lean	One cycle and every generator for the same branch word.
`SingularSectionPointData.exists_branch_component_order`, `ResolutionSectionLevel.singular`	Existing Lean, catalogued	Exact branch data and inherited bounds $2 \le a \le T$.
`Polynomial.resultant_deriv`	Existing Lean, catalogued	Discriminant–resultant sign relation.
`Germ.hasFiniteIntersection_of_resultant_ne_zero`, `Germ.resultant_ne_zero_of_hasFiniteIntersection`, `Germ.intersectionMultiplicity_eq_resultant_order_of_ne_zero`	Existing Lean	Distinguished local resultant formula, with all hypotheses checked.
`PlaneCurve.exists_polar_intersection_identity`	Existing Lean	Teissier identity applied directly to f and parameter ξ.
`PlaneCurve.jacobianColength_add_irreducibleBranchCount`	Existing Lean	Milnor identity for the branch and reduced cases.
Lemmas 6.2.1–6.3.2	New, proved	Continuous arguments, sign, pair cancellation, event resolution, and winding invariance.
Lemma 6.4.1	New, proved	Winding equals analytic vanishing order.
Lemma 6.4.2	New, proved	Distinguished preparation and projection intersection number.
Lemma 6.5.1	New, proved	Equality of the polar pair ideals after a unit factor.
Lemma 6.6.1	New, proved	The normalized a-sheet parameterization gives degree a.
Theorems 6.1.1, 6.1.2, 6.1.3	New, proved	M1155, polar length identity, and branch acceptance.

Declaration or result	Status	Exact use
Corollary 6.1.4	New, proved conditional on the specified J14 word	Reduced full-twist length and tail identity.
J14 transverse full-twist construction	New, in companion document	Existence and positivity of the word representing the same root braid.
Lemmas 6.8.1–6.8.3	New	Finite braid matrices, the cyclic-covering module, and the exact breadth formula over $\mathbb{F}_2$.
Corollary 6.8.4	New	Equal length for ambiently isotopic positive full-twist closures.
Theorem 6.8.5	New	The full positive-word quantifier in M407, with the audit's explicit reduced-link realization input.
`ResolutionSectionAssembly.imageEquation_mem`, `ResolutionSectionAssembly.imageEquation_not_mem`	Existing, audit L431	The assembly's square sum is the exact order of its image equation.

6.11 Precise limits of the conclusion

The counting and analytic identities in Theorems 6.1.1 and 6.1.2 have no remaining mathematical gap. Theorem 6.1.3 assembles the existing branch-word and singular-point inputs with these identities and preserves their chosen word and link representative. The reduced positive-word conclusion uses the specified J14 construction, as required by J15(b). The exact source spelling of the resolution-level structures is not supplied in the attached native listings, so the resolution-level assembly is stated using the existing declarations as quoted in the supplied audit.

Theorem 6.8.5 closes the broader M407 quantifier for positive full-twist words known only through their ambiently isotopic closures. Its additional input is the finite matrix and cyclic-covering argument proved in Section 6.8. The analytic word length still comes from the argument principle, discriminant, resultant, and Teissier identity. No fibre-surface theorem is used. The native sphere identification in the reduced case retains the M403(b) geometric input assigned to J13 by the audit. The definition of its proposed closure predicate belongs to that assembly. This is the same explicit geometric dependency as in J14, rather than an additional word-length gap.

Similarly, the displayed J14 API with only regularity in the fibre variable must be distinguished from its narrative hypothesis of multiplicity S and a transverse projection. Corollary 6.1.4 uses the latter hypotheses, which are explicitly required in the user's reduced-germ task. The tangent smooth example above explains why those hypotheses cannot be silently inferred from projection degree alone.

Chapter 7

J14: Multi-branch positivity and the full-twist prefix

7.1 Statements and the distinction between the two hypotheses

There is a hypothesis error in the displayed proposed API of M405. Regularity of order S in the first coordinate means that the restriction $f(\zeta, 0)$ has order S. It does not say that the two-variable germ has multiplicity S. The API, including its smooth sphere-link conclusion, is false with only its displayed hypotheses. Proposition 7.2.2 gives a counterexample of projection degree two.

The user's geometric statement includes the extra information that the multiplicity is S and that the projection is transverse. That information is sufficient for the full-twist argument. The distinction is retained throughout this document. No assertion with the corrected hypotheses is presented as a proof of the displayed API.

Write σ_i for the positive adjacent generator at native index $i \in \{0, \dots, S-2\}$. Put

$$c_S = \sigma_0\sigma_1 \cdots \sigma_{S-2}, \qquad d_S = c_S^S. \tag{7.1.1}$$

For $S = 1$, both words are empty. These are the precise lists proposed as `coxeterWordS` and `fullTwistWordS` in the audit. The supplied listings define `PositiveAdjacentWord`, its conversion to signed words, its permutation, and its cycle count. They do not include a completed definition of the proposed multi-component smooth model `positiveWordClosedLinkModel`.

Here is the exact analytic statement proved below.

Theorem 7.1.1 (New lemma: multi-sheet positivity)**.** *Let q, S be positive integers. Let $Y_1, \dots, Y_S$ be convergent holomorphic functions on a disc about zero. Assume that their germs are pairwise distinct. Let $c \in \mathbb{C}^*$ satisfy*

$$\operatorname{Re}\big(c(Y_i(0) - Y_j(0))\big) \neq 0 \quad \text{whenever } Y_i(0) \neq Y_j(0).$$

There is $r_0 > 0$ such that, for every $0 < r < r_0$, every projected crossing of the strands

$$K_i(\theta) = cY_i\big(re^{\mathrm{i}\theta/q}\big)$$

is positive. Every pair with $Y_i(0) = Y_j(0)$ has strictly increasing argument of its difference along the parameter interval. Pairs with distinct constant terms have no projected crossing. Every compact parameter interval has finitely many crossing events, with multiple events counted pairwise.

If, in addition, the unordered set of functions is invariant under $t \mapsto e^{2\pi\mathrm{i}/q}t$, these strands define a closed S-braid. On a period with regular endpoints that braid is represented by a word in `PositiveAdjacentWordS`*.*

The monodromy-invariance condition in the last paragraph is necessary for a closed braid. A list consisting only of $\xi^{1/2}$ does not close after one turn in ξ. The root sheets of a reduced equation satisfy this condition. The schematic wording of M1153 must be read as a complete set of root sheets, or must include this condition explicitly.

Theorem 7.1.2 (New lemma: full twist for explicit transverse branch data)**.** *Let $b \geq 1$, let $a_1, \ldots, a_b$ be positive integers with $\sum_j a_j = S$, and let*

$$F_j(s) = (h_j(s), s^{a_j})$$

be convergent injective parametrizations of distinct branch germs, whose punctured images are disjoint in a common neighborhood. Assume $\operatorname{ord} h_j \geq a_j$*, where the zero function is allowed. For all sufficiently small $\eta > 0$, let B_η be their root braid over $\xi = \eta e^{\mathrm{i}\theta}$. There is a fibre rotation $c \in \mathbb{C}^*$ and one positive word $w \in$* `PositiveAdjacentWordS` *such that B_η is isotopic through closed braids to the word $d_S w$. The word w may be chosen as a positive event word of the strict-transform sheets ζ/ξ. Its native permutation has exactly b cycles, with the cycle of branch j of length a_j. The same holds for $d_S w$.*

The theorem is a statement about the same root braid, through a collision-free braid isotopy. It supplies the braid identification needed by the discriminant argument of J15. It is not merely an equality of link types. Section 7.7 constructs the needed finite data directly from the prepared polynomial and its finite root covering. It proves injectivity, the exact kernel ideals, punctured disjointness, and the degree sum without an additional normalization assumption. The sphere conclusion is explicitly conditional on M403(b) and the realization in M419(B), as permitted by audit Section 11.1.

7.2 The displayed API is false

Lemma 7.2.1 (New lemma: positive two-strand closures)**.** *For every integer $r \geq 2$, the closure of the positive two-strand word σ_0^r is not the unknot.*

Proof. The permutation is the rth power of the transposition of two elements. For even r, following the two strands around the closure gives two components, so the link is not a knot.

For odd r, the closure is the curve of slope $(2, r)$ on the torus separating the two standard solid tori of $\mathbb{S}^3$. This follows directly from the sheet formula: each generator rotates the difference of the two fibre points counterclockwise through π, and closure gives the parametrization

$$t \longmapsto (ae^{2\mathrm{i}t}, be^{r\mathrm{i}t}), \qquad a, b > 0, \quad a^2 + b^2 = 1.$$

It is embedded because r is odd. The complement group has presentation

$$\langle u, v \mid u^2 = v^r \rangle. \tag{7.2.1}$$

For completeness, remove an annular neighborhood of this primitive slope curve from the separating torus. The remainder of the torus is an annulus. Thicken the two solid tori slightly into the knot complement. Each thickened piece retracts to its solid-torus core, and their intersection retracts to that annulus. Its core winds twice around one solid-torus core and r times around the other. Applying the van Kampen gluing relation gives exactly (7.2.1).

This group maps onto the dihedral group generated by a reflection s and a rotation v_0 of order r, by $u \mapsto s$, $v \mapsto v_0$. The relation is satisfied since $s^2 = v_0^r = 1$. For odd $r \geq 3$, this quotient is nonabelian: $sv_0s^{-1} = v_0^{-1} \neq v_0$. The unknot complement is a solid torus and has group $\mathbb{Z}$. An ambient diffeomorphism induces an isomorphism of complement groups, which excludes an unknot closure in the odd case as well. □

Proposition 7.2.2 (New lemma: counterexample to the exact M405 API)**.** *The proposed assertion* `Germ.exists_reduced_link_fullTwist_positive_word` *in M405 is false under its displayed hypotheses.*

Proof. Take

$$H(\zeta,\xi)=\zeta^2-\xi,\qquad S=2.$$

The function is analytic and its germ is squarefree, since it is a smooth coordinate germ. Its restriction $H(\zeta,0)=\zeta^2$ has order two, so it satisfies the displayed first-coordinate regularity hypothesis. It has two-variable multiplicity one.

Its small sphere link is the unknot. Indeed, its parametrization is $t\mapsto(t,t^2)$. On the sphere of radius $\epsilon>0$, the parameter has the unique positive modulus r solving $r^2+r^4=\epsilon^2$. After normalization the link is

$$\theta\longmapsto(ae^{\mathrm{i}\theta},be^{2\mathrm{i}\theta}),\qquad a=r/\epsilon>0,\quad b=r^2/\epsilon,\quad a^2+b^2=1.$$

Replace b by $(1-s)b$ and a by $\sqrt{1-(1-s)^2b^2}$ for $0\le s\le 1$. This is a smooth family of embedded graphs over the first circle, ending at $(e^{\mathrm{i}\theta},0)$. A compactly supported fibre vector field extending the velocity of these graphs gives an ambient isotopy, by the explicit construction in Lemma 7.5.3 below.

Every native positive two-strand word is σ_0^{ℓ} for some $\ell\ge 0$. The proposed prefix d_2 is σ_0^2. Thus every candidate model has closure $\sigma_0^{2+\ell}$, which cannot be the unknot by Lemma 7.2.1. This contradicts the displayed sphere conclusion for every possible w. □

The same example gives a word-level obstruction with no knot invariant: the root sheets $\pm\sqrt{\xi}$ give one positive crossing, whereas d_2w has exponent sum $2+\ell$. The discriminant computation of J15 makes that obstruction explicit. The geometric counterexample above also addresses the weaker sphere-link conclusion of the displayed API. On the common cover $\xi=t^2$, the proposed strict-transform sheets would be $\pm t^{-1}$. Their negative order records the failed transverse hypothesis directly. They are outside the nonnegative-order statement of M1153.

7.3 Positive rotation for a finite collection of analytic differences

Lemma 7.3.1 (New lemma: analytic difference estimate)**.** *Let D be a nonzero holomorphic germ at zero. Write $D(t)=t^eV(t)$ with $e\in\mathbb{N}$ and $V(0)\neq 0$. On a sufficiently small punctured disc, D has no zeros and*

$$\frac{tD'(t)}{D(t)}=e+\frac{tV'(t)}{V(t)}.$$

If $e>0$, its real part is greater than $e/2$ on a sufficiently small punctured disc.

Proof. Choose the first nonzero coefficient of the convergent power series of D. Factoring the corresponding power of t produces the holomorphic V with $V(0)\neq 0$. Continuity gives a disc on which V is nonzero. Differentiating the factorization gives the displayed identity there. The continuous function $tV'(t)/V(t)$ tends to zero. Taking its modulus less than $e/2$ proves the last assertion. □

The limiting calculation is also the existing declaration `tendsto_mul_deriv_div_of_analytic_factor` in the supplied `PlaneBranchSheetRotation.lean`. The new use is to every difference in a finite collection, including differences from distinct branches.

Proof of Theorem 7.1.1. For each unordered pair let $D_{ij}=Y_i-Y_j$. Its germ is nonzero by hypothesis. If $D_{ij}(0)=0$, apply Lemma 7.3.1 with $e_{ij}>0$. If $D_{ij}(0)\neq 0$, continuity gives

$$|\mathrm{Re}(cD_{ij}(t))-\mathrm{Re}(cD_{ij}(0))|<\frac{1}{2}\,|\mathrm{Re}(cD_{ij}(0))|$$

on a sufficiently small disc. There are only finitely many pairs, so a common disc works for all estimates and all pairwise nonvanishing conclusions.

Let $t = re^{\mathrm{i}\theta/q}$ and $d(\theta) = cD_{ij}(t)$. For a positive-order difference,

$$\frac{d}{d\theta}\arg d(\theta) = \operatorname{Im}\frac{d'(\theta)}{d(\theta)} = \frac{1}{q}\operatorname{Re}\frac{tD'_{ij}(t)}{D_{ij}(t)} > 0. \tag{7.3.1}$$

The argument on any real interval is defined by continuous lifting since d is nonzero. At a projected crossing write $d(\theta_0) = \mathrm{i}v$ with $v \in \mathbb{R} \setminus \{0\}$. If $a = \operatorname{Re} d'(\theta_0)$, then

$$\frac{d}{d\theta}\arg d(\theta_0) = -\frac{a}{v}, \qquad -av > 0. \tag{7.3.2}$$

This is the native positive projected-crossing determinant convention recorded by `exists_radius_planeBranchAngularDifference_projected_crossings_pos`. In particular $a \neq 0$, so each pairwise projected crossing is simple. The order-zero pairs have no crossing by the preceding real-part estimate.

Each pair has finitely many zeros of its projected real difference on a compact interval. Otherwise a convergent subsequence would accumulate at a zero, contradicting the nonzero derivative there. A finite union over the pairs is finite. Choose the initial angle outside these events whenever a closed period is required.

At an event where several strands have the same real part, their imaginary parts are distinct. Equation (7.3.2) says that the order of their real velocities is the reverse of the order of their imaginary coordinates. Consequently every coincident pair reverses its real order exactly once in a sufficiently small event interval, and every such crossing is positive. Lemma 7.3.2 below resolves the event into positive adjacent letters, preserving the braid. Concatenating the event words gives the required positive adjacent word. Monodromy invariance identifies the endpoint configurations, and therefore makes this a closed braid. □

Lemma 7.3.2 (New lemma: resolution of finite positive events)**.** *Suppose finitely many smooth strands on a compact interval have distinct complex values at each time and distinct projected real coordinates at the endpoints. Suppose their projected real coincidences occur at finitely many times, each pairwise zero is simple, and every crossing satisfies* (7.3.2)*. Their braid is isotopic relative to the endpoints to a concatenation of positive adjacent generators. An event at which a group of r strands has common real coordinate contributes exactly $r(r-1)/2$ letters, one for each pair in that group.*

Proof. Choose pairwise disjoint small intervals around the event times. On the complement, the minimum separation of projected real coordinates is positive. Within an event interval, shorten the interval until every coincident pair has real derivative of a fixed nonzero sign and its imaginary difference stays nonzero with fixed sign. All other pairs stay separated in real coordinate. The implicit function theorem preserves one simple crossing of the same sign under sufficiently small perturbations of real coordinates supported inside the interval.

Choose such perturbations with different small constant offsets for the strands near the event, and multiply them by a bump function supported in the interval. The offsets can be chosen so that no two pairwise crossing times coincide. Here is the finite avoidance argument. Each simple crossing time depends smoothly on the offsets. For two distinct pairs, vary an offset belonging to one pair and not the other. The first crossing time changes with nonzero derivative and the second does not. Thus the equation asserting equality of those two times has empty interior in the finite-dimensional offset space. Avoid the finite union of these sets. The perturbation and its linear interpolation from zero are small enough to preserve complex separation, because the original strands have positive minimum separation.

After perturbation there is one pairwise crossing at each crossing time. That pair is adjacent in real order, since all other projected coordinates are distinct. The sign remains positive by the fixed-sign estimates. Every originally coincident pair crosses once, and all other pairs cross zero times in that interval. This proves the asserted count.

Finally identify the braid with the standard generator model on each small crossing interval. The two crossing strands are separated from all other strands in real coordinate. Their difference travels through one open half-plane to the other with the positive orientation in (7.3.2). Contract its radial coordinate to a fixed positive value and its angular coordinate to a strictly increasing half-turn, keeping its endpoints fixed by short paths inside the initial and final real-order chambers. More explicitly, first straighten the endpoint configurations to distinct real rank positions while keeping their real order. On the small crossing interval the imaginary difference has one fixed nonzero sign. After adjoining the endpoint straightening paths, its argument runs from one real ray to the other through the prescribed upper or lower half-plane. Interpolate that lifted argument with the corresponding monotone half-turn and interpolate the positive modulus with a constant. Both interpolations avoid zero. The midpoint of the crossing pair is put at the midpoint of its two rank positions. All other strands can be kept outside the intervening real rank gap. This realizes the pair homotopy without a collision with a third strand. On a gap with no crossing, the ordered configuration chamber is convex, so a straight homotopy connects to the constant rank model. Choosing collars at interval endpoints makes these homotopies compatible. This constructs the claimed relative braid isotopy. □

7.4 Several tangent points and strict-transform sheets

Lemma 7.4.1 (New lemma: a common Puiseux parameter)**.** *Under the hypotheses of Theorem 7.1.2, let* $q = \operatorname{lcm}(a_1, \dots, a_b)$ *and* $\omega_a = e^{2\pi \mathrm{i}/a}$*. Index sheets by pairs* (j, l) *with* $0 \le l < a_j$*. The functions*

$$Z_{jl}(t) = h_j(\omega_{a_j}^l t^{q/a_j}), \qquad U_{jl}(t) = \frac{Z_{jl}(t)}{t^q} \tag{7.4.1}$$

extend holomorphically across zero and have pairwise distinct germs. Their set is preserved by monodromy, which acts by $(j, l) \mapsto (j, l + 1 \bmod a_j)$*. For fixed* j*, all constants* $U_{jl}(0)$ *are equal to the tangent slope* λ_j *of that branch.*

Proof. Write $h_j(s) = s^{a_j} v_j(s)$ with v_j holomorphic, using the assumed order bound. Since $(\omega_{a_j}^l)^{a_j} = 1$, $U_{jl}(t) = v_j(\omega_{a_j}^l t^{q/a_j})$, proving extension and $U_{jl}(0) = v_j(0) = \lambda_j$.

If two different root labels of the same branch gave equal Z on a punctured neighborhood, their parametrizations would give the same point with different nonzero parameter values, contrary to injectivity. For different branches, equality would contradict disjoint punctured images. Thus the differences are nonzero germs. Dividing by t^q preserves this property. Substitution of $e^{2\pi \mathrm{i}/q} t$ increments l by one in (7.4.1), which proves the monodromy assertion. □

Lemma 7.4.2 (New lemma: one projection separates all tangent clusters)**.** *For any finite set of distinct complex numbers* $\lambda_1, \dots, \lambda_r$*, there is a unit complex number* c *such that* $\operatorname{Re}(c(\lambda_i - \lambda_j)) \neq 0$ *for every distinct pair. For the functions in Lemma 7.4.1, sufficiently small circles have no projected crossings between sheets with distinct tangent slopes.*

Proof. For each nonzero difference $\lambda_i - \lambda_j$, the equation $\operatorname{Re}(c(\lambda_i - \lambda_j)) = 0$ excludes exactly two points of the unit circle. A finite set cannot exhaust that circle. Choose c outside it. The projected constants are then distinct. Continuity and finiteness give disjoint open real intervals containing, respectively, the projected values of all sheets in each tangent cluster on a common sufficiently small disc. Sheets from distinct clusters therefore never have the same projected real coordinate. □

These lemmas address several strict-transform points simultaneously. The points on the exceptional line are $(\lambda_j, 0)$. No hypothesis says that these points coincide. Within a cluster the differences have positive order, so Theorem 7.1.1 applies even when the strict transform is tangent to the exceptional line. Across clusters the differences have order zero and Lemma 7.4.2 excludes crossings. Positivity of a logarithmic derivative is not asserted for an order-zero difference.

7.5 The full turn is the native full-twist word

Let $\mathrm{Conf}_S(\mathbb{C})$ be the space of unordered sets of S distinct complex points. A closed braid is a smooth loop in this space. A collision-free homotopy of loops is a braid isotopy. When a regular projection is chosen, the rank ordering of the real parts identifies a positive crossing with the native adjacent generator at that rank.

Lemma 7.5.1 (New lemma: rigid rotation gives the prescribed word)**.** *Let $S \geq 1$ and let Q be a configuration of S distinct points. The full counterclockwise rotation $R_Q(t) = e^{2\pi \mathrm{i}t}Q$, $0 \leq t \leq 1$, represents d_S of* (7.1.1)*, using real-rank identifications at a regular basepoint. In particular, its chirality is positive, its word length is $S(S-1)$, and its native permutation is the identity.*

Proof. For $S = 1$ the assertion concerns the trivial one-strand braid. Assume $S \geq 2$. First place the points on the upper unit semicircle, including 1 and -1, with strictly increasing arguments $0 = \beta_0 < \cdots < \beta_{S-1} = \pi$. Choose a smooth strictly increasing lift $\phi : \mathbb{R} \to \mathbb{R}$ with $\phi(x + 2\pi) = \phi(x) + 2\pi$ and $\phi(2\pi j/S) = \beta_j$.

The cyclic-shift path

$$Q(t) = \left\{\exp\bigl(\mathrm{i}\phi(2\pi(j+t)/S)\bigr) : 0 \leq j < S\right\}, \qquad 0 \leq t \leq 1,$$

is a loop of unordered configurations. One point travels from -1 to 1 along the lower semicircle. Every other point moves along the upper semicircle to the next point. The lower point has strictly increasing real coordinate, and every upper point has strictly decreasing real coordinate. It therefore crosses each upper point exactly once, in order from the leftmost to the rightmost rank. Its imaginary coordinate is negative and the upper point's is positive at each crossing. Thus the difference has positive real derivative and negative imaginary part. Equation (7.3.2) gives a positive crossing. The word is exactly $\sigma_0\sigma_1 \cdots \sigma_{S-2} = c_S$.

Its Sfold iterate has angles $\phi(\alpha_j + 2\pi t)$ with $\alpha_j = 2\pi j/S$. The following is a based collision-free homotopy from that iterate to rigid rotation of the same initial configuration:

$$\left\{\exp\left(\mathrm{i}\bigl[(1-s)\phi(\alpha_j + 2\pi t) + s(\phi(\alpha_j) + 2\pi t)\bigr]\right) : 0 \leq j < S\right\}, \quad 0 \leq s, t \leq 1. \tag{7.5.1}$$

For $j < l$, each of the two argument differences being combined lies strictly between 0 and 2π. Their convex combination does too, so the points never collide. The endpoints at $t = 0, 1$ are fixed as unordered sets. This proves that rigid rotation at this configuration is c_S^S.

For an arbitrary configuration with distinct projected real coordinates, connect it to the semicircle configuration with the same real-rank labels by a straight homotopy. Every real-coordinate difference keeps its strict sign, so this path has no crossing. Rotating this path through 2π gives a two-parameter homotopy transporting the rotation loop to the one just computed. The transport has no generator letters and preserves the rank identification. A preliminary arbitrarily small change of projection handles a base configuration with a projected coincidence.

There are S copies of a word of length $S - 1$, proving the length formula. Under the native product convention, applying the adjacent swaps in c_S to each index gives the permutation

$$0 \mapsto 1 \mapsto \cdots \mapsto S - 1 \mapsto 0.$$

Its Sth power is the identity. This verifies the permutation assertion directly from `SignedAdjacentWord.permutation`. □

Lemma 7.5.2 (New lemma: separating rotation from a sheet loop)**.** *Let $U : [0,1] \to \mathrm{Conf}_S(\mathbb{C})$ be a smooth loop. The loop $t \mapsto e^{2\pi \mathrm{i}t}U(t)$ is isotopic through configuration loops to the concatenation of $R_{U(0)}$ followed by U.*

Proof. Consider $F(a, b) = e^{2\pi \mathrm{i}a}U(b)$ on the unit square. Every value is a configuration. The diagonal path from $(0, 0)$ to $(1, 1)$ is homotopic relative to its endpoints to the path along the bottom

and right sides. For example, interpolate linearly between (t,t) and $(\min(2t,1),\max(2t-1,0))$. Composing with F gives the desired homotopy. The bottom side is rotation of $U(0)$, and the right side is U because $e^{2\pi \mathrm{i}}=1$. Replace the piecewise linear parameter functions by smooth nondecreasing ones with endpoint collars. This preserves the square, the endpoints, and collision-freeness, and gives a smooth braid isotopy. □

Lemma 7.5.3 (New lemma: a braid isotopy has an ambient realization)**.** *Every smooth collision-free homotopy of finite-sheet closed braids whose images stay in a compact subset of the open solid torus extends to an ambient isotopy supported in that solid torus. The assertion allows any endpoint permutation and any number of components.*

Proof. Use the coordinates (θ,z) of $\mathbb{S}^1\times\mathbb{C}$ on a neighborhood of the compact union of the braids. Locally label the fibre points as $Z_j(s,\theta)$. Their minimum pairwise distance over the compact parameter set is positive. Choose a radius $a>0$ smaller than one third of that distance and smaller than the distance to the edge of the chosen fibre chart. Let χ be a smooth bump function equal to one near zero and zero outside $[0,1)$. Define a vertical vector field by

$$V_s(\theta,z)=\sum_j \chi\left(\frac{|z-Z_j(s,\theta)|^2}{a^2}\right)\partial_s Z_j(s,\theta).$$

For one strand there is no pairwise-distance restriction, and a radius smaller than the distance to the edge of the fibre chart suffices. The discs are disjoint, so this vector field agrees with the prescribed velocity on each strand. Relabeling permutes the summands. Thus it is smooth and periodic in θ, including around the seam of a multi-component closure. It has compact support. Its time-dependent flow exists for the whole compact s interval and takes each initial point along its prescribed strand, by uniqueness for the differential equation. Extending the vector field by zero gives the claimed ambient isotopy. The same construction works in the standard solid-torus chart $(\theta,z)\mapsto(\sqrt{1-|z|^2}e^{\mathrm{i}\theta},z)$ of $\mathbb{S}^3$. □

7.6 Proof of the factorization and the branch cycles

Proof of Theorem 7.1.2. Use the common parameter of Lemma 7.4.1 and choose c by Lemma 7.4.2. The strict-transform functions cU_{jl} satisfy Theorem 7.1.1. Their closed braid therefore has a positive event word w at a fixed sufficiently small reference radius. The same word works at every smaller positive radius in the common disc. Indeed, interpolate two such radii through positive radii. The sheet formulas in (7.4.1) give a smooth family of configurations at every angle, and their differences never vanish in the common punctured disc. This is a closed-braid isotopy from the braid at either radius to the one at the reference radius. Thus the chosen word is independent of the radius as a representative, even if a particular event decomposition changes. The construction includes every tangent cluster and has no crossings between different clusters.

On the circle $\xi=\eta e^{\mathrm{i}\theta}$ the original sheets satisfy

$$c\zeta_{jl}(\theta)=\eta e^{\mathrm{i}\theta}\,cu_{jl}(\theta).$$

Multiplication by the fixed positive number η is a configuration isotopy and creates no crossings. Lemma 7.5.2 splits the remaining loop into a rigid full rotation followed by the strict-transform loop. Lemma 7.5.1 identifies the first loop with the exact word d_S. The event resolution in Lemma 7.3.2 identifies the second with w. Thus the original root braid is isotopic to $d_S w$, with this literal prefix. All homotopies stay in configuration space. After a uniform fibre rescaling their images lie in a compact part of the prescribed solid torus, so Lemma 7.5.3 also realizes them as ambient braid isotopies.

Let τ be the permutation of sheet labels given by $\tau(j,l)=(j,l+1 \bmod a_j)$. Following ξ once counterclockwise has exactly this monodromy, by Lemma 7.4.1. In native real-rank coordinates,

let σ_- and σ_+ enumerate the labels before and after the period. The native convention is

$$\sigma_- \, w^{\mathrm{perm}} = \sigma_+, \qquad \tau\sigma_+ = \sigma_-.$$

The first identity is obtained by composing the adjacent swaps read in time order, as in the existing event-word declarations. The second follows by identifying the endpoint sheets. Solving gives

$$w^{\mathrm{perm}} = \sigma_-^{-1}\tau^{-1}\sigma_-. \tag{7.6.1}$$

This is the same inverse-conjugacy convention as `braid_permutation_eq_conjugate_inverse_rotation` in the branch case. The permutation τ is a disjoint union of b cycles of respective lengths a_j. Taking the inverse and conjugating preserves each orbit and its cardinality. Since `SignedAdjacentWord.cycleCount` counts these orbits, w has cycle count b. Lemma 7.5.1 gives $d_S^{\mathrm{perm}} = 1$, so $d_S w$ has the same permutation and cycle count. □

Corollary 7.6.1 (New lemma: the word-level conclusion M406)**.** *For the words constructed in Theorem 7.1.2,*

$$(d_S w)^{\mathrm{perm}} = w^{\mathrm{perm}}, \qquad \texttt{cycleCount}(d_S w) = \texttt{cycleCount}(w) = b.$$

A smooth branch transverse to the projection contributes one fixed point when its covering degree is one.

Proof. The first two assertions were proved in (7.6.1) and Lemma 7.5.1. A one-element block is a one-element orbit and is counted by the native `SameCycle` quotient, which includes fixed points. □

7.7 Application to every transverse reduced germ

Let f be a nonzero reduced germ of exact multiplicity $S \geq 1$ at the origin. Write its first nonzero homogeneous part as $f_S(\zeta, \xi)$. Transversality of the projection to ξ is the condition $f_S(1, 0) \neq 0$. The existing declaration `Germ.exists_linear_equiv_regularInFirstCoordinateOfOrder`, recorded as L430, chooses such coordinates for a germ of exact order S. Indeed, only the finitely many tangent directions given by the roots of f_S are forbidden as the kernel of the projection. This is a choice of base projection, distinct from the subsequent choice of the real fibre projection $\mathrm{Re}(c\zeta)$.

The following elementary order argument explains exactly where transversality enters.

Lemma 7.7.1 (New lemma: transverse parametrizations have bounded slope)**.** *Suppose f has multiplicity S, $f_S(1, 0) \neq 0$, and $f(h(s), s^a) = 0$ for a holomorphic h with $h(0) = 0$ and an integer $a > 0$. Then* $\operatorname{ord} h \geq a$.

Proof. The conclusion holds if $h = 0$. Otherwise put $r = \operatorname{ord} h$ and suppose $r < a$. For a monomial $\zeta^p \xi^q$ of f, its order after substitution is $pr + qa \geq r(p + q) \geq rS$. Equality holds only when $q = 0$ and $p = S$. The coefficient of ζ^S is $f_S(1, 0) \neq 0$. Thus the substituted series has a unique nonzero term of lowest order rS, which contradicts its vanishing. Hence $r \geq a$. □

Lemma 7.7.2 (New lemma: roots of a holomorphic function persist)**.** *Let $g(\zeta, \xi)$ be holomorphic near $(0, 0)$, and suppose $g(\zeta, 0)$ has finite positive order d at $\zeta = 0$. There are positive radii R, η such that, for every $|\xi| < \eta$, the function $\zeta \mapsto g(\zeta, \xi)$ has exactly d zeros counted with multiplicity in $|\zeta| < R$ and no zero on $|\zeta| = R$.*

Proof. Choose R so that $g(\zeta, 0)$ has no zero other than zero on $|\zeta| \leq R$. Its modulus has a positive minimum on the boundary. Uniform continuity gives η such that the straight interpolation from $g(\zeta, 0)$ to $g(\zeta, \xi)$ is nonzero on that boundary whenever $|\xi| < \eta$. Its boundary winding is therefore

constant and equals d. The local argument principle, proved as the new lemma Lemma 6.4.1, identifies that winding with the sum of the interior zero orders. Its proof factors the finitely many zeros and writes the remaining nonzero function as the exponential of a holomorphic logarithm. It applies on a neighborhood of the closed disc chosen here. This proves the assertion. □

Lemma 7.7.3 (New lemma: finite branch data from kernel parametrizations)**.** *Let $f_1, \dots, f_b$ be pairwise nonassociated irreducible pointed analytic germs in two variables, with $b \geq 1$, and let $f = u \prod_{j=1}^{b} f_j$ for an analytic unit u. For every j, let $a_j \geq 1$ and let $F_j(s) = (h_j(s), s^{a_j})$ be an injective analytic map on a disc about zero, with $F_j(0) = 0$ and pullback kernel exactly (f_j). Then there is a distinguished monic polynomial $Q_j(\zeta, \xi)$ of degree a_j in ζ, with coefficients holomorphic in ξ, such that $(Q_j) = (f_j)$. Moreover, after shrinking neighborhoods, the following conclusions hold:*

1. *The zero set of f is the union of the images of these parametrizations.*
2. *The images are pairwise disjoint away from the origin.*
3. *For each sufficiently small $\xi \neq 0$, the roots belonging to branch j are exactly $h_j(s)$ with $s^{a_j} = \xi$, each occurring once.*
4. *The order of $f(\zeta, 0)$ is $\sum_j a_j$.*

If f has multiplicity S and $f_S(1, 0) \neq 0$, then $\sum_j a_j = S$ and $\operatorname{ord} h_j \geq a_j$ for every j.

Proof. Fix j temporarily and write $a = a_j$, $h = h_j$. Form the product

$$Q(\zeta, s^a) = \prod_{l=0}^{a-1} (\zeta - h(\omega_a^l s)). \tag{7.7.1}$$

Its coefficients are convergent power series in s invariant under $s \mapsto \omega_a s$. If a coefficient is $\sum_{k \geq 0} b_k s^k$, this invariance implies $(\omega_a^k - 1) b_k = 0$. Thus only exponents divisible by a occur, and the coefficient is a convergent power series in $\xi = s^a$. The radius of convergence in ξ may be taken smaller than the ath power of a common radius in s. Every $h(0)$ is zero, so $Q(\zeta, 0) = \zeta^a$. Hence Q is distinguished monic of degree a.

The pullback of Q along F_j vanishes because one factor in (7.7.1) is zero. The kernel assumption gives $Q = f_j v$ for some analytic germ v. We claim that v is a unit. If it were not, then $v(0, 0) = 0$. Restriction of the product identity to $\xi = 0$ gives $\zeta^a = f_j(\zeta, 0) v(\zeta, 0)$, so $v(\zeta, 0)$ is a nonzero germ of positive finite order. Lemma 7.7.2 gives a zero ζ_ξ of $v(\cdot, \xi)$ for every sufficiently small nonzero ξ. Choose all representatives and the radii so that $Q = f_j v$ holds at these zeros and all parameters with $s^a = \xi$ lie in the injectivity disc.

At ζ_ξ the polynomial Q is zero. Formula (7.7.1) implies $\zeta_\xi = h(\omega_a^l s)$ for some s with $s^a = \xi$. Consequently $f_j(\zeta_\xi, \xi) = 0$, since f_j belongs to the pullback kernel. Both factors in $Q = f_j v$ therefore vanish at this point, which gives $\partial_\zeta Q(\zeta_\xi, \xi) = 0$ by the product rule. On the other hand, the a parameters $\omega_a^l s$ are distinct. Injectivity of F_j, together with their common projection coordinate, makes their h values distinct. Formula (7.7.1) therefore gives only simple roots. This is a contradiction. Hence v is a unit and $(Q) = (f_j)$.

Apply this construction to all branches. Absorbing the finitely many units gives $f = v_0 \prod_j Q_j$ for a unit v_0. At a small point of its zero set, some Q_j vanishes. If $\xi \neq 0$, its product formula identifies that point with $F_j(s)$ for a root $s^{a_j} = \xi$. If $\xi = 0$, the identity $Q_j(\zeta, 0) = \zeta^{a_j}$ forces $\zeta = 0$. Conversely, each parametrization lies on $f = 0$ because its pullback annihilates f_j. This proves the local zero-set coverage and the asserted fibre description.

For $j \neq k$, the germ $f_k \circ F_j$ is nonzero. Otherwise the kernel condition would imply $f_j \mid f_k$, and irreducibility would make them associated. A nonzero one-variable holomorphic germ has no punctured zero after shrinking its disc, by its factorization into a power and a unit. Choose a

common radius for the finitely many pairs. Then a nonzero point of the image of F_j cannot lie on $f_k = 0$, and therefore cannot lie in the image of F_k. This proves disjointness.

Finally, restriction of $f = v_0 \prod_j Q_j$ to $\xi = 0$ gives a unit times $\zeta^{\sum_j a_j}$. The order is exactly $\sum_j a_j$. If $f_S(1,0) \neq 0$, that same restriction has exact order S, proving $\sum_j a_j = S$. Each F_j annihilates f, so Lemma 7.7.1 proves $\operatorname{ord} h_j \geq a_j$. □

Corollary 7.7.4 (New lemma: the transverse germ under the precise per-branch normalization input)**.** *Let f be a reduced germ of exact multiplicity $S \geq 1$ with $f_S(1,0) \neq 0$. Assume, for each irreducible factor f_j of f, a pointed analytic map germ β_j with an injective representative and $\ker \beta_j^* = (f_j)$. Then the word-level conclusions of Theorem 7.1.2 hold for the root braid of f, with $b =$* `PlaneCurve.irreducibleBranchCount` *f.*

Proof. Choose pairwise nonassociated irreducible factors of the reduced germ. There is at least one factor because the positive multiplicity makes f a nonunit. The native finite factorization is recorded in the supplied Milnor–delta source by `associated_branchProduct_branchSet` and `irreducible_branchEquation_of_mem`. Its number of factors is the native irreducible branch count.

The pullback of ξ by each β_j is nonzero. Otherwise $\xi \in (f_j)$, so the irreducible germ f_j would be associated to the coordinate ξ. This would make ξ divide f, contrary to $f(\zeta, 0)$ having finite order S. Since β_j is pointed, its projection coordinate has finite positive order a_j. Apply the supplied `PointedMapGerm.exists_injective_power_representative_sheetRotation` to β_j, this coordinate, and this positive order. Its conclusions provide an injective representative with projection s^{a_j}. A source reparametrization by an analytic equivalence preserves the kernel ideal, since its pullback on the one-variable germ ring is an automorphism. Thus these representatives have exactly the hypotheses of Lemma 7.7.3. That lemma supplies coverage, disjointness, $\sum_j a_j = S$, and all order bounds. Theorem 7.1.2 and Corollary 7.6.1 now give the claimed word and the native branch-cycle count. □

Lemma 7.7.5 (New lemma: parametrizations and branch cycles from a finite root covering)**.** *Let $f \in \mathbb{C}\{\zeta, \xi\}$ be a reduced nonunit germ with $f(\zeta, 0)$ nonzero of order $S \geq 1$. Let $f_1, \ldots, f_b$ be pairwise nonassociated irreducible factors of f, each occurring once. There are positive integers $a_1, \ldots, a_b$ with $\sum_j a_j = S$ and convergent holomorphic functions h_j such that the following assertions hold after shrinking all neighborhoods.*

1. *The map $F_j(s) = (h_j(s), s^{a_j})$ is pointed and injective. Its pullback kernel is exactly (f_j).*
2. *The zero set of f is the union of these images, and their punctured images are pairwise disjoint.*
3. *Over each sufficiently small $\xi \neq 0$, the jth branch has exactly the a_j distinct sheets $h_j(s)$ with $s^{a_j} = \xi$. Positive continuation once around $\xi = 0$ permutes them in one a_j-cycle. Consequently the complete root permutation has exactly b cycles.*

If, in addition, f has multiplicity S and $f_S(1,0) \neq 0$, then $\operatorname{ord} h_j \geq a_j$ for every j, with the zero function allowed.

Proof. Lemma 6.4.2, with the coordinates named (ζ, ξ) here, writes

$$f = uP, \qquad P(\zeta, \xi) = \zeta^S + A_1(\xi)\zeta^{S-1} + \cdots + A_S(\xi), \qquad A_l(0) = 0,$$

where u is a unit. It proves also that (f, ξ) has finite colength. The existing declaration `PlaneCurve.finite_first_partial_polar_of_squarefree_axis_finite` therefore makes (f, f_ζ) a finite intersection. The product rule and invertibility of u give $(f, f_\zeta) = (P, P_\zeta)$, as proved in Lemma 6.5.1. The supplied declaration `Germ.resultant_ne_zero_of_hasFiniteIntersection` and the monic identity `Polynomial.resultant_deriv` show that the discriminant $D(\xi)$ of P

is not the zero germ. In the resultant declaration take the degree parameter $S-1$, since P is distinguished monic of positive degree S. Thus $D(\xi)=\xi^d v(\xi)$ with $v(0)\neq 0$ for some $d\geq 0$. There is a disc $|\xi|<\eta$ on which the roots are distinct whenever $\xi\neq 0$.

Each restriction $f_j(\zeta,0)$ is nonzero, because its product with the restrictions of the other factors and a unit is $f(\zeta,0)$. It has positive order a_j, since f_j is pointed. Orders in this one-variable product add, so $\sum_j a_j = S$. Apply the same proved preparation lemma to each factor to obtain $f_j = u_j Q_j$, where Q_j is distinguished monic of degree a_j. For each fixed small ξ, the zeros of P and of $\prod_j Q_j$ in the preparation disc agree with multiplicity. Both polynomials are monic of degree S, so

$$P=\prod_{j=1}^{b} Q_j. \tag{7.7.2}$$

The equality holds coefficientwise as an identity of holomorphic functions. Every Q_j is irreducible as an analytic germ, since it differs from f_j by a unit.

We give the continuation and extension arguments explicitly. A simple root of a holomorphic polynomial has a unique local holomorphic continuation. To prove this assertion near a root (ζ_0,ξ_0), subtract its nonzero ζ derivative from the equation and rewrite it as a fixed point equation for $\zeta-\zeta_0$. On a small closed polydisc its derivative in that variable has norm less than one. Iteration from zero converges uniformly on a smaller polydisc to a holomorphic fixed point, which is unique. All roots remain in a fixed disc, by the preparation construction. Around any punctured base point, the finitely many distinct roots have disjoint such neighborhoods, each mapping bijectively to a smaller base neighborhood. Thus they form a finite covering of the punctured base disc.

Continuation along a path is obtained by subdividing its compact parameter interval into these neighborhoods. Uniqueness on overlaps makes the result independent of the subdivision. Continuation along homotopic paths with fixed endpoints has the same endpoint. Indeed, subdivide the parameter square of a homotopy into sufficiently small rectangles whose images lie in covering neighborhoods. The unique local lift around each rectangle has the same values at its corners, and induction over the rectangles proves equality along the two boundary paths. A loop in a punctured disc is homotopic to the corresponding integer number of turns: divide it by its positive radius and use a continuous argument on its parameter interval. Interpolation of that argument with the linear argument of the same total increase supplies the homotopy. Consequently continuation is specified by one permutation π of the roots, namely continuation around one positive turn.

For each orbit E of π, form the monic polynomial

$$Q_E(\zeta,\xi)=\prod_{y\in E(\xi)}(\zeta-y).$$

Its coefficients are single-valued holomorphic functions on the punctured disc. Local holomorphy follows from the simple-root continuations, and orbit invariance makes the local definitions agree after any loop. They are bounded because all roots are bounded. A bounded holomorphic function on a punctured disc extends across zero. To verify the needed removable-singularity statement, its Laurent coefficient at degree $-k$, for $k\geq 1$, is $\frac{1}{2\pi\mathrm{i}}\int_{|s|=r} h(s)s^{k-1}\,ds$. If $|h|\leq M$, its modulus is at most Mr^k, and the coefficient is independent of r. Letting r decrease to zero makes it zero. The Laurent series therefore has no negative terms.

All roots tend to zero with ξ. If this were false, a sequence of bounded roots over base points tending to zero would have a subsequence converging to a nonzero root of $P(\zeta,0)=\zeta^S$. This is impossible. Hence the extended polynomial Q_E is distinguished, and every orbit of positive size contributes a nonunit analytic factor of P.

The roots of each Q_j form a union of π-orbits. In fact, a continuation of a root of Q_j remains a root of Q_j locally, and uniqueness continues this property along each path. If it contained two or more orbits, the preceding construction would factor Q_j into two distinguished polynomials of positive degree. Their equality with Q_j follows on the punctured disc by comparing their roots

and leading coefficients, then at zero by continuity. Both factors are nonunits, contradicting irreducibility of Q_j. Therefore its a_j roots form precisely one orbit. Different j give disjoint orbits by (7.7.2) and the nonvanishing discriminant. This proves the asserted orbit count without assuming a normalization theorem.

Fix one root in the jth orbit. Pull the punctured covering back by $\xi = s^{a_j}$. A positive turn of s induces a_j turns of ξ, whose permutation is the identity on this orbit. The continuation argument above therefore gives a single-valued holomorphic root $h_j(s)$ on the punctured s disc. It is bounded, so the same Laurent argument extends it holomorphically across zero, with $h_j(0) = 0$. Replacing s by $e^{2\pi\mathrm{i}/a_j}s$ continues that root once around $\xi = 0$. Thus the values $h_j(\omega_{a_j}^l s)$, for $0 \le l < a_j$, are all the distinct roots of Q_j over s^{a_j}. It follows that F_j is injective on its punctured disc. No punctured point maps to the origin because its second coordinate is nonzero, so it is injective on the whole disc.

Equation (7.7.2) now proves local coverage and punctured disjointness. Indeed a zero over $\xi \neq 0$ is one of the listed roots, and a zero over $\xi = 0$ has $\zeta = 0$. Distinct factors have no common root over the punctured disc. The parametrizations annihilate their respective factors.

We also verify the exact pullback kernel. Let g be a holomorphic germ whose pullback by F_j is zero. Shrink a product neighborhood so that all representatives are defined and choose a circle $|v| = R$ surrounding all roots of Q_j with no root on the circle. For $|\zeta| < R$, define

$$H(\zeta,\xi) = \frac{1}{2\pi\mathrm{i}} \int_{|v|=R} \frac{g(v,\xi)}{Q_j(v,\xi)(v-\zeta)}\, dv$$

and

$$R_j(\zeta,\xi) = \frac{1}{2\pi\mathrm{i}} \int_{|v|=R} \frac{g(v,\xi)\big(Q_j(v,\xi) - Q_j(\zeta,\xi)\big)}{Q_j(v,\xi)(v-\zeta)}\, dv.$$

The contour denominators are nonzero, so the functions are jointly holomorphic by differentiation under the compact integral. The quotient $(Q_j(v,\xi) - Q_j(\zeta,\xi))/(v-\zeta)$ is a polynomial in ζ of degree at most $a_j - 1$. Thus R_j has that degree bound. The power-series form of Cauchy's formula, proved in the preparation lemma of J15, gives $g = Q_j H + R_j$. For each punctured ξ, the hypothesis on $g \circ F_j$ says that g vanishes at all a_j roots of Q_j. So does R_j. A polynomial of degree at most $a_j - 1$ with a_j distinct roots is zero. Hence all coefficients of R_j vanish on the punctured disc and then at zero. It follows that $g \in (Q_j) = (f_j)$. The reverse inclusion was already proved by the root equation. This establishes the kernel assertion.

Under the final transversality hypotheses, each F_j annihilates f. Lemma 7.7.1 therefore gives $\operatorname{ord} h_j \ge a_j$. □

Theorem 7.7.6 (New lemma: full J14 word statement for a transverse reduced germ)**.** *Let $f \in \mathbb{C}\{\zeta,\xi\}$ be reduced of exact multiplicity $S \ge 1$, and suppose $f_S(1,0) \neq 0$. Let b denote the number* `PlaneCurve.irreducibleBranchCount` *f. For all sufficiently small positive base radii, the closed root braid of $f = 0$ over the positively oriented ξ circle is represented, through closed braids, by*

$$\texttt{fullTwistWord}(S)\, w, \qquad w \in \texttt{PositiveAdjacentWord}(S).$$

One may choose w as the positive event word of the strict-transform sheets ζ/ξ, with one fibre projection chosen for all tangent clusters. The native permutations of w and of the whole word each have exactly b cycles. The cycle belonging to f_j has length $i_0(f_j,\xi)$.

Proof. The native finite reduced factorization is indexed by the irreducible branch set, by `asso ciated_branchProduct_branchSet` and `irreducible_branchEquation_of_mem`, used in the supplied Milnor–delta source. Its number of factors is the native branch count. Transversality makes $f(\zeta,0)$ have exact order S. Lemma 7.7.5 supplies precisely the injective, punctured-disjoint finite branch data of Theorem 7.1.2, with $\sum_j a_j = S$ and $\operatorname{ord} h_j \ge a_j$. Apply that theorem and Corollary 7.6.1. They give the positive strict-transform word, the full-twist prefix, the identification through the same closed braid, and the stated native cycle counts. The preparation calculation

for f_j gives $\mathbb{C}\{\zeta,\xi\}/(f_j,\xi) \cong \mathbb{C}\{\zeta\}/(\zeta^{a_j})$, whose dimension is a_j. Consequently the cycle length equals $i_0(f_j,\xi)$. □

Theorem 7.7.6 has no extra normalization input. It proves the user's transverse, multiplicity-S word statement for every reduced germ with those hypotheses. The earlier Corollary 7.7.4 remains a useful direct application of an already supplied parametrization. Its kernel and injectivity hypotheses are now constructed in Lemma 7.7.5, so the omitted return type of `exists_intersectionParametrization_of_irreducible` is not needed for this conclusion. The counterexample to the uncorrected M405 API remains unchanged.

For the smooth sphere conclusion, assume additionally M403(b): the normalized small sphere link is smoothly ambiently isotopic, with its component orientations, to this same root braid in the native solid-torus chart. Assume M419(B) realizes the given native word by the standard collared generator sheets used in Lemma 7.3.2. Composing that sphere isotopy with the ambient isotopies of Lemma 7.5.3 proves the corrected sphere conclusion for $d_S w$. The identities specifying `planeSolidTorusMap0` must be checked in those inputs. No identification of its index with the chosen base coordinate is silently assumed here. Those two geometric obligations remain explicit hypotheses and are not replaced by an assertion that an arbitrary path homotopy is an ambient diffeotopy.

7.8 Exceptional cases and source ledger

If $S = 1$ with transverse projection, the only sheet has no pairwise crossing, both words d_1, w are empty, and the permutation has one fixed point. If $S = 2$ and the curve is an ordinary node with distinct slopes, the strict-transform word is empty and the original word is $d_2 = \sigma_0^2$. For the transverse cusp $\zeta^2 = \xi^3$, the strict-transform sheets are $\pm\xi^{1/2}$, so $w = \sigma_0$ and the original word is σ_0^3. Thus the sign agrees with the native positive convention already in the smallest singular examples.

For an ordinary S-fold point, every branch is smooth with degree one and all slopes are distinct. Lemma 7.4.2 removes every strict-transform crossing, so w is empty and $b = S$. Several branches with a common tangent require no additional argument: their strict-transform differences have positive order in the common parameter. Tangency of a strict-transform branch to the exceptional line does not affect Lemma 7.3.1. Events with three or more sheets are treated by Lemma 7.3.2, which counts each pair once. Choosing a regular initial angle prevents an event from lying at the seam of the closed period.

The assertions involve complex analytic functions, integer covering degrees, and native finite words. No Floer chain coefficients or homology theory are introduced. In particular, no holomorphic Heegaard Floer result or fibre-surface argument is a proof step.

Declaration or statement	Status	Exact role
`PositiveAdjacentWord`, `PositiveAdjacentWord.toSigned`	Existing	Finite lists and positive signs.
`adjacentSwap`, `SignedAdjacentWord.permutation`, `SignedAdjacentWord.cycleCount`	Existing	Native permutation multiplication and orbit count.
`tendsto_mul_deriv_div_of_analytic_factor`	Existing	The analytic logarithmic-derivative identity behind Lemma 7.3.1.
`exists_radius_planeBranchAngularDifference_projected_crossings_pos`	Existing, recorded	Native crossing-sign convention, not a multibranch theorem.
Lemma 7.3.1, Theorem 7.1.1	New	Uniform finite-pair estimates, order-zero pairs, and positivity.
Lemma 7.3.2	New	Multiple-event resolution to positive adjacent words.

Declaration or statement	Status	Exact role
Lemmas 7.4.1 and 7.4.2	New	Common ramification, several tangent points, and choice of real projection.
Lemma 7.5.1	New	Literal word d_S, length $S(S-1)$, positive chirality, identity permutation.
Lemmas 7.5.2 and 7.5.3	New	Explicit braid homotopy and its ambient realization.
`braid_permutation_eq_conjugate_inverse_rotation`	Existing	Verification of the inverse-conjugacy convention used in (7.6.1).
Theorem 7.1.2, Corollary 7.6.1	New	Full twist and exact branch cycles for the displayed finite branch data.
`Germ.exists_linear_equiv_regularInFirstCoordinateOfOrder`	Existing, recorded	Choice of transverse coordinates for a germ of exact order S.
Lemma 7.7.1	New	Transversality makes division by ξ holomorphic on the common cover.
Lemma 6.4.1	New, proved in J15	The local argument principle used in Lemma 7.7.2.
Lemmas 7.7.2 and 7.7.3	New	Kernel ideals give the branch polynomials, full local coverage, punctured disjointness, and exact degree sum.
`associated_branchProduct_branchSet`, `irreducible_branchEquation_of_mem`	Existing, used in supplied source	Finite reduced factorization indexed by native branches.
`PointedMapGerm.exists_injective_power_representative_sheetRotation`	Existing	Power normalization preserves injectivity and the kernel after source equivalence.
`exists_intersectionParametrization_of_irreducible`	Existing, recorded	Optional alternate source of parametrizations. It is not needed for Theorem 7.7.6.
Corollary 7.7.4	New	Direct application when a kernel parametrization is already supplied.
Lemma 7.7.5	New	Constructs all parametrizations from the finite root covering, proves their exact kernels, and identifies monodromy orbits with irreducible factors.
Theorem 7.7.6	New	The complete transverse reduced-germ word statement, with no extra normalization assumption.
`PlaneCurve.finite_first_partial_polar_of_squarefree_axis_finite`, `Germ.resultant_ne_zero_of_hasFiniteIntersection`, `Polynomial.resultant_deriv`	Existing	The prepared polynomial has nonzero discriminant.
Preparation and unit lemmas of `J15-braid-length.tex`	New, proved in J15	Produce distinguished polynomials for the equation and its factors, and identify their polar pair ideals.
M403(b), M419(B)	Unproved input	Required only for the stated conditional native sphere conclusion.
Lemma 7.2.1, Proposition 7.2.2	New	Counterexample to the uncorrected exact API, including its sphere formulation.

The supplied reference by Fomin, Pylyavskyy, Shustin, and Thurston, *Morsifications and mutations*, Lemma 13.5, motivates the decomposition by a full twist. Its one-sentence proof

is not used as a replacement for Lemmas 7.4.1–7.5.2. The source conventions and acceptance statement are those of the attached `combined_01-10.md`, J14 and records M405, M406, M1153, together with the supplied native word and analytic listings. No Lean code is written. No finite computation is used as a proof.

References

[1] P. Ozsváth, A. I. Stipsicz, and Z. Szabó, *Grid Homology for Knots and Links*, preliminary version supplied with the request. Definitions 3.1.6 and 3.1.7 and Section 3.2 are used only for the elementary grid-move conventions.

[2] H. R. Morton, *The multivariable Alexander polynomial for a closed braid*, in H. Nencka, editor, *Low-dimensional topology*, Contemporary Mathematics 233, American Mathematical Society (1999), 167–172. Author's manuscript: `https://www.liverpool.ac.uk/~su14/papers/multiburau.pdf`. Accessed 1 November 2026. The Introduction records the classical determinant formula. Lemma 6.8.2 proves its precise coefficient-field version used here.